

Lecture 5 - Experimental Design, Cross-validation Schemes & The Curse of Dimensionality

Strategies to Mitigate Overfitting

1. Add more data! - when possible
 2. Occam's Razor - select a model with lower complexity
 3. Regularization - constrains/regularizes the coefficients of the model
 4. Cross-validation - technique for utilizing the training data to fine-tune hyperparameter values
-

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
plt.style.use('bmh')
```

```
In [2]: def NoisySinusoidalData(N, a, b, sigma):
    '''Generates N data points in the range [a,b) sampled from a sin(2*pi*x)
    with additive zero-mean Gaussian random noise with standard deviation si

    # Fixing random seed for reproducibility
    np.random.seed(42)

    # N input samples, evenly spaced numbers between [a,b) incrementing by 1
    x = np.linspace(a,b,N)

    # draw N sampled from a univariate Gaussian distribution with mean 0, si
    noise = np.random.normal(0,sigma,N,)

    # desired values, noisy sinusoidal
    t = np.sin(2*np.pi*x) + noise

    return x, t
```

```
In [3]: from sklearn.preprocessing import StandardScaler

# Generate input samples and desired values
N_train = 70 # number of data samples for training
```

```

N_test = 20 # number of data samples for test

a, b = [0,1] # data samples interval

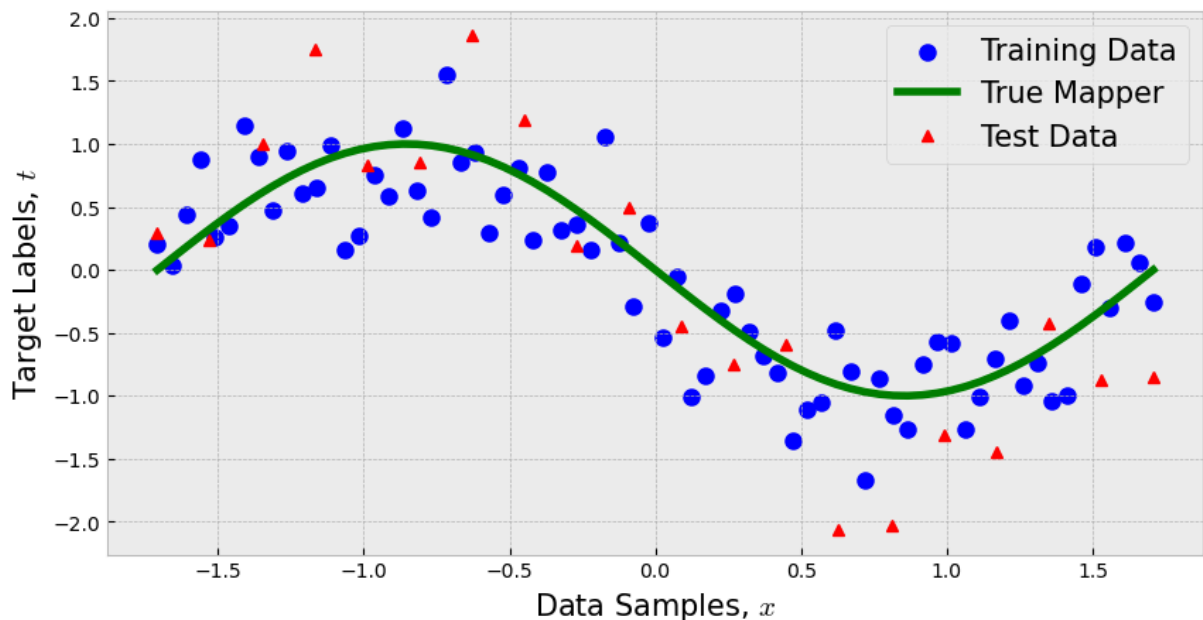
sigma_train = 0.4 # standard deviation of the zero-mean Gaussian noise -- tr
sigma_test = 0.6 # standard deviation of the zero-mean Gaussian noise -- tes

x_train, t_train = NoisySinusoidalData(N_train, a, b, sigma_train) # Trainin
x_true, t_true = NoisySinusoidalData(N_train, a, b, 0) # True Sinusoidal - i
x_test, t_test = NoisySinusoidalData(N_test, a, b, sigma_test) # Test Data -

scaler = StandardScaler()
x_train = scaler.fit_transform(x_train[:,np.newaxis]).ravel()
x_test = scaler.transform(x_test[:,np.newaxis]).ravel()
x_true = scaler.transform(x_true[:,np.newaxis]).ravel()

plt.figure(figsize=(10,5))
plt.scatter(x_train, t_train, c='b', linewidths=3, label = 'Training Data')
plt.plot(x_true, t_true, 'g', linewidth=4, label = 'True Mapper')
plt.plot(x_test, t_test, 'r^', label = 'Test Data')
plt.legend(fontsize=15)
plt.xlabel('Data Samples, $x$',size=15)
plt.ylabel('Target Labels, $t$',size=15);

```



```

In [4]: def PolynomialRegression(x,t,M):
'''Fit a polynomial of order M to the data input data x and desire value
MSE objective function.'''

# Compute feature matrix X with polynomial features
X = np.array([x**i for i in range(M+1)]).T # this creates an N-by-(M+1)

# Compute the solution for the parameters w, using the MSE ojective func
w = np.linalg.inv(X.T@X)@X.T@t

# Compute model prediction
y = X@w

```

```

    return w, y

def PolynomialRegression_test(x_test, w):
    '''Use a trained linear regression model with polynomial features for inference
    # Compute test feature matrix
    X_test = np.array([x_test**i for i in range(len(w))]).T # this is a N_test by M matrix

    # Prediction for test samples using the trained model
    y_test = X_test@w

    return y_test

def PolynomialRegression_reg(x,t,M,lam):
    '''Fit a polynomial of order M to the data input data x and desired value t using the
    MSE objective function with the Ridge regularizer.'''

    # Compute feature matrix X with polynomial features
    X = np.array([x**i for i in range(M+1)]).T # this creates an N-by-(M+1) matrix

    # Compute the solution for the parameters w, using the MSE objective function
    w = np.linalg.inv(X.T@X + lam*np.eye(M+1))@X.T@t

    # Compute model prediction
    y = X@w

    return w, y

```

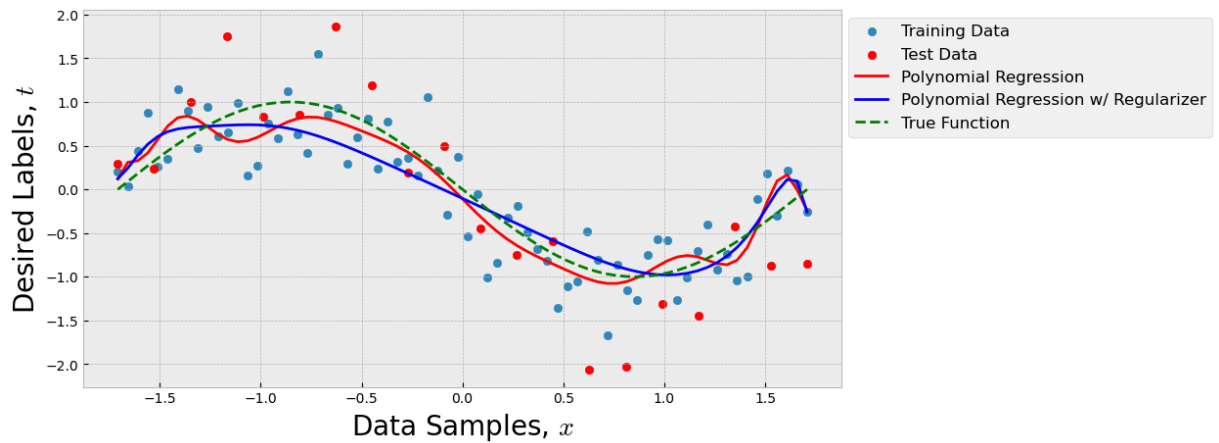
```

In [5]: M = 15
        lam = 2

w, y = PolynomialRegression(x_train,t_train,M)
w_reg, y_reg = PolynomialRegression_reg(x_train,t_train,M,lam)

fig=plt.figure(figsize=(10,5))
plt.scatter(x_train,t_train, label='Training Data')
plt.scatter(x_test,t_test, c='r', label='Test Data')
plt.plot(x_train,y,'r', label = 'Polynomial Regression')
plt.plot(x_train,y_reg, 'b',label = 'Polynomial Regression w/ Regularizer')
plt.plot(x_true,t_true,'--g', label = 'True Function')
plt.legend(bbox_to_anchor=(1.5, 1),fontsize=12,ncol=1)
plt.xlabel('Data Samples, $x$', fontsize=20)
plt.ylabel('Desired Labels, $t$', fontsize=20);

```



Performance Measures for Regression Tasks

In order to determine if the model is able to **generalize** to a **validation set**, we need to determine a **performance measure**.

Which measures would you use to compare model performances?

1. Error measures are always a good start for **regression** tasks. Some examples include:

- Mean Squared Error (MSE) - best when the data does not have outliers. The MSE will penalize outliers heavily.
- Mean Absolute Error (MAE) - best when you suspect data has outliers.
- Normalized mean squared error (NMSE).
- Normalized mean absolute error (NMAE).

2. Coefficient of Determination r^2

- Advantage: it is bounded between $[0,1]$, where values closer to 1 indicate a better prediction.

3. Quantile-Quantile plot (or Q-Q plot)

- Qualitative measurement of the model predictions vs the target values.
- The Q-Q plot is used to answer the following questions:
 - Do two data sets come from populations with a common distribution?
 - Do two data sets have common location and scale?
 - Do two data sets have similar distributional shapes?
 - Do two data sets have similar tail behavior?

```
In [6]: ## Predictions in training/test sets with and without regularization
y = PolynomialRegression_test(x_train, w)
```

```

y_test = PolynomialRegression_test(x_test, w)

y_reg = PolynomialRegression_test(x_train, w_reg)
y_test_reg = PolynomialRegression_test(x_test, w_reg)

## Error Measures
# Residual error for Training data for polynomial regression without and with regularization
error_train = t_train - y
error_train_reg = t_train - y_reg

# Residual error for Test data for polynomial regression without and with regularization
error_test = t_test - y_test
error_test_reg = t_test - y_test_reg

# Error Measures
print('Mean Squared Error \n')
print('Training Set')
print('Without regularizer: ', np.mean(error_train**2))
print('With regularizer: ', np.mean(error_train_reg**2), '\n')
print('Test Set')
print('Without regularizer: ', np.mean(error_test**2))
print('With regularizer: ', np.mean(error_test_reg**2), '\n')
print('-----')
print('Mean Absolute Error \n')
print('Training Set')
print('Without regularizer: ', np.mean(np.abs(error_train)))
print('With regularizer: ', np.mean(np.abs(error_train_reg)), '\n')
print('Test Set')
print('Without regularizer: ', np.mean(np.abs(error_test)))
print('With regularizer: ', np.mean(np.abs(error_test_reg)), '\n')

```

Mean Squared Error

Training Set

Without regularizer: 0.09927043634520141

With regularizer: 0.12894251548077149

Test Set

Without regularizer: 0.36282431622746925

With regularizer: 0.423004972462368

Mean Absolute Error

Training Set

Without regularizer: 0.263595936675582

With regularizer: 0.2839971490402194

Test Set

Without regularizer: 0.48984370556677626

With regularizer: 0.5113613777430893

```

In [7]: ## R2-SCORE - coefficient of determination

# it measures how much variance in one variance is explained by another variable

```

```

from sklearn.metrics import r2_score

print('Test set without regularization:', r2_score(t_test, y_test))

print('Test set with regularization:', r2_score(t_test, y_test_reg))

```

Test set without regularization: 0.7161038177697114

Test set with regularization: 0.6690147507335047

The coefficient of determination r^2 is bounded between $[0,1]$. The closer to 1, the closer the predictions are to the target value.

In [8]: *## QUANTILES*

```

np.quantile(y, 0.5), np.median(y) #, np.mean(y)

# median

```

Out[8]: (0.08015506559222224, 0.08015506559222224)

In [9]: np.quantile(y, 0), np.min(y)

```

# minimum

```

Out[9]: (-1.076338966299511, -1.076338966299511)

In [10]: np.quantile(y, 1), np.max(y)

```

# maximum

```

Out[10]: (0.8394412929669056, 0.8394412929669056)

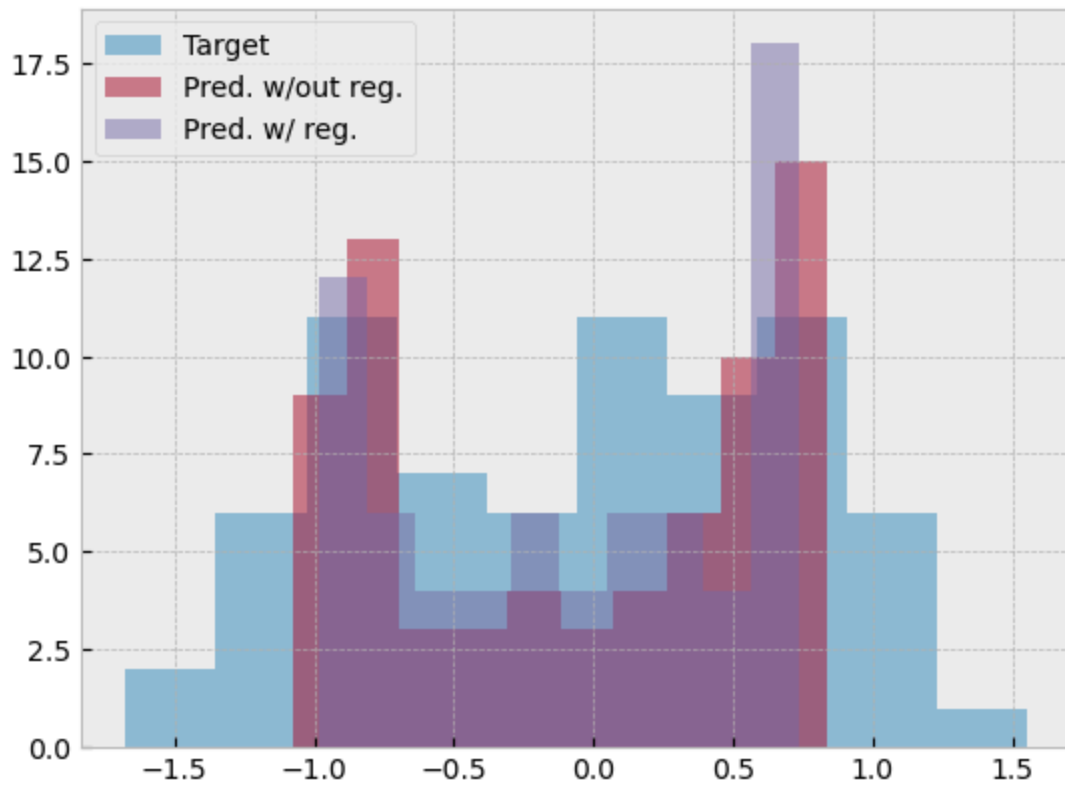
In [11]: np.linspace(0,1,10)

Out[11]: array([0. , 0.11111111, 0.22222222, 0.33333333, 0.44444444,
0.55555556, 0.66666667, 0.77777778, 0.88888889, 1.])

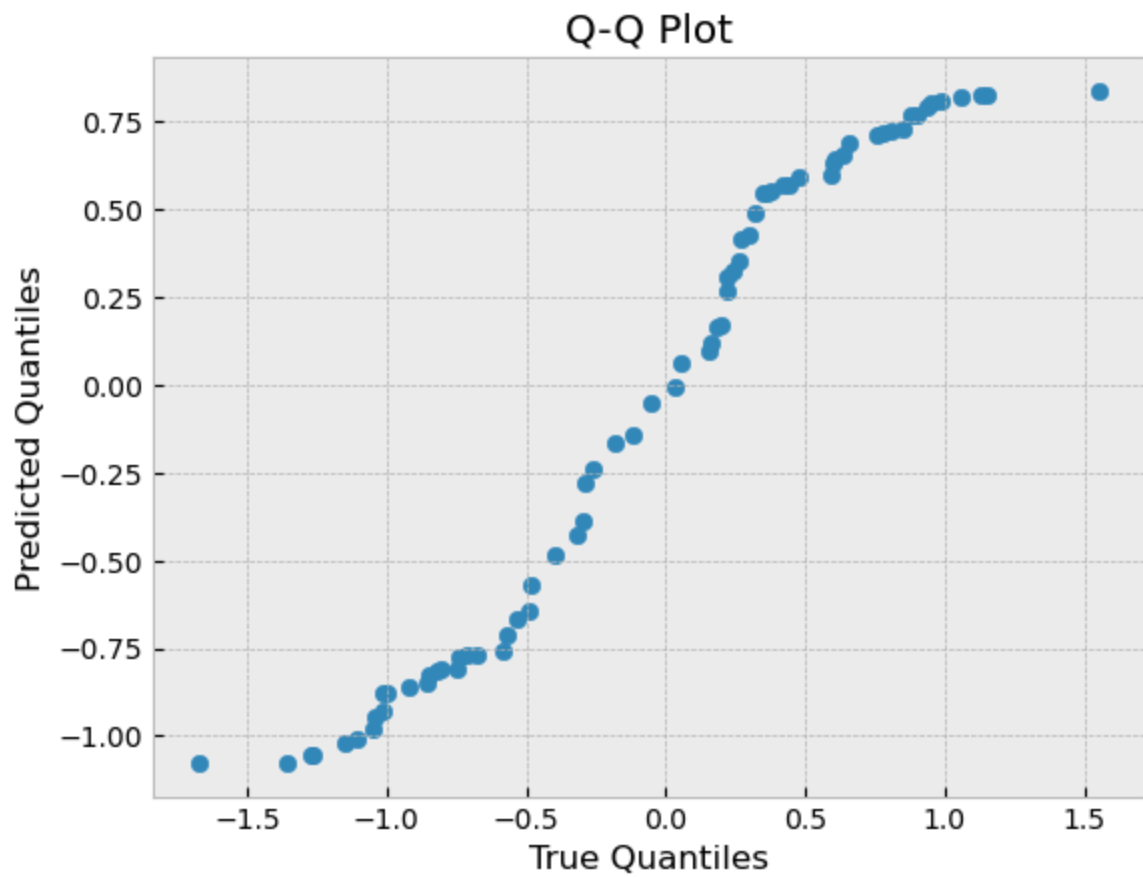
In [12]: np.quantile(y, np.linspace(0,1,10))

Out[12]: array([-1.07633897, -0.93538563, -0.80969823, -0.64607063, -0.15132625,
0.20331689, 0.54479529, 0.63884559, 0.77173963, 0.83944129])

In [13]: plt.hist(t_train, alpha=0.5, label='Target')
plt.hist(y, alpha=0.5, label='Pred. w/out reg.')
plt.hist(y_reg, alpha=0.5, label='Pred. w/ reg.')
plt.legend();

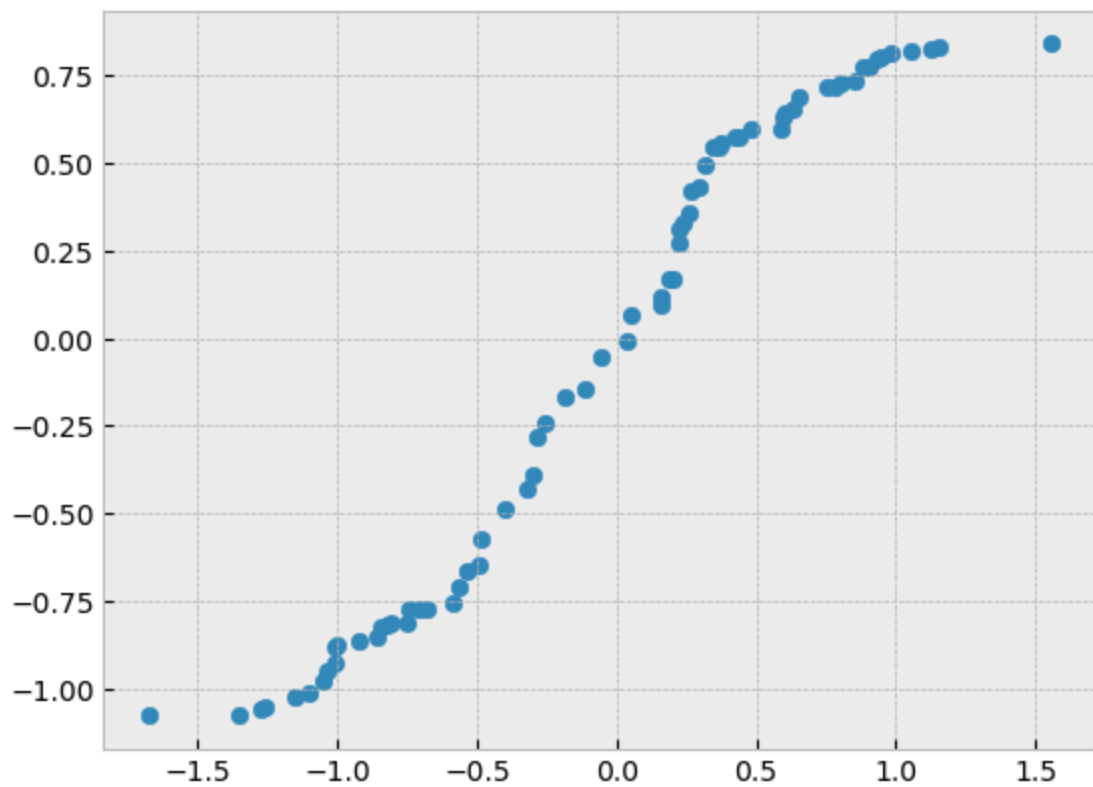


```
In [14]: plt.scatter(np.quantile(t_train, np.linspace(0,1,len(t_train))),
                    np.quantile(y, np.linspace(0,1,len(t_train))))
plt.xlabel('True Quantiles')
plt.ylabel('Predicted Quantiles')
plt.title('Q-Q Plot');
```

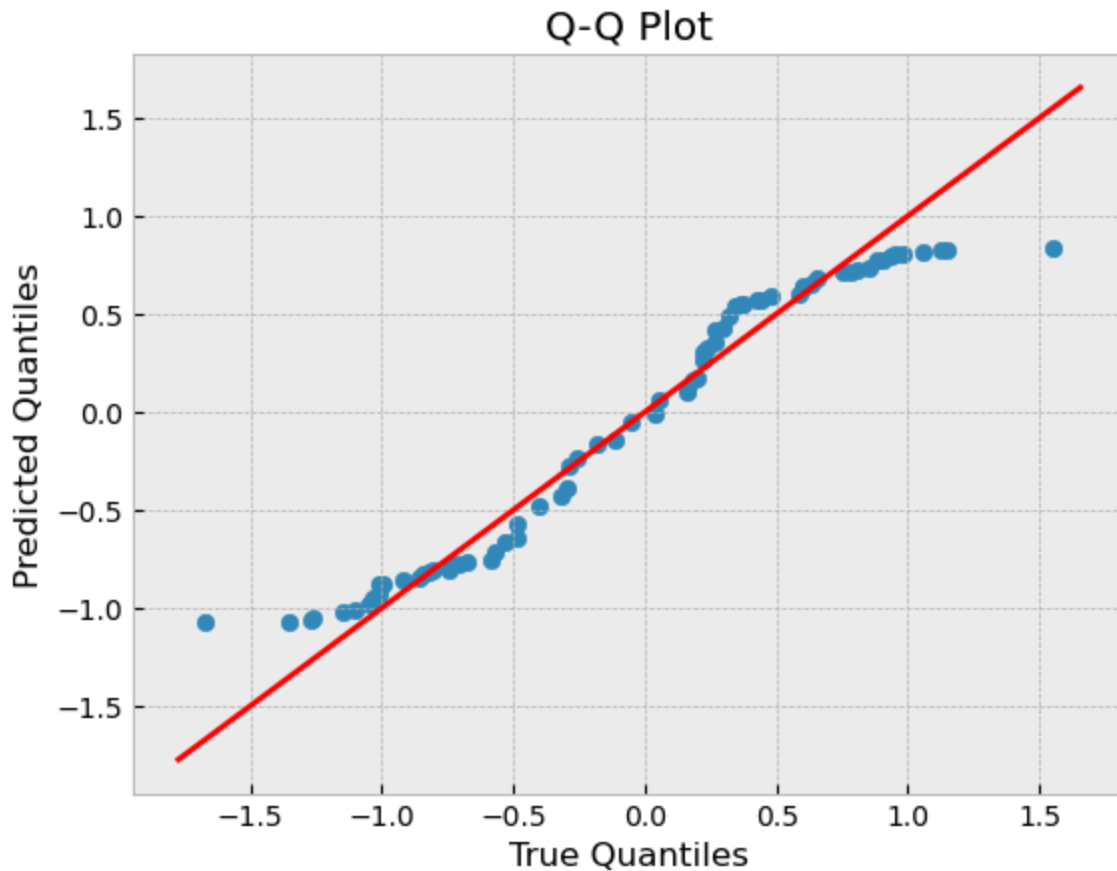


In [15]: *# alternative way to plot the Q-Q plot*

```
plt.scatter(np.sort(t_train), np.sort(y));
```




```
In [16]: base = np.linspace(min(t_train)-0.1,max(t_train)+0.1,100)
plt.plot(base,base,'r')
plt.scatter(np.sort(t_train), np.sort(y))
plt.xlabel('True Quantiles')
plt.ylabel('Predicted Quantiles')
plt.title('Q-Q Plot');
```



```
In [17]: print('R2 Score for Training without regularizer:', r2_score(np.sort(t_train), np.sort(y_train)))
print('R2 Score for Test without regularizer:', r2_score(np.sort(t_test), np.sort(y_test)))
```

R2 Score for Training without regularizer: 0.951826963889152
R2 Score for Test without regularizer: 0.7939864348423977

```
In [18]: print('R2 Score for Training with regularizer:', r2_score(np.sort(t_train), np.sort(y_train)))
print('R2 Score for Test with regularizer:', r2_score(np.sort(t_test), np.sort(y_test)))
```

R2 Score for Training with regularizer: 0.9365956093159628
R2 Score for Test with regularizer: 0.7512121340203461

In practice, we consider a good fit if the r^2 of the Q-Q plot is **greater than 0.9**, ideally >0.99 .

Other Performance Measures

Performance of an algorithm can be determined using a variety of statistical goodness-of-fit measures.

- For regression tasks this includes error-based measurements, hypothesis tests, Q-Q plots.
- For classification tasks this includes error rate, accuracy, ROC curves, performance-recall curves.

But it can also be in terms of:

1. Risk,
 2. Training time,
 3. Training storage/memory,
 4. Testing time,
 5. Testing storage/memory,
 6. Interpretability, namely, whether the method allows knowledge extraction which can be checked and validated by experts, and
 7. computational complexity.
-

Strategies to Mitigate Overfitting

1. Add more data! - when possible
 2. Occam's Razor - select a model with lower complexity
 3. Regularization - constrains/regularizes the coefficients of the model
 4. Cross-validation - technique for utilizing the training data to fine-tune hyperparameter values
-

Cross-Validation

The goal of **cross-validation**, or CV, is to test the model's ability to predict new data that was not used in estimating the model, in order to flag problems like overfitting or selection bias and to give an insight on how the model will generalize to an independent dataset (i.e., an unknown dataset).

Cross-validation allows us to determine the values for the hyperparameter that provide the most **generalization**. Cross-validation mitigates the occurrence of overfitting, but it

does not eliminate it entirely.

Suggested reading:

- [Sections 19.1-19.6](#) from the Alpaydin textbook
- [Section 5.3 Hyperparameters and Model Validation](#) from the "Python Data Science Handbook" textbook.

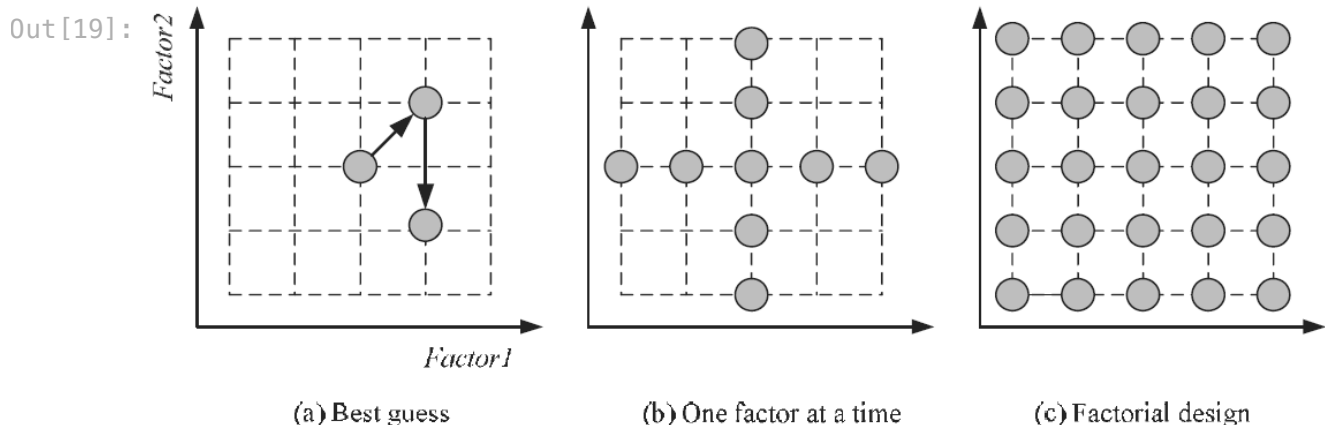
Strategies of Experimentation - Fine-Tuning Hyperparameters

To fully test a system, you want to try to evaluate each of these parameters separately. However, this is often not easily done.

A few strategies for fine-tuning hyperparameters include:

- Best guess
- One factor at a time
- Factorial design or **grid search**
- Randomized search

```
In [19]: from IPython.display import Image
Image('figures/Experimentation.png',width=800)
# Source: Alpaydin, E. (2014) Introduction to Machine Learning. 3rd edition,
```



We first must select a **performance measure**, e.g. mean square error (MSE), coefficient of determination R^2 . There are different strategies for implementing cross-validation:

1. k -times 2 cross-validation
2. k -fold cross-validation
3. Leave-One-Out cross-validation
4. Stratified cross-validation

We also need to make sure that classes are represented in the right proportions when subsets of data are held out, not to disturb the class prior probabilities; this is called

stratification. If a class has 20 percent examples in the whole dataset, in all samples drawn from the dataset, it should also have approximately 20 percent examples.

10-fold cross-validation

Let's consider the case, where we only have a single validation set which resulted from splitting the training data into a training set and a validation set with splitting ratios 70% and 30% respectively.

```
In [20]: from sklearn.model_selection import train_test_split

x_train2, x_validation, t_train2, t_validation = train_test_split(x_train, t_train,
                                                                test_size=0.3,
                                                                shuffle=True,
                                                                random_state=42)

x_train2.shape, x_validation.shape, t_train2.shape, t_validation.shape
```

```
Out[20]: ((49,), (21,), (49,), (21,))
```

Let's now vary the values for M and estimate the associated coefficients $\mathbf{w}$ using the training set. Then, let's evaluate performance measure (e.g. MSE) in both training and validation sets:

```
In [21]: # Range of values for M – Polynomial Regression WITHOUT regularizer
M_vals = range(1,11)

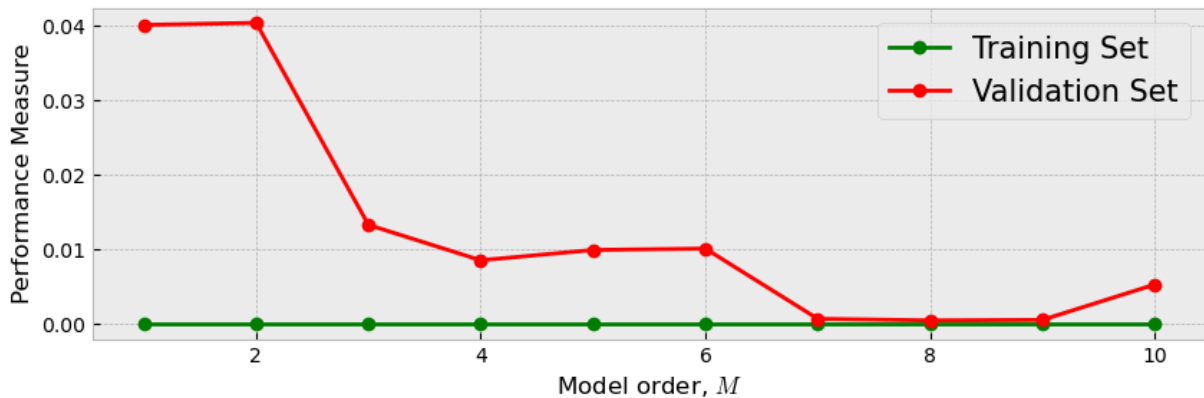
perf_train = []
perf_val = []
for M in M_vals:

    # Train the model (without regularization for simplicity of illustration)
    w, y_train = PolynomialRegression(x_train2, t_train2, M)

    # Make predictions for the training and validation sets
    y_validation = PolynomialRegression_test(x_validation, w)

    # Performance Measure – MSE
    perf_train += [np.mean(t_train2-y_train)**2]
    perf_val += [np.mean(t_validation-y_validation)**2]

# Plotting results
plt.figure(figsize=(10,3))
plt.plot(M_vals, perf_train, '-og', label='Training Set')
plt.plot(M_vals, perf_val, '-or', label='Validation Set')
plt.legend(fontsize=15)
plt.xlabel('Model order, $M$', size=12)
plt.ylabel('Performance Measure', size=12);
```

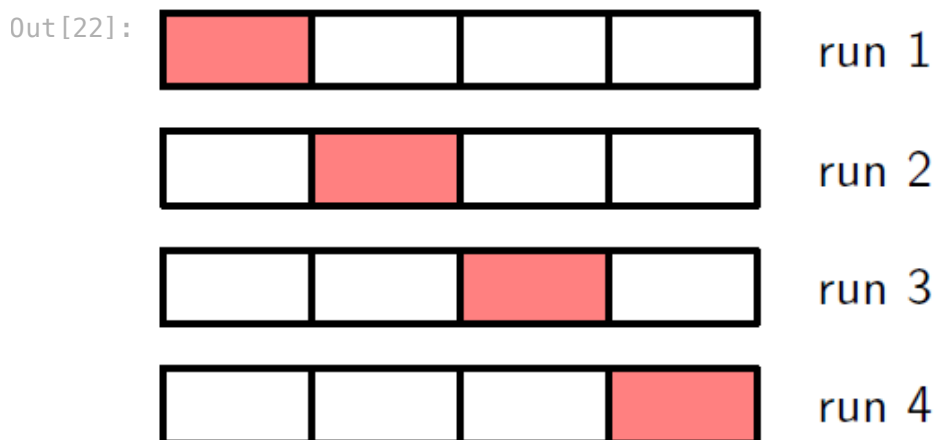


K-Fold Cross-Validation

The technique of k -fold cross-validation, illustrated below for the case of $k=4$, involves taking the available data and partitioning it into k groups (in the simplest case these are of equal size). Then $k-1$ of the groups are used to train a set of models that are then evaluated on the remaining group. This procedure is then repeated for all k possible choices for the held-out group, indicated in the picture below by the red blocks, and the performance scores from the runs are then averaged.

k is typically 10 or 30. As k increases, the percentage of training instances increases and we get more robust estimators, but the validation set becomes smaller. Furthermore, there is the cost of training the classifier k times, which increases as k is increased. As N increases, k can be smaller; if N is small, k should be large to allow large enough training sets.

```
In [22]: Image('figures/Kfold CV.png',width=400)
```



```
In [23]: from sklearn.model_selection import KFold
KFold?
```

Init signature: `KFold(n_splits=5, *, shuffle=False, random_state=None)`

Docstring:

K-Fold cross-validator.

Provides train/test indices to split data in train/test sets. Split dataset into k consecutive folds (without shuffling by default).

Each fold is then used once as a validation while the k - 1 remaining folds form the training set.

Read more in the :ref:`User Guide <k\_fold>`.

For visualisation of cross-validation behaviour and comparison between common scikit-learn split methods refer to :ref:`sphx\_glr\_auto\_examples\_model\_selection\_plot\_cv\_indices.py`

Parameters

`n_splits` : int, default=5

Number of folds. Must be at least 2.

.. versionchanged:: 0.22

``n\_splits`` default value changed from 3 to 5.

`shuffle` : bool, default=False

Whether to shuffle the data before splitting into batches.

Note that the samples within each split will not be shuffled.

`random_state` : int, RandomState instance or None, default=None

When ``shuffle`` is True, ``random\_state`` affects the ordering of the indices, which controls the randomness of each fold. Otherwise, this parameter has no effect.

Pass an int for reproducible output across multiple function calls.

See :term:`Glossary <random\_state>`.

Examples

```
>>> import numpy as np
>>> from sklearn.model_selection import KFold
>>> X = np.array([[1, 2], [3, 4], [1, 2], [3, 4]])
>>> y = np.array([1, 2, 3, 4])
>>> kf = KFold(n_splits=2)
>>> kf.get_n_splits(X)
2
>>> print(kf)
KFold(n_splits=2, random_state=None, shuffle=False)
>>> for i, (train_index, test_index) in enumerate(kf.split(X)):
...     print(f"Fold {i}:")
...     print(f"  Train: index={train_index}")
...     print(f"  Test:  index={test_index}")
Fold 0:
  Train: index=[2 3]
  Test:  index=[0 1]
Fold 1:
  Train: index=[0 1]
  Test:  index=[2 3]
```

Notes

The first ``n\_samples % n\_splits`` folds have size
``n\_samples // n\_splits + 1``, other folds have size
``n\_samples // n\_splits``, where ``n\_samples`` is the number of samples.

Randomized CV splitters may return different results for each call of
split. You can make the results identical by setting ``random\_state``
to an integer.

See Also

StratifiedKFold : Takes class information into account to avoid building
folds with imbalanced class distributions (for binary or multiclass
classification tasks).

GroupKFold : K-fold iterator variant with non-overlapping groups.

RepeatedKFold : Repeats K-Fold n times.

File: ~/anaconda3/envs/ml/lib/python3.12/site-packages/sklearn/model
selection/_split.py

Type: ABCMeta

Subclasses:

Goal: find the best value for the **hyperparameters** M (model order) and λ (regularization trade-off parameter).

Let's use **4-fold cross-validation** on this data:

```
In [24]: k = 4 # number of folds

kf = KFold(n_splits=k, shuffle=True, random_state=0)
kf
```

```
Out[24]: KFold(n_splits=4, random_state=0, shuffle=True)
```

```
In [25]: kf.split(x_train)
```

```
Out[25]: <generator object _BaseKFold.split at 0x15a621690>
```

```
In [26]: # Split training feature matrix into training and validation sets

f=1
for train_index, validation_index in kf.split(x_train):
    print('Fold ', f)
    print('The training set has ', train_index.shape[0], ' samples')
    print('Their index locations are: ', train_index)
    print('The validation set has ', validation_index.shape[0], ' samples')
    print('Their index locations are: ', validation_index, '\n')
    f+=1
```

Fold 1
 The training set has 52 samples
 Their index locations are: [0 1 2 3 4 5 6 8 9 10 11 12 13 14 15 16 17 18 19 20 21 23 24 25 29 32 35 36 37 38 39 40 41 43 44 45 46 47 49 50 52 53 54 57 58 60 61 63 64 65 66 67]
 The validation set has 18 samples
 Their index locations are: [7 22 26 27 28 30 31 33 34 42 48 51 55 56 59 62 68 69]

Fold 2
 The training set has 52 samples
 Their index locations are: [0 1 5 6 7 8 9 12 13 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 33 34 35 36 37 38 39 42 44 46 47 48 51 52 55 56 58 59 60 61 62 63 64 66 68 69]
 The validation set has 18 samples
 Their index locations are: [2 3 4 10 11 14 32 40 41 43 45 49 50 53 54 57 65 67]

Fold 3
 The training set has 53 samples
 Their index locations are: [1 2 3 4 6 7 9 10 11 12 14 21 22 23 24 26 27 28 30 31 32 33 34 36 38 39 40 41 42 43 44 45 46 47 48 49 50 51 53 54 55 56 57 58 59 61 62 63 64 65 67 68 69]
 The validation set has 17 samples
 Their index locations are: [0 5 8 13 15 16 17 18 19 20 25 29 35 37 52 60 66]

Fold 4
 The training set has 53 samples
 Their index locations are: [0 2 3 4 5 7 8 10 11 13 14 15 16 17 18 19 20 22 25 26 27 28 29 30 31 32 33 34 35 37 40 41 42 43 45 48 49 50 51 52 53 54 55 56 57 59 60 62 65 66 67 68 69]
 The validation set has 17 samples
 Their index locations are: [1 6 9 12 21 23 24 36 38 39 44 46 47 58 61 63 64]

```
In [27]: from sklearn.metrics import r2_score

# Set of values for lambda to explore - Polynomial Regression WITH REGULARIZATION
M_vals = np.array(range(2,11))
lam_vals = np.arange(0.01,1.1,0.01)

avg_train = np.zeros((len(M_vals), len(lam_vals)))
avg_valid  = np.zeros((len(M_vals), len(lam_vals)))

for i in range(len(M_vals)):
    for j in range(len(lam_vals)):
        M      = M_vals[i]
        lam     = lam_vals[j]
        print('M Value = ',M)
```



```

print('Lambda Value = ',lam)

# For each training/validation split
f=1

#initialize performance measures
perf_train_avg, perf_val_avg = 0, 0

for train_index, validation_index in kf.split(x_train):
    print('\nFold ',f)

    # Select training set using the indices found from kf.split
    x_train2, x_validation = x_train[train_index], x_train[validation_index]

    # Select validation set using the indices found from kf.split
    t_train2, t_validation = t_train[train_index], t_train[validation_index]

    # Training model with training set
    w, y_train = PolynomialRegression_reg(x_train2, t_train2, M, lam)

    # Evaluate trained model in validation set
    y_val = PolynomialRegression_test(x_validation, w)

    # Performance Measure - coefficient of determination
    perf_train = r2_score(t_train2,y_train) #np.mean((t_train2-y_train)**2)
    perf_val = r2_score(t_validation, y_val) #np.mean((t_validation-y_val)**2)

    # Average performance measure
    perf_train_avg = perf_train_avg + perf_train
    perf_val_avg = perf_val_avg + perf_val
    print('Performance Training = ', perf_train)
    print('Performance Validation = ', perf_val)
    f+=1

# Average Performance
avg_train[i,j] = perf_train_avg/k
avg_valid[i,j] = perf_val_avg/k

print('\nAverage Performance in Training = ', avg_train[i,j])
print('Average Performance in Validation = ', avg_valid[i,j])
print('-----\n')

```

M Value = 2
Lambda Value = 0.01

Fold 1
Performance Training = 0.5151811163712634
Performance Validation = 0.430136073459658

Fold 2
Performance Training = 0.5229009702389276
Performance Validation = 0.34877513250841086

Fold 3
Performance Training = 0.5139715530956834
Performance Validation = 0.3692135559272326

Fold 4
Performance Training = 0.46708466937561444
Performance Validation = 0.5913291090211586

Average Performance in Training = 0.5047845772703722
Average Performance in Validation = 0.434863467729115

M Value = 2
Lambda Value = 0.02

Fold 1
Performance Training = 0.5151810356564245
Performance Validation = 0.43017517737707744

Fold 2
Performance Training = 0.5229008973304188
Performance Validation = 0.3488252572223578

Fold 3
Performance Training = 0.5139714603955554
Performance Validation = 0.36926321943139373

Fold 4
Performance Training = 0.467084603260399
Performance Validation = 0.5912913149343322

Average Performance in Training = 0.5047844991606995
Average Performance in Validation = 0.4348887422412903

M Value = 2
Lambda Value = 0.03

Fold 1
Performance Training = 0.5151809012703839
Performance Validation = 0.43021417484074775

Fold 2
Performance Training = 0.5229007759070499
Performance Validation = 0.3488753046612021

Fold 3
Performance Training = 0.5139713060490858
Performance Validation = 0.36931277046225375

Fold 4
Performance Training = 0.467084493164065
Performance Validation = 0.5912534987193809

Average Performance in Training = 0.5047843690976461
Average Performance in Validation = 0.4349139371708961

M Value = 2
Lambda Value = 0.04

Fold 1
Performance Training = 0.5151807133262374
Performance Validation = 0.43025306607408886

Fold 2
Performance Training = 0.5229006060429584
Performance Validation = 0.34892527491419834

Fold 3
Performance Training = 0.5139710901816914
Performance Validation = 0.3693622092188046

Fold 4
Performance Training = 0.4670843391646925
Performance Validation = 0.5912156604308043

Average Performance in Training = 0.504784187178895
Average Performance in Validation = 0.434939052659474

M Value = 2
Lambda Value = 0.05

Fold 1
Performance Training = 0.5151804719367477
Performance Validation = 0.4302918512998639

Fold 2
Performance Training = 0.5229003878121385
Performance Validation = 0.3489751680704627

Fold 3
Performance Training = 0.513970812918461
Performance Validation = 0.36941153589956977

Fold 4
Performance Training = 0.46708414134017484
Performance Validation = 0.5911778001229593

Average Performance in Training = 0.5047839535018805

Average Performance in Validation = 0.4349640888482139

M Value = 2

Lambda Value = 0.06

Fold 1

Performance Training = 0.5151801772143471

Performance Validation = 0.43033053074018035

Fold 2

Performance Training = 0.5229001212884412

Performance Validation = 0.34902498421897277

Fold 3

Performance Training = 0.5139704743841577

Performance Validation = 0.3694607507026051

Fold 4

Performance Training = 0.46708389976821896

Performance Validation = 0.5911399178500618

Average Performance in Training = 0.5047836681637912

Average Performance in Validation = 0.434989045877955

M Value = 2

Lambda Value = 0.07

Fold 1

Performance Training = 0.515179829271138

Performance Validation = 0.43036910461649414

Fold 2

Performance Training = 0.5228998065455749

Performance Validation = 0.3490747234485685

Fold 3

Performance Training = 0.5139700747032188

Performance Validation = 0.36950985382550006

Fold 4

Performance Training = 0.4670836145263465

Performance Validation = 0.5911020136661855

Average Performance in Training = 0.5047833312615695

Average Performance in Validation = 0.4350139238891871

M Value = 2

Lambda Value = 0.08

Fold 1

Performance Training = 0.5151794282188948

Performance Validation = 0.43040757314961053

Fold 2
Performance Training = 0.5228994436571058
Performance Validation = 0.349124385847952

Fold 3
Performance Training = 0.5139696139997576
Performance Validation = 0.36955884546537987

Fold 4
Performance Training = 0.46708328569189395
Performance Validation = 0.591064087625264

Average Performance in Training = 0.5047829428919131
Average Performance in Validation = 0.4350387230220516

M Value = 2
Lambda Value = 0.09

Fold 1
Performance Training = 0.5151789741690642
Performance Validation = 0.4304459365596879

Fold 2
Performance Training = 0.5228990326964585
Performance Validation = 0.3491739715056885

Fold 3
Performance Training = 0.5139690923975642
Performance Validation = 0.3696077258189061

Fold 4
Performance Training = 0.46708291334201335
Performance Validation = 0.5910261397810901

Average Performance in Training = 0.5047825031512752
Average Performance in Validation = 0.4350634434163432

M Value = 2
Lambda Value = 0.1

Fold 1
Performance Training = 0.5151784672327674
Performance Validation = 0.4304841950662398

Fold 2
Performance Training = 0.522898573736916
Performance Validation = 0.34922348051020613

Fold 3
Performance Training = 0.5139685100201066
Performance Validation = 0.36965649508227927

Fold 4
Performance Training = 0.4670824975536728

Performance Validation = 0.5909881701873174

Average Performance in Training = 0.5047820121358657

Average Performance in Validation = 0.43508808521151066

M Value = 2

Lambda Value = 0.11

Fold 1

Performance Training = 0.5151779075208014

Performance Validation = 0.4305223488881371

Fold 2

Performance Training = 0.5228980668516199

Performance Validation = 0.3492729129497967

Fold 3

Performance Training = 0.513967866990532

Performance Validation = 0.36970515345123856

Fold 4

Performance Training = 0.4670820384036578

Performance Validation = 0.5909501788974595

Average Performance in Training = 0.5047814699416528

Average Performance in Validation = 0.435112648546658

M Value = 2

Lambda Value = 0.12

Fold 1

Performance Training = 0.5151772951436395

Performance Validation = 0.4305603982436109

Fold 2

Performance Training = 0.5228975121135717

Performance Validation = 0.34932226891261564

Fold 3

Performance Training = 0.5139671634316677

Performance Validation = 0.3697537011210653

Fold 4

Performance Training = 0.4670815359685705

Performance Validation = 0.5909121659648914

Average Performance in Training = 0.5047808766643624

Average Performance in Validation = 0.43513713356054584

M Value = 2

Lambda Value = 0.13

Fold 1

Performance Training = 0.5151766302114331
Performance Validation = 0.43059834335025493

Fold 2
Performance Training = 0.5228969095956326
Performance Validation = 0.3493715484866823

Fold 3
Performance Training = 0.513966399466022
Performance Validation = 0.36980213828658237

Fold 4
Performance Training = 0.46708099032483186
Performance Validation = 0.5908741314428497

Average Performance in Training = 0.5047802323994799
Average Performance in Validation = 0.4351615403915924

M Value = 2
Lambda Value = 0.14

Fold 1
Performance Training = 0.515175912834013
Performance Validation = 0.4306361844250274

Fold 2
Performance Training = 0.5228962593705235
Performance Validation = 0.34942075175988085

Fold 3
Performance Training = 0.5139655752157859
Performance Validation = 0.36985046514215714

Fold 4
Performance Training = 0.4670804015486809
Performance Validation = 0.5908360753844332

Average Performance in Training = 0.5047795372422508
Average Performance in Validation = 0.4351858691778746

M Value = 2
Lambda Value = 0.15

Fold 1
Performance Training = 0.5151751431208904
Performance Validation = 0.4306739216842542

Fold 2
Performance Training = 0.5228955615108262
Performance Validation = 0.34946987881996017

Fold 3
Performance Training = 0.5139646908028332
Performance Validation = 0.36989868188170183

Fold 4

Performance Training = 0.46707976971617604

Performance Validation = 0.5907979978426029

Average Performance in Training = 0.5047787912876814

Average Performance in Validation = 0.4352101200571298

M Value = 2

Lambda Value = 0.16

Fold 1

Performance Training = 0.5151743211812585

Performance Validation = 0.4307115553436308

Fold 2

Performance Training = 0.5228948160889832

Performance Validation = 0.34951892975453436

Fold 3

Performance Training = 0.5139637463487225

Performance Validation = 0.3699467886986758

Fold 4

Performance Training = 0.46707909490319544

Performance Validation = 0.5907598988701832

Average Performance in Training = 0.50477799463054

Average Performance in Validation = 0.43523429316675605

M Value = 2

Lambda Value = 0.17

Fold 1

Performance Training = 0.5151734471239928

Performance Validation = 0.43074908561822534

Fold 2

Performance Training = 0.5228940231772985

Performance Validation = 0.34956790465108256

Fold 3

Performance Training = 0.5139627419746979

Performance Validation = 0.36999478578608613

Fold 4

Performance Training = 0.46707837718543754

Performance Validation = 0.5907217785198615

Average Performance in Training = 0.5047771473653567

Average Performance in Validation = 0.4352583886438139

M Value = 2

Lambda Value = 0.18

Fold 1

Performance Training = 0.5151725210576541

Performance Validation = 0.4307865127224797

Fold 2

Performance Training = 0.5228931828479375

Performance Validation = 0.3496168035969498

Fold 3

Performance Training = 0.5139616778016894

Performance Validation = 0.3700426733364891

Fold 4

Performance Training = 0.4670776166384223

Performance Validation = 0.5906836368441898

Average Performance in Training = 0.5047762495864259

Average Performance in Validation = 0.43528240662502704

M Value = 2

Lambda Value = 0.19

Fold 1

Performance Training = 0.515171543090488

Performance Validation = 0.4308238368702133

Fold 2

Performance Training = 0.5228922951729282

Performance Validation = 0.34966562667934775

Fold 3

Performance Training = 0.5139605539503154

Performance Validation = 0.37009045154199294

Fold 4

Performance Training = 0.4670768133374903

Performance Validation = 0.5906454738955838

Average Performance in Training = 0.5047753013878055

Average Performance in Validation = 0.43530634724678446

M Value = 2

Lambda Value = 0.2

Fold 1

Performance Training = 0.5151705133304265

Performance Validation = 0.43086105827462473

Fold 2

Performance Training = 0.5228913602241602

Performance Validation = 0.34971437398535365

Fold 3
Performance Training = 0.5139593705408823
Performance Validation = 0.37013812059425666

Fold 4
Performance Training = 0.4670759673578052
Performance Validation = 0.5906072897263244

Average Performance in Training = 0.5047743028633186
Average Performance in Validation = 0.4353302106451398

M Value = 2
Lambda Value = 0.21

Fold 1
Performance Training = 0.5151694318850902
Performance Validation = 0.43089817714829426

Fold 2
Performance Training = 0.5228903780733866
Performance Validation = 0.34976304560191207

Fold 3
Performance Training = 0.5139581276933864
Performance Validation = 0.37018568068449353

Fold 4
Performance Training = 0.4670750787743527
Performance Validation = 0.5905690843885578

Average Performance in Training = 0.5047732541065539
Average Performance in Validation = 0.43535399695581445

M Value = 2
Lambda Value = 0.22

Fold 1
Performance Training = 0.5151682988617885
Performance Validation = 0.4309351937031862

Fold 2
Performance Training = 0.5228893487922235
Performance Validation = 0.34981164161583433

Fold 3
Performance Training = 0.5139568255275149
Performance Validation = 0.3702331320034723

Fold 4
Performance Training = 0.4670741476619419
Performance Validation = 0.5905308579342959

Average Performance in Training = 0.5047721552108672
Average Performance in Validation = 0.43537770631419714

M Value = 2
Lambda Value = 0.23

Fold 1

Performance Training = 0.5151671143675212
Performance Validation = 0.4309721081506511

Fold 2

Performance Training = 0.5228882724521504
Performance Validation = 0.34986016211379933

Fold 3

Performance Training = 0.513955464162646
Performance Validation = 0.37028047474151704

Fold 4

Performance Training = 0.4670731740952061
Performance Validation = 0.5904926104154171

Average Performance in Training = 0.5047710062693809
Average Performance in Validation = 0.43540133885534615

M Value = 2
Lambda Value = 0.24

Fold 1

Performance Training = 0.5151658785089792
Performance Validation = 0.43100892070142827

Fold 2

Performance Training = 0.5228871491245111
Performance Validation = 0.34990860718235406

Fold 3

Performance Training = 0.5139540437178516
Performance Validation = 0.37032770908850987

Fold 4

Performance Training = 0.4670721581486027
Performance Validation = 0.5904543418836661

Average Performance in Training = 0.5047698073749861
Average Performance in Validation = 0.4354248947139896

M Value = 2
Lambda Value = 0.25

Fold 1

Performance Training = 0.5151645913925469
Performance Validation = 0.431045631565648

Fold 2

Performance Training = 0.5228859788805134
Performance Validation = 0.3499569769079126

Fold 3
Performance Training = 0.5139525643118965
Performance Validation = 0.37037483523389225

Fold 4
Performance Training = 0.46707109989641415
Performance Validation = 0.5904160523906548

Average Performance in Training = 0.5047685586203428
Average Performance in Validation = 0.4354483740245269

M Value = 2
Lambda Value = 0.26

Fold 1
Performance Training = 0.5151632531243018
Performance Validation = 0.4310822409528339

Fold 2
Performance Training = 0.5228847617912298
Performance Validation = 0.35000527137675785

Fold 3
Performance Training = 0.5139510260632407
Performance Validation = 0.37042185336666544

Fold 4
Performance Training = 0.46706999941274896
Performance Validation = 0.5903777419878625

Average Performance in Training = 0.5047672600978803
Average Performance in Validation = 0.4354717769210299

M Value = 2
Lambda Value = 0.27

Fold 1
Performance Training = 0.515161863810017
Performance Validation = 0.4311187490719053

Fold 2
Performance Training = 0.522883497927598
Performance Validation = 0.35005349067504177

Fold 3
Performance Training = 0.5139494290900404
Performance Validation = 0.3704687636753925

Fold 4
Performance Training = 0.46706885677154075
Performance Validation = 0.5903394107266373

Average Performance in Training = 0.504765911899799
Average Performance in Validation = 0.4354951035372442

M Value = 2
Lambda Value = 0.28

Fold 1
Performance Training = 0.5151604235551617
Performance Validation = 0.43115515613117916

Fold 2
Performance Training = 0.5228821873604212
Performance Validation = 0.35010163488878443

Fold 3
Performance Training = 0.5139477735101476
Performance Validation = 0.37051556634820026

Fold 4
Performance Training = 0.46706767204655075
Performance Validation = 0.5903010586581949

Average Performance in Training = 0.5047645141180703
Average Performance in Validation = 0.43551835400658967

M Value = 2
Lambda Value = 0.29

Fold 1
Performance Training = 0.5151589324649026
Performance Validation = 0.4311914623383727

Fold 2
Performance Training = 0.5228808301603685
Performance Validation = 0.35014970410387614

Fold 3
Performance Training = 0.5139460594411134
Performance Validation = 0.37056226157277883

Fold 4
Performance Training = 0.46706644531136676
Performance Validation = 0.5902626858336204

Average Performance in Training = 0.5047630668444378
Average Performance in Validation = 0.43554152846216204

M Value = 2
Lambda Value = 0.3

Fold 1
Performance Training = 0.5151573906441047

Performance Validation = 0.43122766790060596

Fold 2

Performance Training = 0.5228794263979746

Performance Validation = 0.35019769840607584

Fold 3

Performance Training = 0.5139442870001864

Performance Validation = 0.3706088495363845

Fold 4

Performance Training = 0.46706517663940506

Performance Validation = 0.590224292303868

Average Performance in Training = 0.5047615701704177

Average Performance in Validation = 0.4355646270367336

M Value = 2

Lambda Value = 0.31

Fold 1

Performance Training = 0.5151557981973331

Performance Validation = 0.4312637730244029

Fold 2

Performance Training = 0.5228779761436413

Performance Validation = 0.35024561788101294

Fold 3

Performance Training = 0.513942456304316

Performance Validation = 0.37065533042584076

Fold 4

Performance Training = 0.46706386610390993

Performance Validation = 0.5901858781197618

Average Performance in Training = 0.5047600241873

Average Performance in Validation = 0.43558764986275456

M Value = 2

Lambda Value = 0.32

Fold 1

Performance Training = 0.5151541552288539

Performance Validation = 0.4312997779156946

Fold 2

Performance Training = 0.5228764794676369

Performance Validation = 0.35029346261418726

Fold 3

Performance Training = 0.513940567470152

Performance Validation = 0.3707017044275388

Fold 4
Performance Training = 0.4670625137779545
Performance Validation = 0.5901474433319961

Average Performance in Training = 0.5047584289861493
Average Performance in Validation = 0.43561059707235417

M Value = 2
Lambda Value = 0.33

Fold 1
Performance Training = 0.5151524618426349
Performance Validation = 0.43133568277982204

Fold 2
Performance Training = 0.5228749364400973
Performance Validation = 0.35034123269096873

Fold 3
Performance Training = 0.5139386206140462
Performance Validation = 0.37074797172744

Fold 4
Performance Training = 0.4670611197344414
Performance Validation = 0.5901089879911358

Average Performance in Training = 0.5047567846578049
Average Performance in Validation = 0.43563346879734166

M Value = 2
Lambda Value = 0.34

Fold 1
Performance Training = 0.5151507181423469
Performance Validation = 0.4313714878215369

Fold 2
Performance Training = 0.5228733471310263
Performance Validation = 0.35038892819659806

Fold 3
Performance Training = 0.5139366158520531
Performance Validation = 0.3707941325110765

Fold 4
Performance Training = 0.46705968404610354
Performance Validation = 0.5900705121476171

Average Performance in Training = 0.5047550912928824
Average Performance in Validation = 0.43565626516920714

M Value = 2
Lambda Value = 0.35000000000000003

Fold 1
Performance Training = 0.515148924231366
Performance Validation = 0.43140719324500465

Fold 2
Performance Training = 0.5228717116102946
Performance Validation = 0.3504365492161876

Fold 3
Performance Training = 0.5139345532999311
Performance Validation = 0.3708401869635527

Fold 4
Performance Training = 0.467058206785504
Performance Validation = 0.5900320158517475

Average Performance in Training = 0.5047533489817739
Average Performance in Validation = 0.43567898631912305

M Value = 2
Lambda Value = 0.36

Fold 1
Performance Training = 0.5151470802127729
Performance Validation = 0.4314427992538059

Fold 2
Performance Training = 0.522870029947642
Performance Validation = 0.35048409583472084

Fold 3
Performance Training = 0.5139324330731432
Performance Validation = 0.3708861352695463

Fold 4
Performance Training = 0.4670566880250373
Performance Validation = 0.5899934991537065

Average Performance in Training = 0.5047515578146489
Average Performance in Validation = 0.4357016323779449

M Value = 2
Lambda Value = 0.37

Fold 1
Performance Training = 0.5151451861893551
Performance Validation = 0.4314783060509403

Fold 2
Performance Training = 0.5228683022126772
Performance Validation = 0.3505315681370529

Fold 3

Performance Training = 0.513930255286858
Performance Validation = 0.37093197761331

Fold 4

Performance Training = 0.4670551278369294
Performance Validation = 0.5899549621035463

Average Performance in Training = 0.504749717881455
Average Performance in Validation = 0.4357242034762123

M Value = 2

Lambda Value = 0.38

Fold 1

Performance Training = 0.5151432422636083
Performance Validation = 0.4315137138388271

Fold 2

Performance Training = 0.5228665284748777
Performance Validation = 0.35057896620791107

Fold 3

Performance Training = 0.5139280200559515
Performance Validation = 0.3709777141786723

Fold 4

Performance Training = 0.4670535262932384
Performance Validation = 0.5899164047511912

Average Performance in Training = 0.504747829271919
Average Performance in Validation = 0.4357466997441505

M Value = 2

Lambda Value = 0.39

Fold 1

Performance Training = 0.5151412485377365
Performance Validation = 0.4315490228193082

Fold 2

Performance Training = 0.52286470880359
Performance Validation = 0.35062629013189495

Fold 3

Performance Training = 0.5139257274950066
Performance Validation = 0.37102334514904

Fold 4

Performance Training = 0.46705188346585547
Performance Validation = 0.5898778271464395

Average Performance in Training = 0.5047458920755472
Average Performance in Validation = 0.43576912131167067

M Value = 2
Lambda Value = 0.4

Fold 1

Performance Training = 0.5151392051136539
Performance Validation = 0.43158423319364947

Fold 2

Performance Training = 0.5228628432680311
Performance Validation = 0.3506735399934767

Fold 3

Performance Training = 0.5139233777183152
Performance Validation = 0.371068870707397

Fold 4

Performance Training = 0.4670501994265045
Performance Validation = 0.589839229338963

Average Performance in Training = 0.5047439063816261
Average Performance in Validation = 0.43579146830837157

M Value = 2
Lambda Value = 0.41000000000000003

Fold 1

Performance Training = 0.5151371120929855
Performance Validation = 0.43161934516254374

Fold 2

Performance Training = 0.5228609319372877
Performance Validation = 0.3507207158770025

Fold 3

Performance Training = 0.5139209708398789
Performance Validation = 0.3711142910363088

Fold 4

Performance Training = 0.4670484742467437
Performance Validation = 0.5898006113783074

Average Performance in Training = 0.504741872279224
Average Performance in Validation = 0.43581374086354063

M Value = 2
Lambda Value = 0.42

Fold 1

Performance Training = 0.515134969577069
Performance Validation = 0.4316543589261128

Fold 2

Performance Training = 0.5228589748803166

Performance Validation = 0.35076781786668965

Fold 3

Performance Training = 0.5139185069734098

Performance Validation = 0.37115960631792055

Fold 4

Performance Training = 0.4670467079979653

Performance Validation = 0.5897619733138926

Average Performance in Training = 0.5047397898571901

Average Performance in Validation = 0.43583593910615387

M Value = 2

Lambda Value = 0.43

Fold 1

Performance Training = 0.5151327776669548

Performance Validation = 0.4316892746839093

Fold 2

Performance Training = 0.5228569721659462

Performance Validation = 0.35081484604663116

Fold 3

Performance Training = 0.5139159862323313

Performance Validation = 0.371204816733961

Fold 4

Performance Training = 0.4670449007513967

Performance Validation = 0.589723315195014

Average Performance in Training = 0.5047376592041573

Average Performance in Validation = 0.43585806316487885

M Value = 2

Lambda Value = 0.44

Fold 1

Performance Training = 0.5151305364634082

Performance Validation = 0.43172409263491873

Fold 2

Performance Training = 0.5228549238628756

Performance Validation = 0.3508618005007925

Fold 3

Performance Training = 0.5139134087297794

Performance Validation = 0.37124992246574184

Fold 4

Performance Training = 0.46704305257809997

Performance Validation = 0.589684637070842

Average Performance in Training = 0.5047354804085408
Average Performance in Validation = 0.43588011316807374

M Value = 2
Lambda Value = 0.45

Fold 1
Performance Training = 0.5151282460669098
Performance Validation = 0.43175881297756213

Fold 2
Performance Training = 0.5228528300396754
Performance Validation = 0.3509086813130141

Fold 3
Performance Training = 0.5139107745786038
Performance Validation = 0.3712949236941606

Fold 4
Performance Training = 0.46704116354897396
Performance Validation = 0.589645938990423

Average Performance in Training = 0.5047332535585407
Average Performance in Validation = 0.43590208924379

M Value = 2
Lambda Value = 0.46

Fold 1
Performance Training = 0.5151259065776561
Performance Validation = 0.43179343590969743

Fold 2
Performance Training = 0.5228506907647883
Performance Validation = 0.35095548856701053

Fold 3
Performance Training = 0.513908083891368
Performance Validation = 0.37133982059969994

Fold 4
Performance Training = 0.4670392337347534
Performance Validation = 0.5896072210026793

Average Performance in Training = 0.5047309787421415
Average Performance in Validation = 0.4359239915197718

M Value = 2
Lambda Value = 0.47000000000000003

Fold 1
Performance Training = 0.5151235180955622
Performance Validation = 0.4318279616286219

Fold 2

Performance Training = 0.5228485061065289

Performance Validation = 0.3510022223463708

Fold 3

Performance Training = 0.5139053367803514

Performance Validation = 0.37138461336243067

Fold 4

Performance Training = 0.46703726320601013

Performance Validation = 0.5895684831564096

Average Performance in Training = 0.5047286560471131

Average Performance in Validation = 0.4359458201234582

M Value = 2

Lambda Value = 0.48

Fold 1

Performance Training = 0.5151210807202613

Performance Validation = 0.43186239033107476

Fold 2

Performance Training = 0.5228462761330848

Performance Validation = 0.35104888273455914

Fold 3

Performance Training = 0.5139025333575491

Performance Validation = 0.3714293021620123

Fold 4

Performance Training = 0.46703525203315355

Performance Validation = 0.5895297255002907

Average Performance in Training = 0.5047262855610122

Average Performance in Validation = 0.43596757518198426

M Value = 2

Lambda Value = 0.49

Fold 1

Performance Training = 0.5151185945511065

Performance Validation = 0.4318967222132386

Fold 2

Performance Training = 0.5228440009125157

Performance Validation = 0.3510954698149158

Fold 3

Performance Training = 0.5138996737346735

Performance Validation = 0.3714738871776936

Fold 4

Performance Training = 0.4670332002864307
Performance Validation = 0.5894909480828753

Average Performance in Training = 0.5047238673711816
Average Performance in Validation = 0.4359892568221808

M Value = 2
Lambda Value = 0.5

Fold 1
Performance Training = 0.5151160596871719
Performance Validation = 0.4319309574707414

Fold 2
Performance Training = 0.5228416805127553
Performance Validation = 0.35114198367065563

Fold 3
Performance Training = 0.5138967580231555
Performance Validation = 0.37151836858831544

Fold 4
Performance Training = 0.4670311080359276
Performance Validation = 0.5894521509525947

Average Performance in Training = 0.5047214015647525
Average Performance in Validation = 0.4360108651705768

M Value = 2
Lambda Value = 0.51

Fold 1
Performance Training = 0.5151134762272528
Performance Validation = 0.43196509629865865

Fold 2
Performance Training = 0.5228393150016104
Performance Validation = 0.35118842438486986

Fold 3
Performance Training = 0.5138937863341446
Performance Validation = 0.3715627465723099

Fold 4
Performance Training = 0.46702897535156884
Performance Validation = 0.5894133341577585

Average Performance in Training = 0.5047188882286442
Average Performance in Validation = 0.43603240035339924

M Value = 2
Lambda Value = 0.52

Fold 1
Performance Training = 0.5151108442698682
Performance Validation = 0.431999138891516

Fold 2
Performance Training = 0.5228369044467622
Performance Validation = 0.351234792040526

Fold 3
Performance Training = 0.5138907587785102
Performance Validation = 0.37160702130770407

Fold 4
Performance Training = 0.467026802303119
Performance Validation = 0.5893744977465543

Average Performance in Training = 0.504716327449565
Average Performance in Validation = 0.4360538624965751

M Value = 2
Lambda Value = 0.53

Fold 1
Performance Training = 0.5151081639132606
Performance Validation = 0.432033085443291

Fold 2
Performance Training = 0.5228344489157655
Performance Validation = 0.351281086720468

Fold 3
Performance Training = 0.5138876754668424
Performance Validation = 0.37165119297211857

Fold 4
Performance Training = 0.46702458896018206
Performance Validation = 0.5893356417670494

Average Performance in Training = 0.5047137193140127
Average Performance in Validation = 0.43607525172573175

M Value = 2
Lambda Value = 0.54

Fold 1
Performance Training = 0.5151054352553981
Performance Validation = 0.4320669361474143

Fold 2
Performance Training = 0.5228319484760497
Performance Validation = 0.3513273085074162

Fold 3
Performance Training = 0.5138845365094538

Performance Validation = 0.3716952617427709

Fold 4

Performance Training = 0.4670223353922033

Performance Validation = 0.5892967662671902

Average Performance in Training = 0.5047110639082762

Average Performance in Validation = 0.4360965681661979

M Value = 2

Lambda Value = 0.55

Fold 1

Performance Training = 0.5151026583939741

Performance Validation = 0.43210069119677275

Fold 2

Performance Training = 0.5228294031949202

Performance Validation = 0.3513734574839683

Fold 3

Performance Training = 0.5138813420163788

Performance Validation = 0.37173922779647506

Fold 4

Performance Training = 0.467020041668468

Performance Validation = 0.5892578712948027

Average Performance in Training = 0.5047083613184353

Average Performance in Validation = 0.4361178119430047

M Value = 2

Lambda Value = 0.56

Fold 1

Performance Training = 0.5150998334264093

Performance Validation = 0.43213435078371143

Fold 2

Performance Training = 0.5228268131395563

Performance Validation = 0.3514195337325995

Fold 3

Performance Training = 0.5138780920973758

Performance Validation = 0.37178309130964426

Fold 4

Performance Training = 0.46701770785810437

Performance Validation = 0.5892189568975935

Average Performance in Training = 0.5047056116303614

Average Performance in Validation = 0.4361389831808872

M Value = 2
Lambda Value = 0.5700000000000001

Fold 1
Performance Training = 0.5150969604498531
Performance Validation = 0.43216791510003416

Fold 2
Performance Training = 0.5228241783770133
Performance Validation = 0.3514655373356623

Fold 3
Performance Training = 0.5138747868619278
Performance Validation = 0.3718268524582906

Fold 4
Performance Training = 0.46701533403008133
Performance Validation = 0.5891800231231497

Average Performance in Training = 0.5047028149297188
Average Performance in Validation = 0.4361600820042842

M Value = 2
Lambda Value = 0.58

Fold 1
Performance Training = 0.5150940395611832
Performance Validation = 0.43220138433700783

Fold 2
Performance Training = 0.522821498974223
Performance Validation = 0.3515114683753875

Fold 3
Performance Training = 0.5138714264192427
Performance Validation = 0.3718705114180276

Fold 4
Performance Training = 0.4670129202532113
Performance Validation = 0.589141070018939

Average Performance in Training = 0.5046999713019651
Average Performance in Validation = 0.4361811085373405

M Value = 2
Lambda Value = 0.59

Fold 1
Performance Training = 0.515091070857008
Performance Validation = 0.4322347586853622

Fold 2
Performance Training = 0.5228187749979927
Performance Validation = 0.35155732693388386

Fold 3
Performance Training = 0.5138680108782548
Performance Validation = 0.37191406836407026

Fold 4
Performance Training = 0.4670104665961492
Performance Validation = 0.5891020976323111

Average Performance in Training = 0.5046970808323512
Average Performance in Validation = 0.43620206290390684

M Value = 2
Lambda Value = 0.6

Fold 1
Performance Training = 0.5150880544336671
Performance Validation = 0.43226803833529315

Fold 2
Performance Training = 0.5228160065150063
Performance Validation = 0.3516031130931381

Fold 3
Performance Training = 0.5138645403476261
Performance Validation = 0.37195752347123723

Fold 4
Performance Training = 0.4670079731273936
Performance Validation = 0.589063106010497

Average Performance in Training = 0.5046941436059232
Average Performance in Validation = 0.43622294522754135

M Value = 2
Lambda Value = 0.61

Fold 1
Performance Training = 0.5150849903872319
Performance Validation = 0.4323012234764646

Fold 2
Performance Training = 0.5228131935918251
Performance Validation = 0.351648826935017

Fold 3
Performance Training = 0.5138610149357461
Performance Validation = 0.37200087691395134

Fold 4
Performance Training = 0.4670054399152871
Performance Validation = 0.58902409520061

Average Performance in Training = 0.5046911597075225

Average Performance in Validation = 0.43624375563151074

M Value = 2

Lambda Value = 0.62

Fold 1

Performance Training = 0.5150818788135074

Performance Validation = 0.43233431429801006

Fold 2

Performance Training = 0.5228103362948873

Performance Validation = 0.35169446854126485

Fold 3

Performance Training = 0.5138574347507332

Performance Validation = 0.37204412886624083

Fold 4

Performance Training = 0.46700286702801685

Performance Validation = 0.5889850652496461

Average Performance in Training = 0.5046881292217862

Average Performance in Validation = 0.43626449423879043

M Value = 2

Lambda Value = 0.63

Fold 1

Performance Training = 0.5150787198080324

Performance Validation = 0.4323673109885344

Fold 2

Performance Training = 0.5228074346905085

Performance Validation = 0.3517400379935065

Fold 3

Performance Training = 0.5138537999004358

Performance Validation = 0.3720872795017408

Fold 4

Performance Training = 0.46700025453361527

Performance Validation = 0.5889460162044841

Average Performance in Training = 0.504685052233148

Average Performance in Validation = 0.4362851611720664

M Value = 2

Lambda Value = 0.64

Fold 1

Performance Training = 0.5150755134660814

Performance Validation = 0.4324002137361158

Fold 2
Performance Training = 0.5228044888448825
Performance Validation = 0.3517855353732453

Fold 3
Performance Training = 0.5138501104924331
Performance Validation = 0.37213032899369425

Fold 4
Performance Training = 0.4669976024999597
Performance Validation = 0.5889069481118858

Average Performance in Training = 0.5046819288258392
Average Performance in Validation = 0.4363057565537353

M Value = 2
Lambda Value = 0.65

Fold 1
Performance Training = 0.5150722598826647
Performance Validation = 0.4324330227283091

Fold 2
Performance Training = 0.5228014988240812
Performance Validation = 0.3518309607618655

Fold 3
Performance Training = 0.5138463666340354
Performance Validation = 0.3721732775149533

Fold 4
Performance Training = 0.46699491099477375
Performance Validation = 0.5888678610184974

Average Performance in Training = 0.5046787590838888
Average Performance in Validation = 0.4363262805059063

M Value = 2
Lambda Value = 0.66

Fold 1
Performance Training = 0.5150689591525296
Performance Validation = 0.4324657381521456

Fold 2
Performance Training = 0.5227984646940547
Performance Validation = 0.3518763142406308

Fold 3
Performance Training = 0.5138425684322858
Performance Validation = 0.37221612523798

Fold 4
Performance Training = 0.4669921800856276

Performance Validation = 0.5888287549708489

Average Performance in Training = 0.5046755430911245

Average Performance in Validation = 0.4363467331504013

M Value = 2

Lambda Value = 0.67

Fold 1

Performance Training = 0.5150656113701622

Performance Validation = 0.4324983601941367

Fold 2

Performance Training = 0.5227953865206324

Performance Validation = 0.3519215958906853

Fold 3

Performance Training = 0.5138387159939608

Performance Validation = 0.372258872334848

Fold 4

Performance Training = 0.4669894098399383

Performance Validation = 0.5887896300153541

Average Performance in Training = 0.5046722809311734

Average Performance in Validation = 0.43636711460875605

M Value = 2

Lambda Value = 0.68

Fold 1

Performance Training = 0.5150622166297871

Performance Validation = 0.43253088904027426

Fold 2

Performance Training = 0.5227922643695226

Performance Validation = 0.35196680579305406

Fold 3

Performance Training = 0.5138348094255706

Performance Validation = 0.37230151897724373

Fold 4

Performance Training = 0.4669866003249702

Performance Validation = 0.5887504861983128

Average Performance in Training = 0.5046689726874626

Average Performance in Validation = 0.4363874250022212

M Value = 2

Lambda Value = 0.6900000000000001

Fold 1

Performance Training = 0.5150587750253692
Performance Validation = 0.43256332487603444

Fold 2
Performance Training = 0.522789098306313
Performance Validation = 0.35201194402864344

Fold 3
Performance Training = 0.5138308488333605
Performance Validation = 0.37234406533646636

Fold 4
Performance Training = 0.46698375160783556
Performance Validation = 0.5887113235659092

Average Performance in Training = 0.5046656184432196
Average Performance in Validation = 0.4364076644517634

M Value = 2
Lambda Value = 0.7000000000000001

Fold 1
Performance Training = 0.5150552866506148
Performance Validation = 0.43259566788637727

Fold 2
Performance Training = 0.5227858883964711
Performance Validation = 0.35205701067824

Fold 3
Performance Training = 0.5138268343233126
Performance Validation = 0.37238651158343083

Fold 4
Performance Training = 0.4669808637554952
Performance Validation = 0.5886721421642132

Average Performance in Training = 0.5046622182814734
Average Performance in Validation = 0.4364278330780653

M Value = 2
Lambda Value = 0.71

Fold 1
Performance Training = 0.5150517515989718
Performance Validation = 0.4326279182557503

Fold 2
Performance Training = 0.5227826347053448
Performance Validation = 0.3521020058225126

Fold 3
Performance Training = 0.5138227660011441
Performance Validation = 0.37242885788866764

Fold 4

Performance Training = 0.4669779368347585

Performance Validation = 0.5886329420391807

Average Performance in Training = 0.5046587722850548

Average Performance in Validation = 0.43644793100152784

M Value = 2

Lambda Value = 0.72

Fold 1

Performance Training = 0.5150481699636313

Performance Validation = 0.4326600761680903

Fold 2

Performance Training = 0.5227793372981622

Performance Validation = 0.35214692954201166

Fold 3

Performance Training = 0.513818643972311

Performance Validation = 0.37247110442232423

Fold 4

Performance Training = 0.46697497091228457

Performance Validation = 0.5885937232366545

Average Performance in Training = 0.5046552805365974

Average Performance in Validation = 0.43646795834227015

M Value = 2

Lambda Value = 0.73

Fold 1

Performance Training = 0.5150445418375283

Performance Validation = 0.4326921418068247

Fold 2

Performance Training = 0.522775996240032

Performance Validation = 0.3521917819171698

Fold 3

Performance Training = 0.5138144683420076

Performance Validation = 0.37251325135416646

Fold 4

Performance Training = 0.4669719660545818

Performance Validation = 0.588554485802363

Average Performance in Training = 0.5046517431185374

Average Performance in Validation = 0.436487915220131

M Value = 2

Lambda Value = 0.74

Fold 1

Performance Training = 0.5150408673133429

Performance Validation = 0.4327241153548722

Fold 2

Performance Training = 0.5227726115959441

Performance Validation = 0.35223656302830175

Fold 3

Performance Training = 0.5138102392151667

Performance Validation = 0.3725552988535795

Fold 4

Performance Training = 0.46696892232800913

Performance Validation = 0.5885152297819224

Average Performance in Training = 0.5046481601131156

Average Performance in Validation = 0.43650780175466897

M Value = 2

Lambda Value = 0.75

Fold 1

Performance Training = 0.5150371464835007

Performance Validation = 0.43275599699464795

Fold 2

Performance Training = 0.5227691834307697

Performance Validation = 0.3522812729556045

Fold 3

Performance Training = 0.5138059566964621

Performance Validation = 0.3725972470895689

Fold 4

Performance Training = 0.46696583979877626

Performance Validation = 0.5884759552208364

Average Performance in Training = 0.5046445316023772

Average Performance in Validation = 0.4365276180651645

M Value = 2

Lambda Value = 0.76

Fold 1

Performance Training = 0.5150333794401745

Performance Validation = 0.4327877869080615

Fold 2

Performance Training = 0.5227657118092617

Performance Validation = 0.3523259117791584

Fold 3
Performance Training = 0.513801620890308
Performance Validation = 0.37263909623076164

Fold 4
Performance Training = 0.4669627185329439
Performance Validation = 0.5884366621644963

Average Performance in Training = 0.504640857668172
Average Performance in Validation = 0.43654736427061946

M Value = 2
Lambda Value = 0.77

Fold 1
Performance Training = 0.5150295662752844
Performance Validation = 0.4328194852765219

Fold 2
Performance Training = 0.522762196796055
Performance Validation = 0.35237047957892675

Fold 3
Performance Training = 0.5137972319008606
Performance Validation = 0.37268084644540767

Fold 4
Performance Training = 0.4669595585964246
Performance Validation = 0.5883973506581816

Average Performance in Training = 0.5046371383921562
Average Performance in Validation = 0.4365670404897595

M Value = 2
Lambda Value = 0.78

Fold 1
Performance Training = 0.5150257070804996
Performance Validation = 0.4328510922809373

Fold 2
Performance Training = 0.5227586384556667
Performance Validation = 0.35241497643475594

Fold 3
Performance Training = 0.5137927898320187
Performance Validation = 0.37272249790138057

Fold 4
Performance Training = 0.46695636005498276
Performance Validation = 0.5883580207470601

Average Performance in Training = 0.5046333738557919
Average Performance in Validation = 0.43658664684103343

M Value = 2
Lambda Value = 0.79

Fold 1

Performance Training = 0.5150218019472388
Performance Validation = 0.43288260810171775

Fold 2

Performance Training = 0.5227550368524967
Performance Validation = 0.35245940242637663

Fold 3

Performance Training = 0.5137882947874244
Performance Validation = 0.3727640507661788

Fold 4

Performance Training = 0.4669531229742353
Performance Validation = 0.5883186724761889

Average Performance in Training = 0.5046295641403488
Average Performance in Validation = 0.43660618344261554

M Value = 2
Lambda Value = 0.8

Fold 1

Performance Training = 0.5150178509666707
Performance Validation = 0.4329140329187767

Fold 2

Performance Training = 0.5227513920508273
Performance Validation = 0.35250375763340247

Fold 3

Performance Training = 0.5137837468704638
Performance Validation = 0.3728055052069268

Fold 4

Performance Training = 0.46694984741965273
Performance Validation = 0.588279305890514

Average Performance in Training = 0.5046257093269036
Average Performance in Validation = 0.43662565041240503

M Value = 2
Lambda Value = 0.81

Fold 1

Performance Training = 0.5150138542297167
Performance Validation = 0.4329453669115336

Fold 2

Performance Training = 0.5227477041148245
Performance Validation = 0.3525480421353321

Fold 3
Performance Training = 0.513779146184269
Performance Validation = 0.37284686139037626

Fold 4
Performance Training = 0.46694653345655857
Performance Validation = 0.5882399210348714

Average Performance in Training = 0.5046218094963422
Average Performance in Validation = 0.4366450478680283

M Value = 2
Lambda Value = 0.8200000000000001

Fold 1
Performance Training = 0.5150098118270492
Performance Validation = 0.43297661025891454

Fold 2
Performance Training = 0.5227439731085375
Performance Validation = 0.352592256011548

Fold 3
Performance Training = 0.5137744928317171
Performance Validation = 0.3728881194829069

Fold 4
Performance Training = 0.4669431811501302
Performance Validation = 0.5882005179539866

Average Performance in Training = 0.5046178647293584
Average Performance in Validation = 0.436664375926839

M Value = 2
Lambda Value = 0.8300000000000001

Fold 1
Performance Training = 0.5150057238490946
Performance Validation = 0.4330077631393542

Fold 2
Performance Training = 0.5227401990958999
Performance Validation = 0.3526363993413172

Fold 3
Performance Training = 0.5137697869154321
Performance Validation = 0.3729292796505277

Fold 4
Performance Training = 0.4669397905653998
Performance Validation = 0.588161096692476

Average Performance in Training = 0.5046138751064566
Average Performance in Validation = 0.43668363470591876

M Value = 2
Lambda Value = 0.84

Fold 1
Performance Training = 0.5150015903860334
Performance Validation = 0.43303882573079855

Fold 2
Performance Training = 0.5227363821407283
Performance Validation = 0.3526804722037926

Fold 3
Performance Training = 0.5137650285377855
Performance Validation = 0.37297034205887813

Fold 4
Performance Training = 0.46693636176725384
Performance Validation = 0.5881216572948458

Average Performance in Training = 0.5046098407079502
Average Performance in Validation = 0.43670282432207874

M Value = 2
Lambda Value = 0.85

Fold 1
Performance Training = 0.5149974115278013
Performance Validation = 0.4330697982107057

Fold 2
Performance Training = 0.5227325223067245
Performance Validation = 0.35272447467801094

Fold 3
Performance Training = 0.5137602178008975
Performance Validation = 0.3730113068732288

Fold 4
Performance Training = 0.46693289482043454
Performance Validation = 0.5880821998054941

Average Performance in Training = 0.5046057616139645
Average Performance in Validation = 0.4367219448918599

M Value = 2
Lambda Value = 0.86

Fold 1
Performance Training = 0.5149931873640896

Performance Validation = 0.43310068075604846

Fold 2

Performance Training = 0.5227286196574752

Performance Validation = 0.3527684068428959

Fold 3

Performance Training = 0.5137553548066369

Performance Validation = 0.37305217425848347

Fold 4

Performance Training = 0.4669293897895397

Performance Validation = 0.5880427242687102

Average Performance in Training = 0.5046016379044354

Average Performance in Validation = 0.43674099653153453

M Value = 2

Lambda Value = 0.87

Fold 1

Performance Training = 0.5149889179843472

Performance Validation = 0.43313147354331494

Fold 2

Performance Training = 0.5227246742564513

Performance Validation = 0.352812268777255

Fold 3

Performance Training = 0.513750439656623

Performance Validation = 0.37309294437917895

Fold 4

Performance Training = 0.46692584673902293

Performance Validation = 0.5880032307286747

Average Performance in Training = 0.5045974696591111

Average Performance in Validation = 0.4367599793571059

M Value = 2

Lambda Value = 0.88

Fold 1

Performance Training = 0.5149846034777801

Performance Validation = 0.4331621767485113

Fold 2

Performance Training = 0.5227206861670097

Performance Validation = 0.3528560605597839

Fold 3

Performance Training = 0.5137454724522255

Performance Validation = 0.3731336173994866

Fold 4
Performance Training = 0.46692226573319506
Performance Validation = 0.5879637192294607

Average Performance in Training = 0.5045932569575526
Average Performance in Validation = 0.43677889348431065

M Value = 2
Lambda Value = 0.89

Fold 1
Performance Training = 0.5149802439333535
Performance Validation = 0.43319279054716364

Fold 2
Performance Training = 0.5227166554523927
Performance Validation = 0.3528997822690625

Fold 3
Performance Training = 0.5137404532945653
Performance Validation = 0.373174193483214

Fold 4
Performance Training = 0.4669186468362235
Performance Validation = 0.5879241898150336

Average Performance in Training = 0.5045889998791337
Average Performance in Validation = 0.4367977390286184

M Value = 2
Lambda Value = 0.9

Fold 1
Performance Training = 0.5149758394397916
Performance Validation = 0.43322331511431855

Fold 2
Performance Training = 0.5227125821757282
Performance Validation = 0.3529434339835582

Fold 3
Performance Training = 0.5137353822845163
Performance Validation = 0.3732146727938055

Fold 4
Performance Training = 0.4669149901121338
Performance Validation = 0.5878846425292513

Average Performance in Training = 0.5045846985030426
Average Performance in Validation = 0.4368165161052334

M Value = 2
Lambda Value = 0.91

Fold 1
Performance Training = 0.5149713900855797
Performance Validation = 0.43325375062454563

Fold 2
Performance Training = 0.5227084664000305
Performance Validation = 0.3529870157816247

Fold 3
Performance Training = 0.5137302595227048
Performance Validation = 0.3732550554943427

Fold 4
Performance Training = 0.4669112956248085
Performance Validation = 0.5878450774158656

Average Performance in Training = 0.5045803529082808
Average Performance in Validation = 0.43683522482909465

M Value = 2
Lambda Value = 0.92

Fold 1
Performance Training = 0.5149668959589642
Performance Validation = 0.4332840972519395

Fold 2
Performance Training = 0.5227043081882008
Performance Validation = 0.35303052774150345

Fold 3
Performance Training = 0.5137250851095116
Performance Validation = 0.3732953417475464

Fold 4
Performance Training = 0.4669075634379892
Performance Validation = 0.5878054945185205

Average Performance in Training = 0.5045759631736665
Average Performance in Validation = 0.43685386531487747

M Value = 2
Lambda Value = 0.93

Fold 1
Performance Training = 0.5149623571479535
Performance Validation = 0.4333143551701213

Fold 2
Performance Training = 0.5227001076030263
Performance Validation = 0.3530739699413219

Fold 3

Performance Training = 0.513719859145072
Performance Validation = 0.37333553171577794

Fold 4
Performance Training = 0.4669037936152761
Performance Validation = 0.5877658938807553

Average Performance in Training = 0.504571529377832
Average Performance in Validation = 0.43687243767699413

M Value = 2
Lambda Value = 0.9400000000000001

Fold 1
Performance Training = 0.5149577737403191
Performance Validation = 0.4333445245522396

Fold 2
Performance Training = 0.5226958647071817
Performance Validation = 0.3531173424590961

Fold 3
Performance Training = 0.5137145817292765
Performance Validation = 0.37337562556103887

Fold 4
Performance Training = 0.4668999862201286
Performance Validation = 0.5877262755460022

Average Performance in Training = 0.5045670515992264
Average Performance in Validation = 0.43689094202959416

M Value = 2
Lambda Value = 0.9500000000000001

Fold 1
Performance Training = 0.5149531458235967
Performance Validation = 0.43337460557097296

Fold 2
Performance Training = 0.522691579563229
Performance Validation = 0.3531606453727286

Fold 3
Performance Training = 0.5137092529617722
Performance Validation = 0.3734156234449729

Fold 4
Performance Training = 0.466896141315866
Performance Validation = 0.5876866395575886

Average Performance in Training = 0.5045625299161159
Average Performance in Validation = 0.4369093784865658

M Value = 2
Lambda Value = 0.96

Fold 1
Performance Training = 0.5149484734850869
Performance Validation = 0.43340459839853174

Fold 2
Performance Training = 0.5226872522336177
Performance Validation = 0.35320387876001125

Fold 3
Performance Training = 0.5137038729419626
Performance Validation = 0.3734555255288675

Fold 4
Performance Training = 0.46689225896566733
Performance Validation = 0.5876469859587363

Average Performance in Training = 0.5045579644065836
Average Performance in Validation = 0.4369277471615367

M Value = 2
Lambda Value = 0.97

Fold 1
Performance Training = 0.5149437568118553
Performance Validation = 0.43343450320665866

Fold 2
Performance Training = 0.5226828827806855
Performance Validation = 0.3532470426986227

Fold 3
Performance Training = 0.51369844176901
Performance Validation = 0.37349533197365326

Fold 4
Performance Training = 0.46688833923257245
Performance Validation = 0.5876073147925625

Average Performance in Training = 0.5045533551485308
Average Performance in Validation = 0.4369460481678743

M Value = 2
Lambda Value = 0.98

Fold 1
Performance Training = 0.5149389958907347
Performance Validation = 0.4334643201666323

Fold 2
Performance Training = 0.5226784712666586

Performance Validation = 0.35329013726613157

Fold 3

Performance Training = 0.5136929595418342

Performance Validation = 0.3735350429399067

Fold 4

Performance Training = 0.4668843821794818

Performance Validation = 0.5875676261020797

Average Performance in Training = 0.5045487022196773

Average Performance in Validation = 0.43696428161868756

M Value = 2

Lambda Value = 0.99

Fold 1

Performance Training = 0.5149341908083251

Performance Validation = 0.4334940494492666

Fold 2

Performance Training = 0.5226740177536504

Performance Validation = 0.3533331625399939

Fold 3

Performance Training = 0.5136874263591148

Performance Validation = 0.3735746585878502

Fold 4

Performance Training = 0.46688038786915786

Performance Validation = 0.5875279199301966

Average Performance in Training = 0.5045440056975621

Average Performance in Validation = 0.4369824476268268

M Value = 2

Lambda Value = 1.0

Fold 1

Performance Training = 0.5149293416509946

Performance Validation = 0.4335236912249145

Fold 2

Performance Training = 0.5226695223036646

Performance Validation = 0.35337611859755536

Fold 3

Performance Training = 0.5136818423192917

Performance Validation = 0.3736141790773533

Fold 4

Performance Training = 0.4668763563642243

Performance Validation = 0.5874881963197177

Average Performance in Training = 0.5045392656595438
Average Performance in Validation = 0.4370005463048853

M Value = 2
Lambda Value = 1.01

Fold 1
Performance Training = 0.5149244485048802
Performance Validation = 0.4335532456634682

Fold 2
Performance Training = 0.5226649849785929
Performance Validation = 0.35341900551605054

Fold 3
Performance Training = 0.5136762075205649
Performance Validation = 0.37365360456793406

Fold 4
Performance Training = 0.46687228772716716
Performance Validation = 0.587448455313344

Average Performance in Training = 0.5045344821828013
Average Performance in Validation = 0.43701857776519926

M Value = 2
Lambda Value = 1.02

Fold 1
Performance Training = 0.5149195114558891
Performance Validation = 0.43358271293436135

Fold 2
Performance Training = 0.5226604058402171
Performance Validation = 0.3534618233726031

Fold 3
Performance Training = 0.5136705220608968
Performance Validation = 0.37369293521875935

Fold 4
Performance Training = 0.4668681820203353
Performance Validation = 0.5874086969536735

Average Performance in Training = 0.5045296553443346
Average Performance in Validation = 0.4370365421198493

M Value = 2
Lambda Value = 1.03

Fold 1
Performance Training = 0.5149145305896994
Performance Validation = 0.4336120932065708

Fold 2

Performance Training = 0.5226557849502083

Performance Validation = 0.3535045722442276

Fold 3

Performance Training = 0.5136647860380117

Performance Validation = 0.37373217118864643

Fold 4

Performance Training = 0.46686403930594045

Performance Validation = 0.5873689212832012

Average Performance in Training = 0.5045247852209649

Average Performance in Validation = 0.4370544394806615

M Value = 2

Lambda Value = 1.04

Fold 1

Performance Training = 0.5149095059917601

Performance Validation = 0.43364138664861807

Fold 2

Performance Training = 0.5226511223701273

Performance Validation = 0.35354725220782723

Fold 3

Performance Training = 0.5136589995493972

Performance Validation = 0.3737713126360642

Fold 4

Performance Training = 0.4668598596460577

Performance Validation = 0.5873291283443196

Average Performance in Training = 0.5045198718893356

Average Performance in Validation = 0.4370722699592073

M Value = 2

Lambda Value = 1.05

Fold 1

Performance Training = 0.5149044377472933

Performance Validation = 0.43367059342857106

Fold 2

Performance Training = 0.5226464181614259

Performance Validation = 0.3535898633401957

Fold 3

Performance Training = 0.5136531626923043

Performance Validation = 0.37381035971913334

Fold 4

Performance Training = 0.4668556431026266
Performance Validation = 0.5872893181793193

Average Performance in Training = 0.5045149154259125
Average Performance in Validation = 0.4370900336668048

M Value = 2
Lambda Value = 1.06

Fold 1
Performance Training = 0.5148993259412937
Performance Validation = 0.43369971371404503

Fold 2
Performance Training = 0.522641672385445
Performance Validation = 0.35363240571801735

Fold 3
Performance Training = 0.5136472755637491
Performance Validation = 0.37384931259562826

Fold 4
Performance Training = 0.46685138973745033
Performance Validation = 0.587249490830388

Average Performance in Training = 0.5045099159069846
Average Performance in Validation = 0.43710773071451964

M Value = 2
Lambda Value = 1.07

Fold 1
Performance Training = 0.5148941706585308
Performance Validation = 0.4337287476722056

Fold 2
Performance Training = 0.5226368851034175
Performance Validation = 0.35367487941786735

Fold 3
Performance Training = 0.5136413382605127
Performance Validation = 0.37388817142297703

Fold 4
Performance Training = 0.46684709961219717
Performance Validation = 0.5872096463396135

Average Performance in Training = 0.5045048734086646
Average Performance in Validation = 0.43712536121316586

M Value = 2
Lambda Value = 1.08

Fold 1
Performance Training = 0.5148889719835483
Performance Validation = 0.4337576954697687

Fold 2
Performance Training = 0.5226320563764668
Performance Validation = 0.3537172845162109

Fold 3
Performance Training = 0.5136353508791427
Performance Validation = 0.3739269363582638

Fold 4
Performance Training = 0.4668427727884005
Performance Validation = 0.587169784748981

Average Performance in Training = 0.5044997880068895
Average Performance in Validation = 0.43714292527330606

M Value = 2
Lambda Value = 1.09

Fold 1
Performance Training = 0.5148837300006656
Performance Validation = 0.43378655727300386

Fold 2
Performance Training = 0.522627186265608
Performance Validation = 0.35375962108940506

Fold 3
Performance Training = 0.513629313515953
Performance Validation = 0.37396560755822794

Fold 4
Performance Training = 0.4668384093274591
Performance Validation = 0.587129906100375

Average Performance in Training = 0.5044946597774215
Average Performance in Validation = 0.437160423005253

M Value = 3
Lambda Value = 0.01

Fold 1
Performance Training = 0.801826453298675
Performance Validation = 0.6657112792433185

Fold 2
Performance Training = 0.7898242482096233
Performance Validation = 0.720269040457459

Fold 3
Performance Training = 0.7853899063708318

Performance Validation = 0.722688864226273

Fold 4

Performance Training = 0.7600759357734249

Performance Validation = 0.8351260844128408

Average Performance in Training = 0.7842791359131388

Average Performance in Validation = 0.7359488170849728

M Value = 3

Lambda Value = 0.02

Fold 1

Performance Training = 0.8018229332291034

Performance Validation = 0.6659797870953712

Fold 2

Performance Training = 0.7898205788052856

Performance Validation = 0.7204273391072044

Fold 3

Performance Training = 0.7853860649736232

Performance Validation = 0.722811783552145

Fold 4

Performance Training = 0.760072869875394

Performance Validation = 0.8349733686765083

Average Performance in Training = 0.7842756117208516

Average Performance in Validation = 0.7360480696078072

M Value = 3

Lambda Value = 0.03

Fold 1

Performance Training = 0.8018170916696403

Performance Validation = 0.6662447402479492

Fold 2

Performance Training = 0.789814490374378

Performance Validation = 0.7205816385454242

Fold 3

Performance Training = 0.7853796926915485

Performance Validation = 0.7229305031710347

Fold 4

Performance Training = 0.7600677804777614

Performance Validation = 0.8348195363719796

Average Performance in Training = 0.784269763803332

Average Performance in Validation = 0.736144104584097

M Value = 3
Lambda Value = 0.04

Fold 1

Performance Training = 0.8018089490509012
Performance Validation = 0.6665061665165353

Fold 2

Performance Training = 0.7898060049760485
Performance Validation = 0.7207319732922028

Fold 3

Performance Training = 0.7853708138376307
Performance Validation = 0.7230450630203438

Fold 4

Performance Training = 0.7600606841424907
Performance Validation = 0.8346645982122955

Average Performance in Training = 0.7842616130017678
Average Performance in Validation = 0.7362369502603443

M Value = 3
Lambda Value = 0.05

Fold 1

Performance Training = 0.8017985256237187
Performance Validation = 0.6667640934819787

Fold 2

Performance Training = 0.7897951444687881
Performance Validation = 0.7208783775590966

Fold 3

Performance Training = 0.7853594524896096
Performance Validation = 0.7231555026494789

Fold 4

Performance Training = 0.7600515972960951
Performance Validation = 0.834508564818689

Average Performance in Training = 0.7842511799695528
Average Performance in Validation = 0.7363266346273107

M Value = 3
Lambda Value = 0.06

Fold 1

Performance Training = 0.801785841460899
Performance Validation = 0.6670185484927338

Fold 2

Performance Training = 0.7897819305124563
Performance Validation = 0.7210208852522213

Fold 3

Performance Training = 0.7853456324924765

Performance Validation = 0.7232618612240465

Fold 4

Performance Training = 0.7600405362308666

Performance Validation = 0.8343514467214375

Average Performance in Training = 0.7842384851741746

Average Performance in Validation = 0.7364131854226097

M Value = 3

Lambda Value = 0.07

Fold 1

Performance Training = 0.8017709164589595

Performance Validation = 0.6672695586670849

Fold 2

Performance Training = 0.7897663845702823

Performance Validation = 0.7211595299753031

Fold 3

Performance Training = 0.785329377460976

Performance Validation = 0.7233641775299926

Fold 4

Performance Training = 0.7600275171060937

Performance Validation = 0.8341932543607065

Average Performance in Training = 0.7842235488990779

Average Performance in Validation = 0.7364966301332718

M Value = 3

Lambda Value = 0.08

Fold 1

Performance Training = 0.8017537703398463

Performance Validation = 0.6675171508953363

Fold 2

Performance Training = 0.7897485279108432

Performance Validation = 0.7212943450326939

Fold 3

Performance Training = 0.78531071078208

Performance Validation = 0.7234624899776956

Fold 4

Performance Training = 0.7600125559492649

Performance Validation = 0.8340339980873833

Average Performance in Training = 0.7842063912455086

Average Performance in Validation = 0.7365769959982773

M Value = 3

Lambda Value = 0.09

Fold 1

Performance Training = 0.8017344226526347

Performance Validation = 0.6677613518419865

Fold 2

Performance Training = 0.7897283816100209

Performance Validation = 0.7214253634323566

Fold 3

Performance Training = 0.78528965561743

Performance Validation = 0.723556836606007

Fold 4

Performance Training = 0.7599956686572624

Performance Validation = 0.8338736881639043

Average Performance in Training = 0.784187032134337

Average Performance in Validation = 0.7366543100110636

M Value = 3

Lambda Value = 0.1

Fold 1

Performance Training = 0.801712892775209

Performance Validation = 0.6680021879478737

Fold 2

Performance Training = 0.7897059665529365

Performance Validation = 0.7215526178888119

Fold 3

Performance Training = 0.785266234905751

Performance Validation = 0.7236472550862441

Fold 4

Performance Training = 0.7599768709975419

Performance Validation = 0.8337123347650717

Average Performance in Training = 0.7841654913078595

Average Performance in Validation = 0.7367285989220003

M Value = 3

Lambda Value = 0.11

Fold 1

Performance Training = 0.801689199915925

Performance Validation = 0.6682396854323025

Fold 2
Performance Training = 0.7896813034358621
Performance Validation = 0.7216761408260584

Fold 3
Performance Training = 0.7852404713652354
Performance Validation = 0.7237337827261352

Fold 4
Performance Training = 0.7599561786093006
Performance Validation = 0.8335499479788636

Average Performance in Training = 0.7841417883315808
Average Performance in Validation = 0.73679988924084

M Value = 3
Lambda Value = 0.12

Fold 1
Performance Training = 0.8016633631152552
Performance Validation = 0.6684738702951447

Fold 2
Performance Training = 0.7896544127681127
Performance Validation = 0.7217959643804542

Fold 3
Performance Training = 0.7852123874958994
Performance Validation = 0.7238164564737168

Fold 4
Performance Training = 0.7599336070046345
Performance Validation = 0.8333865378072346

Average Performance in Training = 0.7841159425959754
Average Performance in Validation = 0.7368682072391376

M Value = 3
Lambda Value = 0.13

Fold 1
Performance Training = 0.8016354012474146
Performance Validation = 0.6687047683189172

Fold 2
Performance Training = 0.7896253148739152
Performance Validation = 0.7219121204035672

Fold 3
Performance Training = 0.7851820055819103
Performance Validation = 0.7238953129211829

Fold 4
Performance Training = 0.7599091715696823

Performance Validation = 0.8332221141669072

Average Performance in Training = 0.7840879733182307

Average Performance in Validation = 0.7369335789526437

M Value = 3

Lambda Value = 0.14

Fold 1

Performance Training = 0.801605333021969

Performance Validation = 0.6689324050708414

Fold 2

Performance Training = 0.7895940298942574

Performance Validation = 0.7220246404649969

Fold 3

Performance Training = 0.785149347693886

Performance Validation = 0.723970388308691

Fold 4

Performance Training = 0.7598828875657591

Performance Validation = 0.8330566868901587

Average Performance in Training = 0.7840578995439678

Average Performance in Validation = 0.736996030183672

M Value = 3

Lambda Value = 0.15

Fold 1

Performance Training = 0.801573176985427

Performance Validation = 0.6691568059048763

Fold 2

Performance Training = 0.7895605777887161

Performance Validation = 0.7221335558551638

Fold 3

Performance Training = 0.7851144356911672

Performance Validation = 0.7240417185281179

Fold 4

Performance Training = 0.7598547701304783

Performance Validation = 0.8328902657255958

Average Performance in Training = 0.7840257401489472

Average Performance in Validation = 0.7370555865034384

M Value = 3

Lambda Value = 0.16

Fold 1

Performance Training = 0.8015389515228136
Performance Validation = 0.6693779959637303

Fold 2
Performance Training = 0.7895249783372646
Performance Validation = 0.722238897588062

Fold 3
Performance Training = 0.7850772912240621
Performance Validation = 0.7241093391267754

Fold 4
Performance Training = 0.7598248342788612
Performance Validation = 0.8327228603389242

Average Performance in Training = 0.7839915138407504
Average Performance in Validation = 0.7371122732543729

M Value = 3
Lambda Value = 0.17

Fold 1
Performance Training = 0.801502674859227
Performance Validation = 0.669596000180857

Fold 2
Performance Training = 0.7894872511420601
Performance Validation = 0.7223406964039909

Fold 3
Performance Training = 0.7850379357360651
Performance Validation = 0.7241732853110778

Fold 4
Performance Training = 0.7597930949044372
Performance Validation = 0.8325544803137084

Average Performance in Training = 0.7839552391604474
Average Performance in Validation = 0.7371661155524085

M Value = 3
Lambda Value = 0.18

Fold 1
Performance Training = 0.8014643650613793
Performance Validation = 0.6698108432824192

Fold 2
Performance Training = 0.7894474156292107
Performance Validation = 0.7224389827722479

Fold 3
Performance Training = 0.7849963904660481
Performance Validation = 0.7242335919501675

Fold 4

Performance Training = 0.7597595667803321

Performance Validation = 0.8323851351521259

Average Performance in Training = 0.7839169344842425

Average Performance in Validation = 0.7372171382892402

M Value = 3

Lambda Value = 0.19

Fold 1

Performance Training = 0.8014240400391186

Performance Validation = 0.6700225497892449

Fold 2

Performance Training = 0.7894054910505237

Performance Validation = 0.7225337868937965

Fold 3

Performance Training = 0.784952676450426

Performance Validation = 0.7242902935794966

Fold 4

Performance Training = 0.759724264560344

Performance Validation = 0.8322148342757116

Average Performance in Training = 0.7838766180251031

Average Performance in Validation = 0.7372653661345624

M Value = 3

Lambda Value = 0.2

Fold 1

Performance Training = 0.8013817175469377

Performance Validation = 0.6702311440187493

Fold 2

Performance Training = 0.7893614964852331

Performance Validation = 0.7226251387039029

Fold 3

Performance Training = 0.784906814525297

Performance Validation = 0.724343424404367

Fold 4

Performance Training = 0.7596872027800108

Performance Validation = 0.8320435870260959

Average Performance in Training = 0.7838343078343697

Average Performance in Validation = 0.7373108235382787

M Value = 3

Lambda Value = 0.21

Fold 1

Performance Training = 0.8013374151854632

Performance Validation = 0.6704366500868482

Fold 2

Performance Training = 0.7893154508417097

Performance Validation = 0.7227130678747427

Fold 3

Performance Training = 0.7848588253285567

Performance Validation = 0.7243930183034262

Fold 4

Performance Training = 0.759648395857666

Performance Validation = 0.8318714026657342

Average Performance in Training = 0.7837900218033489

Average Performance in Validation = 0.7373535347326878

M Value = 3

Lambda Value = 0.22

Fold 1

Performance Training = 0.8012911504029313

Performance Validation = 0.6706390919098415

Fold 2

Performance Training = 0.7892673728591498

Performance Validation = 0.7227976038179817

Fold 3

Performance Training = 0.7848087293019878

Performance Validation = 0.724439108832124

Fold 4

Performance Training = 0.7596078580954834

Performance Validation = 0.8316982903786312

Average Performance in Training = 0.7837437776648881

Average Performance in Validation = 0.7373935237346446

M Value = 3

Lambda Value = 0.23

Fold 1

Performance Training = 0.8012429404966461

Performance Validation = 0.6708384932062829

Fold 2

Performance Training = 0.7892172811092475

Performance Validation = 0.7228787756873234

Fold 3
Performance Training = 0.7847565466933246
Performance Validation = 0.7244817292261256

Fold 4
Performance Training = 0.7595656036805118
Performance Validation = 0.8315242592710546

Average Performance in Training = 0.7836955929949325
Average Performance in Validation = 0.7374308143476966

M Value = 3
Lambda Value = 0.24

Fold 1
Performance Training = 0.8011928026144226
Performance Validation = 0.6710348774988284

Fold 2
Performance Training = 0.7891651939978466
Performance Validation = 0.7229566123810347

Fold 3
Performance Training = 0.7847022975582937
Performance Validation = 0.7245209124046876

Fold 4
Performance Training = 0.7595216466856995
Performance Validation = 0.8313493183722444

Average Performance in Training = 0.7836454852140656
Average Performance in Validation = 0.7374654301641987

M Value = 3
Lambda Value = 0.25

Fold 1
Performance Training = 0.8011407537560151
Performance Validation = 0.6712282681160628

Fold 2
Performance Training = 0.7891111297665762
Performance Validation = 0.723031142544438

Fold 3
Performance Training = 0.7846460017626298
Performance Validation = 0.7245566909739911

Fold 4
Performance Training = 0.7594760010709085
Performance Validation = 0.8311734766351133

Average Performance in Training = 0.7835934715890324
Average Performance in Validation = 0.7374973945674013

M Value = 3
Lambda Value = 0.26

Fold 1
Performance Training = 0.8010868107745288
Performance Validation = 0.6714186881943103

Fold 2
Performance Training = 0.7890551064944662
Performance Validation = 0.7231023945723783

Fold 3
Performance Training = 0.7845876789840681
Performance Validation = 0.7245890972304367

Fold 4
Performance Training = 0.7594286806839188
Performance Validation = 0.830996742936942

Average Performance in Training = 0.7835395692342455
Average Performance in Validation = 0.7375267307335169

M Value = 3
Lambda Value = 0.27

Fold 1
Performance Training = 0.8010309903778171
Performance Validation = 0.6716061606794235

Fold 2
Performance Training = 0.7889971420995459
Performance Validation = 0.723170396611666

Fold 3
Performance Training = 0.7845273487143138
Performance Validation = 0.7246181631639002

Fold 4
Performance Training = 0.7593796992614215
Performance Validation = 0.8308191260800641

Average Performance in Training = 0.7834837951132745
Average Performance in Validation = 0.7375534616337635

M Value = 3
Lambda Value = 0.28

Fold 1
Performance Training = 0.8009733091298645
Performance Validation = 0.6717907083285539

Fold 2

Performance Training = 0.7889372543404259
Performance Validation = 0.7232351765634857

Fold 3
Performance Training = 0.784465030260988
Performance Validation = 0.724643920460951

Fold 4
Performance Training = 0.7593290704300033
Performance Validation = 0.8306406347925482

Average Performance in Training = 0.7834261660403204
Average Performance in Validation = 0.7375776100363847

M Value = 3
Lambda Value = 0.29

Fold 1
Performance Training = 0.8009137834521539
Performance Validation = 0.6719723537119051

Fold 2
Performance Training = 0.7888754608178604
Performance Validation = 0.7232967620857874

Fold 3
Performance Training = 0.7844007427495501
Performance Validation = 0.724666400508031

Fold 4
Performance Training = 0.759276807707122
Performance Validation = 0.8304612777288691

Average Performance in Training = 0.7833666986816715
Average Performance in Validation = 0.7375991985086481

M Value = 3
Lambda Value = 0.3

Fold 1
Performance Training = 0.8008524296250192
Performance Validation = 0.6721511192144698

Fold 2
Performance Training = 0.7888117789762948
Performance Validation = 0.7233551805956462

Fold 3
Performance Training = 0.7843345051251989
Performance Validation = 0.7246856343945975

Fold 4
Performance Training = 0.7592229245020694
Performance Validation = 0.8302810634705757

Average Performance in Training = 0.7833054095571456
Average Performance in Validation = 0.7376182494188224

M Value = 3
Lambda Value = 0.31

Fold 1
Performance Training = 0.8007892637889849
Performance Validation = 0.6723270270377383

Fold 2
Performance Training = 0.7887462261053944
Performance Validation = 0.7234104592715942

Fold 3
Performance Training = 0.7842663361547505
Performance Validation = 0.7247016529162289

Fold 4
Performance Training = 0.7591674341169279
Performance Validation = 0.8301000005269508

Average Performance in Training = 0.7832423150415144
Average Performance in Validation = 0.737634784938128

M Value = 3
Lambda Value = 0.32

Fold 1
Performance Training = 0.800724301946089
Performance Validation = 0.672500099201403

Fold 2
Performance Training = 0.7886788193415569
Performance Validation = 0.723462625055935

Fold 3
Performance Training = 0.7841962544284953
Performance Validation = 0.7247144865776939

Fold 4
Performance Training = 0.7591103497475146
Performance Validation = 0.829918097335662

Average Performance in Training = 0.783177431365914
Average Performance in Validation = 0.7376488270426734

M Value = 3
Lambda Value = 0.33

Fold 1
Performance Training = 0.8006575599611948

Performance Validation = 0.6726703575450368

Fold 2

Performance Training = 0.7886095756694093

Performance Validation = 0.7235117046570243

Fold 3

Performance Training = 0.7841242783620316

Performance Validation = 0.7247241655959833

Fold 4

Performance Training = 0.7590516844843195

Performance Validation = 0.8297353622634112

Average Performance in Training = 0.7831107746192387

Average Performance in Validation = 0.7376603975153638

M Value = 3

Lambda Value = 0.34

Fold 1

Performance Training = 0.8005890535632864

Performance Validation = 0.6728378237297528

Fold 2

Performance Training = 0.788538511923287

Performance Validation = 0.7235577245515322

Fold 3

Performance Training = 0.7840504261980801

Performance Validation = 0.7247307199033104

Fold 4

Performance Training = 0.7589914513134312

Performance Validation = 0.8295518036065729

Average Performance in Training = 0.7830423607495212

Average Performance in Validation = 0.737669517947792

M Value = 3

Lambda Value = 0.35000000000000003

Fold 1

Performance Training = 0.8005187983467518

Performance Validation = 0.6730025192398531

Fold 2

Performance Training = 0.7884656447886977

Performance Validation = 0.7236007109866758

Fold 3

Performance Training = 0.7839747160082755

Performance Validation = 0.7247341791500721

Fold 4
Performance Training = 0.7589296631174551
Performance Validation = 0.8293674295918274

Average Performance in Training = 0.782972205565295
Average Performance in Validation = 0.737676209742107

M Value = 3
Lambda Value = 0.36

Fold 1
Performance Training = 0.8004468097726515
Performance Validation = 0.6731644653844546

Fold 2
Performance Training = 0.7883909908037686
Performance Validation = 0.7236406899824321

Fold 3
Performance Training = 0.783897165694937
Performance Validation = 0.7247345727077774

Fold 4
Performance Training = 0.758866332676423
Performance Validation = 0.8291822483767906

Average Performance in Training = 0.782900324736945
Average Performance in Validation = 0.7376804941128636

M Value = 3
Lambda Value = 0.37

Fold 1
Performance Training = 0.8003731031699749
Performance Validation = 0.6733236832991008

Fold 2
Performance Training = 0.7883145663606791
Performance Validation = 0.7236776873337236

Fold 3
Performance Training = 0.783817792992821
Performance Validation = 0.7247319296719429

Fold 4
Performance Training = 0.758801472668692
Performance Validation = 0.8289962680506315

Average Performance in Training = 0.7828267337980417
Average Performance in Validation = 0.7376823920888497

M Value = 3
Lambda Value = 0.38

Fold 1
Performance Training = 0.8002976937368824
Performance Validation = 0.6734801939473554

Fold 2
Performance Training = 0.7882363877070758
Performance Validation = 0.7237117286125825

Fold 3
Performance Training = 0.7837366154708497
Performance Validation = 0.7247262788649513

Fold 4
Performance Training = 0.7587350956718361
Performance Validation = 0.8288094966346902

Average Performance in Training = 0.782751448146661
Average Performance in Validation = 0.7376819245148949

M Value = 3
Lambda Value = 0.39

Fold 1
Performance Training = 0.8002205965419349
Performance Validation = 0.6736340181223817

Fold 2
Performance Training = 0.7881564709474749
Performance Validation = 0.7237428391702895

Fold 3
Performance Training = 0.7836536505338231
Performance Validation = 0.7247176488388813

Fold 4
Performance Training = 0.7586672141635289
Performance Validation = 0.8286219420830859

Average Performance in Training = 0.7826744830466905
Average Performance in Validation = 0.7376791120536595

M Value = 3
Lambda Value = 0.4

Fold 1
Performance Training = 0.8001418265253096
Performance Validation = 0.6737851764485014

Fold 2
Performance Training = 0.7880748320446469
Performance Validation = 0.7237710441394908

Fold 3

Performance Training = 0.7835689154241089
Performance Validation = 0.7247060678783004

Fold 4
Performance Training = 0.7585978405224161
Performance Validation = 0.8284336122833167

Average Performance in Training = 0.7825958536291203
Average Performance in Validation = 0.7376739751874024

M Value = 3
Lambda Value = 0.41000000000000003

Fold 1
Performance Training = 0.8000613985000049
Performance Validation = 0.6739336893827417

Fold 2
Performance Training = 0.7879914868209873
Performance Validation = 0.7237963684362934

Fold 3
Performance Training = 0.7834824272233138
Performance Validation = 0.7246915640030285

Fold 4
Performance Training = 0.7585269870289826
Performance Validation = 0.8282445150568616

Average Performance in Training = 0.7825155748933221
Average Performance in Validation = 0.7376665342197313

M Value = 3
Lambda Value = 0.42

Fold 1
Performance Training = 0.7999793271530301
Performance Validation = 0.674079577216364

Fold 2
Performance Training = 0.7879064509598732
Performance Validation = 0.7238188367623346

Fold 3
Performance Training = 0.7833942028539352
Performance Validation = 0.7246741649708676

Fold 4
Performance Training = 0.758454665866407
Performance Validation = 0.8280546581597663

Average Performance in Training = 0.7824336617083114
Average Performance in Validation = 0.737656809277333

M Value = 3
Lambda Value = 0.43

Fold 1

Performance Training = 0.7998956270465848
Performance Validation = 0.6742228600763763

Fold 2

Performance Training = 0.7878197400070035
Performance Validation = 0.7238384736068323

Fold 3

Performance Training = 0.7833042590809951
Performance Validation = 0.724653898280301

Fold 4

Performance Training = 0.7583808891214112
Performance Validation = 0.8278640492832308

Average Performance in Training = 0.7823501288139987
Average Performance in Validation = 0.7376448203116851

M Value = 3
Lambda Value = 0.44

Fold 1

Performance Training = 0.7998103126192245
Performance Validation = 0.6743635579270324

Fold 2

Performance Training = 0.7877313693717268
Performance Validation = 0.7238553032486124

Fold 3

Performance Training = 0.7832126125136528
Performance Validation = 0.724630791173162

Fold 4

Performance Training = 0.7583056687851001
Performance Validation = 0.8276726960541881

Average Performance in Training = 0.7822649908224261
Average Performance in Validation = 0.7376305871007487

M Value = 3
Lambda Value = 0.45

Fold 1

Performance Training = 0.7997233981870152
Performance Validation = 0.6745016905713135

Fold 2

Performance Training = 0.7876413543283524

Performance Validation = 0.7238693497581132

Fold 3

Performance Training = 0.7831192796068023

Performance Validation = 0.7246048706372694

Fold 4

Performance Training = 0.7582290167537935

Performance Validation = 0.8274806060358769

Average Performance in Training = 0.7821782622189909

Average Performance in Validation = 0.7376141292506433

M Value = 3

Lambda Value = 0.46

Fold 1

Performance Training = 0.7996348979446743

Performance Validation = 0.6746372776523955

Fold 2

Performance Training = 0.7875497100174501

Performance Validation = 0.7238806369993702

Fold 3

Performance Training = 0.7830242766626476

Performance Validation = 0.7245761634090353

Fold 4

Performance Training = 0.75815094482985

Performance Validation = 0.8272877867284107

Average Performance in Training = 0.7820899573636555

Average Performance in Validation = 0.737595466197303

M Value = 3

Lambda Value = 0.47000000000000003

Fold 1

Performance Training = 0.7995448259667022

Performance Validation = 0.6747703386550995

Fold 2

Performance Training = 0.7874564514471334

Performance Validation = 0.7238891886319787

Fold 3

Performance Training = 0.7829276198322637

Performance Validation = 0.724544695976042

Fold 4

Performance Training = 0.758071464722482

Performance Validation = 0.8270942455693384

Average Performance in Training = 0.7820000904921454
Average Performance in Validation = 0.7375746172081147

M Value = 3
Lambda Value = 0.48

Fold 1
Performance Training = 0.7994531962084985
Performance Validation = 0.6749008929073317

Fold 2
Performance Training = 0.7873615934943303
Performance Validation = 0.7238950281130362

Fold 3
Performance Training = 0.7828293251171371
Performance Validation = 0.7245104945795902

Fold 4
Performance Training = 0.7579905880485639
Performance Validation = 0.8268999899342008

Average Performance in Training = 0.7819086757171325
Average Performance in Validation = 0.7375516013835397

M Value = 3
Lambda Value = 0.49

Fold 1
Performance Training = 0.7993600225074704
Performance Validation = 0.6750289595815038

Fold 2
Performance Training = 0.7872651509060409
Performance Validation = 0.7238981786990607

Fold 3
Performance Training = 0.7827294083706897
Performance Validation = 0.7244735852172148

Fold 4
Performance Training = 0.7579083263334321
Performance Validation = 0.8267050271370833

Average Performance in Training = 0.7818157270294082
Average Performance in Validation = 0.7375264376587156

M Value = 3
Lambda Value = 0.5

Fold 1
Performance Training = 0.799265318584127
Performance Validation = 0.6751545576959394

Fold 2
Performance Training = 0.78716713830058
Performance Validation = 0.7238986634478957

Fold 3
Performance Training = 0.7826278852997866
Performance Validation = 0.7244339936451774

Fold 4
Performance Training = 0.7578246910116777
Performance Validation = 0.826509364431161

Average Performance in Training = 0.7817212582990427
Average Performance in Validation = 0.7374991448050433

M Value = 3
Lambda Value = 0.51

Fold 1
Performance Training = 0.7991690980431628
Performance Validation = 0.6752777061162715

Fold 2
Performance Training = 0.7870675701688074
Performance Validation = 0.7238965052205876

Fold 3
Performance Training = 0.7825247714662225
Performance Validation = 0.7243917453809248

Fold 4
Performance Training = 0.7577396934279295
Performance Validation = 0.8263130090092374

Average Performance in Training = 0.7816252832765305
Average Performance in Validation = 0.7374697414317553

M Value = 3
Lambda Value = 0.52

Fold 1
Performance Training = 0.7990713743745301
Performance Validation = 0.6753984235568171

Fold 2
Performance Training = 0.7869664608753452
Performance Validation = 0.7238917266832465

Fold 3
Performance Training = 0.7824200822881971
Performance Validation = 0.724346865705523

Fold 4

Performance Training = 0.7576533448376328
Performance Validation = 0.8261159680042823

Average Performance in Training = 0.7815278155939263
Average Performance in Validation = 0.7374382459874672

M Value = 3
Lambda Value = 0.53

Fold 1
Performance Training = 0.7989721609545019
Performance Validation = 0.675516728581945

Fold 2
Performance Training = 0.7868638246597817
Performance Validation = 0.7238843503088884

Fold 3
Performance Training = 0.7823138330417698
Performance Validation = 0.7242993796660634

Fold 4
Performance Training = 0.757565656407817
Performance Validation = 0.8259182484899582

Average Performance in Training = 0.7814288687659675
Average Performance in Validation = 0.7374046767617137

M Value = 3
Lambda Value = 0.54

Fold 1
Performance Training = 0.7988714710467195
Performance Validation = 0.6756326396074266

Fold 2
Performance Training = 0.7867596756378631
Performance Validation = 0.7238743983792542

Fold 3
Performance Training = 0.7822060388622976
Performance Validation = 0.7242493120780391

Fold 4
Performance Training = 0.7574766392178596
Performance Validation = 0.8257198574811486

Average Performance in Training = 0.7813284561911849
Average Performance in Validation = 0.7373690518864671

M Value = 3
Lambda Value = 0.55

Fold 1
Performance Training = 0.7987693178032346
Performance Validation = 0.6757461749017717

Fold 2
Performance Training = 0.7866540278026701
Performance Validation = 0.7238618929866132

Fold 3
Performance Training = 0.7820967147458595
Performance Validation = 0.7241966875276964

Fold 4
Performance Training = 0.7573863042602387
Performance Validation = 0.8255208019344737

Average Performance in Training = 0.7812265911530007
Average Performance in Validation = 0.7373313893376388

M Value = 3
Lambda Value = 0.56

Fold 1
Performance Training = 0.798665714265537
Performance Validation = 0.6758573525875542

Fold 2
Performance Training = 0.7865468950257867
Performance Validation = 0.7238468560355455

Fold 3
Performance Training = 0.781985875550663
Performance Validation = 0.7241415303743592

Fold 4
Performance Training = 0.7572946624412815
Performance Validation = 0.825321088748807

Average Performance in Training = 0.7811232868208171
Average Performance in Validation = 0.7372917069365665

M Value = 3
Lambda Value = 0.5700000000000001

Fold 1
Performance Training = 0.7985606733655727
Performance Validation = 0.6759661906427213

Fold 2
Performance Training = 0.7864382910584513
Performance Validation = 0.7238293092447058

Fold 3
Performance Training = 0.7818735359984338

Performance Validation = 0.7240838647527275

Fold 4

Performance Training = 0.7572017245819048

Performance Validation = 0.8251207247657842

Average Performance in Training = 0.7810185562510906

Average Performance in Validation = 0.7372500223514847

M Value = 3

Lambda Value = 0.58

Fold 1

Performance Training = 0.7984542079267518

Performance Validation = 0.6760727069018901

Fold 2

Performance Training = 0.7863282295326997

Performance Validation = 0.7238092741485695

Fold 3

Performance Training = 0.781759710675793

Performance Validation = 0.7240237145751478

Fold 4

Performance Training = 0.7571075014183477

Performance Validation = 0.8249197167703084

Average Performance in Training = 0.780912412388398

Average Performance in Validation = 0.737206353098979

M Value = 3

Lambda Value = 0.59

Fold 1

Performance Training = 0.7983463306649459

Performance Validation = 0.6761769190576343

Fold 2

Performance Training = 0.7862167239624931

Performance Validation = 0.7237867720991598

Fold 3

Performance Training = 0.7816444140356145

Performance Validation = 0.7239611035338611

Fold 4

Performance Training = 0.757012003602897

Performance Validation = 0.8247180714910473

Average Performance in Training = 0.7808048680664876

Average Performance in Validation = 0.7371607165454256

M Value = 3
Lambda Value = 0.6

Fold 1

Performance Training = 0.7982370541894754
Performance Validation = 0.6762788446617505

Fold 2

Performance Training = 0.7861037877448365
Performance Validation = 0.7237618242677548

Fold 3

Performance Training = 0.7815276603983713
Performance Validation = 0.7238960551032226

Fold 4

Performance Training = 0.7569152417046076
Performance Validation = 0.8245157956009298

Average Performance in Training = 0.7806959360093227
Average Performance in Validation = 0.7371131299084145

M Value = 3
Lambda Value = 0.61

Fold 1

Performance Training = 0.7981263910040859
Performance Validation = 0.6763785011265206

Fold 2

Performance Training = 0.7859894341608837
Performance Validation = 0.7237344516465809

Fold 3

Performance Training = 0.7814094639534638
Performance Validation = 0.7238285925418978

Fold 4

Performance Training = 0.7568172262100132
Performance Validation = 0.8243128957176357

Average Performance in Training = 0.7805856288321117
Average Performance in Validation = 0.7370636102581587

M Value = 3
Lambda Value = 0.62

Fold 1

Performance Training = 0.7980143535079155
Performance Validation = 0.6764759057259544

Fold 2

Performance Training = 0.785873676377031
Performance Validation = 0.7237046750504832

Fold 3
Performance Training = 0.7812898387605341
Performance Validation = 0.7237587388950346

Fold 4
Performance Training = 0.7567179675238334
Performance Validation = 0.8241093784040789

Average Performance in Training = 0.7804739590423284
Average Performance in Validation = 0.7370121745188878

M Value = 3
Lambda Value = 0.63

Fold 1
Performance Training = 0.7979009539964523
Performance Validation = 0.6765710755970238

Fold 2
Performance Training = 0.7857565274459999
Performance Validation = 0.7236725151185867

Fold 3
Performance Training = 0.7811687987507657
Performance Validation = 0.7236865169964088

Fold 4
Performance Training = 0.7566174759696701
Performance Validation = 0.82390525016889

Average Performance in Training = 0.780360939040722
Average Performance in Validation = 0.7369588394702273

M Value = 3
Lambda Value = 0.64

Fold 1
Performance Training = 0.797786204662481
Performance Validation = 0.6766640277408833

Fold 2
Performance Training = 0.7856380003079074
Performance Validation = 0.7236379923159266

Fold 3
Performance Training = 0.7810463577281701
Performance Validation = 0.7236119494705487

Fold 4
Performance Training = 0.7565157617907021
Performance Validation = 0.8237005174668907

Average Performance in Training = 0.7802465811223152

Average Performance in Validation = 0.7369036217485624

M Value = 3

Lambda Value = 0.65

Fold 1

Performance Training = 0.7976701175970208

Performance Validation = 0.6767547790240767

Fold 2

Performance Training = 0.7855181077913269

Performance Validation = 0.7236011269350755

Fold 3

Performance Training = 0.7809225293708545

Performance Validation = 0.7235350587348309

Fold 4

Performance Training = 0.7564128351503685

Performance Validation = 0.8234951866995658

Average Performance in Training = 0.7801308974773927

Average Performance in Validation = 0.7368465378483873

M Value = 3

Lambda Value = 0.66

Fold 1

Performance Training = 0.7975527047902536

Performance Validation = 0.6768433461797351

Fold 2

Performance Training = 0.7853968626143351

Performance Validation = 0.723561939097747

Fold 3

Performance Training = 0.7807973272322808

Performance Validation = 0.7234558670015592

Fold 4

Performance Training = 0.7563087061330481

Performance Validation = 0.8232892642155273

Average Performance in Training = 0.7800139001924794

Average Performance in Validation = 0.7367876041236422

M Value = 3

Lambda Value = 0.67

Fold 1

Performance Training = 0.7974339781324422

Performance Validation = 0.6769297458087588

Fold 2
Performance Training = 0.7852742773855503
Performance Validation = 0.7235204487563812

Fold 3
Performance Training = 0.7806707647425072
Performance Validation = 0.7233743962800147

Fold 4
Performance Training = 0.7562033847447319
Performance Validation = 0.8230827563109795

Average Performance in Training = 0.7798956012513079
Average Performance in Validation = 0.7367268367890335

M Value = 3
Lambda Value = 0.68

Fold 1
Performance Training = 0.7973139494148402
Performance Validation = 0.6770139943809912

Fold 2
Performance Training = 0.78515036460516
Performance Validation = 0.723476675695716

Fold 3
Performance Training = 0.780542855209417
Performance Validation = 0.7232906683784854

Fold 4
Performance Training = 0.7560968809136897
Performance Validation = 0.8228756692301759

Average Performance in Training = 0.7797760125357767
Average Performance in Validation = 0.7366642519213422

M Value = 3
Lambda Value = 0.6900000000000001

Fold 1
Performance Training = 0.7971926303305914
Performance Validation = 0.6770961082363793

Fold 2
Performance Training = 0.7850251366659345
Performance Validation = 0.7234306395343444

Fold 3
Performance Training = 0.780413611819934
Performance Validation = 0.7232047049062742

Fold 4
Performance Training = 0.7559892044911284

Performance Validation = 0.8226680091658701

Average Performance in Training = 0.7796551458268971

Average Performance in Validation = 0.736599865460717

M Value = 3

Lambda Value = 0.7000000000000001

Fold 1

Performance Training = 0.7970700324756208

Performance Validation = 0.677176103586121

Fold 2

Performance Training = 0.7848986058542347

Performance Validation = 0.7233823597262496

Fold 3

Performance Training = 0.7802830476412229

Performance Validation = 0.7231165272756808

Fold 4

Performance Training = 0.7558803652518468

Performance Validation = 0.8224597822597668

Average Performance in Training = 0.7795330128057313

Average Performance in Validation = 0.7365336932119546

M Value = 3

Lambda Value = 0.71

Fold 1

Performance Training = 0.7969461673495171

Performance Validation = 0.6772539965138025

Fold 2

Performance Training = 0.7847707843510054

Performance Validation = 0.7233318555623298

Fold 3

Performance Training = 0.7801511756218787

Performance Validation = 0.7230261567039664

Fold 4

Performance Training = 0.7557703728948817

Performance Validation = 0.8222509946029656

Average Performance in Training = 0.7794096250543208

Average Performance in Validation = 0.7364657508457662

M Value = 3

Lambda Value = 0.72

Fold 1

Performance Training = 0.7968210463564045
Performance Validation = 0.6773298029765271

Fold 2
Performance Training = 0.7846416842327603
Performance Validation = 0.7232791461719057

Fold 3
Performance Training = 0.7800180085931016
Performance Validation = 0.7229336142152929

Fold 4
Performance Training = 0.7556592370441503
Performance Validation = 0.8220416522364012

Average Performance in Training = 0.7792849940566041
Average Performance in Validation = 0.7363960539000318

M Value = 3
Lambda Value = 0.73

Fold 1
Performance Training = 0.7966946808058079
Performance Validation = 0.6774035388060273

Fold 2
Performance Training = 0.7845113174725555
Performance Validation = 0.7232242505242076

Fold 3
Performance Training = 0.7798835592698583
Performance Validation = 0.7228389206426424

Fold 4
Performance Training = 0.755546967249084
Performance Validation = 0.821831761151279

Average Performance in Training = 0.7791591311993264
Average Performance in Validation = 0.7363246177810391

M Value = 3
Lambda Value = 0.74

Fold 1
Performance Training = 0.7965670819135074
Performance Validation = 0.6774752197097688

Fold 2
Performance Training = 0.7843796959409546
Performance Validation = 0.7231671874298564

Fold 3
Performance Training = 0.779747840252032
Performance Validation = 0.7227420966297138

Fold 4

Performance Training = 0.7554335729852582

Performance Validation = 0.8216213272895077

Average Performance in Training = 0.779032047772938

Average Performance in Validation = 0.7362514577647117

M Value = 3

Lambda Value = 0.75

Fold 1

Performance Training = 0.796438260802386

Performance Validation = 0.6775448612720444

Fold 2

Performance Training = 0.7842468314069818

Performance Validation = 0.7231079755423215

Fold 3

Performance Training = 0.779610864025558

Performance Validation = 0.7226431626328015

Fold 4

Performance Training = 0.7553190636550154

Performance Validation = 0.8214103565441263

Average Performance in Training = 0.7789037549724853

Average Performance in Validation = 0.7361765889978235

M Value = 3

Lambda Value = 0.76

Fold 1

Performance Training = 0.7963082285032668

Performance Validation = 0.6776124789550548

Fold 2

Performance Training = 0.7841127355390667

Performance Validation = 0.7230466333593661

Fold 3

Performance Training = 0.7794726429635487

Performance Validation = 0.7225421389226492

Fold 4

Performance Training = 0.7552034485880822

Performance Validation = 0.8211988547597278

Average Performance in Training = 0.778774263898491

Average Performance in Validation = 0.7361000264991995

M Value = 3

Lambda Value = 0.77

Fold 1

Performance Training = 0.7961769959557427

Performance Validation = 0.6776780880999809

Fold 2

Performance Training = 0.783977419905978

Performance Validation = 0.7229831792244775

Fold 3

Performance Training = 0.7793331893274048

Performance Validation = 0.7224390455862868

Fold 4

Performance Training = 0.7550867370421804

Performance Validation = 0.8209868277328781

Average Performance in Training = 0.7786435855578265

Average Performance in Validation = 0.7360217851609059

M Value = 3

Lambda Value = 0.78

Fold 1

Performance Training = 0.7960445740089996

Performance Validation = 0.6777417039280442

Fold 2

Performance Training = 0.7838408959777491

Performance Validation = 0.7229176313282855

Fold 3

Performance Training = 0.7791925152679146

Performance Validation = 0.7223339025288453

Fold 4

Performance Training = 0.7549689382036344

Performance Validation = 0.8207742812125333

Average Performance in Training = 0.7785117308645744

Average Performance in Validation = 0.735941879749427

M Value = 3

Lambda Value = 0.79

Fold 1

Performance Training = 0.7959109734226273

Performance Validation = 0.6778033415415561

Fold 2

Performance Training = 0.7837031751265924

Performance Validation = 0.7228500077099618

Fold 3
Performance Training = 0.7790506328263415
Performance Validation = 0.7222267294753523

Fold 4
Performance Training = 0.7548500611879693
Performance Validation = 0.8205612209004495

Average Performance in Training = 0.7783787106408826
Average Performance in Validation = 0.73586032490683

M Value = 3
Lambda Value = 0.8

Fold 1
Performance Training = 0.7957762048674257
Performance Validation = 0.677863015924958

Fold 2
Performance Training = 0.7835642686278044
Performance Validation = 0.7227803262586052

Fold 3
Performance Training = 0.7789075539354986
Performance Validation = 0.7221175459725058

Fold 4
Performance Training = 0.7547301150405057
Performance Validation = 0.8203476524515908

Average Performance in Training = 0.7782445356178085
Average Performance in Validation = 0.735777135151915

M Value = 3
Lambda Value = 0.81

Fold 1
Performance Training = 0.7956402789262018
Performance Validation = 0.6779207419458513

Fold 2
Performance Training = 0.783424187660664
Performance Validation = 0.722708604714619

Fold 3
Performance Training = 0.7787632904208146
Performance Validation = 0.7220063713904328

Fold 4
Performance Training = 0.7546091087369498
Performance Validation = 0.8201335814745333

Average Performance in Training = 0.7781092164361576
Average Performance in Validation = 0.735692324881359

M Value = 3
Lambda Value = 0.8200000000000001

Fold 1
Performance Training = 0.7955032060945586
Performance Validation = 0.6779765343560182

Fold 2
Performance Training = 0.7832829433093182
Performance Validation = 0.7226348606710654

Fold 3
Performance Training = 0.7786178540013825
Performance Validation = 0.7218932249244252

Fold 4
Performance Training = 0.7544870511839751
Performance Validation = 0.819919013531865

Average Performance in Training = 0.7779727636473086
Average Performance in Validation = 0.7356059083708435

M Value = 3
Lambda Value = 0.8300000000000001

Fold 1
Performance Training = 0.7953649967816753
Performance Validation = 0.6780304077924256

Fold 2
Performance Training = 0.7831405465636597
Performance Validation = 0.7225591115750127

Fold 3
Performance Training = 0.7784712562910031
Performance Validation = 0.7217781255966558

Fold 4
Performance Training = 0.7543639512198013
Performance Validation = 0.8197039541405823

Average Performance in Training = 0.7778351877140348
Average Performance in Validation = 0.7355178997761691

M Value = 3
Lambda Value = 0.84

Fold 1
Performance Training = 0.795225661311082
Performance Validation = 0.6780823767782316

Fold 2

Performance Training = 0.7829970083201973
Performance Validation = 0.7224813747288651

Fold 3
Performance Training = 0.7783235087992129
Performance Validation = 0.7216610922578806

Fold 4
Performance Training = 0.7542398176147664
Performance Validation = 0.8194884087724806

Average Performance in Training = 0.7776964990113147
Average Performance in Validation = 0.7354283131343644

M Value = 3
Lambda Value = 0.85

Fold 1
Performance Training = 0.7950852099214246
Performance Validation = 0.6781324557237713

Fold 2
Performance Training = 0.7828523393829143
Performance Validation = 0.7224016672916813

Fold 3
Performance Training = 0.778174622932301
Performance Validation = 0.721542143589115

Fold 4
Performance Training = 0.7541146590718941
Performance Validation = 0.8192723828545456

Average Performance in Training = 0.7775567078271335
Average Performance in Validation = 0.7353371623647783

M Value = 3
Lambda Value = 0.86

Fold 1
Performance Training = 0.7949436527672225
Performance Validation = 0.6781806589275361

Fold 2
Performance Training = 0.7827065504641206
Performance Validation = 0.7223200062804782

Fold 3
Performance Training = 0.7780246099943187
Performance Validation = 0.7214212981033

Fold 4
Performance Training = 0.7539884842274553
Performance Validation = 0.8190558817693364

Average Performance in Training = 0.7774158243632792
Average Performance in Validation = 0.7352444612701626

M Value = 3
Lambda Value = 0.87

Fold 1
Performance Training = 0.7948009999196194
Performance Validation = 0.6782270005771465

Fold 2
Performance Training = 0.7825596521852954
Performance Validation = 0.7222364085715199

Fold 3
Performance Training = 0.7778734811880719
Performance Validation = 0.7212985741469434

Fold 4
Performance Training = 0.753861301651525
Performance Validation = 0.8188389108553668

Average Performance in Training = 0.7772738587361279
Average Performance in Validation = 0.7351502235377441

M Value = 3
Lambda Value = 0.88

Fold 1
Performance Training = 0.794657261367126
Performance Validation = 0.6782714947503102

Fold 2
Performance Training = 0.7824116550779201
Performance Validation = 0.7221508909015995

Fold 3
Performance Training = 0.7777212476161097
Performance Validation = 0.7211739899017469

Fold 4
Performance Training = 0.7537331198485335
Performance Validation = 0.8186214754074835

Average Performance in Training = 0.7771308209774224
Average Performance in Validation = 0.7350544627402851

M Value = 3
Lambda Value = 0.89

Fold 1
Performance Training = 0.7945124470163559

Performance Validation = 0.6783141554157768

Fold 2

Performance Training = 0.7822625695843056

Performance Validation = 0.722063469869298

Fold 3

Performance Training = 0.7775679202816963

Performance Validation = 0.7210475633862139

Fold 4

Performance Training = 0.7536039472578133

Performance Validation = 0.8184035806772411

Average Performance in Training = 0.7769867210350427

Average Performance in Validation = 0.7349571923371324

M Value = 3

Lambda Value = 0.9

Fold 1

Performance Training = 0.7943665666927535

Performance Validation = 0.6783549964342772

Fold 2

Performance Training = 0.7821124060584059

Performance Validation = 0.7219741619362433

Fold 3

Performance Training = 0.777413510089776

Performance Validation = 0.7209193124572411

Fold 4

Performance Training = 0.7534737922541397

Performance Validation = 0.818185231873271

Average Performance in Training = 0.7768415687737688

Average Performance in Validation = 0.734858425675258

M Value = 3

Lambda Value = 0.91

Fold 1

Performance Training = 0.7942196301413158

Performance Validation = 0.6783940315594581

Fold 2

Performance Training = 0.7819611747666304

Performance Validation = 0.7218829834283429

Fold 3

Performance Training = 0.7772580278479269

Performance Validation = 0.7207892548116921

Fold 4
Performance Training = 0.7533426631482673
Performance Validation = 0.817966434161649

Average Performance in Training = 0.776695373976035
Average Performance in Validation = 0.7347581759902855

M Value = 3
Lambda Value = 0.92

Fold 1
Performance Training = 0.7940716470273048
Performance Validation = 0.678431274438805

Fold 2
Performance Training = 0.7818088858886413
Performance Validation = 0.7217899505370136

Fold 3
Performance Training = 0.7771014842673027
Performance Validation = 0.7206574079879524

Fold 4
Performance Training = 0.7532105681874612
Performance Validation = 0.8177471926662603

Average Performance in Training = 0.7765481463426775
Average Performance in Validation = 0.7346564564075078

M Value = 3
Lambda Value = 0.93

Fold 1
Performance Training = 0.7939226269369564
Performance Validation = 0.6784667386145584

Fold 2
Performance Training = 0.7816555495181479
Performance Validation = 0.7216950793203967

Fold 3
Performance Training = 0.7769438899635674
Performance Validation = 0.7205237893674722

Fold 4
Performance Training = 0.7530775155560242
Performance Validation = 0.8175275124691582

Average Performance in Training = 0.776399895493674
Average Performance in Validation = 0.7345532799428964

M Value = 3
Lambda Value = 0.9400000000000001

Fold 1
Performance Training = 0.7937725793781785
Performance Validation = 0.6785004375246176

Fold 2
Performance Training = 0.7815011756636894
Performance Validation = 0.7215983857045563

Fold 3
Performance Training = 0.7767852554578178
Performance Validation = 0.7203884161762869

Fold 4
Performance Training = 0.7529435133758164
Performance Validation = 0.8173073986109213

Average Performance in Training = 0.7762506309688755
Average Performance in Validation = 0.7344486595040955

M Value = 3
Lambda Value = 0.9500000000000001

Fold 1
Performance Training = 0.793621513781245
Performance Validation = 0.6785323845034421

Fold 2
Performance Training = 0.7813457742494119
Performance Validation = 0.7214998854846739

Fold 3
Performance Training = 0.7766255911774951
Performance Validation = 0.7202513054865242

Fold 4
Performance Training = 0.7528085697067746
Performance Validation = 0.8170868560910073

Average Performance in Training = 0.7761003622287316
Average Performance in Validation = 0.7343426078914118

M Value = 3
Lambda Value = 0.96

Fold 1
Performance Training = 0.7934694394994827
Performance Validation = 0.6785625927829351

Fold 2
Performance Training = 0.7811893551158364
Performance Validation = 0.7213995943262231

Fold 3

Performance Training = 0.7764649074572895
Performance Validation = 0.7201124742178957

Fold 4

Performance Training = 0.7526726925474214
Performance Validation = 0.8168658898681034

Average Performance in Training = 0.7759490986550075
Average Performance in Validation = 0.7342351377987895

M Value = 3
Lambda Value = 0.97

Fold 1

Performance Training = 0.7933163658099481
Performance Validation = 0.6785910754933271

Fold 2

Performance Training = 0.78103192802062
Performance Validation = 0.7212975277661371

Fold 3

Performance Training = 0.7763032145400337
Performance Validation = 0.7199719391391705

Fold 4

Performance Training = 0.7525358898353744
Performance Validation = 0.8166445048604726

Average Performance in Training = 0.775796849551494
Average Performance in Validation = 0.7341262618147768

M Value = 3
Lambda Value = 0.98

Fold 1

Performance Training = 0.7931623019141031
Performance Validation = 0.6786178456640457

Fold 2

Performance Training = 0.7808735026393074
Performance Validation = 0.7211937012139621

Fold 3

Performance Training = 0.7761405225775864
Performance Validation = 0.7198297168696335

Fold 4

Performance Training = 0.7523981694478483
Performance Validation = 0.8164227059462977

Average Performance in Training = 0.7756436241447113
Average Performance in Validation = 0.7340159924234847

M Value = 3
Lambda Value = 0.99

Fold 1

Performance Training = 0.7930072569384792
Performance Validation = 0.6786429162245798

Fold 2

Performance Training = 0.7807140885660775
Performance Validation = 0.7210881299530003

Fold 3

Performance Training = 0.7759768416317059
Performance Validation = 0.719685823880529

Fold 4

Performance Training = 0.752259539202152
Performance Validation = 0.816200497964022

Average Performance in Training = 0.7754894315846037
Average Performance in Validation = 0.7339043420055329

M Value = 3
Lambda Value = 1.0

Fold 1

Performance Training = 0.7928512399353383
Performance Validation = 0.6786663000053343

Fold 2

Performance Training = 0.7805536953144808
Performance Validation = 0.7209808291414428

Fold 3

Performance Training = 0.775812181674918
Performance Validation = 0.719540276496488

Fold 4

Performance Training = 0.7521200068561839
Performance Validation = 0.8159778857126867

Average Performance in Training = 0.7753342809452303
Average Performance in Validation = 0.7337913228389878

M Value = 3
Lambda Value = 1.01

Fold 1

Performance Training = 0.7926942598833253
Performance Validation = 0.6786880097384747

Fold 2

Performance Training = 0.7803923323181695

Performance Validation = 0.720871813813487

Fold 3

Performance Training = 0.775646552591368

Performance Validation = 0.7193930908969393

Fold 4

Performance Training = 0.7519795801089181

Performance Validation = 0.8157548739522636

Average Performance in Training = 0.7751781812254454

Average Performance in Validation = 0.7336769471002911

M Value = 3

Lambda Value = 1.02

Fold 1

Performance Training = 0.7925363256881142

Performance Validation = 0.6787080580587674

Fold 2

Performance Training = 0.7802300089316202

Performance Validation = 0.7207610988804468

Fold 3

Performance Training = 0.7754799641776702

Performance Validation = 0.7192442831175077

Fold 4

Performance Training = 0.7518382666008907

Performance Validation = 0.815531467403988

Average Performance in Training = 0.7750211413495738

Average Performance in Validation = 0.7335612268651776

M Value = 3

Lambda Value = 1.03

Fold 1

Performance Training = 0.7923774461830502

Performance Validation = 0.678726457504411

Fold 2

Performance Training = 0.7800667344308512

Performance Validation = 0.7206486991318508

Fold 3

Performance Training = 0.7753124261437452

Performance Validation = 0.7190938690513954

Fold 4

Performance Training = 0.7516960739146799

Performance Validation = 0.8153076707506864

Average Performance in Training = 0.7748631701680815
Average Performance in Validation = 0.7334441741095858

M Value = 3
Lambda Value = 1.04

Fold 1
Performance Training = 0.7922176301297824
Performance Validation = 0.6787432205178552

Fold 2
Performance Training = 0.7799025180141286
Performance Validation = 0.7205346292365291

Fold 3
Performance Training = 0.775143948113648
Performance Validation = 0.7189418644507506

Fold 4
Performance Training = 0.7515530095753797
Performance Validation = 0.8150834886371

Average Performance in Training = 0.7747042764582346
Average Performance in Validation = 0.7333258007105588

M Value = 3
Lambda Value = 1.05

Fold 1
Performance Training = 0.7920568862188926
Performance Validation = 0.6787583594466184

Fold 2
Performance Training = 0.7797373688026696
Performance Validation = 0.7204189037436868

Fold 3
Performance Training = 0.7749745396263877
Performance Validation = 0.7187882849280172

Fold 4
Performance Training = 0.7514090810510735
Performance Validation = 0.8148589256702076

Average Performance in Training = 0.7745444689247559
Average Performance in Validation = 0.7332061184471325

M Value = 3
Lambda Value = 1.06

Fold 1
Performance Training = 0.7918952230705167
Performance Validation = 0.6787718865440935

Fold 2
Performance Training = 0.7795712958413361
Performance Validation = 0.7203015370839723

Fold 3
Performance Training = 0.7748042101367402
Performance Validation = 0.7186331459572775

Fold 4
Performance Training = 0.7512642957532987
Performance Validation = 0.8146339864195445

Average Performance in Training = 0.774383756200473
Average Performance in Validation = 0.733085139001222

M Value = 3
Lambda Value = 1.07

Fold 1
Performance Training = 0.7917326492349612
Performance Validation = 0.6787838139703448

Fold 2
Performance Training = 0.7794043080993224
Performance Validation = 0.7201825435705314

Fold 3
Performance Training = 0.7746329690160496
Performance Validation = 0.7184764628755721

Fold 4
Performance Training = 0.7511186610375111
Performance Validation = 0.8144086754175188

Average Performance in Training = 0.774222146846961
Average Performance in Validation = 0.7329628739584917

M Value = 3
Lambda Value = 1.08

Fold 1
Performance Training = 0.7915691731933123
Performance Validation = 0.6787941537929045

Fold 2
Performance Training = 0.7792364144708355
Performance Validation = 0.7200619374000505

Fold 3
Performance Training = 0.7744608255530228
Performance Validation = 0.7183182508842126

Fold 4

Performance Training = 0.7509721842035424
Performance Validation = 0.8141829971597228

Average Performance in Training = 0.7740596493551781
Average Performance in Validation = 0.7328393348092226

M Value = 3
Lambda Value = 1.09

Fold 1
Performance Training = 0.7914048033580405
Performance Validation = 0.67880291798755

Fold 2
Performance Training = 0.7790676237757692
Performance Validation = 0.7199397326537897

Fold 3
Performance Training = 0.7742877889545151
Performance Validation = 0.7181585250500773

Fold 4
Performance Training = 0.7508248724960553
Performance Validation = 0.8139569561052453

Average Performance in Training = 0.7738962721460951
Average Performance in Validation = 0.7327145329491657

M Value = 4
Lambda Value = 0.01

Fold 1
Performance Training = 0.8060433114018198
Performance Validation = 0.6415511903384286

Fold 2
Performance Training = 0.7899211634561959
Performance Validation = 0.7164936991133477

Fold 3
Performance Training = 0.7856138184757877
Performance Validation = 0.7260893634904757

Fold 4
Performance Training = 0.7666623359298548
Performance Validation = 0.8123600699953057

Average Performance in Training = 0.7870601573159146
Average Performance in Validation = 0.7241235807343893

M Value = 4
Lambda Value = 0.02

Fold 1
Performance Training = 0.8060392793419121
Performance Validation = 0.6421632122298752

Fold 2
Performance Training = 0.789917437784238
Performance Validation = 0.7166974163367578

Fold 3
Performance Training = 0.7856096897257474
Performance Validation = 0.7262406718334525

Fold 4
Performance Training = 0.7666593077649319
Performance Validation = 0.8122111176673075

Average Performance in Training = 0.7870564286542073
Average Performance in Validation = 0.7243281045168483

M Value = 4
Lambda Value = 0.03

Fold 1
Performance Training = 0.8060325974539292
Performance Validation = 0.6427666787009422

Fold 2
Performance Training = 0.7899112570015747
Performance Validation = 0.7168967324018038

Fold 3
Performance Training = 0.7856028463805135
Performance Validation = 0.7263855513987739

Fold 4
Performance Training = 0.7666542810590895
Performance Validation = 0.8120610578454986

Average Performance in Training = 0.7870502454737767
Average Performance in Validation = 0.7245275050867546

M Value = 4
Lambda Value = 0.04

Fold 1
Performance Training = 0.806023296441565
Performance Validation = 0.6433617100732836

Fold 2
Performance Training = 0.7899026443019548
Performance Validation = 0.7170916874243913

Fold 3
Performance Training = 0.7855933189982826

Performance Validation = 0.7265240817944142

Fold 4

Performance Training = 0.7666472722138289

Performance Validation = 0.8119099014815192

Average Performance in Training = 0.7870416329889078

Average Performance in Validation = 0.7247218451934021

M Value = 4

Lambda Value = 0.05

Fold 1

Performance Training = 0.80601140652796

Performance Validation = 0.6439484243438628

Fold 2

Performance Training = 0.7898916226498561

Performance Validation = 0.7172823211137624

Fold 3

Performance Training = 0.785581137744446

Performance Validation = 0.7266563414933777

Fold 4

Performance Training = 0.7666382974956525

Performance Validation = 0.8117576594278677

Average Performance in Training = 0.7870306161044787

Average Performance in Validation = 0.7249111865947178

M Value = 4

Lambda Value = 0.06

Fold 1

Performance Training = 0.805996957466932

Performance Validation = 0.6445269372434581

Fold 2

Performance Training = 0.789878214783232

Performance Validation = 0.7174686727775827

Fold 3

Performance Training = 0.7855663323982349

Performance Validation = 0.7267824078541237

Fold 4

Performance Training = 0.7666273730373117

Performance Validation = 0.8116043424389298

Average Performance in Training = 0.7870172194214275

Average Performance in Validation = 0.7250955900785235

M Value = 4
Lambda Value = 0.07

Fold 1

Performance Training = 0.8059799785538577
Performance Validation = 0.6450973622933681

Fold 2

Performance Training = 0.789862443216216
Performance Validation = 0.7176507813269581

Fold 3

Performance Training = 0.7855489323592151
Performance Validation = 0.7269023571405265

Fold 4

Performance Training = 0.7666145148390396
Performance Validation = 0.8114499611719875

Average Performance in Training = 0.7870014672420821
Average Performance in Validation = 0.72527511548321

M Value = 4
Lambda Value = 0.08

Fold 1

Performance Training = 0.8059604986362179
Performance Validation = 0.6456598108604104

Fold 2

Performance Training = 0.7898443302417857
Performance Validation = 0.7178286852813641

Fold 3

Performance Training = 0.7855289666536404
Performance Validation = 0.7270162645413996

Fold 4

Performance Training = 0.7665997387697716
Performance Validation = 0.8112945261882208

Average Performance in Training = 0.7869833835753539
Average Performance in Validation = 0.7254498217178488

M Value = 4
Lambda Value = 0.09

Fold 1

Performance Training = 0.8059385461238189
Performance Validation = 0.6462143922102346

Fold 2

Performance Training = 0.7898238979343835
Performance Validation = 0.7180024227734998

Fold 3
Performance Training = 0.7855064639406639
Performance Validation = 0.7271242041895789

Fold 4
Performance Training = 0.7665830605683517
Performance Validation = 0.8111380479536947

Average Performance in Training = 0.7869629921418044
Average Performance in Validation = 0.7256197667817521

M Value = 4
Lambda Value = 0.1

Fold 1
Performance Training = 0.8059141489987008
Performance Validation = 0.6467612135590448

Fold 2
Performance Training = 0.7898011681525009
Performance Validation = 0.7181720315540638

Fold 3
Performance Training = 0.7854814525184143
Performance Validation = 0.7272262491805843

Fold 4
Performance Training = 0.766564495844728
Performance Validation = 0.810980536840333

Average Performance in Training = 0.786940316378586
Average Performance in Validation = 0.7257850077835064

M Value = 4
Lambda Value = 0.11

Fold 1
Performance Training = 0.8058873348247471
Performance Validation = 0.6473003801237764

Fold 2
Performance Training = 0.7897761625412207
Performance Validation = 0.7183375489964574

Fold 3
Performance Training = 0.7854539603299389
Performance Validation = 0.7273224715908729

Fold 4
Performance Training = 0.7665440600811338
Performance Validation = 0.8108220031268827

Average Performance in Training = 0.7869153794442602

Average Performance in Validation = 0.7259456009594973

M Value = 4

Lambda Value = 0.12

Fold 1

Performance Training = 0.8058581307570026

Performance Validation = 0.6478319951707598

Fold 2

Performance Training = 0.7897489025347229

Performance Validation = 0.7184990121014095

Fold 3

Performance Training = 0.7854240149690177

Performance Validation = 0.72741294249569

Fold 4

Performance Training = 0.7665217686332575

Performance Validation = 0.8106624569998631

Average Performance in Training = 0.7868882042235001

Average Performance in Validation = 0.7261016016919306

M Value = 4

Lambda Value = 0.13

Fold 1

Performance Training = 0.8058265635507134

Performance Validation = 0.6483561600629557

Fold 2

Performance Training = 0.7897194093587511

Performance Validation = 0.7186564575015362

Fold 3

Performance Training = 0.785391643685854

Performance Validation = 0.727497731986529

Fold 4

Performance Training = 0.7664976367313987

Performance Validation = 0.8105019085545047

Average Performance in Training = 0.7868588133316794

Average Performance in Validation = 0.7262530645263814

M Value = 4

Lambda Value = 0.14

Fold 1

Performance Training = 0.8057926595700967

Performance Validation = 0.648872974305795

Fold 2
Performance Training = 0.7896877040330449
Performance Validation = 0.7188099214658221

Fold 3
Performance Training = 0.7853568733926399
Performance Validation = 0.7275769091882149

Fold 4
Performance Training = 0.766471679481614
Performance Validation = 0.8103403677956765

Average Performance in Training = 0.786827229119349
Average Performance in Validation = 0.7264000431888772

M Value = 4
Lambda Value = 0.15

Fold 1
Performance Training = 0.8057564447968534
Performance Validation = 0.6493825355916649

Fold 2
Performance Training = 0.7896538073737333
Performance Validation = 0.7189594399040374

Fold 3
Performance Training = 0.7853197306690058
Performance Validation = 0.7276505422756194

Fold 4
Performance Training = 0.7664439118668489
Performance Validation = 0.8101778446388025

Average Performance in Training = 0.7867934736766103
Average Performance in Validation = 0.726542590602531

M Value = 4
Lambda Value = 0.16

Fold 1
Performance Training = 0.8057179448384272
Performance Validation = 0.6498849398431041

Fold 2
Performance Training = 0.789617739995694
Performance Validation = 0.7191050483710879

Fold 3
Performance Training = 0.7852802417673533
Performance Validation = 0.7277186984900077

Fold 4
Performance Training = 0.7664143487480597

Performance Validation = 0.8100143489107636

Average Performance in Training = 0.7867575688373836

Average Performance in Validation = 0.7266807589037408

M Value = 4

Lambda Value = 0.17

Fold 1

Performance Training = 0.805677184936025

Performance Validation = 0.6503802812547398

Fold 2

Performance Training = 0.7895795223148788

Performance Validation = 0.7192467820712904

Fold 3

Performance Training = 0.7852384326180752

Performance Validation = 0.7277814441550468

Fold 4

Performance Training = 0.7663830048653217

Performance Validation = 0.8098498903507935

Average Performance in Training = 0.7867195361835753

Average Performance in Validation = 0.7268145994579676

M Value = 4

Lambda Value = 0.18

Fold 1

Performance Training = 0.8056341899724037

Performance Validation = 0.6508686523340059

Fold 2

Performance Training = 0.7895391745506031

Performance Validation = 0.7193846758625946

Fold 3

Performance Training = 0.7851943288346673

Performance Validation = 0.7278388446924631

Fold 4

Performance Training = 0.7663498948389279

Performance Validation = 0.8096844786113592

Average Performance in Training = 0.7866793970491505

Average Performance in Validation = 0.7269441628751057

M Value = 4

Lambda Value = 0.19

Fold 1

Performance Training = 0.805588984479432
Performance Validation = 0.6513501439406996

Fold 2
Performance Training = 0.7894967167278037
Performance Validation = 0.719518764260728

Fold 3
Performance Training = 0.7851479557187326
Performance Validation = 0.7278909646373688

Fold 4
Performance Training = 0.7663150331704743
Performance Validation = 0.8095181232590336

Average Performance in Training = 0.7866371725241107
Average Performance in Validation = 0.7270694990244575

M Value = 4
Lambda Value = 0.2

Fold 1
Performance Training = 0.8055415926454361
Performance Validation = 0.6518248453253936

Fold 2
Performance Training = 0.7894521686792629
Performance Validation = 0.7196490814432918

Fold 3
Performance Training = 0.7850993382648817
Performance Validation = 0.7279378676532706

Fold 4
Performance Training = 0.7662784342439355
Performance Validation = 0.8093508337753558

Average Performance in Training = 0.7865928834583791
Average Performance in Validation = 0.7271906570493278

M Value = 4
Lambda Value = 0.21

Fold 1
Performance Training = 0.8054920383223358
Performance Validation = 0.6522928441667686

Fold 2
Performance Training = 0.7894055500478004
Performance Validation = 0.7197756612537825

Fold 3
Performance Training = 0.785048501165533
Performance Validation = 0.7279796165467535

Fold 4

Performance Training = 0.7662401123267281

Performance Validation = 0.8091826195576818

Average Performance in Training = 0.7865465504655993

Average Performance in Validation = 0.7273076853812466

M Value = 4

Lambda Value = 0.22

Fold 1

Performance Training = 0.8054403450325791

Performance Validation = 0.6527542266078643

Fold 2

Performance Training = 0.7893568802884342

Performance Validation = 0.7198985372055589

Fold 3

Performance Training = 0.7849954688156141

Performance Validation = 0.7280162732818596

Fold 4

Performance Training = 0.7662000815707634

Performance Validation = 0.809013489920026

Average Performance in Training = 0.7864981939268477

Average Performance in Validation = 0.7274206317538272

M Value = 4

Lambda Value = 0.23

Fold 1

Performance Training = 0.805386535975881

Performance Validation = 0.6532090772913284

Fold 2

Performance Training = 0.7893061786705102

Performance Validation = 0.720017742485753

Fold 3

Performance Training = 0.7849402653171658

Performance Validation = 0.7280478989941722

Fold 4

Performance Training = 0.7661583560134895

Performance Validation = 0.8088434540938884

Average Performance in Training = 0.7864478339942615

Average Performance in Validation = 0.7275295432162855

M Value = 4

Lambda Value = 0.24

Fold 1

Performance Training = 0.8053306340357747

Performance Validation = 0.6536574793936684

Fold 2

Performance Training = 0.7892534642798019

Performance Validation = 0.7201333099591143

Fold 3

Performance Training = 0.7848829144838557

Performance Validation = 0.728074554004599

Fold 4

Performance Training = 0.7661149495789221

Performance Validation = 0.8086725212290778

Average Performance in Training = 0.7863954905945886

Average Performance in Validation = 0.7276344661466149

M Value = 4

Lambda Value = 0.25

Fold 1

Performance Training = 0.8052726617859809

Performance Validation = 0.6540995146585402

Fold 2

Performance Training = 0.7891987560205791

Performance Validation = 0.7202452721718071

Fold 3

Performance Training = 0.7848234398453976

Performance Validation = 0.7280962978328758

Fold 4

Performance Training = 0.766069876078666

Performance Validation = 0.8085007003945217

Average Performance in Training = 0.7863411834326559

Average Performance in Validation = 0.7277354462644361

M Value = 4

Lambda Value = 0.26

Fold 1

Performance Training = 0.805212641496601

Performance Validation = 0.65453526342912

Fold 2

Performance Training = 0.7891420726176485

Performance Validation = 0.7203536613551403

Fold 3
Performance Training = 0.7847618646518842
Performance Validation = 0.728113189210789

Fold 4
Performance Training = 0.7660231492129234
Performance Validation = 0.8083280005790652

Average Performance in Training = 0.7862849319947643
Average Performance in Validation = 0.7278325286435287

M Value = 4
Lambda Value = 0.27

Fold 1
Performance Training = 0.8051505951401404
Performance Validation = 0.6549648046795682

Fold 2
Performance Training = 0.7890834326183653
Performance Validation = 0.720458509429253

Fold 3
Performance Training = 0.7846982118780326
Performance Validation = 0.7281252860951279

Fold 4
Performance Training = 0.7659747825714945
Performance Validation = 0.8081544306922639

Average Performance in Training = 0.7862267555520083
Average Performance in Validation = 0.7279257577240532

M Value = 4
Lambda Value = 0.28

Fold 1
Performance Training = 0.805086544397369
Performance Validation = 0.6553882160456406

Fold 2
Performance Training = 0.7890228543946166
Performance Validation = 0.7205598480067401

Fold 3
Performance Training = 0.7846325042273449
Performance Validation = 0.7281326456803734

Fold 4
Performance Training = 0.7659247896347676
Performance Validation = 0.8079799995651661

Average Performance in Training = 0.7861666731635246
Average Performance in Validation = 0.7280151773244801

M Value = 4
Lambda Value = 0.29

Fold 1
Performance Training = 0.805020510663021
Performance Validation = 0.6558055738544484

Fold 2
Performance Training = 0.7889603561447762
Performance Validation = 0.7206577083962262

Fold 3
Performance Training = 0.7845647641361886
Performance Validation = 0.7281353244111274

Fold 4
Performance Training = 0.7658731837746968
Performance Validation = 0.8078047159510847

Average Performance in Training = 0.7861047036796707
Average Performance in Validation = 0.7281008306532216

M Value = 4
Lambda Value = 0.3

Fold 1
Performance Training = 0.804952515051343
Performance Validation = 0.6562169531534146

Fold 2
Performance Training = 0.7888959558956337
Performance Validation = 0.7207521216058884

Fold 3
Performance Training = 0.7844950137777935
Performance Validation = 0.7281333779942893

Fold 4
Performance Training = 0.7658199782557722
Performance Validation = 0.8076285885263594

Average Performance in Training = 0.7860408657451357
Average Performance in Validation = 0.728182760319988

M Value = 4
Lambda Value = 0.31

Fold 1
Performance Training = 0.8048825784014935
Performance Validation = 0.6566224277384332

Fold 2

Performance Training = 0.7888296715042957
Performance Validation = 0.7208431183469259

Fold 3
Performance Training = 0.7844232750661743
Performance Validation = 0.728126861410992

Fold 4
Performance Training = 0.7657651862359793
Performance Validation = 0.807451625891115

Average Performance in Training = 0.7859751778019857
Average Performance in Validation = 0.7282610083468665

M Value = 4
Lambda Value = 0.32

Fold 1
Performance Training = 0.8048107212827984
Performance Validation = 0.6570220701812783

Fold 2
Performance Training = 0.7887615206600616
Performance Validation = 0.7209307290369817

Fold 3
Performance Training = 0.7843495696599734
Performance Validation = 0.7281158289282901

Fold 4
Performance Training = 0.765708820767748
Performance Validation = 0.8072738365700034

Average Performance in Training = 0.7859076580926454
Average Performance in Validation = 0.7283356161791384

M Value = 4
Lambda Value = 0.33

Fold 1
Performance Training = 0.804736963999869
Performance Validation = 0.6574159518562726

Fold 2
Performance Training = 0.7886915208862726
Performance Validation = 0.721014983803516

Fold 3
Performance Training = 0.7842739189662322
Performance Validation = 0.728100334110622

Fold 4
Performance Training = 0.7656508947988936
Performance Validation = 0.8070952290129428

Average Performance in Training = 0.7858383246628169
Average Performance in Validation = 0.7284066246958384

M Value = 4
Lambda Value = 0.34

Fold 1
Performance Training = 0.8046613265975844
Performance Validation = 0.6578041429662342

Fold 2
Performance Training = 0.7886196895421363
Performance Validation = 0.7210959124871271

Fold 3
Performance Training = 0.7841963441440876
Performance Validation = 0.728080429831041

Fold 4
Performance Training = 0.765591421173546
Performance Validation = 0.8069158115958459

Average Performance in Training = 0.7857671953643386
Average Performance in Validation = 0.7284740742200619

M Value = 4
Lambda Value = 0.35000000000000003

Fold 1
Performance Training = 0.8045838288659454
Performance Validation = 0.6581867125677435

Fold 2
Performance Training = 0.7885460438245265
Performance Validation = 0.7211735446448294

Fold 3
Performance Training = 0.7841168661083997
Performance Validation = 0.7280561682822223

Fold 4
Performance Training = 0.7655304126330714
Performance Validation = 0.8067355926213395

Average Performance in Training = 0.7856942878579857
Average Performance in Validation = 0.7285380045290337

M Value = 4
Lambda Value = 0.36

Fold 1
Performance Training = 0.8045044903448018

Performance Validation = 0.6585637285957308

Fold 2

Performance Training = 0.7884706007697576

Performance Validation = 0.7212479095532814

Fold 3

Performance Training = 0.7840355055333083

Performance Validation = 0.7280276009872539

Fold 4

Performance Training = 0.7654678818169841

Performance Validation = 0.806554580319477

Average Performance in Training = 0.785619619616213

Average Performance in Validation = 0.7285984548639358

M Value = 4

Lambda Value = 0.37

Fold 1

Performance Training = 0.8044233303284586

Performance Validation = 0.6589352578874177

Fold 2

Performance Training = 0.7883933772553354

Performance Validation = 0.7213190362119712

Fold 3

Performance Training = 0.783952282855724

Performance Validation = 0.7279947788102163

Fold 4

Performance Training = 0.7654038412638486

Performance Validation = 0.806372782848441

Average Performance in Training = 0.7855432079258418

Average Performance in Validation = 0.7286554639395115

M Value = 4

Lambda Value = 0.38

Fold 1

Performance Training = 0.8043403678701646

Performance Validation = 0.659301366205634

Fold 2

Performance Training = 0.7883143900016851

Performance Validation = 0.7213869533463555

Fold 3

Performance Training = 0.7838672182787522

Performance Validation = 0.7279577519665538

Fold 4
Performance Training = 0.765338303412173
Performance Validation = 0.8061902082952391

Average Performance in Training = 0.7854650698906938
Average Performance in Validation = 0.7287090699534456

M Value = 4
Lambda Value = 0.39

Fold 1
Performance Training = 0.8042556217864869
Performance Validation = 0.6596621182615097

Fold 2
Performance Training = 0.7882336555738549
Performance Validation = 0.7214516894109504

Fold 3
Performance Training = 0.7837803317750531
Performance Validation = 0.7279165700332431

Fold 4
Performance Training = 0.7652712806012938
Performance Validation = 0.8060068646763928

Average Performance in Training = 0.7853852224341722
Average Performance in Validation = 0.7287593105955239

M Value = 4
Lambda Value = 0.4

Fold 1
Performance Training = 0.8041691106615741
Performance Validation = 0.6600175777366013

Fold 2
Performance Training = 0.7881511903831978
Performance Validation = 0.7215132725923895

Fold 3
Performance Training = 0.7836916430901397
Performance Validation = 0.7278712819587663

Fold 4
Performance Training = 0.7652027850722508
Performance Validation = 0.805822759938615

Average Performance in Training = 0.7853036823017906
Average Performance in Validation = 0.728806223056593

M Value = 4
Lambda Value = 0.41000000000000003

Fold 1
Performance Training = 0.8040808528513149
Performance Validation = 0.6603678073044138

Fold 2
Performance Training = 0.7880670106890301
Performance Validation = 0.7215717308124263

Fold 3
Performance Training = 0.7836011717456138
Performance Validation = 0.7278219360728895

Fold 4
Performance Training = 0.7651328289686543
Performance Validation = 0.8056379019594828

Average Performance in Training = 0.7852204660636533
Average Performance in Validation = 0.728849844037303

M Value = 4
Lambda Value = 0.42

Fold 1
Performance Training = 0.8039908664873902
Performance Validation = 0.660712868651396

Fold 2
Performance Training = 0.7879811326002679
Performance Validation = 0.7216270917309052

Fold 3
Performance Training = 0.7835089370423414
Performance Validation = 0.7277685800962546

Fold 4
Performance Training = 0.7650614243375432
Performance Validation = 0.8054522985481025

Average Performance in Training = 0.7851355901168856
Average Performance in Validation = 0.7288902097566645

M Value = 4
Lambda Value = 0.43

Fold 1
Performance Training = 0.8038991694812275
Performance Validation = 0.6610528224973768

Fold 2
Performance Training = 0.7878935720770439
Performance Validation = 0.7216793827486869

Fold 3

Performance Training = 0.7834149580635726
Performance Validation = 0.7277112611497838

Fold 4

Performance Training = 0.7649885831302339
Performance Validation = 0.8052659574457652

Average Performance in Training = 0.7850490706880195
Average Performance in Validation = 0.7289273559604031

M Value = 4
Lambda Value = 0.44

Fold 1

Performance Training = 0.8038057795278564
Performance Validation = 0.6613877286154854

Fold 2

Performance Training = 0.7878043449322996
Performance Validation = 0.7217286310105318

Fold 3

Performance Training = 0.7833192536780014
Performance Validation = 0.7276500257639079

Fold 4

Performance Training = 0.7649143172031619
Performance Validation = 0.805078886326596

Average Performance in Training = 0.7849609238353298
Average Performance in Validation = 0.7289613179291303

M Value = 4
Lambda Value = 0.45

Fold 1

Performance Training = 0.8037107141096702
Performance Validation = 0.6617176458515723

Fold 2

Performance Training = 0.7877134668333589
Performance Validation = 0.7217748634079477

Fold 3

Performance Training = 0.7832218425427707
Performance Validation = 0.7275849198876195

Fold 4

Performance Training = 0.764838638318714
Performance Validation = 0.804891092798198

Average Performance in Training = 0.7848711654511284
Average Performance in Validation = 0.7289921304863344

M Value = 4
Lambda Value = 0.46

Fold 1
Performance Training = 0.803613990500097
Performance Validation = 0.662042632143126

Fold 2
Performance Training = 0.7876209533034813
Performance Validation = 0.7218181065819953

Fold 3
Performance Training = 0.7831227431064243
Performance Validation = 0.7275159888973503

Fold 4
Performance Training = 0.7647615581460531
Performance Validation = 0.8047025844022848

Average Performance in Training = 0.7847798112640139
Average Performance in Validation = 0.729019828006189

M Value = 4
Lambda Value = 0.47000000000000003

Fold 1
Performance Training = 0.803515625767181
Performance Validation = 0.6623627445377132

Fold 2
Performance Training = 0.7875268197233924
Performance Validation = 0.7218583869260593

Fold 3
Performance Training = 0.7830219736118031
Performance Validation = 0.7274432776056842

Fold 4
Performance Training = 0.7646830882619331
Performance Validation = 0.8045133686153083

Average Performance in Training = 0.7846868768410774
Average Performance in Validation = 0.7290444444211912

M Value = 4
Lambda Value = 0.48

Fold 1
Performance Training = 0.8034156367770791
Performance Validation = 0.6626780392109723

Fold 2
Performance Training = 0.7874310813327982

Performance Validation = 0.7218957305885751

Fold 3

Performance Training = 0.7829195520988924

Performance Validation = 0.7273668302699077

Fold 4

Performance Training = 0.7646032401515087

Performance Validation = 0.8043234528490812

Average Performance in Training = 0.7845923775900696

Average Performance in Validation = 0.729066013229634

M Value = 4

Lambda Value = 0.49

Fold 1

Performance Training = 0.8033140401974729

Performance Validation = 0.6629885714841395

Fold 2

Performance Training = 0.7873337532318755

Performance Validation = 0.7219301634757247

Fold 3

Performance Training = 0.7828154964076152

Performance Validation = 0.7272866906003944

Fold 4

Performance Training = 0.7645220252091343

Performance Validation = 0.804132844451389

Average Performance in Training = 0.7844963287615244

Average Performance in Validation = 0.7290845675029118

M Value = 4

Lambda Value = 0.5

Fold 1

Performance Training = 0.8032108525009019

Performance Validation = 0.6632943958411416

Fold 2

Performance Training = 0.7872348503827474

Performance Validation = 0.7219617112540959

Fold 3

Performance Training = 0.7827098241805786

Performance Validation = 0.7272029017688433

Fold 4

Performance Training = 0.7644394547391556

Performance Validation = 0.8039415507065971

Average Performance in Training = 0.7843987454508459
Average Performance in Validation = 0.7291001398926695

M Value = 4
Lambda Value = 0.51

Fold 1
Performance Training = 0.8031060899680155
Performance Validation = 0.663595565945275

Fold 2
Performance Training = 0.787134387610936
Performance Validation = 0.721990399353299

Fold 3
Performance Training = 0.7826025528657694
Performance Validation = 0.7271155064163518

Fold 4
Performance Training = 0.7643555399566958
Performance Validation = 0.8037495788362522

Average Performance in Training = 0.7842996426003542
Average Performance in Validation = 0.7291127626377945

M Value = 4
Lambda Value = 0.52

Fold 1
Performance Training = 0.8029997686907521
Performance Validation = 0.663892134655466

Fold 2
Performance Training = 0.7870323796067994
Performance Validation = 0.7220162529685578

Fold 3
Performance Training = 0.7824936997192047
Performance Validation = 0.7270245466613485

Fold 4
Performance Training = 0.7642702919884297
Performance Validation = 0.8035569359996733

Average Performance in Training = 0.7841990350012964
Average Performance in Validation = 0.7291224675712614

M Value = 4
Lambda Value = 0.53

Fold 1
Performance Training = 0.8028919045754419
Performance Validation = 0.6641841540421323

Fold 2
Performance Training = 0.7869288409269485
Performance Validation = 0.7220392970632563

Fold 3
Performance Training = 0.782383281807534
Performance Validation = 0.7269300641073759

Fold 4
Performance Training = 0.7641837218733543
Performance Validation = 0.8033636292945414

Average Performance in Training = 0.7840969372958198
Average Performance in Validation = 0.7291292861268265

M Value = 4
Lambda Value = 0.54

Fold 1
Performance Training = 0.8027825133458395
Performance Validation = 0.6644716754026598

Fold 2
Performance Training = 0.7868237859956464
Performance Validation = 0.722059556371462

Fold 3
Performance Training = 0.7822713160105964
Performance Validation = 0.7268320998507325

Fold 4
Performance Training = 0.7640958405635498
Performance Validation = 0.8031696657574782

Average Performance in Training = 0.7839933639789081
Average Performance in Validation = 0.7291332493455831

M Value = 4
Lambda Value = 0.55

Fold 1
Performance Training = 0.8026716105460865
Performance Validation = 0.6647547492764962

Fold 2
Performance Training = 0.7867172291061901
Performance Validation = 0.7220770554004046

Fold 3
Performance Training = 0.7821578190239339
Performance Validation = 0.7267306944879703

Fold 4

Performance Training = 0.7640066589249329
Performance Validation = 0.802975052364621

Average Performance in Training = 0.7838883294002859
Average Performance in Validation = 0.7291343878823731

M Value = 4
Lambda Value = 0.56

Fold 1
Performance Training = 0.8025592115436067
Performance Validation = 0.6650334254598949

Fold 2
Performance Training = 0.786609184422275
Performance Validation = 0.7220918184329309

Fold 3
Performance Training = 0.782042807361262
Performance Validation = 0.7266258881232627

Fold 4
Performance Training = 0.7639161877380036
Performance Validation = 0.8027797960321886

Average Performance in Training = 0.7837818477662868
Average Performance in Validation = 0.7291327320120693

M Value = 4
Lambda Value = 0.5700000000000001

Fold 1
Performance Training = 0.8024453315319359
Performance Validation = 0.665307753020282

Fold 2
Performance Training = 0.7864996659793411
Performance Validation = 0.7221038695299213

Fold 3
Performance Training = 0.7819262973568959
Performance Validation = 0.7265177203756328

Fold 4
Performance Training = 0.7638244376985847
Performance Validation = 0.8025839036170469

Average Performance in Training = 0.7836739331416893
Average Performance in Validation = 0.7291283116357207

M Value = 4
Lambda Value = 0.58

Fold 1
Performance Training = 0.8023299855334874
Performance Validation = 0.6655777803102938

Fold 2
Performance Training = 0.7863886876859034
Performance Validation = 0.7221132325326791

Fold 3
Performance Training = 0.7818083051681368
Performance Validation = 0.7264062303860536

Fold 4
Performance Training = 0.7637314194185527
Performance Validation = 0.8023873819172608

Average Performance in Training = 0.7835645994515201
Average Performance in Validation = 0.7291211562865718

M Value = 4
Lambda Value = 0.59

Fold 1
Performance Training = 0.8022131884022566
Performance Validation = 0.6658435549814746

Fold 2
Performance Training = 0.7862762633248654
Performance Validation = 0.7221199310652815

Fold 3
Performance Training = 0.781688846777617
Performance Validation = 0.7262914568244209

Fold 4
Performance Training = 0.763637143426564
Performance Validation = 0.8021902376726436

Average Performance in Training = 0.7834538604828257
Average Performance in Validation = 0.7291112951359552

M Value = 4
Lambda Value = 0.6

Fold 1
Performance Training = 0.8020949548264655
Performance Validation = 0.6661051239976443

Fold 2
Performance Training = 0.7861624065548156
Performance Validation = 0.7221239885369093

Fold 3
Performance Training = 0.7815679379956058

Performance Validation = 0.7261734378963998

Fold 4

Performance Training = 0.763541620168771

Performance Validation = 0.8019924775653031

Average Performance in Training = 0.7833417298864145

Average Performance in Validation = 0.7290987569990641

M Value = 4

Lambda Value = 0.61

Fold 1

Performance Training = 0.8019752993311475

Performance Validation = 0.6663625336479604

Fold 2

Performance Training = 0.7860471309113088

Performance Validation = 0.7221254281441387

Fold 3

Performance Training = 0.7814455944622748

Performance Validation = 0.7260522113501469

Fold 4

Performance Training = 0.7634448600095345

Performance Validation = 0.8017941082201768

Average Performance in Training = 0.7832282211785664

Average Performance in Validation = 0.7290835703406058

M Value = 4

Lambda Value = 0.62

Fold 1

Performance Training = 0.8018542362806766

Performance Validation = 0.666615829559657

Fold 2

Performance Training = 0.7859304498081302

Performance Validation = 0.7221242728732078

Fold 3

Performance Training = 0.7813218316499281

Performance Validation = 0.72592781448292

Fold 4

Performance Training = 0.7633468732321256

Performance Validation = 0.8015951362055642

Average Performance in Training = 0.7831133477427151

Average Performance in Validation = 0.7290657632803372

M Value = 4
Lambda Value = 0.63

Fold 1
Performance Training = 0.8017317798812424
Performance Validation = 0.666865056710496

Fold 2
Performance Training = 0.7858123765385452
Performance Validation = 0.7221205455022481

Fold 3
Performance Training = 0.7811966648651917
Performance Validation = 0.7258002841475609

Fold 4
Performance Training = 0.7632476700394241
Performance Validation = 0.8013955680336525

Average Performance in Training = 0.7829971228311009
Average Performance in Validation = 0.7290453635984894

M Value = 4
Lambda Value = 0.64

Fold 1
Performance Training = 0.8016079441832697
Performance Validation = 0.6671102594409237

Fold 2
Performance Training = 0.7856929242765333
Performance Validation = 0.7221142686034943

Fold 3
Performance Training = 0.7810701092511698
Performance Validation = 0.7256696567588716

Fold 4
Performance Training = 0.7631472605546086
Performance Validation = 0.8011954101610389

Average Performance in Training = 0.7828795595663954
Average Performance in Validation = 0.7290223987410821

M Value = 4
Lambda Value = 0.65

Fold 1
Performance Training = 0.8014827430837865
Performance Validation = 0.6673514814659423

Fold 2
Performance Training = 0.7855721060780058
Performance Validation = 0.7221054645454608

Fold 3

Performance Training = 0.7809421797895633

Performance Validation = 0.7255359682998723

Fold 4

Performance Training = 0.7630456548218393

Performance Validation = 0.8009946689892423

Average Performance in Training = 0.7827606709432987

Average Performance in Validation = 0.7289968958251295

M Value = 4

Lambda Value = 0.66

Fold 1

Performance Training = 0.8013561903287422

Performance Validation = 0.6675887658867135

Fold 2

Performance Training = 0.785449934882011

Performance Validation = 0.7220941554950928

Fold 3

Performance Training = 0.7808128913027532

Performance Validation = 0.7253992543279549

Fold 4

Performance Training = 0.7629428628069356

Performance Validation = 0.8007933508652145

Average Performance in Training = 0.7826404698301105

Average Performance in Validation = 0.7289688816437438

M Value = 4

Lambda Value = 0.67

Fold 1

Performance Training = 0.8012282995152757

Performance Validation = 0.667822155201879

Fold 2

Performance Training = 0.785326423511922

Performance Validation = 0.7220803634198854

Fold 3

Performance Training = 0.7806822584558517

Performance Validation = 0.7252595499809253

Fold 4

Performance Training = 0.7628388943980454

Performance Validation = 0.8005914620818437

Average Performance in Training = 0.7825189689702737

Average Performance in Validation = 0.7289383826711333

M Value = 4

Lambda Value = 0.68

Fold 1

Performance Training = 0.8010990840939366

Performance Validation = 0.6680516913186396

Fold 2

Performance Training = 0.7852015846766109

Performance Validation = 0.7220641100899843

Fold 3

Performance Training = 0.7805502957587178

Performance Validation = 0.725116889982942

Fold 4

Performance Training = 0.7627337594063106

Performance Validation = 0.8003890088784515

Average Performance in Training = 0.782396180983894

Average Performance in Validation = 0.7289054250675044

M Value = 4

Lambda Value = 0.6900000000000001

Fold 1

Performance Training = 0.800968557370858

Performance Validation = 0.6682774155635722

Fold 2

Performance Training = 0.7850754309716107

Performance Validation = 0.7220454170802477

Fold 3

Performance Training = 0.780417017567941

Performance Validation = 0.7249713086503493

Fold 4

Performance Training = 0.7626274675665234

Performance Validation = 0.800185997441289

Average Performance in Training = 0.7822721183692334

Average Performance in Validation = 0.7288700346838646

M Value = 4

Lambda Value = 0.7000000000000001

Fold 1

Performance Training = 0.8008367325098857

Performance Validation = 0.6684993686931913

Fold 2
Performance Training = 0.7849479748802592
Performance Validation = 0.7220243057722957

Fold 3
Performance Training = 0.7802824380887929
Performance Validation = 0.7248228398974152

Fold 4
Performance Training = 0.7625200285377777
Performance Validation = 0.7999824339040214

Average Performance in Training = 0.7821467935041789
Average Performance in Validation = 0.7288322370667308

M Value = 4
Lambda Value = 0.71

Fold 1
Performance Training = 0.8007036225346633
Performance Validation = 0.66871759090429

Fold 2
Performance Training = 0.784819228774833
Performance Validation = 0.7220007973565206

Fold 3
Performance Training = 0.780146571377147
Performance Validation = 0.7246715172419642

Fold 4
Performance Training = 0.7624114519041144
Performance Validation = 0.7997783243482119

Average Performance in Training = 0.7820202186476894
Average Performance in Validation = 0.7287920574627467

M Value = 4
Lambda Value = 0.72

Fold 1
Performance Training = 0.8005692403306727
Performance Validation = 0.6689321218440323

Fold 2
Performance Training = 0.7846892049176655
Performance Validation = 0.7219749128340827

Fold 3
Performance Training = 0.7800094313413677
Performance Validation = 0.7245173738109166

Fold 4
Performance Training = 0.7623017471751599

Performance Validation = 0.7995736748038003

Average Performance in Training = 0.7818924059412165

Average Performance in Validation = 0.728749520823208

M Value = 4

Lambda Value = 0.73

Fold 1

Performance Training = 0.8004335986472336

Performance Validation = 0.6691430006198235

Fold 2

Performance Training = 0.7845579154622518

Performance Validation = 0.7219466730188726

Fold 3

Performance Training = 0.7798710317441689

Performance Validation = 0.7243604423457335

Fold 4

Performance Training = 0.7621909237867586

Performance Validation = 0.7993684912495742

Average Performance in Training = 0.7817633674101032

Average Performance in Validation = 0.7287046518085009

M Value = 4

Lambda Value = 0.74

Fold 1

Performance Training = 0.8002967100994645

Performance Validation = 0.6693502658089614

Fold 2

Performance Training = 0.7844253724543411

Performance Validation = 0.7219160985394548

Fold 3

Performance Training = 0.7797313862044442

Performance Validation = 0.724200755207768

Fold 4

Performance Training = 0.7620789911015995

Performance Validation = 0.7991627796136361

Average Performance in Training = 0.7816331149649622

Average Performance in Validation = 0.7286574747924551

M Value = 4

Lambda Value = 0.75

Fold 1

Performance Training = 0.8001585871702005
Performance Validation = 0.6695539554680654

Fold 2
Performance Training = 0.7842915878330154
Performance Validation = 0.7218832098409852

Fold 3
Performance Training = 0.7795905081990661
Performance Validation = 0.724038344383525

Fold 4
Performance Training = 0.7619659584098373
Performance Validation = 0.7989565457738694

Average Performance in Training = 0.7815016604030298
Average Performance in Validation = 0.7286080138666113

M Value = 4
Lambda Value = 0.76

Fold 1
Performance Training = 0.8000192422118746
Performance Validation = 0.6697541071423093

Fold 2
Performance Training = 0.7841565734317559
Performance Validation = 0.7218480271871025

Fold 3
Performance Training = 0.7794484110646582
Performance Validation = 0.7238732414898332

Fold 4
Performance Training = 0.7618518349297063
Performance Validation = 0.7987497955583907

Average Performance in Training = 0.7813690154094988
Average Performance in Validation = 0.7285562928444089

M Value = 4
Lambda Value = 0.77

Fold 1
Performance Training = 0.7998786874483615
Performance Validation = 0.6699507578744353

Fold 2
Performance Training = 0.7840203409794958
Performance Validation = 0.721810570661797

Fold 3
Performance Training = 0.7793051079993389
Performance Validation = 0.7237054777789294

Fold 4

Performance Training = 0.7617366298081291

Performance Validation = 0.7985425347460063

Average Performance in Training = 0.7812351915588314

Average Performance in Validation = 0.7285023352652921

M Value = 4

Lambda Value = 0.78

Fold 1

Performance Training = 0.7997369349767847

Performance Validation = 0.6701439442135746

Fold 2

Performance Training = 0.7838829021016616

Performance Validation = 0.7217708601712626

Fold 3

Performance Training = 0.7791606120644385

Performance Validation = 0.7235350841434578

Fold 4

Performance Training = 0.7616203521213213

Performance Validation = 0.7983347690666628

Average Performance in Training = 0.7811002003160514

Average Performance in Validation = 0.7284461643987395

M Value = 4

Lambda Value = 0.79

Fold 1

Performance Training = 0.7995939967692862

Performance Validation = 0.6703337022238857

Fold 2

Performance Training = 0.7837442683212014

Performance Validation = 0.7217289154457123

Fold 3

Performance Training = 0.7790149361861891

Performance Validation = 0.7233620911213852

Fold 4

Performance Training = 0.7615030108753861

Performance Validation = 0.7981265042018835

Average Performance in Training = 0.7809640530380157

Average Performance in Validation = 0.7283878032482167

M Value = 4

Lambda Value = 0.8

Fold 1

Performance Training = 0.7994498846747646

Performance Validation = 0.6705200674929828

Fold 2

Performance Training = 0.7836044510596015

Performance Validation = 0.7216847560411861

Fold 3

Performance Training = 0.778868093157387

Performance Validation = 0.7231865289008339

Fold 4

Performance Training = 0.7613846150069082

Performance Validation = 0.7979177457852146

Average Performance in Training = 0.7808267609746653

Average Performance in Validation = 0.7283272745550543

M Value = 4

Lambda Value = 0.81

Fold 1

Performance Training = 0.7993046104205774

Performance Validation = 0.6707030751401966

Fold 2

Performance Training = 0.7834634616378916

Performance Validation = 0.7216384013413255

Fold 3

Performance Training = 0.7787200956390317

Performance Validation = 0.7230084273248351

Fold 4

Performance Training = 0.7612651733835385

Performance Validation = 0.7977084994026546

Average Performance in Training = 0.7806883352702598

Average Performance in Validation = 0.7282646008022529

M Value = 4

Lambda Value = 0.8200000000000001

Fold 1

Performance Training = 0.7991581856142105

Performance Validation = 0.6708827598246567

Fold 2

Performance Training = 0.7833213112776363

Performance Validation = 0.7215898705591346

Fold 3
Performance Training = 0.7785709561619385
Performance Validation = 0.7228278158960033

Fold 4
Performance Training = 0.7611446948045755
Performance Validation = 0.7974987705930883

Average Performance in Training = 0.7805487869645902
Average Performance in Validation = 0.7281998042182207

M Value = 4
Lambda Value = 0.8300000000000001

Fold 1
Performance Training = 0.7990106217449151
Performance Validation = 0.6710591557531883

Fold 2
Performance Training = 0.783178011101917
Performance Validation = 0.7215391827387103

Fold 3
Performance Training = 0.7784206871283257
Performance Validation = 0.7226447237811329

Fold 4
Performance Training = 0.7610231880015388
Performance Validation = 0.7972885648487085

Average Performance in Training = 0.7804081269941741
Average Performance in Validation = 0.728132906780435

M Value = 4
Lambda Value = 0.84

Fold 1
Performance Training = 0.7988619301853158
Performance Validation = 0.6712322966880525

Fold 2
Performance Training = 0.7830335721363014
Performance Validation = 0.7214863567569638

Fold 3
Performance Training = 0.7782693008133786
Performance Validation = 0.7224591798157194

Fold 4
Performance Training = 0.76090066163874
Performance Validation = 0.7970778876154385

Average Performance in Training = 0.780266366193434
Average Performance in Validation = 0.7280639302190435

M Value = 4
Lambda Value = 0.85

Fold 1
Performance Training = 0.7987121221929847
Performance Validation = 0.6714022159545068

Fold 2
Performance Training = 0.7828880053098026
Performance Validation = 0.7214314113253077

Fold 3
Performance Training = 0.7781168093667894
Performance Validation = 0.722271212508409

Fold 4
Performance Training = 0.760777124313847
Performance Validation = 0.7968667442933506

Average Performance in Training = 0.7801235152958559
Average Performance in Validation = 0.7279928960203935

M Value = 4
Lambda Value = 0.86

Fold 1
Performance Training = 0.7985612089119879
Performance Validation = 0.6715689464482195

Fold 2
Performance Training = 0.7827413214558248
Performance Validation = 0.7213743649913331

Fold 3
Performance Training = 0.7779632248142732
Performance Validation = 0.7220808500453697

Fold 4
Performance Training = 0.7606525845584409
Performance Validation = 0.7966551402370747

Average Performance in Training = 0.7799795849351316
Average Performance in Validation = 0.7279198254304993

M Value = 4
Lambda Value = 0.87

Fold 1
Performance Training = 0.7984092013744024
Performance Validation = 0.6717325206425161

Fold 2

Performance Training = 0.7825935313131014
Performance Validation = 0.7213152361404618

Fold 3
Performance Training = 0.7778085590590622
Performance Validation = 0.7218881202945986

Fold 4
Performance Training = 0.760527050838572
Performance Validation = 0.7964430807562101

Average Performance in Training = 0.7798345856462845
Average Performance in Validation = 0.7278447394584466

M Value = 4
Lambda Value = 0.88

Fold 1
Performance Training = 0.7982561105018053
Performance Validation = 0.6718929705954794

Fold 2
Performance Training = 0.7824446455266191
Performance Validation = 0.7212540429975767

Fold 3
Performance Training = 0.7776528238833756
Performance Validation = 0.7216930508101522

Fold 4
Performance Training = 0.7604005315553064
Performance Validation = 0.7962305711157276

Average Performance in Training = 0.7796885278667767
Average Performance in Validation = 0.7277676588797339

M Value = 4
Lambda Value = 0.89

Fold 1
Performance Training = 0.7981019471067338
Performance Validation = 0.6720503279568921

Fold 2
Performance Training = 0.7822946746485341
Performance Validation = 0.7211908036286377

Fold 3
Performance Training = 0.7774960309498706
Performance Validation = 0.7214956688363134

Fold 4
Performance Training = 0.7602730350452706
Performance Validation = 0.79601761653637

Average Performance in Training = 0.7795414219376022
Average Performance in Validation = 0.7276886042395533

M Value = 4
Lambda Value = 0.9

Fold 1
Performance Training = 0.7979467218941201
Performance Validation = 0.6722046239750457

Fold 2
Performance Training = 0.7821436291390752
Performance Validation = 0.7211255359422768

Fold 3
Performance Training = 0.7773381918030678
Performance Validation = 0.7212960013116905

Fold 4
Performance Training = 0.760144569581188
Performance Validation = 0.7958042221950473

Average Performance in Training = 0.7793932781043628
Average Performance in Validation = 0.727607595856015

M Value = 4
Lambda Value = 0.91

Fold 1
Performance Training = 0.7977904454626992
Performance Validation = 0.6723558895033943

Fold 2
Performance Training = 0.7819915193674388
Performance Validation = 0.7210582576913677

Fold 3
Performance Training = 0.7771793178707589
Performance Validation = 0.7210940748732466

Fold 4
Performance Training = 0.7600151433724135
Performance Validation = 0.7955903932252287

Average Performance in Training = 0.7792441065183277
Average Performance in Validation = 0.7275246538233093

M Value = 4
Lambda Value = 0.92

Fold 1
Performance Training = 0.7976331283063895

Performance Validation = 0.6725041550070814

Fold 2

Performance Training = 0.7818383556126718

Performance Validation = 0.7209889864745869

Fold 3

Performance Training = 0.77701942046539

Performance Validation = 0.7208899158602697

Fold 4

Performance Training = 0.759884764565461

Performance Validation = 0.7953761347173312

Average Performance in Training = 0.779093917237478

Average Performance in Validation = 0.7274397980148173

M Value = 4

Lambda Value = 0.93

Fold 1

Performance Training = 0.797474780815653

Performance Validation = 0.6726494505693232

Fold 2

Performance Training = 0.7816841480645462

Performance Validation = 0.7209177397379489

Fold 3

Performance Training = 0.7768585107854298

Performance Validation = 0.7206835503182756

Fold 4

Performance Training = 0.7597534412445264

Performance Validation = 0.7951614517191014

Average Performance in Training = 0.7789427202275387

Average Performance in Validation = 0.7273530480861623

M Value = 4

Lambda Value = 0.9400000000000001

Fold 1

Performance Training = 0.7973154132788249

Performance Validation = 0.672791805897667

Fold 2

Performance Training = 0.7815289068244222

Performance Validation = 0.7208445347763263

Fold 3

Performance Training = 0.7766965999167115

Performance Validation = 0.7204750040028515

Fold 4
Performance Training = 0.7596211814320063
Performance Validation = 0.7949463492359983

Average Performance in Training = 0.7787905253629912
Average Performance in Validation = 0.7272644234782107

M Value = 4
Lambda Value = 0.9500000000000001

Fold 1
Performance Training = 0.797155035883423
Performance Validation = 0.6729312503301147

Fold 2
Performance Training = 0.7813726419061024
Performance Validation = 0.7207693887349487

Fold 3
Performance Training = 0.7765336988337591
Performance Validation = 0.7202643023834363

Fold 4
Performance Training = 0.759487993089011
Performance Validation = 0.7947308322315666

Average Performance in Training = 0.7786373424280739
Average Performance in Validation = 0.7271739434200166

M Value = 4
Lambda Value = 0.96

Fold 1
Performance Training = 0.7969936587174351
Performance Validation = 0.6730678128411275

Fold 2
Performance Training = 0.7812153632366764
Performance Validation = 0.7206923186108889

Fold 3
Performance Training = 0.7763698184010934
Performance Validation = 0.7200514706470423

Fold 4
Performance Training = 0.759353884115874
Performance Validation = 0.7945149056278106

Average Performance in Training = 0.7784831811177697
Average Performance in Validation = 0.7270816269317173

M Value = 4
Lambda Value = 0.97

Fold 1
Performance Training = 0.7968312917705798
Performance Validation = 0.6732015220475042

Fold 2
Performance Training = 0.7810570806573548
Performance Validation = 0.7206133412545283

Fold 3
Performance Training = 0.7762049693745188
Performance Validation = 0.7198365337019188

Fold 4
Performance Training = 0.7592188623526548
Performance Validation = 0.7942985743055622

Average Performance in Training = 0.778328051038777
Average Performance in Validation = 0.7269874928273784

M Value = 4
Lambda Value = 0.98

Fold 1
Performance Training = 0.7966679449355475
Performance Validation = 0.6733324062141406

Fold 2
Performance Training = 0.7808978039242953
Performance Validation = 0.7205324733710046

Fold 3
Performance Training = 0.7760391624023909
Performance Validation = 0.7196195161811567

Fold 4
Performance Training = 0.759082935579639
Performance Validation = 0.794081843104846

Average Performance in Training = 0.7781719617104682
Average Performance in Validation = 0.726891559717787

M Value = 4
Lambda Value = 0.99

Fold 1
Performance Training = 0.7965036280092195
Performance Validation = 0.6734604932596735

Fold 2
Performance Training = 0.7807375427094182
Performance Validation = 0.7204497315216444

Fold 3

Performance Training = 0.7758724080268672
Performance Validation = 0.7194004424462382

Fold 4
Performance Training = 0.7589461115178326
Performance Validation = 0.793864716825239

Average Performance in Training = 0.7780149225658344
Average Performance in Validation = 0.7267938460131987

M Value = 4
Lambda Value = 1.0

Fold 1
Performance Training = 0.7963383506938662
Performance Validation = 0.6735858107620112

Fold 2
Performance Training = 0.7805763066012144
Performance Validation = 0.7203651321253768

Fold 3
Performance Training = 0.7757047166851387
Performance Validation = 0.719179336590531

Fold 4
Performance Training = 0.7588083978294529
Performance Validation = 0.7936472002262317

Average Performance in Training = 0.777856942952418
Average Performance in Validation = 0.7266943699260376

M Value = 4
Lambda Value = 1.01

Fold 1
Performance Training = 0.7961721225983224
Performance Validation = 0.6737083859637549

Fold 2
Performance Training = 0.7804141051055418
Performance Validation = 0.7202786914601338

Fold 3
Performance Training = 0.7755360987106448
Performance Validation = 0.7189562224427261

Fold 4
Performance Training = 0.7586698021184122
Performance Validation = 0.7934292980275777

Average Performance in Training = 0.7776980321332303
Average Performance in Validation = 0.726593149473548

M Value = 4
Lambda Value = 1.02

Fold 1

Performance Training = 0.7960049532391458
Performance Validation = 0.6738282457775047

Fold 2

Performance Training = 0.7802509476464152
Performance Validation = 0.7201904256642306

Fold 3

Performance Training = 0.7753665643342688
Performance Validation = 0.7187311235702254

Fold 4

Performance Training = 0.758530331930801
Performance Validation = 0.7932110149096468

Average Performance in Training = 0.7775381992876577
Average Performance in Validation = 0.7264902024804019

M Value = 4
Lambda Value = 1.03

Fold 1

Performance Training = 0.7958368520417534
Performance Validation = 0.6739454167910717

Fold 2

Performance Training = 0.780086843566787
Performance Validation = 0.7201003507377337

Fold 3

Performance Training = 0.7751961236855196
Performance Validation = 0.7185040632824748

Fold 4

Performance Training = 0.7583899947553632
Performance Validation = 0.7929923555137719

Average Performance in Training = 0.7773774535123558
Average Performance in Validation = 0.726385546581263

M Value = 4
Lambda Value = 1.04

Fold 1

Performance Training = 0.7956678283415382
Performance Validation = 0.6740599252725761

Fold 2

Performance Training = 0.7799218021293174

Performance Validation = 0.7200084825438089

Fold 3

Performance Training = 0.7750247867936941

Performance Validation = 0.7182750646342468

Fold 4

Performance Training = 0.7582487980239686

Performance Validation = 0.792773324442591

Average Performance in Training = 0.7772158038221295

Average Performance in Validation = 0.7262791992233056

M Value = 4

Lambda Value = 1.05

Fold 1

Performance Training = 0.7954978913849693

Performance Validation = 0.6741717971754511

Fold 2

Performance Training = 0.7797558325171379

Performance Validation = 0.719914836810058

Fold 3

Performance Training = 0.7748525635890244

Performance Validation = 0.7180441504288723

Fold 4

Performance Training = 0.758106749112081

Performance Validation = 0.7925539262603877

Average Performance in Training = 0.7770532591508031

Average Performance in Validation = 0.7261711776686923

M Value = 4

Lambda Value = 1.06

Fold 1

Performance Training = 0.7953270503306715

Performance Validation = 0.674281058143345

Fold 2

Performance Training = 0.7795889438346071

Performance Validation = 0.719819429129837

Fold 3

Performance Training = 0.7746794639038076

Performance Validation = 0.7178113432214233

Fold 4

Performance Training = 0.7579638553392216

Performance Validation = 0.7923341654934285

Average Performance in Training = 0.7768898283520769
Average Performance in Validation = 0.7260614989970086

M Value = 4
Lambda Value = 1.07

Fold 1
Performance Training = 0.7951553142504882
Performance Validation = 0.6743877335149331

Fold 2
Performance Training = 0.7794211451080547
Performance Validation = 0.7197222749635594

Fold 3
Performance Training = 0.7745054974735222
Performance Validation = 0.7175766653218476

Fold 4
Performance Training = 0.7578201239694289
Performance Validation = 0.7921140466302972

Average Performance in Training = 0.7767255202003736
Average Performance in Validation = 0.7259501801076593

M Value = 4
Lambda Value = 1.08

Fold 1
Performance Training = 0.7949826921305247
Performance Validation = 0.6744918483286239

Fold 2
Performance Training = 0.7792524452865216
Performance Validation = 0.7196233896399871

Fold 3
Performance Training = 0.7743306739379254
Performance Validation = 0.7173401387980539

Fold 4
Performance Training = 0.757675562211712
Performance Validation = 0.7918935741222211

Average Performance in Training = 0.776560343391671
Average Performance in Validation = 0.7258372377222215

M Value = 4
Lambda Value = 1.09

Fold 1
Performance Training = 0.7948091928721774
Performance Validation = 0.6745934273271912

Fold 2
Performance Training = 0.7790828532424885
Performance Validation = 0.7195227883575042

Fold 3
Performance Training = 0.77415500284214
Performance Validation = 0.7171017854789521

Fold 4
Performance Training = 0.7575301772205036
Performance Validation = 0.7916727523834013

Average Performance in Training = 0.7763943065443274
Average Performance in Validation = 0.7257226883867622

M Value = 5
Lambda Value = 0.01

Fold 1
Performance Training = 0.809165762115958
Performance Validation = 0.6945939582635435

Fold 2
Performance Training = 0.7980830195671905
Performance Validation = 0.7378726132644602

Fold 3
Performance Training = 0.7930714623815032
Performance Validation = 0.7460827595217262

Fold 4
Performance Training = 0.7937885227843677
Performance Validation = 0.7668886698273089

Average Performance in Training = 0.798527191712255
Average Performance in Validation = 0.7363595002192598

M Value = 5
Lambda Value = 0.02

Fold 1
Performance Training = 0.8091377405658288
Performance Validation = 0.6928658743380154

Fold 2
Performance Training = 0.7980552407412478
Performance Validation = 0.7377144405426044

Fold 3
Performance Training = 0.79304356230865
Performance Validation = 0.7457857195930533

Fold 4

Performance Training = 0.7937420088823309
Performance Validation = 0.7684944684007595

Average Performance in Training = 0.7984946381245143
Average Performance in Validation = 0.7362151257186081

M Value = 5
Lambda Value = 0.03

Fold 1
Performance Training = 0.8090925618235538
Performance Validation = 0.6911427457222661

Fold 2
Performance Training = 0.7980101875797214
Performance Validation = 0.7375395136486966

Fold 3
Performance Training = 0.7929983521504105
Performance Validation = 0.7454763903295265

Fold 4
Performance Training = 0.793666887458714
Performance Validation = 0.7700244628465179

Average Performance in Training = 0.7984419972530999
Average Performance in Validation = 0.7360457781367518

M Value = 5
Lambda Value = 0.04

Fold 1
Performance Training = 0.8090313776834893
Performance Validation = 0.6894258673235951

Fold 2
Performance Training = 0.7979488148060901
Performance Validation = 0.7373488339094784

Fold 3
Performance Training = 0.7929368189428913
Performance Validation = 0.7451556290332895

Fold 4
Performance Training = 0.7935649824982615
Performance Validation = 0.7714823540753983

Average Performance in Training = 0.798370498482683
Average Performance in Validation = 0.7358531710854403

M Value = 5
Lambda Value = 0.05

Fold 1
Performance Training = 0.8089552631105454
Performance Validation = 0.6877164095494974

Fold 2
Performance Training = 0.7978720247249249
Performance Validation = 0.737143346185172

Fold 3
Performance Training = 0.7928598936870285
Performance Validation = 0.7448242408568012

Fold 4
Performance Training = 0.7934380032946081
Performance Validation = 0.7728716334516482

Average Performance in Training = 0.7982812962042768
Average Performance in Validation = 0.7356389075107798

M Value = 5
Lambda Value = 0.06

Fold 1
Performance Training = 0.8088652218624527
Performance Validation = 0.6860154288356358

Fold 2
Performance Training = 0.7977806703856534
Performance Validation = 0.7369239423237697

Fold 3
Performance Training = 0.7927684548473835
Performance Validation = 0.7444829821762198

Fold 4
Performance Training = 0.7932875523644466
Performance Validation = 0.7741955964672462

Average Performance in Training = 0.798175474864984
Average Performance in Validation = 0.7354044874507177

M Value = 5
Lambda Value = 0.07

Fold 1
Performance Training = 0.8087621916545991
Performance Validation = 0.6843238772473685

Fold 2
Performance Training = 0.7976755585336109
Performance Validation = 0.736691464381246

Fold 3
Performance Training = 0.792663331607577

Performance Validation = 0.7441325637250475

Fold 4

Performance Training = 0.793115132754168

Performance Validation = 0.7754573554023351

Average Performance in Training = 0.7980540536374887

Average Performance in Validation = 0.7351513151889992

M Value = 5

Lambda Value = 0.08

Fold 1

Performance Training = 0.8086470489086558

Performance Validation = 0.6826426112428017

Fold 2

Performance Training = 0.7975574523642468

Performance Validation = 0.7364467076251502

Fold 3

Performance Training = 0.7925453069011422

Performance Validation = 0.7437736535068379

Fold 4

Performance Training = 0.7929221547906569

Performance Validation = 0.7766598510560396

Average Performance in Training = 0.7979179907411755

Average Performance in Validation = 0.7348807058577074

M Value = 5

Lambda Value = 0.09

Fold 1

Performance Training = 0.8085206131221259

Performance Validation = 0.6809723996758379

Fold 2

Performance Training = 0.797427074095053

Performance Validation = 0.7361904233377438

Fold 3

Performance Training = 0.7924151202349794

Performance Validation = 0.7434068795040985

Fold 4

Performance Training = 0.7927099423230851

Performance Validation = 0.7778058636240989

Average Performance in Training = 0.7977681874438108

Average Performance in Validation = 0.7345938915354447

M Value = 5
Lambda Value = 0.1

Fold 1

Performance Training = 0.8083836508923062
Performance Validation = 0.6793139311100147

Fold 2

Performance Training = 0.7972851073685866
Performance Validation = 0.7359233214335048

Fold 3

Performance Training = 0.7922734703211441
Performance Validation = 0.7430328321991208

Fold 4

Performance Training = 0.7924797384982032
Performance Validation = 0.778898022792917

Average Performance in Training = 0.79760549177006
Average Performance in Validation = 0.7342920268838893

M Value = 5
Lambda Value = 0.11

Fold 1

Performance Training = 0.8082368796249081
Performance Validation = 0.677667820506699

Fold 2

Performance Training = 0.7971321994988955
Performance Validation = 0.7356460729044902

Fold 3

Performance Training = 0.7921210175313982
Performance Validation = 0.7426520669210566

Fold 4

Performance Training = 0.792232711107735
Performance Validation = 0.7799388171131438

Average Performance in Training = 0.7974307019407341
Average Performance in Validation = 0.7339761943613474

M Value = 5
Lambda Value = 0.12

Fold 1

Performance Training = 0.8080809709546688
Performance Validation = 0.676034615344905

Fold 2

Performance Training = 0.796968963572647
Performance Validation = 0.7353593121062092

Fold 3
Performance Training = 0.7919583861877462
Performance Validation = 0.7422651060324095

Fold 4
Performance Training = 0.7919699575429655
Performance Validation = 0.7809306027104926

Average Performance in Training = 0.797244569564507
Average Performance in Validation = 0.7336474090485041

M Value = 5
Lambda Value = 0.13

Fold 1
Performance Training = 0.8079165539026858
Performance Validation = 0.6744148012242949

Fold 2
Performance Training = 0.7967959804153809
Performance Validation = 0.7350636388953642

Fold 3
Performance Training = 0.7917861667010987
Performance Validation = 0.7418724409670605

Fold 4
Performance Training = 0.7916925093884473
Performance Validation = 0.7818756113862356

Average Performance in Training = 0.7970478026019032
Average Performance in Validation = 0.7333066231182388

M Value = 5
Lambda Value = 0.14

Fold 1
Performance Training = 0.8077442177928765
Performance Validation = 0.6728088069979765

Fold 2
Performance Training = 0.7966138004324753
Performance Validation = 0.73475962063015

Fold 3
Performance Training = 0.7916049175692197
Performance Validation = 0.7414745341308875

Fold 4
Performance Training = 0.7914013366839016
Performance Validation = 0.7827759581552831

Average Performance in Training = 0.7968410681196183

Average Performance in Validation = 0.7329547299785744

M Value = 5

Lambda Value = 0.15

Fold 1

Performance Training = 0.8075645149478643

Performance Validation = 0.6712170094770216

Fold 2

Performance Training = 0.7964229453336666

Performance Validation = 0.7344477940428102

Fold 3

Performance Training = 0.7914151672442051

Performance Validation = 0.7410718206750893

Fold 4

Performance Training = 0.7910973518808176

Performance Validation = 0.7836336482655959

Average Performance in Training = 0.7966249948516384

Average Performance in Validation = 0.7325925681151293

M Value = 5

Lambda Value = 0.16

Fold 1

Performance Training = 0.8073779631827226

Performance Validation = 0.6696397377448008

Fold 2

Performance Training = 0.7962239097492794

Performance Validation = 0.7341286669935236

Fold 3

Performance Training = 0.7912174158789274

Performance Validation = 0.7406647101516804

Fold 4

Performance Training = 0.7907814135179245

Performance Validation = 0.7844505837388521

Average Performance in Training = 0.7964001755822134

Average Performance in Validation = 0.7322209246572142

M Value = 5

Lambda Value = 0.17

Fold 1

Performance Training = 0.8071850481133124

Performance Validation = 0.6680772771153711

Fold 2
Performance Training = 0.7960171627456891
Performance Validation = 0.7338027201138189

Fold 3
Performance Training = 0.7910121369611229
Performance Validation = 0.7402535880596475

Fold 4
Performance Training = 0.7904543296376074
Performance Validation = 0.7852285694690182

Average Performance in Training = 0.796167169364433
Average Performance in Validation = 0.7318405386894639

M Value = 5
Lambda Value = 0.18

Fold 1
Performance Training = 0.8069862252944284
Performance Validation = 0.6665298727671091

Fold 2
Performance Training = 0.7958031492469759
Performance Validation = 0.7334704083473085

Fold 3
Performance Training = 0.7907997788431226
Performance Validation = 0.7398388172897515

Fold 4
Performance Training = 0.7901168609634411
Performance Validation = 0.7859693189122113

Average Performance in Training = 0.795926503586992
Average Performance in Validation = 0.7314521043290951

M Value = 5
Lambda Value = 0.19

Fold 1
Performance Training = 0.8067819222016012
Performance Validation = 0.6649977330797114

Fold 2
Performance Training = 0.7955822913691886
Performance Validation = 0.7331321623946743

Fold 3
Performance Training = 0.7905807661745929
Performance Validation = 0.7394207394752123

Fold 4
Performance Training = 0.789769723857285

Performance Validation = 0.7866744593985437

Average Performance in Training = 0.7956786759006669

Average Performance in Validation = 0.7310562735870354

M Value = 5

Lambda Value = 0.2

Fold 1

Performance Training = 0.8065725400691611

Performance Validation = 0.6634810327001427

Fold 2

Performance Training = 0.7953549896731569

Performance Validation = 0.732788390069625

Fold 3

Performance Training = 0.7903555012450869

Performance Validation = 0.7389996762550142

Fold 4

Performance Training = 0.7894135930728169

Performance Validation = 0.7873455370941029

Average Performance in Training = 0.7954241560150553

Average Performance in Validation = 0.7306536590297212

M Value = 5

Lambda Value = 0.21

Fold 1

Performance Training = 0.8063584555960643

Performance Validation = 0.6619799153606815

Fold 2

Performance Training = 0.7951216243413537

Performance Validation = 0.7324394775717167

Fold 3

Performance Training = 0.7901243652426807

Performance Validation = 0.7385759304560251

Fold 4

Performance Training = 0.7890491043209721

Performance Validation = 0.7879840216388225

Average Performance in Training = 0.7951633873752677

Average Performance in Validation = 0.7302448362568115

M Value = 5

Lambda Value = 0.22

Fold 1

Performance Training = 0.806140022529959
Performance Validation = 0.6604944964702131

Fold 2
Performance Training = 0.7948825562838966
Performance Validation = 0.7320857906817361

Fold 3
Performance Training = 0.7898877194344914
Performance Validation = 0.7381497871996154

Fold 4
Performance Training = 0.788676856661463
Performance Validation = 0.7885913104839652

Average Performance in Training = 0.7948967887274525
Average Performance in Validation = 0.7298303462088824

M Value = 5
Lambda Value = 0.23

Fold 1
Performance Training = 0.8059175731390714
Performance Validation = 0.6590248654977863

Fold 2
Performance Training = 0.7946381281783921
Performance Validation = 0.731727675884827

Fold 3
Performance Training = 0.7896459062744262
Performance Validation = 0.7377215149380738

Fold 4
Performance Training = 0.7882974147333714
Performance Validation = 0.7891687329510169

Average Performance in Training = 0.7946247555813153
Average Performance in Validation = 0.7294106973179261

M Value = 5
Lambda Value = 0.24

Fold 1
Performance Training = 0.805691419580657
Performance Validation = 0.6575710881659592

Fold 2
Performance Training = 0.7943886654480066
Performance Validation = 0.7313654614261197

Fold 3
Performance Training = 0.7893992504431231
Performance Validation = 0.7372913664256604

Fold 4
Performance Training = 0.7879113108367568
Performance Validation = 0.7897175540319687

Average Performance in Training = 0.7943476615771359
Average Performance in Validation = 0.728986367512427

M Value = 5
Lambda Value = 0.25

Fold 1
Performance Training = 0.8054618551740206
Performance Validation = 0.6561332084697664

Fold 2
Performance Training = 0.7941344771818082
Performance Validation = 0.7309994583033584

Fold 3
Performance Training = 0.7891480598246581
Performance Validation = 0.7368595796288513

Fold 4
Performance Training = 0.787519046876243
Performance Validation = 0.7902389779494939

Average Performance in Training = 0.7940658597641824
Average Performance in Validation = 0.7285578060878676

M Value = 5
Lambda Value = 0.26

Fold 1
Performance Training = 0.8052291555854225
Performance Validation = 0.6547112505356772

Fold 2
Performance Training = 0.7938758570011366
Performance Validation = 0.7306299612006798

Fold 3
Performance Training = 0.788892626424263
Performance Validation = 0.7364263785798385

Fold 4
Performance Training = 0.7871210961766545
Performance Validation = 0.7907341514939473

Average Performance in Training = 0.7937796837968693
Average Performance in Validation = 0.7281254354525357

M Value = 5

Lambda Value = 0.27

Fold 1

Performance Training = 0.8049935799315825

Performance Validation = 0.6533052203337987

Fold 2

Performance Training = 0.7936130838755007

Performance Validation = 0.7302572493673198

Fold 3

Performance Training = 0.788633227230984

Performance Validation = 0.7359919741772676

Fold 4

Performance Training = 0.7867179051799849

Performance Validation = 0.7912041671528629

Average Performance in Training = 0.793489449054513

Average Performance in Validation = 0.7276896527578123

M Value = 5

Lambda Value = 0.28

Fold 1

Performance Training = 0.8047553718079258

Performance Validation = 0.6519151072552933

Fold 2

Performance Training = 0.7933464228912445

Performance Validation = 0.7298815874448534

Fold 3

Performance Training = 0.7883701250289242

Performance Validation = 0.7355565649376481

Fold 4

Performance Training = 0.7863098950322278

Performance Validation = 0.7916500660474062

Average Performance in Training = 0.7931954536900805

Average Performance in Validation = 0.7272508314213002

M Value = 5

Lambda Value = 0.29

Fold 1

Performance Training = 0.8045147602472127

Performance Validation = 0.6505408855659863

Fold 2

Performance Training = 0.7930761259759871

Performance Validation = 0.7295032262462886

Fold 3
Performance Training = 0.7881035691604414
Performance Validation = 0.7351203377008506

Fold 4
Performance Training = 0.7858974630679285
Performance Validation = 0.7920728406890724

Average Performance in Training = 0.7928979796128923
Average Performance in Validation = 0.7268093225505494

M Value = 5
Lambda Value = 0.3

Fold 1
Performance Training = 0.8042719606137234
Performance Validation = 0.6491825157461331

Fold 2
Performance Training = 0.7928024325816565
Performance Validation = 0.7291224034900436

Fold 3
Performance Training = 0.7878337962444458
Performance Validation = 0.7346834682926496

Fold 4
Performance Training = 0.7854809841997059
Performance Validation = 0.7924734375689526

Average Performance in Training = 0.7925972934098829
Average Performance in Validation = 0.7263654562744448

M Value = 5
Lambda Value = 0.31

Fold 1
Performance Training = 0.8040271754377573
Performance Validation = 0.6478399457255977

Fold 2
Performance Training = 0.7925255703287108
Performance Validation = 0.7287393444917132

Fold 3
Performance Training = 0.7875610308526956
Performance Validation = 0.7342461221472537

Fold 4
Performance Training = 0.7850608122194412
Performance Validation = 0.7928527595909722

Average Performance in Training = 0.7922936472096512
Average Performance in Validation = 0.7259195429888842

M Value = 5
Lambda Value = 0.32

Fold 1
Performance Training = 0.8037805951948183
Performance Validation = 0.646513112022696

Fold 2
Performance Training = 0.7922457556139804
Performance Validation = 0.7283542628162782

Fold 3
Performance Training = 0.7872854861468033
Performance Validation = 0.7338084548923807

Fold 4
Performance Training = 0.7846372810172764
Performance Validation = 0.793211668359573

Average Performance in Training = 0.7919872794932195
Average Performance in Validation = 0.7254718745227319

M Value = 5
Lambda Value = 0.33

Fold 1
Performance Training = 0.8035323990334985
Performance Validation = 0.6452019407944624

Fold 2
Performance Training = 0.7919631941843935
Performance Validation = 0.7279673608932129

Fold 3
Performance Training = 0.7870073644784595
Performance Validation = 0.7333706128993376

Fold 4
Performance Training = 0.7842107057241461
Performance Validation = 0.7935509863316171

Average Performance in Training = 0.7916784158551244
Average Performance in Validation = 0.7250227252296575

M Value = 5
Lambda Value = 0.34

Fold 1
Performance Training = 0.8032827554557742
Performance Validation = 0.6439063488052925

Fold 2

Performance Training = 0.791678081678693
Performance Validation = 0.7275788305968304

Fold 3
Performance Training = 0.7867268579552111
Performance Validation = 0.7329327338004263

Fold 4
Performance Training = 0.7837813837830957
Performance Validation = 0.7938714988414947

Average Performance in Training = 0.7913672697181936
Average Performance in Validation = 0.7245723530110109

M Value = 5
Lambda Value = 0.35000000000000003

Fold 1
Performance Training = 0.8030318229531094
Performance Validation = 0.6426262443204414

Fold 2
Performance Training = 0.7913906041391015
Performance Validation = 0.7271888537940052

Fold 3
Performance Training = 0.7864441489739674
Performance Validation = 0.7324949469757189

Fold 4
Performance Training = 0.7833495959542505
Performance Validation = 0.7941739560078225

Average Performance in Training = 0.7910540430051072
Average Performance in Validation = 0.724121000274497

M Value = 5
Lambda Value = 0.36

Fold 1
Performance Training = 0.8027797506015184
Performance Validation = 0.6413615279302092

Fold 2
Performance Training = 0.7911009384947763
Performance Validation = 0.7267976028612793

Fold 3
Performance Training = 0.7861594107242588
Performance Validation = 0.7320573740111989

Fold 4
Performance Training = 0.7829156072579495
Performance Validation = 0.794459074529453

Average Performance in Training = 0.7907389267696256
Average Performance in Validation = 0.7236688948330351

M Value = 5
Lambda Value = 0.37

Fold 1
Performance Training = 0.8025266786184811
Performance Validation = 0.6401120933102924

Fold 2
Performance Training = 0.7908092530187625
Performance Validation = 0.7264052411731997

Fold 3
Performance Training = 0.7858728076631271
Performance Validation = 0.731620129130109

Fold 4
Performance Training = 0.7824796678602075
Performance Validation = 0.7947275393779741

Average Performance in Training = 0.7904221017901445
Average Performance in Validation = 0.7232162507478939

M Value = 5
Lambda Value = 0.38

Fold 1
Performance Training = 0.8022727388843842
Performance Validation = 0.6388778279231946

Fold 2
Performance Training = 0.7905157077600296
Performance Validation = 0.726011923563693

Fold 3
Performance Training = 0.7855844959634113
Performance Validation = 0.7311833195991666

Fold 4
Performance Training = 0.7820420139043632
Performance Validation = 0.7949800053933883

Average Performance in Training = 0.7901037391280471
Average Performance in Validation = 0.7227632691198606

M Value = 5
Lambda Value = 0.39

Fold 1
Performance Training = 0.8020180554309588

Performance Validation = 0.6376586136652989

Fold 2

Performance Training = 0.7902204549520977

Performance Validation = 0.7256177967620612

Fold 3

Performance Training = 0.785294623937057

Performance Validation = 0.7307470461112547

Fold 4

Performance Training = 0.7816028682925065

Performance Validation = 0.7952170987891614

Average Performance in Training = 0.7897840006531549

Average Performance in Validation = 0.7223101388319441

M Value = 5

Lambda Value = 0.4

Fold 1

Performance Training = 0.8017627448989999

Performance Validation = 0.6364543274637824

Fold 2

Performance Training = 0.7899236393996373

Performance Validation = 0.7252229998050952

Fold 3

Performance Training = 0.7850033324349779

Performance Validation = 0.7303114031460592

Fold 4

Performance Training = 0.7811624414199846

Performance Validation = 0.7954394185723709

Average Performance in Training = 0.7894630395383999

Average Performance in Validation = 0.7218570372468269

M Value = 5

Lambda Value = 0.41000000000000003

Fold 1

Performance Training = 0.8015069169674643

Performance Validation = 0.635264841827193

Fold 2

Performance Training = 0.789625398844347

Performance Validation = 0.7248276644268308

Fold 3

Performance Training = 0.7847107552249005

Performance Validation = 0.729876479310021

Fold 4
Performance Training = 0.7807209318660822
Performance Validation = 0.7956475378843202

Average Performance in Training = 0.7891410007256985
Average Performance in Validation = 0.7214041308620913

M Value = 5
Lambda Value = 0.42

Fold 1
Performance Training = 0.8012506747559165
Performance Validation = 0.6340900253532369

Fold 2
Performance Training = 0.7893258643113238
Performance Validation = 0.724431915427151

Fold 3
Performance Training = 0.7844170193485067
Performance Validation = 0.7294423576568823

Fold 4
Performance Training = 0.7802785270437235
Performance Validation = 0.7958420052666246

Average Performance in Training = 0.7888180213648676
Average Performance in Validation = 0.7209515759259737

M Value = 5
Lambda Value = 0.43

Fold 1
Performance Training = 0.8009941152021108
Performance Validation = 0.6329297431970973

Fold 2
Performance Training = 0.7890251604370748
Performance Validation = 0.7240358710205813

Fold 3
Performance Training = 0.7841222454591334
Performance Validation = 0.7290091159900262

Fold 4
Performance Training = 0.7798354038108595
Performance Validation = 0.7960233458573431

Average Performance in Training = 0.7884942312272947
Average Performance in Validation = 0.720499519016262

M Value = 5
Lambda Value = 0.44

Fold 1
Performance Training = 0.8007373294163938
Performance Validation = 0.6317838575031511

Fold 2
Performance Training = 0.7887234057802274
Performance Validation = 0.7236396431663679

Fold 3
Performance Training = 0.7838265481411725
Performance Validation = 0.7285768271477236

Fold 4
Performance Training = 0.7793917290459954
Performance Validation = 0.7961920625215096

Average Performance in Training = 0.7881697530959473
Average Performance in Validation = 0.7200480975846881

M Value = 5
Lambda Value = 0.45

Fold 1
Performance Training = 0.8004804030144714
Performance Validation = 0.6306522278029649

Fold 2
Performance Training = 0.7884207131159378
Performance Validation = 0.7232433378809926

Fold 3
Performance Training = 0.7835300362122701
Performance Validation = 0.7281455592723229

Fold 4
Performance Training = 0.7789476601901646
Performance Validation = 0.7963486369200978

Average Performance in Training = 0.787844703133211
Average Performance in Validation = 0.7195974404690946

M Value = 5
Lambda Value = 0.46

Fold 1
Performance Training = 0.8002234164299811
Performance Validation = 0.6295347113820826

Fold 2
Performance Training = 0.7881171897149439
Performance Validation = 0.7228470555340758

Fold 3

Performance Training = 0.7832328130093279
Performance Validation = 0.7277153760643643

Fold 4
Performance Training = 0.7785033457574717
Performance Validation = 0.7964935305211109

Average Performance in Training = 0.7875191912279311
Average Performance in Validation = 0.7191476683754084

M Value = 5
Lambda Value = 0.47000000000000003

Fold 1
Performance Training = 0.7999664452081957
Performance Validation = 0.6284311636178999

Fold 2
Performance Training = 0.7878129376081224
Performance Validation = 0.7224508911287135

Fold 3
Performance Training = 0.7829349766592602
Performance Validation = 0.7272863370225144

Fold 4
Performance Training = 0.7780589258161993
Performance Validation = 0.7966271855563556

Average Performance in Training = 0.7871933213229444
Average Performance in Validation = 0.7186988943313709

M Value = 5
Lambda Value = 0.48

Fold 1
Performance Training = 0.7997095602821026
Performance Validation = 0.6273414382908263

Fold 2
Performance Training = 0.7875080538373891
Performance Validation = 0.7220549345670288

Fold 3
Performance Training = 0.7826366203353972
Performance Validation = 0.7268584976701804

Fold 4
Performance Training = 0.7776145324423163
Performance Validation = 0.7967500259271175

Average Performance in Training = 0.7868671917243012
Average Performance in Validation = 0.7182512241137884

M Value = 5
Lambda Value = 0.49

Fold 1

Performance Training = 0.7994528282320027
Performance Validation = 0.6262653878707601

Fold 2

Performance Training = 0.7872026306937032
Performance Validation = 0.7216592709018789

Fold 3

Performance Training = 0.7823378325003598
Performance Validation = 0.7264319097695906

Fold 4

Performance Training = 0.7771702901471143
Performance Validation = 0.7968624580617921

Average Performance in Training = 0.786540895393295
Average Performance in Validation = 0.7178047566510054

M Value = 5
Lambda Value = 0.5

Fold 1

Performance Training = 0.7991963115297006
Performance Validation = 0.6252028637806688

Fold 2

Performance Training = 0.7868967559429024
Performance Validation = 0.721263980575473

Fold 3

Performance Training = 0.7820386971361907
Performance Validation = 0.7260066215240895

Fold 4

Performance Training = 0.7767263162805894
Performance Validation = 0.7969648717283501

Average Performance in Training = 0.7862145202223457
Average Performance in Validation = 0.7173595844021453

M Value = 5
Lambda Value = 0.51

Fold 1

Performance Training = 0.7989400687682684
Performance Validation = 0.6241537166389873

Fold 2

Performance Training = 0.7865905130400483

Performance Validation = 0.7208691396455803

Fold 3

Performance Training = 0.7817392939624613

Performance Validation = 0.7255826777693413

Fold 4

Performance Training = 0.776282721412037

Performance Validation = 0.7970576408042382

Average Performance in Training = 0.7858881492957037

Average Performance in Validation = 0.7169157937145368

M Value = 5

Lambda Value = 0.52

Fold 1

Performance Training = 0.7986841548783151

Performance Validation = 0.6231177964823984

Fold 2

Performance Training = 0.786283981332919

Performance Validation = 0.7204748200001461

Fold 3

Performance Training = 0.7814396986430583

Performance Validation = 0.7251601201540805

Fold 4

Performance Training = 0.7758396096892833

Performance Validation = 0.7971411240062498

Average Performance in Training = 0.785561861135894

Average Performance in Validation = 0.7164734651607187

M Value = 5

Lambda Value = 0.53

Fold 1

Performance Training = 0.7984286213316223

Performance Validation = 0.6220949529704991

Fold 2

Performance Training = 0.7859772362552513

Performance Validation = 0.7200810895608363

Fold 3

Performance Training = 0.7811399829822682

Performance Validation = 0.7247389873110481

Fold 4

Performance Training = 0.7753970791778443

Performance Validation = 0.7972156655826554

Average Performance in Training = 0.7852357299367465
Average Performance in Validation = 0.7160326738562597

M Value = 5
Lambda Value = 0.54

Fold 1
Performance Training = 0.7981735163329263
Performance Validation = 0.6210850355735897

Fold 2
Performance Training = 0.7856703495102878
Performance Validation = 0.719688012476205

Fold 3
Performance Training = 0.7808402151107791
Performance Validation = 0.7243193150186438

Fold 4
Performance Training = 0.7749552221812187
Performance Validation = 0.7972815959697994

Average Performance in Training = 0.784909825783803
Average Performance in Validation = 0.7155934897595595

M Value = 5
Lambda Value = 0.55

Fold 1
Performance Training = 0.7979188850006151
Performance Validation = 0.6200878937449738

Fold 2
Performance Training = 0.7853633892451672
Performance Validation = 0.7192956493050393

Fold 3
Performance Training = 0.7805404596621544
Performance Validation = 0.7239011363538714

Fold 4
Performance Training = 0.774514125543474
Performance Validation = 0.7973392324151807

Average Performance in Training = 0.7845842148628528
Average Performance in Validation = 0.7151559779547663

M Value = 5
Lambda Value = 0.56

Fold 1
Performance Training = 0.7976647695370161
Performance Validation = 0.6191033770787381

Fold 2
Performance Training = 0.7850564202166511
Performance Validation = 0.7189040571904128

Fold 3
Performance Training = 0.7802407779403181
Performance Validation = 0.7234844818370474

Fold 4
Performance Training = 0.7740738709351664
Performance Validation = 0.7973888795689278

Average Performance in Training = 0.784258959657288
Average Performance in Validation = 0.7147201989187815

M Value = 5
Lambda Value = 0.5700000000000001

Fold 1
Performance Training = 0.7974112093889276
Performance Validation = 0.6181313354542255

Fold 2
Performance Training = 0.7847495039486551
Performance Validation = 0.7185132900249392

Fold 3
Performance Training = 0.7799412280785388
Performance Validation = 0.7230693795687496

Fold 4
Performance Training = 0.7736345351235927
Performance Validation = 0.797430830045452

Average Performance in Training = 0.7839341191349285
Average Performance in Validation = 0.7142862087733415

M Value = 5
Lambda Value = 0.58

Fold 1
Performance Training = 0.7971582413989948
Performance Validation = 0.6171716191681307

Fold 2
Performance Training = 0.7844426988820243
Performance Validation = 0.7181233986077691

Fold 3
Performance Training = 0.7796418651903905
Performance Validation = 0.7226558553594729

Fold 4

Performance Training = 0.7731961902283069
Performance Validation = 0.7974653649569785

Average Performance in Training = 0.7836097489249291
Average Performance in Validation = 0.7138540595230878

M Value = 5
Lambda Value = 0.59

Fold 1
Performance Training = 0.7969058999484948
Performance Validation = 0.6162240790550768

Fold 2
Performance Training = 0.7841360605169708
Performance Validation = 0.7177344307936887

Fold 3
Performance Training = 0.7793427415131261
Performance Validation = 0.7222439328523677

Fold 4
Performance Training = 0.7727589039627678
Performance Validation = 0.797492754420495

Average Performance in Training = 0.7832859014853399
Average Performance in Validation = 0.713423799280407

M Value = 5
Lambda Value = 0.6

Fold 1
Performance Training = 0.7966542170920525
Performance Validation = 0.6152885665976114

Fold 2
Performance Training = 0.7838296415485648
Performance Validation = 0.7173464316348254

Fold 3
Performance Training = 0.7790439065438755
Performance Validation = 0.7218336336394748

Fold 4
Performance Training = 0.7723227398629223
Performance Validation = 0.7975132580396078

Average Performance in Training = 0.7829626262618538
Average Performance in Validation = 0.7129954724778799

M Value = 5
Lambda Value = 0.61

Fold 1
Performance Training = 0.7964032226847766
Performance Validation = 0.61436493402633

Fold 2
Performance Training = 0.7835234919956362
Performance Validation = 0.7169594435153028

Fold 3
Performance Training = 0.7787454071690624
Performance Validation = 0.7214249773718193

Fold 4
Performance Training = 0.7718877575034984
Performance Validation = 0.7975271253627103

Average Performance in Training = 0.7826399698382434
Average Performance in Validation = 0.7125691200690405

M Value = 5
Lambda Value = 0.62

Fold 1
Performance Training = 0.7961529445022805
Performance Validation = 0.613453034410872

Fold 2
Performance Training = 0.7832176593234561
Performance Validation = 0.7165735062792347

Fold 3
Performance Training = 0.778447287787398
Performance Validation = 0.7210179818637028

Fold 4
Performance Training = 0.7714540127027099
Performance Validation = 0.7975345963187077

Average Performance in Training = 0.782317976078961
Average Performance in Validation = 0.7121447797181293

M Value = 5
Lambda Value = 0.63

Fold 1
Performance Training = 0.7959034083540039
Performance Validation = 0.6125527217424666

Fold 2
Performance Training = 0.7829121885604944
Performance Validation = 0.7161886573524432

Fold 3
Performance Training = 0.7781495904268105

Performance Validation = 0.7206126631915118

Fold 4

Performance Training = 0.7710215577160475

Performance Validation = 0.797535901631597

Average Performance in Training = 0.781996686264339

Average Performance in Validation = 0.7117224859795046

M Value = 5

Lambda Value = 0.64

Fold 1

Performance Training = 0.7956546381902452

Performance Validation = 0.6116638510086496

Fold 2

Performance Training = 0.7826071224095889

Performance Validation = 0.7158049318581929

Fold 3

Performance Training = 0.7778523548556134

Performance Validation = 0.7202090357873661

Fold 4

Performance Training = 0.7705904414197795

Performance Validation = 0.7975312632149812

Average Performance in Training = 0.7816761392188067

Average Performance in Validation = 0.7113022704672974

M Value = 5

Lambda Value = 0.65

Fold 1

Performance Training = 0.7954066562032795

Performance Validation = 0.6107862782607187

Fold 2

Performance Training = 0.7823025013538047

Performance Validation = 0.7154223627272532

Fold 3

Performance Training = 0.7775556186882431

Performance Validation = 0.7198071125278606

Fold 4

Performance Training = 0.770160709484744

Performance Validation = 0.7975208945476211

Average Performance in Training = 0.7813563714325179

Average Performance in Validation = 0.7108841620158635

M Value = 5
Lambda Value = 0.66

Fold 1
Performance Training = 0.7951594829228985
Performance Validation = 0.609919860674466

Fold 2
Performance Training = 0.7819983637572546
Performance Validation = 0.7150409808026343

Fold 3
Performance Training = 0.7772594174858325
Performance Validation = 0.7194069048182012

Fold 4
Performance Training = 0.7697324045410001
Performance Validation = 0.7975050010310669

Average Performance in Training = 0.7810374171767465
Average Performance in Validation = 0.7104681868315921

M Value = 5
Lambda Value = 0.67

Fold 1
Performance Training = 0.7949131373067181
Performance Validation = 0.6090644566047001

Fold 2
Performance Training = 0.7816947459611467
Performance Validation = 0.7146608149392158

Fold 3
Performance Training = 0.7769637848519013
Performance Validation = 0.7190084226719671

Fold 4
Performance Training = 0.7693055663338304
Performance Validation = 0.7974837803302771

Average Performance in Training = 0.7807193086133992
Average Performance in Validation = 0.71005436863654

M Value = 5
Lambda Value = 0.68

Fold 1
Performance Training = 0.794667636825547
Performance Validation = 0.6082199256339677

Fold 2
Performance Training = 0.7813916823753028
Performance Validation = 0.7142818920985835

Fold 3
Performance Training = 0.7766687525234286
Performance Validation = 0.7186116747867412

Fold 4
Performance Training = 0.7688802318716075
Performance Validation = 0.7974574226981599

Average Performance in Training = 0.7804020758989714
Average Performance in Validation = 0.709642728804363

M Value = 5
Lambda Value = 0.6900000000000001

Fold 1
Performance Training = 0.794422997544109
Performance Validation = 0.6073861286159835

Fold 2
Performance Training = 0.7810892055653695
Performance Validation = 0.7139042374392803

Fold 3
Performance Training = 0.7763743504575233
Performance Validation = 0.7182166686158407

Fold 4
Performance Training = 0.7684564355659572
Performance Validation = 0.7974261112848667

Average Performance in Training = 0.7800857472832398
Average Performance in Validation = 0.7092332864889929

M Value = 5
Lambda Value = 0.7000000000000001

Fold 1
Performance Training = 0.7941792341973911
Performance Validation = 0.6065629277140827

Fold 2
Performance Training = 0.7807873463359566
Performance Validation = 0.7135278744027157

Fold 3
Performance Training = 0.7760806069139465
Performance Validation = 0.7178234104363481

Fold 4
Performance Training = 0.7680342093646547
Performance Validation = 0.7973900224326308

Average Performance in Training = 0.7797703492029873

Average Performance in Validation = 0.7088260587464443

M Value = 5

Lambda Value = 0.71

Fold 1

Performance Training = 0.7939363602628617

Performance Validation = 0.6057501864351295

Fold 2

Performance Training = 0.7804861338098871

Performance Validation = 0.7131528247950087

Fold 3

Performance Training = 0.7757875485336867

Performance Validation = 0.7174319054136444

Fold 4

Performance Training = 0.767613582877668

Performance Validation = 0.7973493259569279

Average Performance in Training = 0.7794559063710258

Average Performance in Validation = 0.7084210606501776

M Value = 5

Lambda Value = 0.72

Fold 1

Performance Training = 0.7936943880288039

Performance Validation = 0.6049477696591716

Fold 2

Performance Training = 0.7801855955037778

Performance Validation = 0.7127791088648788

Fold 3

Performance Training = 0.7754952004137945

Performance Validation = 0.7170421576626442

Fold 4

Performance Training = 0.767194583496704

Performance Validation = 0.7973041854146252

Average Performance in Training = 0.77914244186077

Average Performance in Validation = 0.70801830540033

M Value = 5

Lambda Value = 0.73

Fold 1

Performance Training = 0.7934533286589787

Performance Validation = 0.6041555436652082

Fold 2
Performance Training = 0.7798857574001121
Performance Validation = 0.7124067453778538

Fold 3
Performance Training = 0.7752035861786625
Performance Validation = 0.7166541703058821

Fold 4
Performance Training = 0.7667772365086345
Performance Validation = 0.7972547583598362

Average Performance in Training = 0.778829977186597
Average Performance in Validation = 0.707617804427195

M Value = 5
Lambda Value = 0.74

Fold 1
Performance Training = 0.7932131922538378
Performance Validation = 0.6033733761533118

Fold 2
Performance Training = 0.779586644015991
Performance Validation = 0.7120357516869869

Fold 3
Performance Training = 0.7749127280479455
Performance Validation = 0.7162679455286467

Fold 4
Performance Training = 0.7663615652031179
Performance Validation = 0.7972011965880518

Average Performance in Training = 0.7785185323802231
Average Performance in Validation = 0.7072195674892493

M Value = 5
Lambda Value = 0.75

Fold 1
Performance Training = 0.7929739879084645
Performance Validation = 0.6026011362634336

Fold 2
Performance Training = 0.7792882784687356
Performance Validation = 0.7116661438001741

Fold 3
Performance Training = 0.7746226469012734
Performance Validation = 0.7158834846313031

Fold 4
Performance Training = 0.7659475909747555

Performance Validation = 0.797143646369214

Average Performance in Training = 0.7782081260633072

Average Performance in Validation = 0.7068236027660312

M Value = 5

Lambda Value = 0.76

Fold 1

Performance Training = 0.7927357237674433

Performance Validation = 0.6018386945911233

Fold 2

Performance Training = 0.7789906825384654

Performance Validation = 0.7112979364443702

Fold 3

Performance Training = 0.7743333623399308

Performance Validation = 0.7155007880789478

Fold 4

Performance Training = 0.7655353334200501

Performance Validation = 0.7970822486702216

Average Performance in Training = 0.7778987755164724

Average Performance in Validation = 0.7064299169461656

M Value = 5

Lambda Value = 0.77

Fold 1

Performance Training = 0.7924984070768207

Performance Validation = 0.601085923200384

Fold 2

Performance Training = 0.778693876727843

Performance Validation = 0.7109311431267404

Fold 3

Performance Training = 0.7740448927456548

Performance Validation = 0.7151198555485525

Fold 4

Performance Training = 0.765124810429467

Performance Validation = 0.7970171393674451

Average Performance in Training = 0.7775904967449463

Average Performance in Validation = 0.7060385153107804

M Value = 5

Lambda Value = 0.78

Fold 1

Performance Training = 0.7922620442333228
Performance Validation = 0.6003426956338905

Fold 2
Performance Training = 0.778397880319089
Performance Validation = 0.7105657761929649

Fold 3
Performance Training = 0.7737572553366894
Performance Validation = 0.7147406859737151

Fold 4
Performance Training = 0.7647160382748608
Performance Validation = 0.79694844944974

Average Performance in Training = 0.7772833045409904
Average Performance in Validation = 0.7056494018125776

M Value = 5
Lambda Value = 0.79

Fold 1
Performance Training = 0.79202664083098
Performance Validation = 0.5996088869208109

Fold 2
Performance Training = 0.7781027114284162
Performance Validation = 0.7102018468828291

Fold 3
Performance Training = 0.7734704662212382
Performance Validation = 0.7143632775871431

Fold 4
Performance Training = 0.7643090316924908
Performance Validation = 0.7968763052124036

Average Performance in Training = 0.7769772125432813
Average Performance in Validation = 0.7052625791507966

M Value = 5
Lambda Value = 0.8

Fold 1
Performance Training = 0.7917922017053085
Performance Validation = 0.5988843735823737

Fold 2
Performance Training = 0.7778083870580076
Performance Validation = 0.709839365383214

Fold 3
Performance Training = 0.773184540448442
Performance Validation = 0.7139876279609962

Fold 4
Performance Training = 0.7639038039618944
Performance Validation = 0.7968008284425723

Average Performance in Training = 0.7766722332934132
Average Performance in Validation = 0.704878048842289

M Value = 5
Lambda Value = 0.81

Fold 1
Performance Training = 0.7915587309751744
Performance Validation = 0.598169033635391

Fold 2
Performance Training = 0.7775149231456531
Performance Validation = 0.7094783408786334

Fold 3
Performance Training = 0.7728994920570078
Performance Validation = 0.7136137340451958

Fold 4
Performance Training = 0.7635003669808159
Performance Validation = 0.7967221365964502

Average Performance in Training = 0.7763683782896629
Average Performance in Validation = 0.7044958112889176

M Value = 5
Lambda Value = 0.8200000000000001

Fold 1
Performance Training = 0.7913262320824713
Performance Validation = 0.5974627465938713

Fold 2
Performance Training = 0.7772223346121577
Performance Validation = 0.7091187815994274

Fold 3
Performance Training = 0.7726153341215946
Performance Validation = 0.7132415922037945

Fold 4
Performance Training = 0.7630987313363968
Performance Validation = 0.79664034296875

Average Performance in Training = 0.7760656580381551
Average Performance in Validation = 0.7041158658414608

M Value = 5

Lambda Value = 0.8300000000000001

Fold 1

Performance Training = 0.7910947078297341

Performance Validation = 0.5967653934688995

Fold 2

Performance Training = 0.77693063540663

Performance Validation = 0.7087606948677302

Fold 3

Performance Training = 0.7723320787970804

Performance Validation = 0.7128711982495226

Fold 4

Performance Training = 0.7626989063728359

Performance Validation = 0.796555556854775

Average Performance in Training = 0.7757640821015701

Average Performance in Validation = 0.7037382108602318

M Value = 5

Lambda Value = 0.84

Fold 1

Performance Training = 0.7908641604157993

Performance Validation = 0.5960768567669357

Fold 2

Performance Training = 0.7766398385497546

Performance Validation = 0.7084040871413158

Fold 3

Performance Training = 0.7720497373608012

Performance Validation = 0.7125025474765994

Fold 4

Performance Training = 0.7623009002557003

Performance Validation = 0.7964678837054608

Average Performance in Training = 0.7754636591455139

Average Performance in Validation = 0.7033628437725779

M Value = 5

Lambda Value = 0.85

Fold 1

Performance Training = 0.7906345914696099

Performance Validation = 0.5953970204866197

Fold 2

Performance Training = 0.7763499561751315

Performance Validation = 0.7080489640554415

Fold 3

Performance Training = 0.7717683202528658

Performance Validation = 0.7121356346918917

Fold 4

Performance Training = 0.7619047200330502

Performance Validation = 0.796377425275717

Average Performance in Training = 0.7751643969826643

Average Performance in Validation = 0.7029897611274175

M Value = 5

Lambda Value = 0.86

Fold 1

Performance Training = 0.7904060020822796

Performance Validation = 0.5947257701142517

Fold 2

Performance Training = 0.7760609995687942

Performance Validation = 0.7076953304627542

Fold 3

Performance Training = 0.7714878371146423

Performance Validation = 0.7117704542445076

Fold 4

Performance Training = 0.7615103716935636

Performance Validation = 0.79628427976642

Average Performance in Training = 0.7748663026148199

Average Performance in Validation = 0.7026189586469833

M Value = 5

Lambda Value = 0.87

Fold 1

Performance Training = 0.7901783928375008

Performance Validation = 0.5940629926180514

Fold 2

Performance Training = 0.7757729792069735

Performance Validation = 0.7073431904713849

Fold 3

Performance Training = 0.7712082968254952

Performance Validation = 0.7114070000539126

Fold 4

Performance Training = 0.7611178602217931

Performance Validation = 0.7961885419603159

Average Performance in Training = 0.7745693822729407

Average Performance in Validation = 0.7022504312759161

M Value = 5
Lambda Value = 0.88

Fold 1
Performance Training = 0.7899517638403802
Performance Validation = 0.5934085764412658

Fold 2
Performance Training = 0.7754859047921991
Performance Validation = 0.7069925474812989

Fold 3
Performance Training = 0.7709297075378716
Performance Validation = 0.7110452656366335

Fold 4
Performance Training = 0.7607271896507188
Performance Validation = 0.7960903033521581

Average Performance in Training = 0.7742736414552924
Average Performance in Validation = 0.701884173227839

M Value = 5
Lambda Value = 0.89

Fold 1
Performance Training = 0.7897261147447998
Performance Validation = 0.5927624114942867

Fold 2
Performance Training = 0.775199785287819
Performance Validation = 0.7066434042189778

Fold 3
Performance Training = 0.7706520767107947
Performance Validation = 0.7106852441316176

Fold 4
Performance Training = 0.760338363111731
Performance Validation = 0.7959896522733395

Average Performance in Training = 0.7739790849637861
Average Performance in Validation = 0.7015201780295554

M Value = 5
Lambda Value = 0.9

Fold 1
Performance Training = 0.7895014447793688
Performance Validation = 0.592124389145803

Fold 2

Performance Training = 0.7749146289509972
Performance Validation = 0.7062957627705542

Fold 3
Performance Training = 0.7703754111418666
Performance Validation = 0.7103269283243292

Fold 4
Performance Training = 0.7599513828821687
Performance Validation = 0.7958866740112667

Average Performance in Training = 0.7736857169386002
Average Performance in Validation = 0.7011584385629883

M Value = 5
Lambda Value = 0.91

Fold 1
Performance Training = 0.7892777527720497
Performance Validation = 0.5914944022131496

Fold 2
Performance Training = 0.7746304433642779
Performance Validation = 0.7059496246134079

Fold 3
Performance Training = 0.7700997169978275
Performance Validation = 0.7099703106696392

Fold 4
Performance Training = 0.759566250430548
Performance Validation = 0.7957814509237475

Average Performance in Training = 0.7733935408911758
Average Performance in Validation = 0.700798947104986

M Value = 5
Lambda Value = 0.92

Fold 1
Performance Training = 0.7890550371735201
Performance Validation = 0.5908723449518322

Fold 2
Performance Training = 0.7743472354657679
Performance Validation = 0.7056049906463612

Fold 3
Performance Training = 0.7698249998437501
Performance Validation = 0.7096153833135614

Fold 4
Performance Training = 0.759182966459585
Performance Validation = 0.7956740625485865

Average Performance in Training = 0.7731025597356558
Average Performance in Validation = 0.7004416953650854

M Value = 5
Lambda Value = 0.93

Fold 1
Performance Training = 0.7888332960793475
Performance Validation = 0.5902581130444216

Fold 2
Performance Training = 0.7740650115780111
Performance Validation = 0.7052618612184758

Fold 3
Performance Training = 0.7695512646709382
Performance Validation = 0.7092621381139108

Fold 4
Performance Training = 0.7588015309471356
Performance Validation = 0.7955645857086436

Average Performance in Training = 0.7728127758188581
Average Performance in Validation = 0.7000866745213629

M Value = 5
Lambda Value = 0.9400000000000001

Fold 1
Performance Training = 0.7886125272510286
Performance Validation = 0.5896516035887738

Fold 2
Performance Training = 0.773783777435605
Performance Validation = 0.7049202361565876

Fold 3
Performance Training = 0.7692785159235767
Performance Validation = 0.7089105666599111

Fold 4
Performance Training = 0.7584219431851552
Performance Validation = 0.7954530946125611

Average Performance in Training = 0.7725241909488414
Average Performance in Validation = 0.6997338752544584

M Value = 5
Lambda Value = 0.9500000000000001

Fold 1
Performance Training = 0.7883927281359566

Performance Validation = 0.589052715085701

Fold 2

Performance Training = 0.7735035382116301

Performance Validation = 0.7045801147915659

Fold 3

Performance Training = 0.7690067575242064

Performance Validation = 0.7085606602908356

Fold 4

Performance Training = 0.7580442018167675

Performance Validation = 0.7953396609513268

Average Performance in Training = 0.7722368064221401

Average Performance in Validation = 0.6993832877798574

M Value = 5

Lambda Value = 0.96

Fold 1

Performance Training = 0.7881738958863767

Performance Validation = 0.5884613474261491

Fold 2

Performance Training = 0.7732242985429285

Performance Validation = 0.7042414959834209

Fold 3

Performance Training = 0.7687359928980655

Performance Validation = 0.7082124101136981

Fold 4

Performance Training = 0.757668304871552

Performance Validation = 0.795224353990905

Average Performance in Training = 0.7719506230497306

Average Performance in Validation = 0.6990349018785433

M Value = 5

Lambda Value = 0.97

Fold 1

Performance Training = 0.787956027377374

Performance Validation = 0.5878774018779269

Fold 2

Performance Training = 0.7729460625543046

Performance Validation = 0.703904378145252

Fold 3

Performance Training = 0.7684662249963616

Performance Validation = 0.7078658070200721

Fold 4
Performance Training = 0.7572942497991243
Performance Validation = 0.7951072406610739

Average Performance in Training = 0.7716656411817912
Average Performance in Validation = 0.6986887069260812

M Value = 5
Lambda Value = 0.98

Fold 1
Performance Training = 0.7877391192239533
Performance Validation = 0.5873007810720325

Fold 2
Performance Training = 0.7726688338816784
Performance Validation = 0.7035687592661499

Fold 3
Performance Training = 0.7681974563185119
Performance Validation = 0.7075208417020444

Fold 4
Performance Training = 0.7569220335011067
Performance Validation = 0.7949883856406816

Average Performance in Training = 0.7713818607313127
Average Performance in Validation = 0.6983446919202272

M Value = 5
Lambda Value = 0.99

Fold 1
Performance Training = 0.7875231677972392
Performance Validation = 0.5867313889886068

Fold 2
Performance Training = 0.7723926156942459
Performance Validation = 0.7032346369330549

Fold 3
Performance Training = 0.7679296889334095
Performance Validation = 0.7071775046673849

Fold 4
Performance Training = 0.7565516523615571
Performance Validation = 0.7948678514394412

Average Performance in Training = 0.771099281196613
Average Performance in Validation = 0.6980028455071219

M Value = 5
Lambda Value = 1.0

Fold 1
Performance Training = 0.787308169239875
Performance Validation = 0.5861691309426336

Fold 2
Performance Training = 0.7721174107157014
Performance Validation = 0.7029020083516462

Fold 3
Performance Training = 0.7676629244997575
Performance Validation = 0.7068357862539412

Fold 4
Performance Training = 0.7561831022759427
Performance Validation = 0.7947456984764484

Average Performance in Training = 0.7708179016828192
Average Performance in Validation = 0.6976631560061674

M Value = 5
Lambda Value = 1.01

Fold 1
Performance Training = 0.7870941194806232
Performance Validation = 0.5856139135693255

Fold 2
Performance Training = 0.7718432212445382
Performance Validation = 0.7025708703662812

Fold 3
Performance Training = 0.7673971642855076
Performance Validation = 0.7064956766433094

Fold 4
Performance Training = 0.755816378678719
Performance Validation = 0.7946219851555509

Average Performance in Training = 0.770537720922347
Average Performance in Validation = 0.6973256114336168

M Value = 5
Lambda Value = 1.02

Fold 1
Performance Training = 0.7868810142482319
Performance Validation = 0.585065644809295

Fold 2
Performance Training = 0.7715700491734976
Performance Validation = 0.7022412194790679

Fold 3

Performance Training = 0.7671324091864538
Performance Validation = 0.7061571658738128

Fold 4

Performance Training = 0.7554514765695908
Performance Validation = 0.7944967679377194

Average Performance in Training = 0.7702587372944435
Average Performance in Validation = 0.6969901995249738

M Value = 5

Lambda Value = 1.03

Fold 1

Performance Training = 0.7866688490846029
Performance Validation = 0.5845242338935457

Fold 2

Performance Training = 0.7712978960081831
Performance Validation = 0.7019130518680723

Fold 3

Performance Training = 0.7668686597440082
Performance Validation = 0.7058202438528298

Fold 4

Performance Training = 0.7550883905385243
Performance Validation = 0.7943701014105575

Average Performance in Training = 0.7699809488438297
Average Performance in Validation = 0.6966569077562513

M Value = 5

Lambda Value = 1.04

Fold 1

Performance Training = 0.7864576193572872
Performance Validation = 0.5839895913282762

Fold 2

Performance Training = 0.7710267628848909
Performance Validation = 0.7015863634047006

Fold 3

Performance Training = 0.7666059161622139
Performance Validation = 0.7054849003684855

Fold 4

Performance Training = 0.7547271147895457
Performance Validation = 0.794242038355041

Average Performance in Training = 0.7697043532984844
Average Performance in Validation = 0.6963257233641258

M Value = 5
Lambda Value = 1.05

Fold 1

Performance Training = 0.7862473202713488
Performance Validation = 0.5834616288795744

Fold 2

Performance Training = 0.7707566505876737
Performance Validation = 0.7012611496703451

Fold 3

Performance Training = 0.7663441783240015
Performance Validation = 0.7051511251007594

Fold 4

Performance Training = 0.7543676431634196
Performance Validation = 0.7941126298096592

Average Performance in Training = 0.7694289480866109
Average Performance in Validation = 0.6959966333650845

M Value = 5
Lambda Value = 1.06

Fold 1

Performance Training = 0.7860379468806274
Performance Validation = 0.5829402595579998

Fold 2

Performance Training = 0.7704875595646977
Performance Validation = 0.7009374059722469

Fold 3

Performance Training = 0.766083445806758
Performance Validation = 0.7048189076320175

Fold 4

Performance Training = 0.754009969159241
Performance Validation = 0.7939819251320381

Average Performance in Training = 0.769154730352831
Average Performance in Validation = 0.6956696245735756

M Value = 5
Lambda Value = 1.07

Fold 1

Performance Training = 0.7858294940984336
Performance Validation = 0.5824253976030773

Fold 2

Performance Training = 0.7702194899438982

Performance Validation = 0.7006151273586735

Fold 3

Performance Training = 0.7658237178972039

Performance Validation = 0.7044882374570123

Fold 4

Performance Training = 0.753654085954987

Performance Validation = 0.7938499720581471

Average Performance in Training = 0.7688816969736307

Average Performance in Validation = 0.6953446836192275

M Value = 5

Lambda Value = 1.08

Fold 1

Performance Training = 0.7856219567076901

Performance Validation = 0.5819169584677402

Fold 2

Performance Training = 0.7699524415479744

Performance Validation = 0.7002943086334213

Fold 3

Performance Training = 0.7655649936056389

Performance Validation = 0.7041591039923705

Fold 4

Performance Training = 0.7532999864270999

Performance Validation = 0.7937168167592272

Average Performance in Training = 0.7686098445721008

Average Performance in Validation = 0.6950217969631898

M Value = 5

Lambda Value = 1.09

Fold 1

Performance Training = 0.7854153293705768

Performance Validation = 0.5814148588027644

Fold 2

Performance Training = 0.7696864139087625

Performance Validation = 0.6999749443696719

Fold 3

Performance Training = 0.7653072716795628

Performance Validation = 0.7038314965855856

Fold 4

Performance Training = 0.7529476631691342

Performance Validation = 0.7935825038965154

Average Performance in Training = 0.7683391695320091
Average Performance in Validation = 0.6947009509136344

M Value = 6
Lambda Value = 0.01

Fold 1
Performance Training = 0.8092393362228084
Performance Validation = 0.7030098134228018

Fold 2
Performance Training = 0.7989152429671738
Performance Validation = 0.7459806428108269

Fold 3
Performance Training = 0.8042705190187347
Performance Validation = 0.7099385997354801

Fold 4
Performance Training = 0.7960446955088814
Performance Validation = 0.7717184725866486

Average Performance in Training = 0.8021174484293995
Average Performance in Validation = 0.7326618821389395

M Value = 6
Lambda Value = 0.02

Fold 1
Performance Training = 0.8092057809657407
Performance Validation = 0.6999658144125304

Fold 2
Performance Training = 0.7988859880918083
Performance Validation = 0.7456563713663515

Fold 3
Performance Training = 0.8042270511169112
Performance Validation = 0.7103115849401311

Fold 4
Performance Training = 0.7959921103241601
Performance Validation = 0.7733600615810097

Average Performance in Training = 0.8020777326246551
Average Performance in Validation = 0.7323234580750058

M Value = 6
Lambda Value = 0.03

Fold 1
Performance Training = 0.8091524968519143
Performance Validation = 0.6969702538770142

Fold 2
Performance Training = 0.7988386620884576
Performance Validation = 0.7453222744714869

Fold 3
Performance Training = 0.804158318538474
Performance Validation = 0.7106041234141225

Fold 4
Performance Training = 0.7959075361324489
Performance Validation = 0.7749199378414078

Average Performance in Training = 0.8020142534028236
Average Performance in Validation = 0.7319541474010078

M Value = 6
Lambda Value = 0.04

Fold 1
Performance Training = 0.8090813393305248
Performance Validation = 0.6940236128675803

Fold 2
Performance Training = 0.7987743469894755
Performance Validation = 0.7449789282810719

Fold 3
Performance Training = 0.8040668776710337
Performance Validation = 0.7108258005771844

Fold 4
Performance Training = 0.7957932582091376
Performance Validation = 0.7764023121248221

Average Performance in Training = 0.8019289555500428
Average Performance in Validation = 0.7315576634626646

M Value = 6
Lambda Value = 0.05

Fold 1
Performance Training = 0.808993943700808
Performance Validation = 0.6911260244770916

Fold 2
Performance Training = 0.7986940536239865
Performance Validation = 0.7446268852097351

Fold 3
Performance Training = 0.803954943382176
Performance Validation = 0.7109848395679959

Fold 4

Performance Training = 0.7956513855629714
Performance Validation = 0.7778111288346672

Average Performance in Training = 0.8018235815674855
Average Performance in Validation = 0.7311372195223724

M Value = 6
Lambda Value = 0.06

Fold 1
Performance Training = 0.8088917599788594
Performance Validation = 0.6882773589835227

Fold 2
Performance Training = 0.7985987277423553
Performance Validation = 0.7442666744861015

Fold 3
Performance Training = 0.803824447769813
Performance Validation = 0.7110883315718233

Fold 4
Performance Training = 0.7954838678825493
Performance Validation = 0.7791500874355652

Average Performance in Training = 0.8016997008433943
Average Performance in Validation = 0.7306956131192532

M Value = 6
Lambda Value = 0.07

Fold 1
Performance Training = 0.8087760809604876
Performance Validation = 0.6854772886411553

Fold 2
Performance Training = 0.7984892554181199
Performance Validation = 0.7438988027450517

Fold 3
Performance Training = 0.803677087039783
Performance Validation = 0.7111424214701243

Fold 4
Performance Training = 0.795292510408764
Performance Validation = 0.7804226616386363

Average Performance in Training = 0.8015587334567886
Average Performance in Validation = 0.730235293623742

M Value = 6
Lambda Value = 0.08

Fold 1
Performance Training = 0.8086480650045345
Performance Validation = 0.6827253372146925

Fold 2
Performance Training = 0.7983664678391034
Performance Validation = 0.7435237546525797

Fold 3
Performance Training = 0.8035143591896654
Performance Validation = 0.7111524584920151

Fold 4
Performance Training = 0.7950789870464613
Performance Validation = 0.7816321166544916

Average Performance in Training = 0.8014019697699412
Average Performance in Validation = 0.7297584167534448

M Value = 6
Lambda Value = 0.09

Fold 1
Performance Training = 0.80850875468405
Performance Validation = 0.6800209180025292

Fold 2
Performance Training = 0.7982311455785676
Performance Validation = 0.7431419935567254

Fold 3
Performance Training = 0.803337594518819
Performance Validation = 0.7111231192463283

Fold 4
Performance Training = 0.7948448519729575
Performance Validation = 0.782781524762723

Average Performance in Training = 0.8012305866885985
Average Performance in Validation = 0.7292668888920765

M Value = 6
Lambda Value = 0.1

Fold 1
Performance Training = 0.8083590921765178
Performance Validation = 0.6773633631240861

Fold 2
Performance Training = 0.7980840224208208
Performance Validation = 0.7427539621581183

Fold 3
Performance Training = 0.8031479805001275

Performance Validation = 0.7110585088052861

Fold 4

Performance Training = 0.7945915499565539

Performance Validation = 0.7838737794069979

Average Performance in Training = 0.8010456612635051

Average Performance in Validation = 0.7287624033736221

M Value = 6

Lambda Value = 0.11

Fold 1

Performance Training = 0.8081999320601715

Performance Validation = 0.6747519461402914

Fold 2

Performance Training = 0.797925788802651

Performance Validation = 0.7423600831938422

Fold 3

Performance Training = 0.8029465821895465

Performance Validation = 0.7109622442309673

Fold 4

Performance Training = 0.7943204255624086

Performance Validation = 0.7849116079930687

Average Performance in Training = 0.8008481821536944

Average Performance in Validation = 0.7282464703895424

M Value = 6

Lambda Value = 0.12

Fold 1

Performance Training = 0.8080320520312079

Performance Validation = 0.6721858995612775

Fold 2

Performance Training = 0.7977570949215212

Performance Validation = 0.7419607601290351

Fold 3

Performance Training = 0.8027343590808405

Performance Validation = 0.7108375239691601

Fold 4

Performance Training = 0.7940327313941411

Performance Validation = 0.7858975835404323

Average Performance in Training = 0.8006390593569277

Average Performance in Validation = 0.7277204417999763

M Value = 6
Lambda Value = 0.13

Fold 1

Performance Training = 0.8078561619422603
Performance Validation = 0.6696644284146895

Fold 2

Performance Training = 0.7975785535530915
Performance Validation = 0.741556377850823

Fold 3

Performance Training = 0.8025121791103952
Performance Validation = 0.7106871857995436

Fold 4

Performance Training = 0.7937296354959903
Performance Validation = 0.7868341353169994

Average Performance in Training = 0.8004191325254343
Average Performance in Validation = 0.7271855318455138

M Value = 6
Lambda Value = 0.14

Fold 1

Performance Training = 0.8076729114757688
Performance Validation = 0.6671867207659059

Fold 2

Performance Training = 0.797390742613842
Performance Validation = 0.7411473033603868

Fold 3

Performance Training = 0.8022808303632907
Performance Validation = 0.7105137554674954

Fold 4

Performance Training = 0.7934122280210382
Performance Validation = 0.7877235585678541

Average Performance in Training = 0.8001891781184849
Average Performance in Validation = 0.7266428345404106

M Value = 6
Lambda Value = 0.15

Fold 1

Performance Training = 0.8074828966996905
Performance Validation = 0.6647519558710727

Fold 2

Performance Training = 0.7971942074990599
Performance Validation = 0.7407338864590614

Fold 3

Performance Training = 0.8020410309143335

Performance Validation = 0.7103194876870749

Fold 4

Performance Training = 0.7930815272551548

Performance Validation = 0.7885680234343575

Average Performance in Training = 0.7999499155920597

Average Performance in Validation = 0.7260933383628916

M Value = 6

Lambda Value = 0.16

Fold 1

Performance Training = 0.8072866657020852

Performance Validation = 0.6623593104855321

Fold 2

Performance Training = 0.796989463221929

Performance Validation = 0.7403164604253368

Fold 3

Performance Training = 0.8017934371473572

Performance Validation = 0.710106400866465

Fold 4

Performance Training = 0.7927384850731958

Performance Validation = 0.7893695831472095

Average Performance in Training = 0.7997020127861418

Average Performance in Validation = 0.7255379387311358

M Value = 6

Lambda Value = 0.17

Fold 1

Performance Training = 0.8070847234617078

Performance Validation = 0.6600079637300644

Fold 2

Performance Training = 0.796776996375737

Performance Validation = 0.7398953426800138

Fold 3

Performance Training = 0.801538650826109

Performance Validation = 0.7098763066426894

Fold 4

Performance Training = 0.7923839918930651

Performance Validation = 0.7901301815664307

Average Performance in Training = 0.7994460906391547

Average Performance in Validation = 0.7249774486547996

M Value = 6

Lambda Value = 0.18

Fold 1

Performance Training = 0.8068775360810339

Performance Validation = 0.6576971008274989

Fold 2

Performance Training = 0.7965572669381361

Performance Validation = 0.7394708354372777

Fold 3

Performance Training = 0.8012772251355127

Performance Validation = 0.7096308351041036

Fold 4

Performance Training = 0.7920188811841722

Performance Validation = 0.7908516601322508

Average Performance in Training = 0.7991827273347136

Average Performance in Validation = 0.7244126078752828

M Value = 6

Lambda Value = 0.19

Fold 1

Performance Training = 0.8066655344840526

Performance Validation = 0.6554259159515359

Fold 2

Performance Training = 0.7963307099338278

Performance Validation = 0.7390432263399626

Fold 3

Performance Training = 0.8010096698694038

Performance Validation = 0.7093714564144296

Fold 4

Performance Training = 0.7916439335791554

Performance Validation = 0.791535764283167

Average Performance in Training = 0.7989124619666099

Average Performance in Validation = 0.7238440907472738

M Value = 6

Lambda Value = 0.2

Fold 1

Performance Training = 0.8064491176621694

Performance Validation = 0.6531936143775656

Fold 2
Performance Training = 0.796097736969879
Performance Validation = 0.738612789077671

Fold 3
Performance Training = 0.8007364559071724
Performance Validation = 0.7090994994212612

Fold 4
Performance Training = 0.7912598806312992
Performance Validation = 0.7921841493909623

Average Performance in Training = 0.79863579779263
Average Performance in Validation = 0.7232725130668651

M Value = 6
Lambda Value = 0.21

Fold 1
Performance Training = 0.8062286555364646
Performance Validation = 0.6509994140833861

Fold 2
Performance Training = 0.7958587376560898
Performance Validation = 0.7381797839865885

Fold 3
Performance Training = 0.8004580190951023
Performance Validation = 0.7088161677270886

Fold 4
Performance Training = 0.7908674082546496
Performance Validation = 0.7927983862566589

Average Performance in Training = 0.7983532051355766
Average Performance in Validation = 0.7226984380134305

M Value = 6
Lambda Value = 0.22

Fold 1
Performance Training = 0.8060044914925318
Performance Validation = 0.6488425469162074

Fold 2
Performance Training = 0.7956140809212777
Performance Validation = 0.7377444586304556

Fold 3
Performance Training = 0.8001747636269958
Performance Validation = 0.7085225536165949

Fold 4
Performance Training = 0.7904671598791679

Performance Validation = 0.7933799662067015

Average Performance in Training = 0.7980651239799934

Average Performance in Validation = 0.7221223813424898

M Value = 6

Lambda Value = 0.23

Fold 1

Performance Training = 0.8057769446344386

Performance Validation = 0.6467222594178907

Fold 2

Performance Training = 0.7953641162350625

Performance Validation = 0.7373070483620172

Fold 3

Performance Training = 0.7998870650016955

Performance Validation = 0.7082196501662541

Fold 4

Performance Training = 0.7900597393493245

Performance Validation = 0.7939303058242368

Average Performance in Training = 0.7977719663051304

Average Performance in Validation = 0.7215448159425997

M Value = 6

Lambda Value = 0.24

Fold 1

Performance Training = 0.8055463117965568

Performance Validation = 0.6446378133805819

Fold 2

Performance Training = 0.7951091747436119

Performance Validation = 0.7368677768647909

Fold 3

Performance Training = 0.7995952726214861

Performance Validation = 0.7079083618069008

Fold 4

Performance Training = 0.7896457135911447

Performance Validation = 0.7944507513467938

Average Performance in Training = 0.7974741181881999

Average Performance in Validation = 0.7209661758497667

M Value = 6

Lambda Value = 0.25

Fold 1

Performance Training = 0.8053128693456921
Performance Validation = 0.6425884861901032

Fold 2
Performance Training = 0.7948495703268601
Performance Validation = 0.736426856675066

Fold 3
Performance Training = 0.7992997120843467
Performance Validation = 0.7075895135649704

Fold 4
Performance Training = 0.7892256150698075
Performance Validation = 0.7949425827584389

Average Performance in Training = 0.7971719417066766
Average Performance in Validation = 0.7203868597971446

M Value = 6
Lambda Value = 0.26

Fold 1
Performance Training = 0.8050768748007708
Performance Validation = 0.6405735710028984

Fold 2
Performance Training = 0.7945856005838822
Performance Validation = 0.7359844896840952

Fold 3
Performance Training = 0.7990006872141071
Performance Validation = 0.7072638591716371

Fold 4
Performance Training = 0.7887999440573639
Performance Validation = 0.7954070176015969

Average Performance in Training = 0.796865776664031
Average Performance in Validation = 0.7198072343650569

M Value = 6
Lambda Value = 0.27

Fold 1
Performance Training = 0.8048385682931165
Performance Validation = 0.6385923767922357

Fold 2
Performance Training = 0.7943175477524043
Performance Validation = 0.735540867620792

Fold 3
Performance Training = 0.7986984818652738
Performance Validation = 0.7069320881985346

Fold 4
Performance Training = 0.7883691707279777
Performance Validation = 0.7958452145312508

Average Performance in Training = 0.796555942159693
Average Performance in Validation = 0.7192276367857032

M Value = 6
Lambda Value = 0.28

Fold 1
Performance Training = 0.8045981738868792
Performance Validation = 0.6366442282928011

Fold 2
Performance Training = 0.7940456795678118
Performance Validation = 0.7350961725150502

Fold 3
Performance Training = 0.7983933615333599
Performance Validation = 0.7065948323540862

Fold 4
Performance Training = 0.7879337370962062
Performance Validation = 0.7962582766321418

Average Performance in Training = 0.7962427380210643
Average Performance in Validation = 0.7186483774485197

M Value = 6
Lambda Value = 0.29

Fold 1
Performance Training = 0.8043559007762824
Performance Validation = 0.6347284658663979

Fold 2
Performance Training = 0.7937702500664525
Performance Validation = 0.7346505771422209

Fold 3
Performance Training = 0.7980855747966785
Performance Validation = 0.7062526710536647

Fold 4
Performance Training = 0.7874940588121414
Performance Validation = 0.7966472545174836

Average Performance in Training = 0.7959264461128888
Average Performance in Validation = 0.7180697421449418

M Value = 6

Lambda Value = 0.3

Fold 1

Performance Training = 0.804111944373979

Performance Validation = 0.6328444453072204

Fold 2

Performance Training = 0.7934915003376031

Performance Validation = 0.734204245449029

Fold 3

Performance Training = 0.7977753546115061

Performance Validation = 0.7059061363598497

Fold 4

Performance Training = 0.7870505268258451

Performance Validation = 0.7970131492261304

Average Performance in Training = 0.7956073315372333

Average Performance in Validation = 0.7174919940855573

M Value = 6

Lambda Value = 0.31

Fold 1

Performance Training = 0.8038664873027943

Performance Validation = 0.6309915376011285

Fold 2

Performance Training = 0.793209659228006

Performance Validation = 0.7337573329614705

Fold 3

Performance Training = 0.7974629194792282

Performance Validation = 0.7055557173746914

Fold 4

Performance Training = 0.7866035089321959

Performance Validation = 0.7973569149335018

Average Performance in Training = 0.7952856437355561

Average Performance in Validation = 0.716915375717698

M Value = 6

Lambda Value = 0.32

Fold 1

Performance Training = 0.8036197003014669

Performance Validation = 0.6291691286505022

Fold 2

Performance Training = 0.7929249440025462

Performance Validation = 0.7333099871751916

Fold 3
Performance Training = 0.7971484745012887
Performance Validation = 0.7052018641538047

Fold 4
Performance Training = 0.7861533512061651
Performance Validation = 0.7976794614902407

Average Performance in Training = 0.7949616175028666
Average Performance in Validation = 0.7163401103674348

M Value = 6
Lambda Value = 0.33

Fold 1
Performance Training = 0.8033717430535888
Performance Validation = 0.6273766189743792

Fold 2
Performance Training = 0.7926375609642954
Performance Validation = 0.7328623479289017

Fold 3
Performance Training = 0.7968322123354755
Performance Validation = 0.7048449912023628

Fold 4
Performance Training = 0.7857003793375623
Performance Validation = 0.7979816568013569

Average Performance in Training = 0.7946354739227306
Average Performance in Validation = 0.7157664037267502

M Value = 6
Lambda Value = 0.34

Fold 1
Performance Training = 0.8031227649477538
Performance Validation = 0.6256134233906978

Fold 2
Performance Training = 0.79234770603688
Performance Validation = 0.7324145477613155

Fold 3
Performance Training = 0.7965143140651315
Performance Validation = 0.7044854806044154

Fold 4
Performance Training = 0.7852448998733974
Performance Validation = 0.7982643290574855

Average Performance in Training = 0.7943074212307906
Average Performance in Validation = 0.7151944452034785

M Value = 6
Lambda Value = 0.35000000000000003

Fold 1
Performance Training = 0.8028729057759063
Performance Validation = 0.6238789706869159

Fold 2
Performance Training = 0.79205556531184
Performance Validation = 0.7319667122522974

Fold 3
Performance Training = 0.7961949499912772
Performance Validation = 0.70412368482983

Fold 4
Performance Training = 0.7847872013752533
Performance Validation = 0.7985282688288958

Average Performance in Training = 0.7939776556135691
Average Performance in Validation = 0.7146244091494848

M Value = 6
Lambda Value = 0.36

Fold 1
Performance Training = 0.8026222963760229
Performance Validation = 0.6221727032835669

Fold 2
Performance Training = 0.7917613155634572
Performance Validation = 0.7315189603487057

Fold 3
Performance Training = 0.7958742803562592
Performance Validation = 0.7037599292572171

Fold 4
Performance Training = 0.7843275554983613
Performance Validation = 0.7987742310320609

Average Performance in Training = 0.7936463619485251
Average Performance in Validation = 0.7140564559803877

M Value = 6
Lambda Value = 0.37

Fold 1
Performance Training = 0.8023710592245314
Performance Validation = 0.6204940768943741

Fold 2

Performance Training = 0.7914651247332913
Performance Validation = 0.7310714046755253

Fold 3
Performance Training = 0.7955524560063814
Performance Validation = 0.703394514446047

Fold 4
Performance Training = 0.7838662179984635
Performance Validation = 0.7990029367776826

Average Performance in Training = 0.7933137144906669
Average Performance in Validation = 0.7134907331984073

M Value = 6
Lambda Value = 0.38

Fold 1
Performance Training = 0.8021193089832166
Performance Validation = 0.6188425601859704

Fold 2
Performance Training = 0.7911671523864932
Performance Validation = 0.730624151832882

Fold 3
Performance Training = 0.7952296190000135
Performance Validation = 0.7030277181866242

Fold 4
Performance Training = 0.7834034296719918
Performance Validation = 0.7992150751084441

Average Performance in Training = 0.7929798775104289
Average Performance in Validation = 0.7129273763284802

M Value = 6
Lambda Value = 0.39

Fold 1
Performance Training = 0.8018671530048365
Performance Validation = 0.6172176344393392

Fold 2
Performance Training = 0.7908675501417898
Performance Validation = 0.7301773026795089

Fold 3
Performance Training = 0.7949059031668317
Performance Validation = 0.7026597973531559

Fold 4
Performance Training = 0.7829394172346136
Performance Validation = 0.7994113046340269

Average Performance in Training = 0.792645005887018
Average Performance in Validation = 0.7123665097765077

M Value = 6
Lambda Value = 0.4

Fold 1
Performance Training = 0.8016146918011975
Performance Validation = 0.6156187932148174

Fold 2
Performance Training = 0.7905664620768971
Performance Validation = 0.7297309526031701

Fold 3
Performance Training = 0.7945814346231271
Performance Validation = 0.7022909895816623

Fold 4
Performance Training = 0.7824743941427436
Performance Validation = 0.7995922550704022

Average Performance in Training = 0.7923092456609913
Average Performance in Validation = 0.711808247617513

M Value = 6
Lambda Value = 0.41000000000000003

Fold 1
Performance Training = 0.801362019477007
Performance Validation = 0.6140455420221359

Fold 2
Performance Training = 0.7902640251109625
Performance Validation = 0.7292851917786745

Fold 3
Performance Training = 0.7942563322475429
Performance Validation = 0.7019215147918776

Fold 4
Performance Training = 0.78200856136223
Performance Validation = 0.7997585286897848

Average Performance in Training = 0.7919727345494356
Average Performance in Validation = 0.7112526943206181

M Value = 6
Lambda Value = 0.42

Fold 1
Performance Training = 0.8011092241325017

Performance Validation = 0.6124973979962038

Fold 2

Performance Training = 0.7899603693655309

Performance Validation = 0.7288401054139002

Fold 3

Performance Training = 0.7939307081210296

Performance Validation = 0.7015515765699698

Fold 4

Performance Training = 0.7815421080880622

Performance Validation = 0.7999107016872095

Average Performance in Training = 0.791635602426781

Average Performance in Validation = 0.7106999454168208

M Value = 6

Lambda Value = 0.43

Fold 1

Performance Training = 0.8008563882375006

Performance Validation = 0.6109738895797419

Fold 2

Performance Training = 0.7896556185054064

Performance Validation = 0.7283957739844722

Fold 3

Performance Training = 0.7936046679344176

Performance Validation = 0.7011813634266879

Fold 4

Performance Training = 0.7810752124186366

Performance Validation = 0.800049325469199

Average Performance in Training = 0.7912979717739903

Average Performance in Validation = 0.7101500881150252

M Value = 6

Lambda Value = 0.44

Fold 1

Performance Training = 0.8006035889792894

Performance Validation = 0.6094745562131303

Fold 2

Performance Training = 0.7893498900606855

Performance Validation = 0.7279522734574738

Fold 3

Performance Training = 0.7932783113665712

Performance Validation = 0.7008110499439919

Fold 4
Performance Training = 0.780608041987807
Performance Validation = 0.8001749278695993

Average Performance in Training = 0.7909599580985882
Average Performance in Validation = 0.7096032018710487

M Value = 6
Lambda Value = 0.45

Fold 1
Performance Training = 0.8003508985864736
Performance Validation = 0.6079989480319015

Fold 2
Performance Training = 0.7890432957311353
Performance Validation = 0.7275096755047612

Fold 3
Performance Training = 0.7929517324357763
Performance Validation = 0.7004407978215078

Fold 4
Performance Training = 0.7801407545577056
Performance Validation = 0.8002880142972801

Average Performance in Training = 0.7906216703277726
Average Performance in Validation = 0.7090593589138627

M Value = 6
Lambda Value = 0.46

Fold 1
Performance Training = 0.8000983846307634
Performance Validation = 0.6065466255719782

Fold 2
Performance Training = 0.7887359416740207
Performance Validation = 0.7270680477063247

Fold 3
Performance Training = 0.7926250198267152
Performance Validation = 0.700070756833004

Fold 4
Performance Training = 0.7796734985750602
Performance Validation = 0.8003890688200344

Average Performance in Training = 0.7902832111766399
Average Performance in Validation = 0.7085186247328352

M Value = 6
Lambda Value = 0.47000000000000003

Fold 1
Performance Training = 0.7998461103084256
Performance Validation = 0.6051171594832224

Fold 2
Performance Training = 0.788427928776384
Performance Validation = 0.7266274537441015

Fold 3
Performance Training = 0.7922982571951356
Performance Validation = 0.6997010657017236

Fold 4
Performance Training = 0.7792064136935394
Performance Validation = 0.8004785551887373

Average Performance in Training = 0.7899446774933712
Average Performance in Validation = 0.7079810585294463

M Value = 6
Lambda Value = 0.48

Fold 1
Performance Training = 0.7995941347030039
Performance Validation = 0.6037101302506974

Fold 2
Performance Training = 0.7881193529127345
Performance Validation = 0.7261879535867864

Fold 3
Performance Training = 0.7919715234520889
Performance Validation = 0.6993318529025938

Fold 4
Performance Training = 0.7787396312644281
Performance Validation = 0.8005569178054722

Average Performance in Training = 0.7896061605830639
Average Performance in Validation = 0.7074467136363874

M Value = 6
Lambda Value = 0.49

Fold 1
Performance Training = 0.7993425130307475
Performance Validation = 0.6023251279242013

Fold 2
Performance Training = 0.787810305189012
Performance Validation = 0.7257496036659115

Fold 3

Performance Training = 0.7916448930294225
Performance Validation = 0.6989632373983112

Fold 4
Performance Training = 0.7782732747977981
Performance Validation = 0.8006245826391568

Average Performance in Training = 0.789267746511745
Average Performance in Validation = 0.7069156379068953

M Value = 6
Lambda Value = 0.5

Fold 1
Performance Training = 0.7990912968700568
Performance Validation = 0.6009617518557042

Fold 2
Performance Training = 0.7875008721736624
Performance Validation = 0.7253124570437358

Fold 3
Performance Training = 0.7913184361280449
Performance Validation = 0.6985953293156174

Fold 4
Performance Training = 0.7778074603961396
Performance Validation = 0.8006819580918361

Average Performance in Training = 0.7889295163919758
Average Performance in Validation = 0.7063878740767234

M Value = 6
Lambda Value = 0.51

Fold 1
Performance Training = 0.7988405343761442
Performance Validation = 0.5996196104446531

Fold 2
Performance Training = 0.787191136116567
Performance Validation = 0.7248765635732153

Fold 3
Performance Training = 0.7909922189503263
Performance Validation = 0.6982282305673304

Fold 4
Performance Training = 0.7773422971622752
Performance Validation = 0.8007294358187083

Average Performance in Training = 0.7885915466513282
Average Performance in Validation = 0.7058634601009768

M Value = 6
Lambda Value = 0.52

Fold 1

Performance Training = 0.7985902704819968
Performance Validation = 0.5982983208909995

Fold 2

Performance Training = 0.7868811751565487
Performance Validation = 0.7244419700505045

Fold 3

Performance Training = 0.7906663039178715
Performance Validation = 0.6978620354251532

Fold 4

Performance Training = 0.7768778875832665
Performance Validation = 0.8007673915046462

Average Performance in Training = 0.7882539092849208
Average Performance in Validation = 0.7053424294678259

M Value = 6
Lambda Value = 0.53

Fold 1

Performance Training = 0.7983405470866434
Performance Validation = 0.596997508955772

Fold 2

Performance Training = 0.7865710635181238
Performance Validation = 0.7240087203602923

Fold 3

Performance Training = 0.7903407498757693
Performance Validation = 0.6974968310477706

Fold 4

Performance Training = 0.7764143278918719
Performance Validation = 0.8007961855998491

Average Performance in Training = 0.7879166720931021
Average Performance in Validation = 0.704824811490921

M Value = 6
Lambda Value = 0.54

Fold 1

Performance Training = 0.7980914032316235
Performance Validation = 0.5957168087291148

Fold 2

Performance Training = 0.786260871698103

Performance Validation = 0.7235768556143223

Fold 3

Performance Training = 0.7900156122843509

Performance Validation = 0.6971326979681745

Fold 4

Performance Training = 0.7759517084069965

Performance Validation = 0.800816164017049

Average Performance in Training = 0.7875798989052685

Average Performance in Validation = 0.7043106315821652

M Value = 6

Lambda Value = 0.55

Fold 1

Performance Training = 0.7978428752665093

Performance Validation = 0.5944558624053375

Fold 2

Performance Training = 0.7859506666426451

Performance Validation = 0.7231464142834103

Fold 3

Performance Training = 0.7896909433993485

Performance Validation = 0.6967697105438744

Fold 4

Performance Training = 0.7754901138545037

Performance Validation = 0.8008276587925364

Average Performance in Training = 0.7872436497907517

Average Performance in Validation = 0.7037999115062897

M Value = 6

Lambda Value = 0.56

Fold 1

Performance Training = 0.7975949970042399

Performance Validation = 0.5932143200651969

Fold 2

Performance Training = 0.7856405119152821

Performance Validation = 0.7227174323233196

Fold 3

Performance Training = 0.7893667924413206

Performance Validation = 0.6964079373732663

Fold 4

Performance Training = 0.7750296236696219

Performance Validation = 0.800830988713136

Average Performance in Training = 0.7869079812576161
Average Performance in Validation = 0.7032926696187296

M Value = 6
Lambda Value = 0.5700000000000001

Fold 1
Performance Training = 0.7973477998669755
Performance Validation = 0.5919918394646679

Fold 2
Performance Training = 0.785330467856443
Performance Validation = 0.7222899432946802

Fold 3
Performance Training = 0.7890432057550831
Performance Validation = 0.6960474416809885

Fold 4
Performance Training = 0.7745703122821076
Performance Validation = 0.8008264599110897

Average Performance in Training = 0.7865729464401523
Average Performance in Validation = 0.7027889210878566

M Value = 6
Lambda Value = 0.58

Fold 1
Performance Training = 0.7971013130231117
Performance Validation = 0.5907880858305268

Fold 2
Performance Training = 0.7850205917349419
Performance Validation = 0.7218639784773428

Fold 3
Performance Training = 0.7887202269598605
Performance Validation = 0.6956882816749613

Fold 4
Performance Training = 0.7741122493852582
Performance Validation = 0.8008143664287402

Average Performance in Training = 0.7862385952757932
Average Performance in Validation = 0.7022886781028927

M Value = 6
Lambda Value = 0.59

Fold 1
Performance Training = 0.7968555635160643
Performance Validation = 0.5896027316621961

Fold 2
Performance Training = 0.7847109378918795
Performance Validation = 0.7214395669793816

Fold 3
Performance Training = 0.7883978970907809
Performance Validation = 0.6953305108774981

Fold 4
Performance Training = 0.7736555001897665
Performance Validation = 0.8007949907547032

Average Performance in Training = 0.7859049746721227
Average Performance in Validation = 0.7017919500684446

M Value = 6
Lambda Value = 0.6

Fold 1
Performance Training = 0.7966105763853727
Performance Validation = 0.588435456539748

Fold 2
Performance Training = 0.7844015578773799
Performance Validation = 0.7210167358409676

Fold 3
Performance Training = 0.7880762547323213
Performance Validation = 0.6949741784325947

Fold 4
Performance Training = 0.7732001256633538
Performance Validation = 0.8007686043331538

Average Performance in Training = 0.7855721286646069
Average Performance in Validation = 0.701298743786616

M Value = 6
Lambda Value = 0.61

Fold 1
Performance Training = 0.7963663747806173
Performance Validation = 0.5872859469378615

Fold 2
Performance Training = 0.7840925005805478
Performance Validation = 0.7205955101334423

Fold 3
Performance Training = 0.7877553361442194
Performance Validation = 0.6946193293913547

Fold 4

Performance Training = 0.7727461827570588
Performance Validation = 0.8007354680477953

Average Performance in Training = 0.7852400985656107
Average Performance in Validation = 0.7008090636276134

M Value = 6
Lambda Value = 0.62

Fold 1
Performance Training = 0.796122980068648
Performance Validation = 0.586153896045575

Fold 2
Performance Training = 0.7837838123530233
Performance Validation = 0.7201759130537295

Fold 3
Performance Training = 0.7874351753803568
Performance Validation = 0.6942660049773751

Fold 4
Performance Training = 0.7722937246189856
Performance Validation = 0.8006958326818182

Average Performance in Training = 0.7849089231052534
Average Performance in Validation = 0.7003229116896245

M Value = 6
Lambda Value = 0.63

Fold 1
Performance Training = 0.7958804119345324
Performance Validation = 0.5850390035915789

Fold 2
Performance Training = 0.7834755371264751
Performance Validation = 0.7197579660143918

Fold 3
Performance Training = 0.7871158044010729
Performance Validation = 0.6939142428336049

Fold 4
Performance Training = 0.7718428007962763
Performance Validation = 0.8006499393553406

Average Performance in Training = 0.7845786385645892
Average Performance in Validation = 0.6998402879487291

M Value = 6
Lambda Value = 0.64

Fold 1
Performance Training = 0.795638688476648
Performance Validation = 0.5839409756747992

Fold 2
Performance Training = 0.7831677165243587
Performance Validation = 0.719341688729447

Fold 3
Performance Training = 0.7867972531793237
Performance Validation = 0.69356407725219

Fold 4
Performance Training = 0.771393457426006
Performance Validation = 0.8005980199414056

Average Performance in Training = 0.784249278901584
Average Performance in Validation = 0.6993611903994604

M Value = 6
Lambda Value = 0.65

Fold 1
Performance Training = 0.7953978262962749
Performance Validation = 0.5828595246003758

Fold 2
Performance Training = 0.7828603899682474
Performance Validation = 0.7189270992962309

Fold 3
Performance Training = 0.7864795498010722
Performance Validation = 0.6932155393886577

Fold 4
Performance Training = 0.7709457374156695
Performance Validation = 0.8005402974618565

Average Performance in Training = 0.783920875870316
Average Performance in Validation = 0.6988856151867802

M Value = 6
Lambda Value = 0.66

Fold 1
Performance Training = 0.7951578405820583
Performance Validation = 0.5817943687203978

Fold 2
Performance Training = 0.7825535947790279
Performance Validation = 0.7185142142734479

Fold 3
Performance Training = 0.7861627205602835

Performance Validation = 0.6928686574615668

Fold 4

Performance Training = 0.7704996806138675

Performance Validation = 0.8004769864640823

Average Performance in Training = 0.7835934591338093

Average Performance in Validation = 0.6984135567298737

M Value = 6

Lambda Value = 0.67

Fold 1

Performance Training = 0.7949187451896379

Performance Validation = 0.5807452322796194

Fold 2

Performance Training = 0.782247366273217

Performance Validation = 0.7181030487555954

Fold 3

Performance Training = 0.7858467900488355

Performance Validation = 0.6925234569388199

Fold 4

Performance Training = 0.770055323971775

Performance Validation = 0.8004082933797393

Average Performance in Training = 0.7832670563708664

Average Performance in Validation = 0.6979450078384435

M Value = 6

Lambda Value = 0.68

Fold 1

Performance Training = 0.7946805527167666

Performance Validation = 0.5797118452658437

Fold 2

Performance Training = 0.7819417378546765

Performance Validation = 0.7176936164439509

Fold 3

Performance Training = 0.7855317812416798

Performance Validation = 0.6921799607115695

Fold 4

Performance Training = 0.7696127016959249

Performance Validation = 0.8003344168663686

Average Performance in Training = 0.7829416933772619

Average Performance in Validation = 0.6974799598219332

M Value = 6
Lambda Value = 0.6900000000000001

Fold 1
Performance Training = 0.7944432745741832
Performance Validation = 0.5786939432647551

Fold 2
Performance Training = 0.7816367411019485
Performance Validation = 0.7172859297142656

Fold 3
Performance Training = 0.78521771557751
Performance Validation = 0.6918381892567051

Fold 4
Performance Training = 0.7691718453928134
Performance Validation = 0.8002555481328426

Average Performance in Training = 0.7826173941616137
Average Performance in Validation = 0.6970184025921421

M Value = 6
Lambda Value = 0.7000000000000001

Fold 1
Performance Training = 0.7942069210525017
Performance Validation = 0.5776912673191186

Fold 2
Performance Training = 0.781332405851443
Performance Validation = 0.7168799996813322

Fold 3
Performance Training = 0.7849046130352316
Performance Validation = 0.6914981607887261

Fold 4
Performance Training = 0.7687327842058053
Performance Validation = 0.8001718712495148

Average Performance in Training = 0.7822941810362454
Average Performance in Validation = 0.6965603247596729

M Value = 6
Lambda Value = 0.71

Fold 1
Performance Training = 0.7939715013853681
Performance Validation = 0.5767035637921958

Fold 2
Performance Training = 0.7810287602767022
Performance Validation = 0.7164758362605708

Fold 3
Performance Training = 0.784592492206469
Performance Validation = 0.6911598914017667

Fold 4
Performance Training = 0.7682955449447714
Performance Validation = 0.800083563443856

Average Performance in Training = 0.7819720747033276
Average Performance in Validation = 0.6961057137245973

M Value = 6
Lambda Value = 0.72

Fold 1
Performance Training = 0.7937370238090968
Performance Validation = 0.5757305842352014

Fold 2
Performance Training = 0.7807258309639251
Performance Validation = 0.7160734482267705

Fold 3
Performance Training = 0.78428137036433
Performance Validation = 0.6908233952025213

Fold 4
Performance Training = 0.7678601522088762
Performance Validation = 0.799990795382324

Average Performance in Training = 0.7816510943365571
Average Performance in Validation = 0.6956545557617043

M Value = 6
Lambda Value = 0.73

Fold 1
Performance Training = 0.7935034956190057
Performance Validation = 0.5747720852586591

Fold 2
Performance Training = 0.7804236429839552
Performance Validation = 0.7156728432701354

Fold 3
Performance Training = 0.783971263528665
Performance Validation = 0.6904886844346987

Fold 4
Performance Training = 0.7674266285029059
Performance Validation = 0.7998937314392326

Average Performance in Training = 0.7813312576586329

Average Performance in Validation = 0.6952068361006815

M Value = 6

Lambda Value = 0.74

Fold 1

Performance Training = 0.7932709232226445

Performance Validation = 0.5738278284075642

Fold 2

Performance Training = 0.7801222199609107

Performance Validation = 0.7152740280497416

Fold 3

Performance Training = 0.7836621865280047

Performance Validation = 0.6901557695956013

Fold 4

Performance Training = 0.7669949943475054

Performance Validation = 0.7997925299532567

Average Performance in Training = 0.7810125810147663

Average Performance in Validation = 0.694762539001541

M Value = 6

Lambda Value = 0.75

Fold 1

Performance Training = 0.7930393121901037

Performance Validation = 0.5728975800401555

Fold 2

Performance Training = 0.7798215841376172

Performance Validation = 0.7148770082445458

Fold 3

Performance Training = 0.783354153058373

Performance Validation = 0.6898246595453909

Fold 4

Performance Training = 0.766565268383654

Performance Validation = 0.7996873434721964

Average Performance in Training = 0.780695079442437

Average Performance in Validation = 0.6943216478255722

M Value = 6

Lambda Value = 0.76

Fold 1

Performance Training = 0.7928086673015636

Performance Validation = 0.5719811112103035

Fold 2
Performance Training = 0.7795217564380172
Performance Validation = 0.71448178860203

Fold 3
Performance Training = 0.78304717573914
Performance Validation = 0.6894953616095377

Fold 4
Performance Training = 0.7661374674717061
Performance Validation = 0.7995783189866335

Average Performance in Training = 0.7803787667376068
Average Performance in Validation = 0.6938841451021261

M Value = 6
Lambda Value = 0.77

Fold 1
Performance Training = 0.792578992592262
Performance Validation = 0.5710781975531516

Fold 2
Performance Training = 0.7792227565266904
Performance Validation = 0.7140883729846605

Fold 3
Performance Training = 0.782741266166096
Performance Validation = 0.6891678816749376

Fold 4
Performance Training = 0.765711606785311
Performance Validation = 0.7994655981530387

Average Performance in Training = 0.7800636555175898
Average Performance in Validation = 0.6934500125914471

M Value = 6
Lambda Value = 0.78

Fold 1
Performance Training = 0.7923502913950118
Performance Validation = 0.5701886191741949

Fold 2
Performance Training = 0.7789246028656501
Performance Validation = 0.7136967644141615

Fold 3
Performance Training = 0.7824364349618828
Performance Validation = 0.688842224280122

Fold 4
Performance Training = 0.7652876999004765

Performance Validation = 0.7993493175068438

Average Performance in Training = 0.7797497572807552

Average Performance in Validation = 0.6930192313438306

M Value = 6

Lambda Value = 0.79

Fold 1

Performance Training = 0.7921225663804291

Performance Validation = 0.5693121605414675

Fold 2

Performance Training = 0.7786273127685265

Performance Validation = 0.7133069651138071

Fold 3

Performance Training = 0.7821326918239347

Performance Validation = 0.6885183926999345

Fold 4

Performance Training = 0.7648657588800465

Performance Validation = 0.7992296086659791

Average Performance in Training = 0.7794370824632342

Average Performance in Validation = 0.6925917817552971

M Value = 6

Lambda Value = 0.8

Fold 1

Performance Training = 0.7918958195949835

Performance Validation = 0.5684486103808393

Fold 2

Performance Training = 0.7783309024522863

Performance Validation = 0.7129189765487566

Fold 3

Performance Training = 0.7818300455700695

Performance Validation = 0.6881963890250954

Fold 4

Performance Training = 0.7644457943538541

Performance Validation = 0.7991065985253955

Average Performance in Training = 0.7791256404927984

Average Performance in Validation = 0.6921676436200217

M Value = 6

Lambda Value = 0.81

Fold 1

Performance Training = 0.7916700524970063
Performance Validation = 0.5675977615742046

Fold 2
Performance Training = 0.778035387086596
Performance Validation = 0.7125327994645482

Fold 3
Performance Training = 0.7815285041818391
Performance Validation = 0.6878762142369521

Fold 4
Performance Training = 0.7640278155947783
Performance Validation = 0.7989804094429653

Average Performance in Training = 0.778815439840055
Average Performance in Validation = 0.6917467961796675

M Value = 6
Lambda Value = 0.8200000000000001

Fold 1
Performance Training = 0.7914452659907765
Performance Validation = 0.5667594110606584

Fold 2
Performance Training = 0.7777407808409486
Performance Validation = 0.7121484339238513

Fold 3
Performance Training = 0.7812280748457779
Performance Validation = 0.6875578682777332

Fold 4
Performance Training = 0.7636118305909174
Performance Validation = 0.7988511594172099

Average Performance in Training = 0.7785064880671051
Average Performance in Validation = 0.6913292181698631

M Value = 6
Lambda Value = 0.8300000000000001

Fold 1
Performance Training = 0.7912214604587755
Performance Validation = 0.5659333597403995

Fold 2
Performance Training = 0.7774470969296614
Performance Validation = 0.71176587934152

Fold 3
Performance Training = 0.7809287639926513
Performance Validation = 0.6872413501166282

Fold 4
Performance Training = 0.7631978461141035
Performance Validation = 0.7987189622572725

Average Performance in Training = 0.7781987918737979
Average Performance in Validation = 0.690914887863955

M Value = 6
Lambda Value = 0.84

Fold 1
Performance Training = 0.7909986357922323
Performance Validation = 0.565119412381257

Fold 2
Performance Training = 0.7771543476548548
Performance Validation = 0.7113851345180613

Fold 3
Performance Training = 0.7806305773348075
Performance Validation = 0.6869266578119381

Fold 4
Performance Training = 0.7627858677849488
Performance Validation = 0.7985839277454816

Average Performance in Training = 0.7778923571417109
Average Performance in Validation = 0.6905037831141845

M Value = 6
Lambda Value = 0.85

Fold 1
Performance Training = 0.7907767914200442
Performance Validation = 0.564317377527928

Fold 2
Performance Training = 0.77686254444749
Performance Validation = 0.711006197671574

Fold 3
Performance Training = 0.7803335199017326
Performance Validation = 0.6866137885695389

Fold 4
Performance Training = 0.762375900134597
Performance Validation = 0.7984461617928784

Average Performance in Training = 0.777587188975966
Average Performance in Validation = 0.6900958813904798

M Value = 6

Lambda Value = 0.86

Fold 1

Performance Training = 0.7905559263361571

Performance Validation = 0.563527067413695

Fold 2

Performance Training = 0.7765716979065822

Performance Validation = 0.7106290664682556

Fold 3

Performance Training = 0.7800375960739117

Performance Validation = 0.6863027387979437

Fold 4

Performance Training = 0.7619679466633726

Performance Validation = 0.79830576658806

Average Performance in Training = 0.7772832917450059

Average Performance in Validation = 0.6896911598169886

M Value = 6

Lambda Value = 0.87

Fold 1

Performance Training = 0.7903360391255161

Performance Validation = 0.562748297874504

Fold 2

Performance Training = 0.776281817836663

Performance Validation = 0.7102537380514684

Fold 3

Performance Training = 0.7797428096150697

Performance Validation = 0.6859935041601093

Fold 4

Performance Training = 0.7615620098964848

Performance Validation = 0.7981628407396354

Average Performance in Training = 0.7769806691184333

Average Performance in Validation = 0.6892895952064293

M Value = 6

Lambda Value = 0.88

Fold 1

Performance Training = 0.7901171279886267

Performance Validation = 0.561980888265526

Fold 2

Performance Training = 0.7759929132835727

Performance Validation = 0.709880209069552

Fold 3
Performance Training = 0.779449163702878
Performance Validation = 0.6856860796222427

Fold 4
Performance Training = 0.7611580914369352
Performance Validation = 0.79801747941263

Average Performance in Training = 0.7766793241030032
Average Performance in Validation = 0.6888911640924877

M Value = 6
Lambda Value = 0.89

Fold 1
Performance Training = 0.7898991907648364
Performance Validation = 0.5612246613798704

Fold 2
Performance Training = 0.7757049925686729
Performance Validation = 0.7095084757023109

Fold 3
Performance Training = 0.7791566609582194
Performance Validation = 0.6853804594997732

Fold 4
Performance Training = 0.7607561920157808
Performance Validation = 0.7978697744590704

Average Performance in Training = 0.7763792590768774
Average Performance in Validation = 0.6884958427602562

M Value = 6
Lambda Value = 0.9

Fold 1
Performance Training = 0.7896822249543894
Performance Validation = 0.5604794433696141

Fold 2
Performance Training = 0.77541806332154
Performance Validation = 0.7091385336863492

Fold 3
Performance Training = 0.7788653034730574
Performance Validation = 0.6850766375006692

Fold 4
Performance Training = 0.7603563115398948
Performance Validation = 0.797719814543107

Average Performance in Training = 0.7760804758222204
Average Performance in Validation = 0.6881036072749349

M Value = 6
Lambda Value = 0.91

Fold 1
Performance Training = 0.789466227739323
Performance Validation = 0.5597450636689172

Fold 2
Performance Training = 0.7751321325112256
Performance Validation = 0.7087703783392085

Fold 3
Performance Training = 0.7785750928370062
Performance Validation = 0.6847746067662607

Fold 4
Performance Training = 0.7599584491373474
Performance Validation = 0.7975676852608512

Average Performance in Training = 0.7757829755562256
Average Performance in Validation = 0.6877144335088093

M Value = 6
Lambda Value = 0.92

Fold 1
Performance Training = 0.7892511960032701
Performance Validation = 0.5590213549192162

Fold 2
Performance Training = 0.7748472064761368
Performance Validation = 0.7084040045824437

Fold 3
Performance Training = 0.778286030162651
Performance Validation = 0.6844743599097115

Fold 4
Performance Training = 0.7595626032005337
Performance Validation = 0.7974134692552164

Average Performance in Training = 0.7754867589606479
Average Performance in Validation = 0.687328297166647

M Value = 6
Lambda Value = 0.93

Fold 1
Performance Training = 0.7890371263502287
Performance Validation = 0.5583081528964036

Fold 2

Performance Training = 0.7745632909526092
Performance Validation = 0.7080394069636384

Fold 3
Performance Training = 0.7779981161096845
Performance Validation = 0.684175889052312

Fold 4
Performance Training = 0.7591687714271611
Performance Validation = 0.7972572463259889

Average Performance in Training = 0.7751918262099209
Average Performance in Validation = 0.6869451738095858

M Value = 6
Lambda Value = 0.9400000000000001

Fold 1
Performance Training = 0.7888240151223467
Performance Validation = 0.5576052964400071

Fold 2
Performance Training = 0.7742803911022332
Performance Validation = 0.7076765796774209

Fold 3
Performance Training = 0.7777113509079249
Performance Validation = 0.6838791858576849

Fold 4
Performance Training = 0.7587769508592138
Performance Validation = 0.7970990935353406

Average Performance in Training = 0.7748981769979296
Average Performance in Validation = 0.6865650388776134

M Value = 6
Lambda Value = 0.9500000000000001

Fold 1
Performance Training = 0.7886118584167807
Performance Validation = 0.55691262738422

Fold 2
Performance Training = 0.7739985115379792
Performance Validation = 0.7073155165855227

Fold 3
Performance Training = 0.7774257343792497
Performance Validation = 0.6835842415640547

Fold 4
Performance Training = 0.7583871379199858
Performance Validation = 0.7969390853089997

Average Performance in Training = 0.7746058105634988
Average Performance in Validation = 0.6861878677106993

M Value = 6
Lambda Value = 0.96

Fold 1
Performance Training = 0.7884006521016762
Performance Validation = 0.5562299904907448

Fold 2
Performance Training = 0.7737176563492022
Performance Validation = 0.7069562112359395

Fold 3
Performance Training = 0.7771412659585333
Performance Validation = 0.6832910470146836

Fold 4
Performance Training = 0.7579993284493007
Performance Validation = 0.7967772935332889

Average Performance in Training = 0.7743147257146781
Average Performance in Validation = 0.6858136355686641

M Value = 6
Lambda Value = 0.97

Fold 1
Performance Training = 0.7881903918313181
Performance Validation = 0.5555572333834279

Fold 2
Performance Training = 0.7734378291255468
Performance Validation = 0.7065986568811924

Fold 3
Performance Training = 0.7768579447135955
Performance Validation = 0.6829995926865942

Fold 4
Performance Training = 0.7576135177369797
Performance Validation = 0.796613787648177

Average Performance in Training = 0.77402492085186
Average Performance in Validation = 0.6854423176498479

M Value = 6
Lambda Value = 0.98

Fold 1
Performance Training = 0.7879810730604858

Performance Validation = 0.5548942064846032

Fold 2

Performance Training = 0.773159032979823

Performance Validation = 0.7062428464958113

Fold 3

Performance Training = 0.776575769364246

Performance Validation = 0.6827098687176543

Fold 4

Performance Training = 0.757229700554678

Performance Validation = 0.7964486347365891

Average Performance in Training = 0.7737363939898082

Average Performance in Validation = 0.6850738891086645

M Value = 6

Lambda Value = 0.99

Fold 1

Performance Training = 0.7877726910580652

Performance Validation = 0.5542407629531001

Fold 2

Performance Training = 0.7728812705698943

Performance Validation = 0.7058887727929599

Fold 3

Performance Training = 0.7762947383004399

Performance Validation = 0.6824218649321536

Fold 4

Performance Training = 0.756847871186151

Performance Validation = 0.7962818996100779

Average Performance in Training = 0.7734491427786376

Average Performance in Validation = 0.6847083250720729

M Value = 6

Lambda Value = 1.0

Fold 1

Performance Training = 0.7875652409199562

Performance Validation = 0.553596758623901

Fold 2

Performance Training = 0.772604544119623

Performance Validation = 0.7055364282403591

Fold 3

Performance Training = 0.7760148495996124

Performance Validation = 0.6821355708649434

Fold 4
Performance Training = 0.7564680234560384
Performance Validation = 0.7961136448910766

Average Performance in Training = 0.7731631645238075
Average Performance in Validation = 0.68434560065507

M Value = 6
Lambda Value = 1.01

Fold 1
Performance Training = 0.7873587175813019
Performance Validation = 0.5529620519492475

Fold 2
Performance Training = 0.7723288554389102
Performance Validation = 0.7051858050754473

Fold 3
Performance Training = 0.7757361010432017
Performance Validation = 0.6818509757841926

Fold 4
Performance Training = 0.7560901507572374
Performance Validation = 0.7959439310918395

Average Performance in Training = 0.7728784562051627
Average Performance in Validation = 0.6839856909751818

M Value = 6
Lambda Value = 1.02

Fold 1
Performance Training = 0.787153115828083
Performance Validation = 0.5523365039414513

Fold 2
Performance Training = 0.7720542059428803
Performance Validation = 0.7048368953198737

Fold 3
Performance Training = 0.7754584901324345
Performance Validation = 0.6815680687128975

Fold 4
Performance Training = 0.7557142460769356
Performance Validation = 0.7957728166902528

Average Performance in Training = 0.7725950144950834
Average Performance in Validation = 0.6836285711661189

M Value = 6
Lambda Value = 1.03

Fold 1
Performance Training = 0.7869484303081153
Performance Validation = 0.5517199781169984

Fold 2
Performance Training = 0.7717805966702422
Performance Validation = 0.7044896907933309

Fold 3
Performance Training = 0.7751820141033704
Performance Validation = 0.6812868384491739

Fold 4
Performance Training = 0.7553403020213654
Performance Validation = 0.7956003582026152

Average Performance in Training = 0.7723128357757734
Average Performance in Validation = 0.6832742163905297

M Value = 6
Lambda Value = 1.04

Fold 1
Performance Training = 0.7867446555414669
Performance Validation = 0.5511123404422062

Fold 2
Performance Training = 0.7715080283008626
Performance Validation = 0.704144183126733

Fold 3
Performance Training = 0.7749066699412792
Performance Validation = 0.6810072735854118

Fold 4
Performance Training = 0.7549683108393604
Performance Validation = 0.7954266102535836

Average Performance in Training = 0.7720319161557423
Average Performance in Validation = 0.6829226018519836

M Value = 6
Lambda Value = 1.05

Fold 1
Performance Training = 0.7865417859303339
Performance Validation = 0.5505134592801941

Fold 2
Performance Training = 0.7712365011725901
Performance Validation = 0.7038003637748191

Fold 3

Performance Training = 0.7746324543943491
Performance Validation = 0.6807293625263551

Fold 4

Performance Training = 0.754598264444745
Performance Validation = 0.7952516256433246

Average Performance in Training = 0.7717522514855045
Average Performance in Validation = 0.6825737028061732

M Value = 6

Lambda Value = 1.06

Fold 1

Performance Training = 0.7863398157684051
Performance Validation = 0.5499232053392292

Fold 2

Performance Training = 0.7709660152973723
Performance Validation = 0.7034582240281695

Fold 3

Performance Training = 0.7743593639867733
Performance Validation = 0.680453093506171

Fold 4

Performance Training = 0.7542301544376422
Performance Validation = 0.7950754554120687

Average Performance in Training = 0.7714738373725483
Average Performance in Validation = 0.6822274945714095

M Value = 6

Lambda Value = 1.07

Fold 1

Performance Training = 0.7861387392497367
Performance Validation = 0.5493414516223998

Fold 2

Performance Training = 0.7706965703766666
Performance Validation = 0.7031177550246643

Fold 3

Performance Training = 0.7740873950312522
Performance Validation = 0.6801784546045706

Fold 4

Performance Training = 0.7538639721247355
Performance Validation = 0.7948981489021385

Average Performance in Training = 0.7711966691955978
Average Performance in Validation = 0.6818839525384432

M Value = 6
Lambda Value = 1.08

Fold 1

Performance Training = 0.7859385504771573
Performance Validation = 0.5487680733785095

Fold 2

Performance Training = 0.7704281658162153
Performance Validation = 0.7027789477604302

Fold 3

Performance Training = 0.773816543640913
Performance Validation = 0.6799054337620172

Fold 4

Performance Training = 0.7534997085385389
Performance Validation = 0.7947197538175719

Average Performance in Training = 0.770920742118206
Average Performance in Validation = 0.6815430521796322

M Value = 6
Lambda Value = 1.09

Fold 1

Performance Training = 0.7857392434702424
Performance Validation = 0.5482029480542686

Fold 2

Performance Training = 0.7701608007401882
Performance Validation = 0.7024417931002748

Fold 3

Performance Training = 0.7735468057406979
Performance Validation = 0.6796340187940719

Fold 4

Performance Training = 0.7531373544557316
Performance Validation = 0.7945403162814333

Average Performance in Training = 0.770646051101715
Average Performance in Validation = 0.6812047690575121

M Value = 7
Lambda Value = 0.01

Fold 1

Performance Training = 0.82252279376623
Performance Validation = 0.5362297967114453

Fold 2

Performance Training = 0.8069186826384194

Performance Validation = 0.7437735756669543

Fold 3

Performance Training = 0.814202057385051

Performance Validation = 0.669050709428761

Fold 4

Performance Training = 0.8003815304003821

Performance Validation = 0.781668003912525

Average Performance in Training = 0.8110062660475206

Average Performance in Validation = 0.6826805214299214

M Value = 7

Lambda Value = 0.02

Fold 1

Performance Training = 0.821911624966813

Performance Validation = 0.5570591062869419

Fold 2

Performance Training = 0.8064886162376033

Performance Validation = 0.7449998422619357

Fold 3

Performance Training = 0.8137126511957428

Performance Validation = 0.6771713896888734

Fold 4

Performance Training = 0.8000041761517421

Performance Validation = 0.7824955483617325

Average Performance in Training = 0.8105292671379752

Average Performance in Validation = 0.6904314716498708

M Value = 7

Lambda Value = 0.03

Fold 1

Performance Training = 0.8211122078935364

Performance Validation = 0.5729227892157011

Fold 2

Performance Training = 0.8059118930421182

Performance Validation = 0.7456384235129122

Fold 3

Performance Training = 0.813053841634055

Performance Validation = 0.6834536225558083

Fold 4

Performance Training = 0.7994829981888634

Performance Validation = 0.7830764186042689

Average Performance in Training = 0.8098902351896432
Average Performance in Validation = 0.6962728134721726

M Value = 7
Lambda Value = 0.04

Fold 1
Performance Training = 0.8202197400156198
Performance Validation = 0.5852136867126537

Fold 2
Performance Training = 0.8052538418721041
Performance Validation = 0.7458795459762135

Fold 3
Performance Training = 0.8122994768000635
Performance Validation = 0.6883802627271769

Fold 4
Performance Training = 0.7988723204307121
Performance Validation = 0.7834815951409111

Average Performance in Training = 0.8091613447796249
Average Performance in Validation = 0.7007387726392389

M Value = 7
Lambda Value = 0.05

Fold 1
Performance Training = 0.8192902274176076
Performance Validation = 0.5948711312530779

Fold 2
Performance Training = 0.8045552370626906
Performance Validation = 0.7458494601470913

Fold 3
Performance Training = 0.8114960290169366
Performance Validation = 0.6922882176502076

Fold 4
Performance Training = 0.7982083322933772
Performance Validation = 0.7837602898490124

Average Performance in Training = 0.808387456447653
Average Performance in Validation = 0.7041922747248472

M Value = 7
Lambda Value = 0.06

Fold 1
Performance Training = 0.8183567964444955
Performance Validation = 0.6025468142186692

Fold 2
Performance Training = 0.8038417172274412
Performance Validation = 0.7456336256860949

Fold 3
Performance Training = 0.8106729846679201
Performance Validation = 0.6954179375680544

Fold 4
Performance Training = 0.7975153619278851
Performance Validation = 0.7839471784525278

Average Performance in Training = 0.8075967150669355
Average Performance in Validation = 0.7068863889813366

M Value = 7
Lambda Value = 0.07

Fold 1
Performance Training = 0.81743889236027
Performance Validation = 0.6087040585068708

Fold 2
Performance Training = 0.8031293964335053
Performance Validation = 0.7452908423364557

Fold 3
Performance Training = 0.8098490716068596
Performance Validation = 0.697944456242061

Fold 4
Performance Training = 0.7968098427662855
Performance Validation = 0.7840670356821897

Average Performance in Training = 0.8068068007917302
Average Performance in Validation = 0.7090015981918942

M Value = 7
Lambda Value = 0.08

Fold 1
Performance Training = 0.8165476311803854
Performance Validation = 0.6136793634718223

Fold 2
Performance Training = 0.8024282960829757
Performance Validation = 0.7448621094712888

Fold 3
Performance Training = 0.8090360898957737
Performance Validation = 0.6999974255890153

Fold 4

Performance Training = 0.7961028760609099
Performance Validation = 0.7841377738230189

Average Performance in Training = 0.8060287233050112
Average Performance in Validation = 0.7106691680887863

M Value = 7
Lambda Value = 0.09

Fold 1
Performance Training = 0.8156889954377071
Performance Validation = 0.6177217462220772

Fold 2
Performance Training = 0.8017444914944821
Performance Validation = 0.7443763195919222

Fold 3
Performance Training = 0.8082413212133256
Performance Validation = 0.7016743768830603

Fold 4
Performance Training = 0.7954019177947794
Performance Validation = 0.7841724749738321

Average Performance in Training = 0.8052691814850735
Average Performance in Validation = 0.7119862294177229

M Value = 7
Lambda Value = 0.1

Fold 1
Performance Training = 0.8148657840352879
Performance Validation = 0.6210185743035319

Fold 2
Performance Training = 0.8010814804184528
Performance Validation = 0.7438539975014837

Fold 3
Performance Training = 0.8074690733792124
Performance Validation = 0.7030496971863884

Fold 4
Performance Training = 0.7947119075965536
Performance Validation = 0.7841807748093402

Average Performance in Training = 0.8045320613573768
Average Performance in Validation = 0.7130257609501861

M Value = 7
Lambda Value = 0.11

Fold 1
Performance Training = 0.8140788229908238
Performance Validation = 0.6237129361390232

Fold 2
Performance Training = 0.8004410696956743
Performance Validation = 0.7433098035588419

Fold 3
Performance Training = 0.806721686598793
Performance Validation = 0.7041808281627168

Fold 4
Performance Training = 0.7940360351914884
Performance Validation = 0.7841698200038433

Average Performance in Training = 0.8038194036191948
Average Performance in Validation = 0.7138433469661063

M Value = 7
Lambda Value = 0.12

Fold 1
Performance Training = 0.8133277277170412
Performance Validation = 0.625915571287667

Fold 2
Performance Training = 0.7998239574931375
Performance Validation = 0.7427542383004904

Fold 3
Performance Training = 0.8060001980020407
Performance Validation = 0.7051126243970316

Fold 4
Performance Training = 0.7933762673410636
Performance Validation = 0.7841449402905843

Average Performance in Training = 0.8031320376383206
Average Performance in Validation = 0.7144818435689433

M Value = 7
Lambda Value = 0.13

Fold 1
Performance Training = 0.8126113878714003
Performance Validation = 0.6277132214563912

Fold 2
Performance Training = 0.7992301198269713
Performance Validation = 0.7421948209499682

Fold 3
Performance Training = 0.8053047854582169

Performance Validation = 0.7058804678594629

Fold 4

Performance Training = 0.7927337140776498

Performance Validation = 0.7841101263874218

Average Performance in Training = 0.8024700018085595

Average Performance in Validation = 0.714974659163311

M Value = 7

Lambda Value = 0.14

Fold 1

Performance Training = 0.8119282777267889

Performance Validation = 0.6291745771271624

Fold 2

Performance Training = 0.7986590693264779

Performance Validation = 0.7416369152809877

Fold 3

Performance Training = 0.8046350666198545

Performance Validation = 0.7065125265706929

Fold 4

Performance Training = 0.7921088856241904

Performance Validation = 0.7840683738903909

Average Performance in Training = 0.8018328248243279

Average Performance in Validation = 0.7153480982173084

M Value = 7

Lambda Value = 0.15

Fold 1

Performance Training = 0.8112766552798848

Performance Validation = 0.630354578371878

Fold 2

Performance Training = 0.7981100294845085

Performance Validation = 0.7410843153800507

Fold 3

Performance Training = 0.8039903017363943

Performance Validation = 0.7070314148045777

Fold 4

Performance Training = 0.7915018740382331

Performance Validation = 0.7840219333765946

Average Performance in Training = 0.8012197151347552

Average Performance in Validation = 0.7156230604832753

M Value = 7
Lambda Value = 0.16

Fold 1

Performance Training = 0.8106546896003414
Performance Validation = 0.6312975696988112

Fold 2

Performance Training = 0.7975820523582748
Performance Validation = 0.7405396656067698

Fold 3

Performance Training = 0.8033695317564569
Performance Validation = 0.7074554285147454

Fold 4

Performance Training = 0.7909124824489726
Performance Validation = 0.7839724940648891

Average Performance in Training = 0.8006296890410114
Average Performance in Validation = 0.7158162894713038

M Value = 7
Lambda Value = 0.17

Fold 1

Performance Training = 0.8100605414867846
Performance Validation = 0.6320396444386399

Fold 2

Performance Training = 0.7970740980635913
Performance Validation = 0.7400047645801325

Fold 3

Performance Training = 0.8027716724795584
Performance Validation = 0.7077994751252421

Fold 4

Performance Training = 0.7903403174504919
Performance Validation = 0.7839213198754394

Average Performance in Training = 0.8000616573701067
Average Performance in Validation = 0.7159413010048634

M Value = 7
Lambda Value = 0.18

Fold 1

Performance Training = 0.8094924135485634
Performance Validation = 0.6326104077107708

Fold 2

Performance Training = 0.7965850882486893
Performance Validation = 0.7394807870918347

Fold 3

Performance Training = 0.8021955786073294

Performance Validation = 0.70807578063161

Fold 4

Performance Training = 0.7897848553676459

Performance Validation = 0.783869351031114

Average Performance in Training = 0.799514483943057

Average Performance in Validation = 0.7160090816163324

M Value = 7

Lambda Value = 0.19

Fold 1

Performance Training = 0.8089485801981035

Performance Validation = 0.6330343167704431

Fold 2

Performance Training = 0.7961139417353857

Performance Validation = 0.738968447319809

Fold 3

Performance Training = 0.8016400870402166

Performance Validation = 0.7082944325556151

Fold 4

Performance Training = 0.7892454898528934

Performance Validation = 0.7838172804777829

Average Performance in Training = 0.7989870247066498

Average Performance in Validation = 0.7160286192809125

M Value = 7

Lambda Value = 0.2

Fold 1

Performance Training = 0.808427404440287

Performance Validation = 0.6333317104218237

Fold 2

Performance Training = 0.7956595978842199

Performance Validation = 0.7384681196508277

Fold 3

Performance Training = 0.8011040457902672

Performance Validation = 0.7084638006010707

Fold 4

Performance Training = 0.7887215660554122

Performance Validation = 0.783765611739882

Average Performance in Training = 0.7984781535425466

Average Performance in Validation = 0.7160073106034011

M Value = 7

Lambda Value = 0.21

Fold 1

Performance Training = 0.8079273460175356

Performance Validation = 0.6335196070871378

Fold 2

Performance Training = 0.7952210314874522

Performance Validation = 0.737979928620117

Fold 3

Performance Training = 0.8005863328914354

Performance Validation = 0.7085908652754395

Fold 4

Performance Training = 0.7882124050785579

Performance Validation = 0.7837147029788772

Average Performance in Training = 0.7979867788687453

Average Performance in Validation = 0.7159512759903929

M Value = 7

Lambda Value = 0.22

Fold 1

Performance Training = 0.8074469639467194

Performance Validation = 0.6336123289492845

Fold 2

Performance Training = 0.7947972618134219

Performance Validation = 0.7375038161726426

Fold 3

Performance Training = 0.8000858683443212

Performance Validation = 0.7086814766073962

Fold 4

Performance Training = 0.7877173213813612

Performance Validation = 0.783664800720999

Average Performance in Training = 0.797511853871456

Average Performance in Validation = 0.7158656056125805

M Value = 7

Lambda Value = 0.23

Fold 1

Performance Training = 0.8069849154781032

Performance Validation = 0.6336219940605694

Fold 2
Performance Training = 0.7943873576231724
Performance Validation = 0.7370395921489639

Fold 3
Performance Training = 0.7996016212140629
Performance Validation = 0.7087405593063434

Fold 4
Performance Training = 0.7872356350357412
Performance Validation = 0.7836160657966795

Average Performance in Training = 0.79705238233777
Average Performance in Validation = 0.715754552828139

M Value = 7
Lambda Value = 0.24

Fold 1
Performance Training = 0.8065399528365212
Performance Validation = 0.6335589073116245

Fold 2
Performance Training = 0.7939904394293059
Performance Validation = 0.7365869722826945

Fold 3
Performance Training = 0.7991326133680212
Performance Validation = 0.7087722765542709

Fold 4
Performance Training = 0.7867666802243514
Performance Validation = 0.7835685933708847

Average Performance in Training = 0.79660742146455
Average Performance in Validation = 0.7156216873798686

M Value = 7
Lambda Value = 0.25

Fold 1
Performance Training = 0.8061109186553159
Performance Validation = 0.6334318732647201

Fold 2
Performance Training = 0.7936056798862603
Performance Validation = 0.7361456068449137

Fold 3
Performance Training = 0.7986779209002786
Performance Validation = 0.7087801616016807

Fold 4
Performance Training = 0.7863098109881416

Performance Validation = 0.7835224284639934

Average Performance in Training = 0.7961760826074991

Average Performance in Validation = 0.7154700175438271

M Value = 7

Lambda Value = 0.26

Fold 1

Performance Training = 0.805696740710202

Performance Validation = 0.6332484481470515

Fold 2

Performance Training = 0.7932323029357886

Performance Validation = 0.7357151022487229

Fold 3

Performance Training = 0.7982366739832781

Performance Validation = 0.7087672241266998

Fold 4

Performance Training = 0.7858644049625402

Performance Validation = 0.7834775780130543

Average Performance in Training = 0.7957575306479522

Average Performance in Validation = 0.7153020881338822

M Value = 7

Lambda Value = 0.27

Fold 1

Performance Training = 0.8052964263545248

Performance Validation = 0.6330151441134189

Fold 2

Performance Training = 0.7928695821456013

Performance Validation = 0.7352950373300589

Fold 3

Performance Training = 0.7978080556701563

Performance Validation = 0.7087360366790436

Fold 4

Performance Training = 0.7854298656456175

Performance Validation = 0.7834340202657841

Average Performance in Training = 0.795350982453975

Average Performance in Validation = 0.7151200595970764

M Value = 7

Lambda Value = 0.28

Fold 1

Performance Training = 0.804909056917701
Performance Validation = 0.6327375958024961

Fold 2
Performance Training = 0.7925168385482427
Performance Validation = 0.7348849755871897

Fold 3
Performance Training = 0.7973913000187072
Performance Validation = 0.708688805308782

Fold 4
Performance Training = 0.7850056235991923
Performance Validation = 0.7833917121088538

Average Performance in Training = 0.7949557047709608
Average Performance in Validation = 0.7149257722018303

M Value = 7
Lambda Value = 0.29

Fold 1
Performance Training = 0.8045337822334612
Performance Validation = 0.6324206969067201

Fold 2
Performance Training = 0.7921734381946346
Performance Validation = 0.734484474341794

Fold 3
Performance Training = 0.7969856897987041
Performance Validation = 0.7086274275605553

Fold 4
Performance Training = 0.7845911368795091
Performance Validation = 0.7833505947898505

Average Performance in Training = 0.7945710117765772
Average Performance in Validation = 0.7147207983997299

M Value = 7
Lambda Value = 0.3

Fold 1
Performance Training = 0.8041698154001032
Performance Validation = 0.6320687127468142

Fold 2
Performance Training = 0.7918387895711387
Performance Validation = 0.7340930915499595

Fold 3
Performance Training = 0.7965905539668239
Performance Validation = 0.7085535403172296

Fold 4

Performance Training = 0.7841858909175318

Performance Validation = 0.7833105983861823

Average Performance in Training = 0.7941962624638994

Average Performance in Validation = 0.7145064857500465

M Value = 7

Lambda Value = 0.31

Fold 1

Performance Training = 0.8038164278310775

Performance Validation = 0.6316853735316886

Fold 2

Performance Training = 0.7915123409821877

Performance Validation = 0.7337103908144023

Fold 3

Performance Training = 0.7962052650378554

Performance Validation = 0.7084685594451137

Fold 4

Performance Training = 0.7837893980119025

Performance Validation = 0.7832716452921302

Average Performance in Training = 0.7938308579657558

Average Performance in Validation = 0.7142839922708337

M Value = 7

Lambda Value = 0.32

Fold 1

Performance Training = 0.80347294462498

Performance Validation = 0.6312739519829418

Fold 2

Performance Training = 0.7911935779674271

Performance Validation = 0.7333359450195798

Fold 3

Performance Training = 0.7958292364407474

Performance Validation = 0.7083737127834819

Fold 4

Performance Training = 0.7834011965557464

Performance Validation = 0.7832336529358886

Average Performance in Training = 0.7934742388972252

Average Performance in Validation = 0.7140543156804731

M Value = 7

Lambda Value = 0.33

Fold 1

Performance Training = 0.8031387402645874

Performance Validation = 0.630837328237283

Fold 2

Performance Training = 0.7908820207985942

Performance Validation = 0.7329693389112364

Fold 3

Performance Training = 0.7954619199198205

Performance Validation = 0.708270067705683

Fold 4

Performance Training = 0.783020850086954

Performance Validation = 0.7831965358900184

Average Performance in Training = 0.793125882767489

Average Performance in Validation = 0.7138183176860552

M Value = 7

Lambda Value = 0.34

Fold 1

Performance Training = 0.8028132346417021

Performance Validation = 0.6303780443427887

Fold 2

Performance Training = 0.7905772220846911

Performance Validation = 0.732610170867902

Fold 3

Performance Training = 0.7951028030206599

Performance Validation = 0.7081585542316352

Fold 4

Performance Training = 0.7826479462282216

Performance Validation = 0.7831602075036976

Average Performance in Training = 0.7927853014938187

Average Performance in Validation = 0.713576744236506

M Value = 7

Lambda Value = 0.35000000000000003

Fold 1

Performance Training = 0.8024958893968245

Performance Validation = 0.6298983502046088

Fold 2

Performance Training = 0.7902787645021028

Performance Validation = 0.7322580540547071

Fold 3
Performance Training = 0.7947514066858123
Performance Validation = 0.7080399844795086

Fold 4
Performance Training = 0.7822820955656105
Performance Validation = 0.7831245811576824

Average Performance in Training = 0.7924520390375875
Average Performance in Validation = 0.7133302424741267

M Value = 7
Lambda Value = 0.36

Fold 1
Performance Training = 0.8021862045576593
Performance Validation = 0.6294002424712726

Fold 2
Performance Training = 0.7899862586580844
Performance Validation = 0.7319126171064073

Fold 3
Performance Training = 0.7944072829747636
Performance Validation = 0.707915069092377

Fold 4
Performance Training = 0.7819229305010987
Performance Validation = 0.7830895712202715

Average Performance in Training = 0.7921256691729015
Average Performance in Validation = 0.7130793749725821

M Value = 7
Lambda Value = 0.37

Fold 1
Performance Training = 0.8018837154579831
Performance Validation = 0.6288854975680407

Fold 2
Performance Training = 0.7896993410902243
Performance Validation = 0.7315735044536671

Fold 3
Performance Training = 0.7940700129155323
Performance Validation = 0.7077844311549233

Fold 4
Performance Training = 0.7815701041049887
Performance Validation = 0.7830550937675785

Average Performance in Training = 0.791805793392182
Average Performance in Validation = 0.7128246317360525

M Value = 7
Lambda Value = 0.38

Fold 1
Performance Training = 0.8015879899171553
Performance Validation = 0.6283556998571715

Fold 2
Performance Training = 0.7894176724006423
Performance Validation = 0.7312403763800641

Fold 3
Performance Training = 0.7937392044899331
Performance Validation = 0.7076486180205641

Fold 4
Performance Training = 0.7812232889862809
Performance Validation = 0.7830210671165878

Average Performance in Training = 0.7914920389485028
Average Performance in Validation = 0.712566440343597

M Value = 7
Lambda Value = 0.39

Fold 1
Performance Training = 0.801298625660484
Performance Validation = 0.6278122657251887

Fold 2
Performance Training = 0.789140935521006
Performance Validation = 0.730912908878945

Fold 3
Performance Training = 0.7934144907510274
Performance Validation = 0.7075081113917581

Fold 4
Performance Training = 0.7808821761939462
Performance Validation = 0.7829874122110194

Average Performance in Training = 0.7911840570316159
Average Performance in Validation = 0.7123051745517277

M Value = 7
Lambda Value = 0.4

Fold 1
Performance Training = 0.801015247960995
Performance Validation = 0.6272564642523173

Fold 2

Performance Training = 0.788688341028327
Performance Validation = 0.7305907933622222

Fold 3
Performance Training = 0.7930955280690691
Performance Validation = 0.707363335935942

Fold 4
Performance Training = 0.7805464741575453
Performance Validation = 0.7829540528902204

Average Performance in Training = 0.7908815210726106
Average Performance in Validation = 0.7120411616101755

M Value = 7
Lambda Value = 0.41000000000000003

Fold 1
Performance Training = 0.8007375074840672
Performance Validation = 0.6266894350041586

Fold 2
Performance Training = 0.788601091026515
Performance Validation = 0.7302737362631189

Fold 3
Performance Training = 0.7927819945005091
Performance Validation = 0.7072146666693363

Fold 4
Performance Training = 0.7802159076727797
Performance Validation = 0.7829209160666297

Average Performance in Training = 0.7905841251709678
Average Performance in Validation = 0.7117746885008109

M Value = 7
Lambda Value = 0.42

Fold 1
Performance Training = 0.8004650783174998
Performance Validation = 0.626112203391483

Fold 2
Performance Training = 0.7883374470220238
Performance Validation = 0.7299614585636458

Fold 3
Performance Training = 0.7924735882738334
Performance Validation = 0.7070624353008954

Fold 4
Performance Training = 0.779890216935148
Performance Validation = 0.7828879318311597

Average Performance in Training = 0.7902915826371263
Average Performance in Validation = 0.711506007271796

M Value = 7
Lambda Value = 0.43

Fold 1
Performance Training = 0.8001976561706581
Performance Validation = 0.6255256939679834

Fold 2
Performance Training = 0.7880776593939856
Performance Validation = 0.7296536952722049

Fold 3
Performance Training = 0.7921700263855732
Performance Validation = 0.7069069356961886

Fold 4
Performance Training = 0.7795691566231633
Performance Validation = 0.7828550335020437

Average Performance in Training = 0.7900036246433451
Average Performance in Validation = 0.7112353396096052

M Value = 7
Lambda Value = 0.44

Fold 1
Performance Training = 0.799934956727654
Performance Validation = 0.6249307419747917

Fold 2
Performance Training = 0.787821500843975
Performance Validation = 0.7293501948698289

Fold 3
Performance Training = 0.7918710432994132
Performance Validation = 0.7067484285941278

Fold 4
Performance Training = 0.7792524950315061
Performance Validation = 0.7828221576304938

Average Performance in Training = 0.789719998975637
Average Performance in Validation = 0.7109628807673106

M Value = 7
Lambda Value = 0.45

Fold 1
Performance Training = 0.799676714140735

Performance Validation = 0.6243281033883394

Fold 2

Performance Training = 0.7875687583829281

Performance Validation = 0.7290507187396841

Fold 3

Performance Training = 0.7915763897416233

Performance Validation = 0.7065871456878154

Fold 4

Performance Training = 0.7789400132533632

Performance Validation = 0.7827892439723092

Average Performance in Training = 0.7894404688796623

Average Performance in Validation = 0.7106888029470371

M Value = 7

Lambda Value = 0.46

Fold 1

Performance Training = 0.7994226796511634

Performance Validation = 0.6237184636883302

Fold 2

Performance Training = 0.7873192323269609

Performance Validation = 0.7287550405906725

Fold 3

Performance Training = 0.7912858315859432

Performance Validation = 0.7064232931625721

Fold 4

Performance Training = 0.7786315044108376

Performance Validation = 0.782756235434229

Average Performance in Training = 0.7891648119937262

Average Performance in Validation = 0.710413258218951

M Value = 7

Lambda Value = 0.47000000000000003

Fold 1

Performance Training = 0.7991726203260308

Performance Validation = 0.6231024455271166

Fold 2

Performance Training = 0.7870727353702077

Performance Validation = 0.728462945883888

Fold 3

Performance Training = 0.7909991488215641

Performance Validation = 0.7062570547697452

Fold 4
Performance Training = 0.7783267729316277
Performance Validation = 0.7827230780008717

Average Performance in Training = 0.7888928193623577
Average Performance in Validation = 0.7101363810454054

M Value = 7
Lambda Value = 0.48

Fold 1
Performance Training = 0.7989263179005246
Performance Validation = 0.6224806154535911

Fold 2
Performance Training = 0.7868290917286558
Performance Validation = 0.7281742312674458

Fold 3
Performance Training = 0.7907161345979538
Performance Validation = 0.7060885945021493

Fold 4
Performance Training = 0.7780256338701325
Performance Validation = 0.7826897206478096

Average Performance in Training = 0.7886242945243166
Average Performance in Validation = 0.7098582904677488

M Value = 7
Lambda Value = 0.49

Fold 1
Performance Training = 0.7986835677160348
Performance Validation = 0.6218534898218491

Fold 2
Performance Training = 0.7865881363493554
Performance Validation = 0.7278887040250489

Fold 3
Performance Training = 0.7904365943406636
Performance Validation = 0.7059180589272747

Fold 4
Performance Training = 0.777727912270816
Performance Validation = 0.7826561152446705

Average Performance in Training = 0.7883590526692174
Average Performance in Validation = 0.7095790920047108

M Value = 7
Lambda Value = 0.5

Fold 1
Performance Training = 0.7984441777454756
Performance Validation = 0.6212215399944451

Fold 2
Performance Training = 0.7863497141797903
Performance Validation = 0.7276061815407704

Fold 3
Performance Training = 0.7901603449328468
Performance Validation = 0.7057455792254292

Fold 4
Performance Training = 0.7774334425716042
Performance Validation = 0.7826222164513866

Average Performance in Training = 0.7880969198574292
Average Performance in Validation = 0.7092988793030078

M Value = 7
Lambda Value = 0.51

Fold 1
Performance Training = 0.7982079676979208
Performance Validation = 0.6205851969339604

Fold 2
Performance Training = 0.7861136794926085
Performance Validation = 0.727326490782823

Fold 3
Performance Training = 0.7898872139572175
Performance Validation = 0.7055712729732905

Fold 4
Performance Training = 0.7771420680451171
Performance Validation = 0.7825879816103549

Average Performance in Training = 0.787837732298216
Average Performance in Validation = 0.7090177355751073

M Value = 7
Lambda Value = 0.52

Fold 1
Performance Training = 0.7979747681955076
Performance Validation = 0.6199448552634073

Fold 2
Performance Training = 0.785879895261172
Performance Validation = 0.7270494678074739

Fold 3

Performance Training = 0.789617038993835
Performance Validation = 0.7053952457071038

Fold 4
Performance Training = 0.776853640275533
Performance Validation = 0.7825533706364766

Average Performance in Training = 0.7875813356815119
Average Performance in Validation = 0.7087357348536154

M Value = 7
Lambda Value = 0.53

Fold 1
Performance Training = 0.7977444200161106
Performance Validation = 0.6193008768633947

Fold 2
Performance Training = 0.7856482325819174
Performance Validation = 0.7267749572840829

Fold 3
Performance Training = 0.7893496669693215
Performance Validation = 0.7052175922948538

Fold 4
Performance Training = 0.7765680186688924
Performance Validation = 0.7825183459065346

Average Performance in Training = 0.7873275845590605
Average Performance in Validation = 0.7084529430872165

M Value = 7
Lambda Value = 0.54

Fold 1
Performance Training = 0.7975167733959805
Performance Validation = 0.6186535940652698

Fold 2
Performance Training = 0.785418570139648
Performance Validation = 0.726502812041568

Fold 3
Performance Training = 0.7890849535534736
Performance Validation = 0.7050383981425477

Fold 4
Performance Training = 0.7762850699947577
Performance Validation = 0.7824828721492533

Average Performance in Training = 0.7870763417709649
Average Performance in Validation = 0.7081694190996597

M Value = 7
Lambda Value = 0.55

Fold 1
Performance Training = 0.7972916873870686
Performance Validation = 0.6180033124909305

Fold 2
Performance Training = 0.7851907937123725
Performance Validation = 0.7262328926363504

Fold 3
Performance Training = 0.7888227625995403
Performance Validation = 0.7048577402561162

Fold 4
Performance Training = 0.7760046679572645
Performance Validation = 0.782446916336933

Average Performance in Training = 0.7868274779140615
Average Performance in Validation = 0.7078852154300825

M Value = 7
Lambda Value = 0.56

Fold 1
Performance Training = 0.7970690292642533
Performance Validation = 0.617350313582314

Fold 2
Performance Training = 0.7849647957124518
Performance Validation = 0.7259650669416307

Fold 3
Performance Training = 0.7885629656247594
Performance Validation = 0.7046756881775555

Fold 4
Performance Training = 0.7757266927936013
Performance Validation = 0.782410447579382

Average Performance in Training = 0.7865808708487665
Average Performance in Validation = 0.7076003790702206

M Value = 7
Lambda Value = 0.5700000000000001

Fold 1
Performance Training = 0.7968486739781768
Performance Validation = 0.6166948568589593

Fold 2
Performance Training = 0.7847404747611797

Performance Validation = 0.7256992097571746

Fold 3

Performance Training = 0.7883054413279975

Performance Validation = 0.7044923048110974

Fold 4

Performance Training = 0.7754510308981764

Performance Validation = 0.7823734370206946

Average Performance in Training = 0.7863364052413826

Average Performance in Validation = 0.7073149521119815

M Value = 7

Lambda Value = 0.58

Fold 1

Performance Training = 0.7966305036496901

Performance Validation = 0.6160371819358395

Fold 2

Performance Training = 0.7845177352940739

Performance Validation = 0.7254352024397108

Fold 3

Performance Training = 0.7880500751415476

Performance Validation = 0.7043076471533188

Fold 4

Performance Training = 0.7751775744706952

Performance Validation = 0.7823358577391999

Average Performance in Training = 0.7860939721390017

Average Performance in Validation = 0.7070289723170172

M Value = 7

Lambda Value = 0.59

Fold 1

Performance Training = 0.7964144071024628

Performance Validation = 0.6153775103299377

Fold 2

Performance Training = 0.7842964871944644

Performance Validation = 0.7251729325526189

Fold 3

Performance Training = 0.7877967588145243

Performance Validation = 0.7041217669391651

Fold 4

Performance Training = 0.774906221186599

Performance Validation = 0.782297684650831

Average Performance in Training = 0.7858534685745125
Average Performance in Validation = 0.7067424736181382

M Value = 7
Lambda Value = 0.6

Fold 1
Performance Training = 0.7962002794304485
Performance Validation = 0.6147160470802644

Fold 2
Performance Training = 0.784076645453168
Performance Validation = 0.7249122935342891

Fold 3
Performance Training = 0.7875453900252948
Performance Validation = 0.7039347112139818

Fold 4
Performance Training = 0.7746368738883631
Performance Validation = 0.782258894416275

Average Performance in Training = 0.7856147971993186
Average Performance in Validation = 0.7064554865612025

M Value = 7
Lambda Value = 0.61

Fold 1
Performance Training = 0.7959880215972911
Performance Validation = 0.6140529822026711

Fold 2
Performance Training = 0.7838581298521331
Performance Validation = 0.7246531843846753

Fold 3
Performance Training = 0.7872958720208021
Performance Validation = 0.7037465228407984

Fold 4
Performance Training = 0.7743694402961955
Performance Validation = 0.7822194653516593

Average Performance in Training = 0.7853778659416054
Average Performance in Validation = 0.706168038694951

M Value = 7
Lambda Value = 0.62

Fold 1
Performance Training = 0.7957775400650939
Performance Validation = 0.6133884919984482

Fold 2
Performance Training = 0.7836408646702047
Performance Validation = 0.7243955093689513

Fold 3
Performance Training = 0.7870481132806568
Performance Validation = 0.7035572409504851

Fold 4
Performance Training = 0.7741038327368784
Performance Validation = 0.7821793773431911

Average Performance in Training = 0.7851425876882084
Average Performance in Validation = 0.705880154915269

M Value = 7
Lambda Value = 0.63

Fold 1
Performance Training = 0.7955687464500025
Performance Validation = 0.6127227402327282

Fold 2
Performance Training = 0.7834247784093072
Performance Validation = 0.7241391777374098

Fold 3
Performance Training = 0.7868020272041453
Performance Validation = 0.7033669013416108

Fold 4
Performance Training = 0.7738399678895166
Performance Validation = 0.7821386117654481

Average Performance in Training = 0.784908879988243
Average Performance in Validation = 0.7055918577692992

M Value = 7
Lambda Value = 0.64

Fold 1
Performance Training = 0.795361557202563
Performance Validation = 0.6120558791975543

Fold 2
Performance Training = 0.7832098035394682
Performance Validation = 0.7238841034609242

Fold 3
Performance Training = 0.786557531818382
Performance Validation = 0.7031755368360143

Fold 4

Performance Training = 0.7735777665470045
Performance Validation = 0.7820971514033537

Average Performance in Training = 0.7846766647768544
Average Performance in Validation = 0.7053031677244617

M Value = 7
Lambda Value = 0.65

Fold 1
Performance Training = 0.7951558933107012
Performance Validation = 0.6113880506718598

Fold 2
Performance Training = 0.7829958762611854
Performance Validation = 0.7236302049811791

Fold 3
Performance Training = 0.7863145495060628
Performance Validation = 0.7029831775951441

Fold 4
Performance Training = 0.7733171533921929
Performance Validation = 0.7820549803778895

Average Performance in Training = 0.7844458681175356
Average Performance in Validation = 0.7050141034065182

M Value = 7
Lambda Value = 0.66

Fold 1
Performance Training = 0.7949516800236375
Performance Validation = 0.6107193867896523

Fold 2
Performance Training = 0.7827829362838991
Performance Validation = 0.7233774049747372

Fold 3
Performance Training = 0.7860730067513105
Performance Validation = 0.702789851401798

Fold 4
Performance Training = 0.7730580567877867
Performance Validation = 0.782012084075317

Average Performance in Training = 0.7842164199616585
Average Performance in Validation = 0.7047246818103761

M Value = 7
Lambda Value = 0.67

Fold 1
Performance Training = 0.7947488465950004
Performance Validation = 0.6100500108262498

Fold 2
Performance Training = 0.7825709266192602
Performance Validation = 0.7231256301304357

Fold 3
Performance Training = 0.7858328339022718
Performance Validation = 0.7025955839111839

Fold 4
Performance Training = 0.7728004085789659
Performance Validation = 0.7819684490797992

Average Performance in Training = 0.7839882539238746
Average Performance in Validation = 0.7044349184869172

M Value = 7
Lambda Value = 0.68

Fold 1
Performance Training = 0.7945473260436309
Performance Validation = 0.6093800379111007

Fold 2
Performance Training = 0.782359793388195
Performance Validation = 0.7228748109390077

Fold 3
Performance Training = 0.7855939649492867
Performance Validation = 0.7024003988747947

Fold 4
Performance Training = 0.7725441439079689
Performance Validation = 0.7819240631094198

Average Performance in Training = 0.7837613070722704
Average Performance in Validation = 0.7041448277085807

M Value = 7
Lambda Value = 0.6900000000000001

Fold 1
Performance Training = 0.7943470549307512
Performance Validation = 0.608709575674578

Fold 2
Performance Training = 0.7821494856406287
Performance Validation = 0.7226248814946169

Fold 3
Performance Training = 0.7853563373173911

Performance Validation = 0.7022043183402589

Fold 4

Performance Training = 0.7722892010397917

Performance Validation = 0.7818789149553789

Average Performance in Training = 0.7835355197321406

Average Performance in Validation = 0.7038544226162082

M Value = 7

Lambda Value = 0.7000000000000001

Fold 1

Performance Training = 0.7941479731521769

Performance Validation = 0.6080387248364777

Fold 2

Performance Training = 0.7819399551869871

Performance Validation = 0.7223757793074597

Fold 3

Performance Training = 0.7851198916722163

Performance Validation = 0.702007362829657

Fold 4

Performance Training = 0.7720355211982537

Performance Validation = 0.781832994424188

Average Performance in Training = 0.7833108353024085

Average Performance in Validation = 0.7035637153494456

M Value = 7

Lambda Value = 0.71

Fold 1

Performance Training = 0.7939500237444463

Performance Validation = 0.6073675797406415

Fold 2

Performance Training = 0.7817311564406463

Performance Validation = 0.7221274451267047

Fold 3

Performance Training = 0.7848845717382208

Performance Validation = 0.70180955149891

Fold 4

Performance Training = 0.7717830484118153

Performance Validation = 0.7817862922830049

Average Performance in Training = 0.7830872000837821

Average Performance in Validation = 0.7032727171623152

M Value = 7
Lambda Value = 0.72

Fold 1

Performance Training = 0.7937531527038126
Performance Validation = 0.6066962288426312

Fold 2

Performance Training = 0.7815230462704431
Performance Validation = 0.7218798227734895

Fold 3

Performance Training = 0.7846503241284853
Performance Validation = 0.7016109022801702

Fold 4

Performance Training = 0.7715317293683823
Performance Validation = 0.7817388002075151

Average Performance in Training = 0.7828645631177809
Average Performance in Validation = 0.7029814385259515

M Value = 7
Lambda Value = 0.73

Fold 1

Performance Training = 0.7935573088171172
Performance Validation = 0.6060247551541796

Fold 2

Performance Training = 0.7813155838626202
Performance Validation = 0.7216328589828915

Fold 3

Performance Training = 0.784417098185124
Performance Validation = 0.7014114320092311

Fold 4

Performance Training = 0.771281513278644
Performance Validation = 0.7816905107327113

Average Performance in Training = 0.7826428760358763
Average Performance in Validation = 0.7026898892197535

M Value = 7
Lambda Value = 0.74

Fold 1

Performance Training = 0.7933624435037061
Performance Validation = 0.6053532366489224

Fold 2

Performance Training = 0.781108730591473
Performance Validation = 0.7213865032548252

Fold 3
Performance Training = 0.7841848458296563
Performance Validation = 0.7012111565393566

Fold 4
Performance Training = 0.7710323517473192
Performance Validation = 0.7816414172061943

Average Performance in Training = 0.7824220929180387
Average Performance in Validation = 0.7023980784123247

M Value = 7
Lambda Value = 0.75

Fold 1
Performance Training = 0.793168510667498
Performance Validation = 0.6046817466333443

Fold 2
Performance Training = 0.7809024498980502
Performance Validation = 0.7211407077133034

Fold 3
Performance Training = 0.7839535214226252
Performance Validation = 0.7010100908432597

Fold 4
Performance Training = 0.7707841986517755
Performance Validation = 0.781591513743817

Average Performance in Training = 0.7822021701599872
Average Performance in Validation = 0.702106014733431

M Value = 7
Lambda Value = 0.76

Fold 1
Performance Training = 0.7929754665585217
Performance Validation = 0.6040103540859421

Fold 2
Performance Training = 0.780696707176471
Performance Validation = 0.7208954269730301

Fold 3
Performance Training = 0.7837230816317589
Performance Validation = 0.7008082491042972

Fold 4
Performance Training = 0.7705370100275952
Performance Validation = 0.7815407951876624

Average Performance in Training = 0.7819830663485867

Average Performance in Validation = 0.701813706337733

M Value = 7

Lambda Value = 0.77

Fold 1

Performance Training = 0.7927832696432355

Performance Validation = 0.6033391239680146

Fold 2

Performance Training = 0.7804914696671459

Performance Validation = 0.7206506180136476

Fold 3

Performance Training = 0.7834934853081976

Performance Validation = 0.7006056447981692

Fold 4

Performance Training = 0.7702907439606574

Performance Validation = 0.7814892570663898

Average Performance in Training = 0.7817647421448091

Average Performance in Validation = 0.7015211609615553

M Value = 7

Lambda Value = 0.78

Fold 1

Performance Training = 0.7925918804829657

Performance Validation = 0.602668117508401

Fold 2

Performance Training = 0.7802867063565954

Performance Validation = 0.720406240060715

Fold 3

Performance Training = 0.7832646933701266

Performance Validation = 0.7004022907659858

Fold 4

Performance Training = 0.7700453604852454

Performance Validation = 0.7814368955573938

Average Performance in Training = 0.7815471601737333

Average Performance in Validation = 0.7012283859731239

M Value = 7

Lambda Value = 0.79

Fold 1

Performance Training = 0.7924012616199203

Performance Validation = 0.6019973924649201

Fold 2
Performance Training = 0.7800823878832778
Performance Validation = 0.7201622544730996

Fold 3
Performance Training = 0.7830366686933996
Performance Validation = 0.7001981992798176

Fold 4
Performance Training = 0.7698008214878822
Performance Validation = 0.7813837074511584

Average Performance in Training = 0.7813302849211199
Average Performance in Validation = 0.700935388417249

M Value = 7
Lambda Value = 0.8

Fold 1
Performance Training = 0.7922113774701999
Performance Validation = 0.6013270033643254

Fold 2
Performance Training = 0.7798784864491007
Performance Validation = 0.719918624636606

Fold 3
Performance Training = 0.7828093760085932
Performance Validation = 0.6999933821012805

Fold 4
Performance Training = 0.7695570906164848
Performance Validation = 0.7813296901173883

Average Performance in Training = 0.7811140826360947
Average Performance in Validation = 0.7006421750549001

M Value = 7
Lambda Value = 0.81

Fold 1
Performance Training = 0.7920221942233795
Performance Validation = 0.6006570017229017

Fold 2
Performance Training = 0.7796749757362192
Performance Validation = 0.719675315863088

Fold 3
Performance Training = 0.782582781804183
Performance Validation = 0.6997878505340938

Fold 4
Performance Training = 0.7693141331945209

Performance Validation = 0.7812748414729282

Average Performance in Training = 0.7808985212395756

Average Performance in Validation = 0.700348752398253

M Value = 7

Lambda Value = 0.8200000000000001

Fold 1

Performance Training = 0.791833679748164

Performance Validation = 0.5999874362496744

Fold 2

Performance Training = 0.7794718308287223

Performance Validation = 0.7194322952951948

Fold 3

Performance Training = 0.7823568542353014

Performance Validation = 0.6995816154710646

Fold 4

Performance Training = 0.7690719161398767

Performance Validation = 0.7812191599514124

Average Performance in Training = 0.780683570238016

Average Performance in Validation = 0.7000551267418365

M Value = 7

Lambda Value = 0.8300000000000001

Fold 1

Performance Training = 0.7916458035036821

Performance Validation = 0.599318353033236

Fold 2

Performance Training = 0.7792690281389126

Performance Validation = 0.719189531816095

Fold 3

Performance Training = 0.7821315630378222

Performance Validation = 0.6993746874362393

Fold 4

Performance Training = 0.7688304078881011

Performance Validation = 0.7811626444744293

Average Performance in Training = 0.7804692006421294

Average Performance in Validation = 0.6997613041899999

M Value = 7

Lambda Value = 0.84

Fold 1

Performance Training = 0.7914585364561104
Performance Validation = 0.5986497957142993

Fold 2
Performance Training = 0.7790665453379286
Performance Validation = 0.7189469959639505

Fold 3
Performance Training = 0.7819068794473657
Performance Validation = 0.6991670766226106

Fold 4
Performance Training = 0.7685895783197996
Performance Validation = 0.7811052944242475

Average Performance in Training = 0.7802553848903011
Average Performance in Validation = 0.699467290681277

M Value = 7
Lambda Value = 0.85

Fold 1
Performance Training = 0.7912718510001708
Performance Validation = 0.597981805644536

Fold 2
Performance Training = 0.7788643612902979
Performance Validation = 0.7187046598509685

Fold 3
Performance Training = 0.7816827761229108
Performance Validation = 0.6989587929257847

Fold 4
Performance Training = 0.7683493986919281
Performance Validation = 0.781047109617975

Average Performance in Training = 0.7800420967763269
Average Performance in Validation = 0.699173092009816

M Value = 7
Lambda Value = 0.86

Fold 1
Performance Training = 0.7910857208852666
Performance Validation = 0.5973144220335933

Fold 2
Performance Training = 0.7786624559923012
Performance Validation = 0.7184624970866933

Fold 3
Performance Training = 0.7814592270747471
Performance Validation = 0.6987498459742585

Fold 4
Performance Training = 0.7681098415727069
Performance Validation = 0.7809880902830417

Average Performance in Training = 0.7798293113812554
Average Performance in Validation = 0.6988787138443967

M Value = 7
Lambda Value = 0.87

Fold 1
Performance Training = 0.7909001211458689
Performance Validation = 0.596647682084714

Fold 2
Performance Training = 0.7784608105137968
Performance Validation = 0.7182204827052916

Fold 3
Performance Training = 0.7812362075964719
Performance Validation = 0.6985402451563474

Fold 4
Performance Training = 0.767870880780007
Performance Validation = 0.7809282370339902

Average Performance in Training = 0.7796170050090362
Average Performance in Validation = 0.6985841617450858

M Value = 7
Lambda Value = 0.88

Fold 1
Performance Training = 0.7907150280359666
Performance Validation = 0.5959816211203415

Fold 2
Performance Training = 0.7782594069433798
Performance Validation = 0.7179785930965574

Fold 3
Performance Training = 0.7810136942007534
Performance Validation = 0.6983299996443437

Fold 4
Performance Training = 0.7676324913229127
Performance Validation = 0.7808675508504144

Average Performance in Training = 0.779405155125753
Average Performance in Validation = 0.6982894411779142

M Value = 7

Lambda Value = 0.89

Fold 1

Performance Training = 0.7905304189671869

Performance Validation = 0.595316272698228

Fold 2

Performance Training = 0.7780582283365565

Performance Validation = 0.717736805940668

Fold 3

Performance Training = 0.7807916645586701

Performance Validation = 0.6981191184160913

Fold 4

Performance Training = 0.7673946493463809

Performance Validation = 0.7808060330561456

Average Performance in Training = 0.7791937403021986

Average Performance in Validation = 0.6979945575277832

M Value = 7

Lambda Value = 0.9

Fold 1

Performance Training = 0.7903462724504966

Performance Validation = 0.5946516687191754

Fold 2

Performance Training = 0.777857258666855

Performance Validation = 0.7174951001460332

Fold 3

Performance Training = 0.7805700974424021

Performance Validation = 0.6979076102742029

Fold 4

Performance Training = 0.7671573320787184

Performance Validation = 0.7807436852994282

Average Performance in Training = 0.778982740159618

Average Performance in Validation = 0.6976995161097099

M Value = 7

Lambda Value = 0.91

Fold 1

Performance Training = 0.7901625680411477

Performance Validation = 0.593987839526785

Fold 2

Performance Training = 0.7776564827796025

Performance Validation = 0.7172534557905614

Fold 3
Performance Training = 0.7803489726709805
Performance Validation = 0.6976954838632394

Fold 4
Performance Training = 0.7669205177817828
Performance Validation = 0.7806805095341834

Average Performance in Training = 0.7787721353183783
Average Performance in Validation = 0.6974043221786924

M Value = 7
Lambda Value = 0.92

Fold 1
Performance Training = 0.7899792862866827
Performance Validation = 0.5933248140001185

Fold 2
Performance Training = 0.7774558863482365
Performance Validation = 0.7170118540658037

Fold 3
Performance Training = 0.7801282710590628
Performance Validation = 0.6974827476850277

Fold 4
Performance Training = 0.7666841857036983
Performance Validation = 0.7806165080022047

Average Performance in Training = 0.7785619073494201
Average Performance in Validation = 0.6971089809382887

M Value = 7
Lambda Value = 0.93

Fold 1
Performance Training = 0.7897964086778555
Performance Validation = 0.5926626196396761

Fold 2
Performance Training = 0.7772554558330298
Performance Validation = 0.716770277223895

Fold 3
Performance Training = 0.7799079743683905
Performance Validation = 0.6972694101122474

Fold 4
Performance Training = 0.7664483160339766
Performance Validation = 0.7805516832162782

Average Performance in Training = 0.7783520387283132
Average Performance in Validation = 0.6968134975480242

M Value = 7
Lambda Value = 0.9400000000000001

Fold 1
Performance Training = 0.7896139176022147
Performance Validation = 0.5920012826472493

Fold 2
Performance Training = 0.7770551784419883
Performance Validation = 0.7165287085272782

Fold 3
Performance Training = 0.7796880652618785
Performance Validation = 0.697055479400607

Fold 4
Performance Training = 0.7662128898609031
Performance Validation = 0.7804860379442089

Average Performance in Training = 0.7781425127917462
Average Performance in Validation = 0.6965178771298358

M Value = 7
Lambda Value = 0.9500000000000001

Fold 1
Performance Training = 0.7894317963002518
Performance Validation = 0.5913408280004266

Fold 2
Performance Training = 0.7768550420939246
Performance Validation = 0.7162871322008415

Fold 3
Performance Training = 0.7794685272601154
Performance Validation = 0.6968409636995239

Fold 4
Performance Training = 0.765977889131031
Performance Validation = 0.7804195751936028

Average Performance in Training = 0.7779333136963308
Average Performance in Validation = 0.6962221247735987

M Value = 7
Lambda Value = 0.96

Fold 1
Performance Training = 0.7892500288238842
Performance Validation = 0.590681279521581

Fold 2

Performance Training = 0.7766550353834146
Performance Validation = 0.7160455333865093

Fold 3
Performance Training = 0.779249344700194
Performance Validation = 0.6966258710617437

Fold 4
Performance Training = 0.7657432966107113
Performance Validation = 0.7803522981975062

Average Performance in Training = 0.777724426379551
Average Performance in Validation = 0.6959262455418351

M Value = 7
Lambda Value = 0.97

Fold 1
Performance Training = 0.7890685999972248
Performance Validation = 0.5900226599423968

Fold 2
Performance Training = 0.77645514754769
Performance Validation = 0.7158038981000806

Fold 3
Performance Training = 0.779030502696644
Performance Validation = 0.6964102094517312

Fold 4
Performance Training = 0.7655090958494899
Performance Validation = 0.7802842104007008

Average Performance in Training = 0.7775158365227621
Average Performance in Validation = 0.6956302444737273

M Value = 7
Lambda Value = 0.98

Fold 1
Performance Training = 0.7888874953793723
Performance Validation = 0.5893649909639158

Fold 2
Performance Training = 0.7762553684352146
Performance Validation = 0.7155622131901407

Fold 3
Performance Training = 0.7788119871044403
Performance Validation = 0.6961939867531641

Fold 4
Performance Training = 0.7652752711453332
Performance Validation = 0.7802153154467752

Average Performance in Training = 0.77730753051609
Average Performance in Validation = 0.6953341265884989

M Value = 7
Lambda Value = 0.99

Fold 1
Performance Training = 0.7887067012292421
Performance Validation = 0.5887082933124766

Fold 2
Performance Training = 0.7760556884759457
Performance Validation = 0.7153204662990644

Fold 3
Performance Training = 0.7785937844838928
Performance Validation = 0.6959772107755118

Fold 4
Performance Training = 0.765041807511511
Performance Validation = 0.7801456171658041

Average Performance in Training = 0.777099495425148
Average Performance in Validation = 0.6950378968882143

M Value = 7
Lambda Value = 1.0

Fold 1
Performance Training = 0.7885262044722209
Performance Validation = 0.5880525867921145

Fold 2
Performance Training = 0.7758560986531087
Performance Validation = 0.7150786458258883

Fold 3
Performance Training = 0.778375882067365
Performance Validation = 0.6957598892598252

Fold 4
Performance Training = 0.7648086906451149
Performance Validation = 0.780075119562683

Average Performance in Training = 0.7768917189594523
Average Performance in Validation = 0.6947415603601277

M Value = 7
Lambda Value = 1.01

Fold 1
Performance Training = 0.7883459926685381

Performance Validation = 0.5873978903332165

Fold 2

Performance Training = 0.7756565904764428

Performance Validation = 0.7148367408909968

Fold 3

Performance Training = 0.7781582677276482

Performance Validation = 0.6955420298837858

Fold 4

Performance Training = 0.7645759068970569

Performance Validation = 0.7800038268060293

Average Performance in Training = 0.7766841894424215

Average Performance in Validation = 0.694445121978507

M Value = 7

Lambda Value = 1.02

Fold 1

Performance Training = 0.7881660539833213

Performance Validation = 0.5867442220383632

Fold 2

Performance Training = 0.7754571559568333

Performance Validation = 0.7145947413024903

Fold 3

Performance Training = 0.7779409299479889

Performance Validation = 0.6953236402661955

Fold 4

Performance Training = 0.7643434432435293

Performance Validation = 0.7799317432176565

Average Performance in Training = 0.7764768957829182

Average Performance in Validation = 0.6941485867061764

M Value = 7

Lambda Value = 1.03

Fold 1

Performance Training = 0.7879863771581787

Performance Validation = 0.5860915992247857

Fold 2

Performance Training = 0.7752577875821928

Performance Validation = 0.7143526375242619

Fold 3

Performance Training = 0.7777238577935865

Performance Validation = 0.69510472797081

Fold 4
Performance Training = 0.7641112872588113
Performance Validation = 0.7798588732626406

Average Performance in Training = 0.7762698274481923
Average Performance in Validation = 0.6938519594956245

M Value = 7
Lambda Value = 1.04

Fold 1
Performance Training = 0.7878069514842068
Performance Validation = 0.5854400384645276

Fold 2
Performance Training = 0.7750584782946361
Performance Validation = 0.7141104206454294

Fold 3
Performance Training = 0.7775070408845437
Performance Validation = 0.6948853005097047

Fold 4
Performance Training = 0.7638794270893503
Performance Validation = 0.7797852215397689

Average Performance in Training = 0.7760629744381843
Average Performance in Validation = 0.6935552452898577

M Value = 7
Lambda Value = 1.05

Fold 1
Performance Training = 0.7876277667764378
Performance Validation = 0.5847895556217276

Fold 2
Performance Training = 0.77485922146871
Performance Validation = 0.7138680823514847

Fold 3
Performance Training = 0.7772904693701481
Performance Validation = 0.6946653653461796

Fold 4
Performance Training = 0.7636478514290628
Performance Validation = 0.7797107927725727

Average Performance in Training = 0.7758563272610897
Average Performance in Validation = 0.6932584490229912

M Value = 7
Lambda Value = 1.06

Fold 1
Performance Training = 0.787448813349507
Performance Validation = 0.5841401658877066

Fold 2
Performance Training = 0.7746600108908279
Performance Validation = 0.7136256148964835

Fold 3
Performance Training = 0.777074133904408
Performance Validation = 0.6944449298972416

Fold 4
Performance Training = 0.7634165494957968
Performance Validation = 0.7796355918007687

Average Performance in Training = 0.775649876910135
Average Performance in Validation = 0.69296157562055

M Value = 7
Lambda Value = 1.07

Fold 1
Performance Training = 0.787270081994581
Performance Validation = 0.5834918838138852

Fold 2
Performance Training = 0.7744608407396106
Performance Validation = 0.7133830110769278

Fold 3
Performance Training = 0.7768580256228439
Performance Validation = 0.6942240015357235

Fold 4
Performance Training = 0.763185511008883
Performance Validation = 0.779559623572155

Average Performance in Training = 0.7754436148414796
Average Performance in Validation = 0.6926646299996728

M Value = 7
Lambda Value = 1.08

Fold 1
Performance Training = 0.7870915639574387
Performance Validation = 0.5828447233426464

Fold 2
Performance Training = 0.7742617055672858
Performance Validation = 0.7131402642065187

Fold 3

Performance Training = 0.7766421361203721
Performance Validation = 0.6940025875920992

Fold 4
Performance Training = 0.7629547261677121
Performance Validation = 0.7794828931349029

Average Performance in Training = 0.7752375329532022
Average Performance in Validation = 0.6923676170690418

M Value = 7
Lambda Value = 1.09

Fold 1
Performance Training = 0.7869132509176133
Performance Validation = 0.582198697836259

Fold 2
Performance Training = 0.7740626002819777
Performance Validation = 0.7128973680923314

Fold 3
Performance Training = 0.7764264574302646
Performance Validation = 0.6937806953559631

Fold 4
Performance Training = 0.7627241856312994
Performance Validation = 0.7794054056302052

Average Performance in Training = 0.7750316235652888
Average Performance in Validation = 0.6920705417286896

M Value = 8
Lambda Value = 0.01

Fold 1
Performance Training = 0.8227550052591857
Performance Validation = 0.5900248590084021

Fold 2
Performance Training = 0.8083507178923683
Performance Validation = 0.7569916316484295

Fold 3
Performance Training = 0.8166416642064815
Performance Validation = 0.6501475804108253

Fold 4
Performance Training = 0.8100263385632748
Performance Validation = 0.7610821670848604

Average Performance in Training = 0.8144434314803276
Average Performance in Validation = 0.6895615595381293

M Value = 8
Lambda Value = 0.02

Fold 1

Performance Training = 0.822171396979051
Performance Validation = 0.6106124384488779

Fold 2

Performance Training = 0.8078399005432457
Performance Validation = 0.7576716101586644

Fold 3

Performance Training = 0.8161821129423975
Performance Validation = 0.6579741459800155

Fold 4

Performance Training = 0.8092495766681632
Performance Validation = 0.7651283235451409

Average Performance in Training = 0.8138607467832144
Average Performance in Validation = 0.6978466295331747

M Value = 8
Lambda Value = 0.03

Fold 1

Performance Training = 0.8214097966077716
Performance Validation = 0.6255085864227108

Fold 2

Performance Training = 0.8071816036163479
Performance Validation = 0.7578739505328372

Fold 3

Performance Training = 0.815565099207215
Performance Validation = 0.6640363435711639

Fold 4

Performance Training = 0.8083128748527225
Performance Validation = 0.7678626292343427

Average Performance in Training = 0.8131173435710143
Average Performance in Validation = 0.7038203774402636

M Value = 8
Lambda Value = 0.04

Fold 1

Performance Training = 0.8205599235459123
Performance Validation = 0.6366064617235768

Fold 2

Performance Training = 0.8064499927458758

Performance Validation = 0.7577689652936471

Fold 3

Performance Training = 0.8148599487539294

Performance Validation = 0.6687984488515177

Fold 4

Performance Training = 0.8073192323462453

Performance Validation = 0.7697914226362726

Average Performance in Training = 0.8122972743479906

Average Performance in Validation = 0.7082413246262537

M Value = 8

Lambda Value = 0.05

Fold 1

Performance Training = 0.8196744005862351

Performance Validation = 0.6450432824602141

Fold 2

Performance Training = 0.8056873893004486

Performance Validation = 0.757463224256275

Fold 3

Performance Training = 0.8141101226527733

Performance Validation = 0.6725815608700276

Fold 4

Performance Training = 0.8063166695906335

Performance Validation = 0.7711951846252565

Average Performance in Training = 0.8114471455325226

Average Performance in Validation = 0.7115708130529433

M Value = 8

Lambda Value = 0.06

Fold 1

Performance Training = 0.8187843963617982

Performance Validation = 0.6515497913010023

Fold 2

Performance Training = 0.8049188257819471

Performance Validation = 0.7570263621744211

Fold 3

Performance Training = 0.81334312995882

Performance Validation = 0.6756139753897876

Fold 4

Performance Training = 0.805329129780803

Performance Validation = 0.7722416634112708

Average Performance in Training = 0.8105938704708421
Average Performance in Validation = 0.7141079480691206

M Value = 8
Lambda Value = 0.07

Fold 1
Performance Training = 0.8179082720222566
Performance Validation = 0.6566181508799994

Fold 2
Performance Training = 0.8041592646054482
Performance Validation = 0.7565050768644669

Fold 3
Performance Training = 0.8125764009754112
Performance Validation = 0.678061700080647

Fold 4
Performance Training = 0.8043688246603916
Performance Validation = 0.7730370469559389

Average Performance in Training = 0.8097531905658769
Average Performance in Validation = 0.716055493695263

M Value = 8
Lambda Value = 0.08

Fold 1
Performance Training = 0.817056579845862
Performance Validation = 0.6605917217232438

Fold 2
Performance Training = 0.8034175491810474
Performance Validation = 0.7559312086948766

Fold 3
Performance Training = 0.8118208830644383
Performance Validation = 0.6800477727464259

Fold 4
Performance Training = 0.8034417916485878
Performance Validation = 0.7736515326664808

Average Performance in Training = 0.8089342009349839
Average Performance in Validation = 0.7175555589577568

M Value = 8
Lambda Value = 0.09

Fold 1
Performance Training = 0.8162350454228163
Performance Validation = 0.6637167991210555

Fold 2
Performance Training = 0.8026987067510836
Performance Validation = 0.7553266968534393

Fold 3
Performance Training = 0.8110832981440522
Performance Validation = 0.6816649048879972

Fold 4
Performance Training = 0.8025506414122557
Performance Validation = 0.774133127512876

Average Performance in Training = 0.8081419229325519
Average Performance in Validation = 0.7187103820938421

M Value = 8
Lambda Value = 0.1

Fold 1
Performance Training = 0.8154463899898418
Performance Validation = 0.6661741173720317

Fold 2
Performance Training = 0.8020053482932622
Performance Validation = 0.7547067445135325

Fold 3
Performance Training = 0.8103675890238339
Performance Validation = 0.6829839825983361

Fold 4
Performance Training = 0.8016960197703005
Performance Validation = 0.7745155653092081

Average Performance in Training = 0.8073788367693095
Average Performance in Validation = 0.7195951024482771

M Value = 8
Lambda Value = 0.11

Fold 1
Performance Training = 0.8146914650245598
Performance Validation = 0.6680988658901814

Fold 2
Performance Training = 0.801338544989175
Performance Validation = 0.754081899287725

Fold 3
Performance Training = 0.809675861650062
Performance Validation = 0.6840599135035701

Fold 4

Performance Training = 0.8008774306754118
Performance Validation = 0.774823078799338

Average Performance in Training = 0.8066458255848021
Average Performance in Validation = 0.7202659393702036

M Value = 8
Lambda Value = 0.12

Fold 1
Performance Training = 0.8139699689060611
Performance Validation = 0.6695938455488033

Fold 2
Performance Training = 0.8006983887307505
Performance Validation = 0.7534594512078804

Fold 3
Performance Training = 0.8090090074503566
Performance Validation = 0.6849357283127677

Fold 4
Performance Training = 0.8000937195131993
Performance Validation = 0.7750733949479109

Average Performance in Training = 0.8059427711500918
Average Performance in Validation = 0.7207656050043405

M Value = 8
Lambda Value = 0.13

Fold 1
Performance Training = 0.8132809043828668
Performance Validation = 0.6707383615008984

Fold 2
Performance Training = 0.8000843560689833
Performance Validation = 0.7528443888906312

Fold 3
Performance Training = 0.8083671189892999
Performance Validation = 0.6856455084554008

Fold 4
Performance Training = 0.7993433656606161
Performance Validation = 0.7752796794906468

Average Performance in Training = 0.8052689362754415
Average Performance in Validation = 0.7211269845843943

M Value = 8
Lambda Value = 0.14

Fold 1
Performance Training = 0.812622872542293
Performance Validation = 0.6715943777440039

Fold 2
Performance Training = 0.7994955470180705
Performance Validation = 0.7522400631671304

Fold 3
Performance Training = 0.8077497699959922
Performance Validation = 0.6862165085141191

Fold 4
Performance Training = 0.7986246631466275
Performance Validation = 0.7754518357476862

Average Performance in Training = 0.8046232131757458
Average Performance in Validation = 0.7213756962932348

M Value = 8
Lambda Value = 0.15

Fold 1
Performance Training = 0.8119942621793834
Performance Validation = 0.6722108639862381

Fold 2
Performance Training = 0.7989308427164721
Performance Validation = 0.7516486533977395

Fold 3
Performance Training = 0.8071562051928758
Performance Validation = 0.6866707166303407

Fold 4
Performance Training = 0.797935833361296
Performance Validation = 0.7755973934154323

Average Performance in Training = 0.8040042858625069
Average Performance in Validation = 0.7215319068574377

M Value = 8
Lambda Value = 0.16

Fold 1
Performance Training = 0.8113933714727097
Performance Validation = 0.6726269207382636

Fold 2
Performance Training = 0.7983890097094859
Performance Validation = 0.7510714985920159

Fold 3
Performance Training = 0.8065854694431728

Performance Validation = 0.6870260164129921

Fold 4

Performance Training = 0.7972750953831915

Performance Validation = 0.77572212960008

Average Performance in Training = 0.80341073650214

Average Performance in Validation = 0.7216116413358379

M Value = 8

Lambda Value = 0.17

Fold 1

Performance Training = 0.8108184854571102

Performance Validation = 0.6728740615865638

Fold 2

Performance Training = 0.7978687687161851

Performance Validation = 0.7505093346310907

Fold 3

Performance Training = 0.8060364956729082

Performance Validation = 0.687297062249725

Fold 4

Performance Training = 0.7966407093381696

Performance Validation = 0.7758305106992553

Average Performance in Training = 0.8028411147960932

Average Performance in Validation = 0.7216277422916587

M Value = 8

Lambda Value = 0.18

Fold 1

Performance Training = 0.810267924446208

Performance Validation = 0.6729779035071475

Fold 2

Performance Training = 0.7973688395602934

Performance Validation = 0.7499624654975972

Fold 3

Performance Training = 0.8055081645565967

Performance Validation = 0.6874959458021248

Fold 4

Performance Training = 0.7960310023537704

Performance Validation = 0.7759260118448786

Average Performance in Training = 0.8022939827292173

Average Performance in Validation = 0.721590581662937

M Value = 8
Lambda Value = 0.19

Fold 1

Performance Training = 0.8097400732931551
Performance Validation = 0.6729594346920988

Fold 2

Performance Training = 0.7968879700054754
Performance Validation = 0.7494308876336951

Fold 3

Performance Training = 0.8049993447369741
Performance Validation = 0.6876327085314866

Fold 4

Performance Training = 0.7954443831847613
Performance Validation = 0.776011351092795

Average Performance in Training = 0.8017679428050914
Average Performance in Validation = 0.7215085954875189

M Value = 8
Lambda Value = 0.2

Fold 1

Performance Training = 0.8092333980053484
Performance Validation = 0.6728359764745832

Fold 2

Performance Training = 0.7964249536834557
Performance Validation = 0.7489143807009377

Fold 3

Performance Training = 0.8045089195603061
Performance Validation = 0.6877157394403193

Fold 4

Performance Training = 0.7948793494422997
Performance Validation = 0.7760886632601761

Average Performance in Training = 0.8012616551728524
Average Performance in Validation = 0.7213886899690041

M Value = 8
Lambda Value = 0.21

Fold 1

Performance Training = 0.8087464540409441
Performance Validation = 0.6726219208545097

Fold 2

Performance Training = 0.7959786406256575
Performance Validation = 0.7484125740559482

Fold 3
Performance Training = 0.8040358044424919
Performance Validation = 0.6877520863653972

Fold 4
Performance Training = 0.7943344900097643
Performance Validation = 0.7761596304053029

Average Performance in Training = 0.8007738472797145
Average Performance in Validation = 0.7212365529202895

M Value = 8
Lambda Value = 0.22

Fold 1
Performance Training = 0.8082778891794127
Performance Validation = 0.6723293014484619

Fold 2
Performance Training = 0.7955479427921338
Performance Validation = 0.7479249955465708

Fold 3
Performance Training = 0.8035789577191387
Performance Validation = 0.6877477015455544

Fold 4
Performance Training = 0.7938084843605532
Performance Validation = 0.776225580743707

Average Performance in Training = 0.8003033185128096
Average Performance in Validation = 0.7210568948210735

M Value = 8
Lambda Value = 0.23

Fold 1
Performance Training = 0.8078264429070023
Performance Validation = 0.6719682394286728

Fold 2
Performance Training = 0.7951318362396345
Performance Validation = 0.7474511073562233

Fold 3
Performance Training = 0.8031373869699798
Performance Validation = 0.6877076367760803

Fold 4
Performance Training = 0.7933000999254853
Performance Validation = 0.776287564311636

Average Performance in Training = 0.7998489415105254

Average Performance in Validation = 0.7208536369681531

M Value = 8

Lambda Value = 0.24

Fold 1

Performance Training = 0.8073909436225657

Performance Validation = 0.6715472947030339

Fold 2

Performance Training = 0.7947293610599586

Performance Validation = 0.7469903323063309

Fold 3

Performance Training = 0.8027101522139187

Performance Validation = 0.6876361995743523

Fold 4

Performance Training = 0.7928081882810099

Performance Validation = 0.7763464113135184

Average Performance in Training = 0.7994096612943632

Average Performance in Validation = 0.7206300594743089

M Value = 8

Lambda Value = 0.25

Fold 1

Performance Training = 0.8069703045410517

Performance Validation = 0.671073744593612

Fold 2

Performance Training = 0.7943396198695296

Performance Validation = 0.7465420730981165

Fold 3

Performance Training = 0.8022963669576891

Performance Validation = 0.6875370789600479

Fold 4

Performance Training = 0.7923316806764353

Performance Validation = 0.7764027774476422

Average Performance in Training = 0.7989844930111765

Average Performance in Validation = 0.7203889185248546

M Value = 8

Lambda Value = 0.26

Fold 1

Performance Training = 0.8065635188826228

Performance Validation = 0.6705538065661848

Fold 2
Performance Training = 0.7939617753898272
Performance Validation = 0.7461057263085862

Fold 3
Performance Training = 0.8018951977924451
Performance Validation = 0.6874134473808063

Fold 4
Performance Training = 0.7918695832472159
Performance Validation = 0.7764571793505155

Average Performance in Training = 0.7985725188280277
Average Performance in Validation = 0.7201325399015233

M Value = 8
Lambda Value = 0.27

Fold 1
Performance Training = 0.8061696547381456
Performance Validation = 0.6699928174334299

Fold 2
Performance Training = 0.7935950474906001
Performance Validation = 0.7456806924788324

Fold 3
Performance Training = 0.8015058630288974
Performance Validation = 0.6872680437839436

Fold 4
Performance Training = 0.7914209721446867
Performance Validation = 0.776510022482798

Average Performance in Training = 0.7981728843505824
Average Performance in Validation = 0.7198628940447509

M Value = 8
Lambda Value = 0.28

Fold 1
Performance Training = 0.8057878498676854
Performance Validation = 0.6693953784440003

Fold 2
Performance Training = 0.7932387099511073
Performance Validation = 0.7452663832855453

Fold 3
Performance Training = 0.80112763071786
Performance Validation = 0.6871032416896293

Fold 4
Performance Training = 0.7909849887324187

Performance Validation = 0.7765616231881205

Average Performance in Training = 0.7977847948172678

Average Performance in Validation = 0.7195816566518238

M Value = 8

Lambda Value = 0.29

Fold 1

Performance Training = 0.805417306596778

Performance Validation = 0.6687654734436388

Fold 2

Performance Training = 0.7928920871128763

Performance Validation = 0.7448622265332193

Fold 3

Performance Training = 0.8007598163005218

Performance Validation = 0.6869211052600808

Fold 4

Performance Training = 0.7905608349446361

Performance Validation = 0.7766122262302827

Average Performance in Training = 0.7974075112387031

Average Performance in Validation = 0.7192902578668054

M Value = 8

Lambda Value = 0.3

Fold 1

Performance Training = 0.8050572869127475

Performance Validation = 0.6681065656404933

Fold 2

Performance Training = 0.7925545505402337

Performance Validation = 0.7444676695188504

Fold 3

Performance Training = 0.8004017800598869

Performance Validation = 0.6867234357063061

Fold 4

Performance Training = 0.7901477688642963

Performance Validation = 0.7766620187976073

Average Performance in Training = 0.7970403465942911

Average Performance in Validation = 0.7189899224158144

M Value = 8

Lambda Value = 0.31

Fold 1

Performance Training = 0.8047071078208645
Performance Validation = 0.667421677265532

Fold 2
Performance Training = 0.7922255157648176
Performance Validation = 0.7440821811825598

Fold 3
Performance Training = 0.8000529244925296
Performance Validation = 0.6865118098748719

Fold 4
Performance Training = 0.7897451005528597
Performance Validation = 0.7767111417324606

Average Performance in Training = 0.7966826621577678
Average Performance in Validation = 0.7186817025138561

M Value = 8
Lambda Value = 0.32

Fold 1
Performance Training = 0.8043661369912933
Performance Validation = 0.6667134554756698

Fold 2
Performance Training = 0.7919044391620609
Performance Validation = 0.7437052533563753

Fold 3
Performance Training = 0.7997126916824981
Performance Validation = 0.686287612472749

Fold 4
Performance Training = 0.7893521881459653
Performance Validation = 0.7767596985675285

Average Performance in Training = 0.7963338639954545
Average Performance in Validation = 0.7183665049680806

M Value = 8
Lambda Value = 0.33

Fold 1
Performance Training = 0.804033788708634
Performance Validation = 0.6659842271323908

Fold 2
Performance Training = 0.7915908149882758
Performance Validation = 0.7433364013457059

Fold 3
Performance Training = 0.7993805607322433
Performance Validation = 0.6860520630910094

Fold 4
Performance Training = 0.7889684342178948
Performance Validation = 0.7768077628215857

Average Performance in Training = 0.795993399661762
Average Performance in Validation = 0.718045113597673

M Value = 8
Lambda Value = 0.34

Fold 1
Performance Training = 0.8037095201233077
Performance Validation = 0.6652360445371004

Fold 2
Performance Training = 0.7912841725931041
Performance Validation = 0.7429751640216617

Fold 3
Performance Training = 0.7990560452865929
Performance Validation = 0.685806238957444

Fold 4
Performance Training = 0.7885932824096297
Performance Validation = 0.7768553839063814

Average Performance in Training = 0.7956607551031586
Average Performance in Validation = 0.717718207855647

M Value = 8
Lambda Value = 0.35000000000000003

Fold 1
Performance Training = 0.8033928277959521
Performance Validation = 0.6644707237773798

Fold 2
Performance Training = 0.7909840738131495
Performance Validation = 0.742621103558337

Fold 3
Performance Training = 0.7987386911717814
Performance Validation = 0.6855510941643854

Fold 4
Performance Training = 0.7882262143106833
Performance Validation = 0.7769025919208772

Average Performance in Training = 0.7953354517728916
Average Performance in Validation = 0.7173863783552448

M Value = 8

Lambda Value = 0.36

Fold 1

Performance Training = 0.8030832445210068

Performance Validation = 0.6636898770088

Fold 2

Performance Training = 0.7906901105459069

Performance Validation = 0.7422738049179167

Fold 3

Performance Training = 0.7984280741621697

Performance Validation = 0.6852874759774557

Fold 4

Performance Training = 0.7878667465817668

Performance Validation = 0.7769494015506986

Average Performance in Training = 0.7950170439527126

Average Performance in Validation = 0.7170501398637177

M Value = 8

Lambda Value = 0.37

Fold 1

Performance Training = 0.8027803364130537

Performance Validation = 0.6628949397354913

Fold 2

Performance Training = 0.7904019024994624

Performance Validation = 0.7419328751589578

Fold 3

Performance Training = 0.7981237978802944

Performance Validation = 0.6850161387151258

Fold 4

Performance Training = 0.7875144283038541

Performance Validation = 0.776995815245041

Average Performance in Training = 0.7947051162741662

Average Performance in Validation = 0.716709942213654

M Value = 8

Lambda Value = 0.38

Fold 1

Performance Training = 0.8024837002379703

Performance Validation = 0.6620871939492592

Fold 2

Performance Training = 0.7901190951103619

Performance Validation = 0.7415979426277022

Fold 3

Performance Training = 0.7978254918312014

Performance Validation = 0.6847377556006127

Fold 4

Performance Training = 0.7871688385384401

Performance Validation = 0.7770418258084256

Average Performance in Training = 0.7943992814294936

Average Performance in Validation = 0.7163661794964999

M Value = 8

Lambda Value = 0.39

Fold 1

Performance Training = 0.8021929609707518

Performance Validation = 0.6612677878251135

Fold 2

Performance Training = 0.7898413576207653

Performance Validation = 0.7412686560751335

Fold 3

Performance Training = 0.7975328095688329

Performance Validation = 0.6844529289134567

Fold 4

Performance Training = 0.7868295840838027

Performance Validation = 0.777087418516812

Average Performance in Training = 0.7940991780610381

Average Performance in Validation = 0.7160191978326289

M Value = 8

Lambda Value = 0.4

Fold 1

Performance Training = 0.8019077695621327

Performance Validation = 0.6604377525426067

Fold 2

Performance Training = 0.7895683813051114

Performance Validation = 0.7409446837330114

Fold 3

Performance Training = 0.7972454269900722

Performance Validation = 0.6841621987115044

Fold 4

Performance Training = 0.7864962974124213

Performance Validation = 0.7771325728465467

Average Performance in Training = 0.7938044688174344

Average Performance in Validation = 0.7156693019584173

M Value = 8
Lambda Value = 0.41000000000000003

Fold 1
Performance Training = 0.801627800896668
Performance Validation = 0.6595980166980229

Fold 2
Performance Training = 0.7892998778362268
Performance Validation = 0.7406257123730434

Fold 3
Performance Training = 0.79696304075067
Performance Validation = 0.6838660503462419

Fold 4
Performance Training = 0.7861686347752426
Performance Validation = 0.7771772638863519

Average Performance in Training = 0.7935148385647018
Average Performance in Validation = 0.715316760825915

M Value = 8
Lambda Value = 0.42

Fold 1
Performance Training = 0.8013527519261188
Performance Validation = 0.6587494186914433

Fold 2
Performance Training = 0.7890355777808147
Performance Validation = 0.7403114463676027

Fold 3
Performance Training = 0.7966853667965711
Performance Validation = 0.683564920956033

Fold 4
Performance Training = 0.785846274459329
Performance Validation = 0.777221463490599

Average Performance in Training = 0.7932299927407084
Average Performance in Validation = 0.7149618123764194

M Value = 8
Lambda Value = 0.43

Fold 1
Performance Training = 0.8010823399627832
Performance Validation = 0.657892717405425

Fold 2

Performance Training = 0.7887752292146483
Performance Validation = 0.7400016067642299

Fold 3
Performance Training = 0.7964121390036618
Performance Validation = 0.6832592050918956

Fold 4
Performance Training = 0.7855289151862601
Performance Validation = 0.7772651412194747

Average Performance in Training = 0.7929496558418383
Average Performance in Validation = 0.7146046676202563

M Value = 8
Lambda Value = 0.44

Fold 1
Performance Training = 0.8008163011187495
Performance Validation = 0.6570286014357929

Fold 2
Performance Training = 0.7885185964480359
Performance Validation = 0.7396959303844399

Fold 3
Performance Training = 0.7961431079189039
Performance Validation = 0.6829492596035187

Fold 4
Performance Training = 0.7852162746395102
Performance Validation = 0.7773082651045122

Average Performance in Training = 0.7926735700312999
Average Performance in Validation = 0.7142455141320659

M Value = 8
Lambda Value = 0.45

Fold 1
Performance Training = 0.8005543888779783
Performance Validation = 0.6561576970951833

Fold 2
Performance Training = 0.788265458852915
Performance Validation = 0.7393941689517398

Fold 3
Performance Training = 0.7958780395958837
Performance Validation = 0.6826354078938597

Fold 4
Performance Training = 0.7849080881099144
Performance Validation = 0.7773508022702964

Average Performance in Training = 0.7924014938591728
Average Performance in Validation = 0.7138845190527698

M Value = 8
Lambda Value = 0.46

Fold 1
Performance Training = 0.8002963727892993
Performance Validation = 0.6552805753688581

Fold 2
Performance Training = 0.7880156097831321
Performance Validation = 0.739096088254836

Fold 3
Performance Training = 0.7956167145180604
Performance Validation = 0.6823179436323122

Fold 4
Performance Training = 0.7846041072492402
Performance Validation = 0.7773927194374329

Average Performance in Training = 0.7921332010849331
Average Performance in Validation = 0.7135218316733598

M Value = 8
Lambda Value = 0.47000000000000003

Fold 1
Performance Training = 0.8000420372694128
Performance Validation = 0.6543977579769538

Fold 2
Performance Training = 0.7877688555803841
Performance Validation = 0.7388014673469161

Fold 3
Performance Training = 0.795358926603147
Performance Validation = 0.6819971340031152

Fold 4
Performance Training = 0.7843040989225851
Performance Validation = 0.7774339833275258

Average Performance in Training = 0.7918684795938822
Average Performance in Validation = 0.7131575856636277

M Value = 8
Lambda Value = 0.48

Fold 1
Performance Training = 0.7997911805059761

Performance Validation = 0.6535097226708994

Fold 2

Performance Training = 0.7875250146586263

Performance Validation = 0.7385100977838446

Fold 3

Performance Training = 0.7951044822825736

Performance Validation = 0.6816732225531072

Fold 4

Performance Training = 0.7840078441512583

Performance Validation = 0.7774745609875898

Average Performance in Training = 0.7916071303996086

Average Performance in Validation = 0.7127919009988603

M Value = 8

Lambda Value = 0.49

Fold 1

Performance Training = 0.7995436134517291

Performance Validation = 0.6526169078711126

Fold 2

Performance Training = 0.7872839166603786

Performance Validation = 0.7382217829008775

Fold 3

Performance Training = 0.7948531996502137

Performance Validation = 0.6813464316939939

Fold 4

Performance Training = 0.7837151371383363

Performance Validation = 0.7775144200472675

Average Performance in Training = 0.7913489667251643

Average Performance in Validation = 0.7124248856283129

M Value = 8

Lambda Value = 0.5

Fold 1

Performance Training = 0.7992991589014139

Performance Validation = 0.651719716738784

Fold 2

Performance Training = 0.7870454016790025

Performance Validation = 0.7379363371277079

Fold 3

Performance Training = 0.794604907675043

Performance Validation = 0.6810169649051173

Fold 4
Performance Training = 0.7834257843699556
Performance Validation = 0.7775535289213005

Average Performance in Training = 0.7910938131563537
Average Performance in Validation = 0.7120566369232275

M Value = 8
Lambda Value = 0.51

Fold 1
Performance Training = 0.799057650644108
Performance Validation = 0.6508185207572834

Fold 2
Performance Training = 0.7868093195412708
Performance Validation = 0.7376535853415978

Fold 3
Performance Training = 0.7943594454726103
Performance Validation = 0.6806850086768411

Fold 4
Performance Training = 0.7831396037858221
Performance Validation = 0.7775918569661837

Average Performance in Training = 0.7908415048609528
Average Performance in Validation = 0.7116872429354766

M Value = 8
Lambda Value = 0.52

Fold 1
Performance Training = 0.7988189326842
Performance Validation = 0.6499136628908917

Fold 2
Performance Training = 0.7865755291452665
Performance Validation = 0.7373733622569726

Fold 3
Performance Training = 0.7941166616308193
Performance Validation = 0.6803507342281216

Fold 4
Performance Training = 0.782856424013151
Performance Validation = 0.77762937459925

Average Performance in Training = 0.7905918868683592
Average Performance in Validation = 0.711316783493809

M Value = 8
Lambda Value = 0.53

Fold 1
Performance Training = 0.7985828585248675
Performance Validation = 0.6490054603748234

Fold 2
Performance Training = 0.7863438978489782
Performance Validation = 0.7370955118503626

Fold 3
Performance Training = 0.7938764135855712
Performance Validation = 0.6800142990274478

Fold 4
Performance Training = 0.782576083658656
Performance Validation = 0.7776660533867121

Average Performance in Training = 0.7903448134045182
Average Performance in Validation = 0.7109453311598364

M Value = 8
Lambda Value = 0.54

Fold 1
Performance Training = 0.7983492905085816
Performance Validation = 0.6480942071867423

Fold 2
Performance Training = 0.7861143009052516
Performance Validation = 0.7368198868194908

Fold 3
Performance Training = 0.7936385670424276
Performance Validation = 0.6796758481422019

Fold 4
Performance Training = 0.7822984306537735
Performance Validation = 0.7777018661063221

Average Performance in Training = 0.7901001472775085
Average Performance in Validation = 0.7105729520636893

M Value = 8
Lambda Value = 0.55

Fold 1
Performance Training = 0.7981180992095414
Performance Validation = 0.6471801762386241

Fold 2
Performance Training = 0.7858866209393931
Performance Validation = 0.7365463480741815

Fold 3

Performance Training = 0.793402995440576
Performance Validation = 0.6793355154376018

Fold 4
Performance Training = 0.7820233216486244
Performance Validation = 0.7777367867887965

Average Performance in Training = 0.7898577593095337
Average Performance in Validation = 0.7101997066348009

M Value = 8
Lambda Value = 0.56

Fold 1
Performance Training = 0.7978891628736089
Performance Validation = 0.6462636213258603

Fold 2
Performance Training = 0.7856607474657672
Performance Validation = 0.7362747642583578

Fold 3
Performance Training = 0.7931695794557481
Performance Validation = 0.6789934246441228

Fold 4
Performance Training = 0.7817506214507609
Performance Validation = 0.7777707907424125

Average Performance in Training = 0.7896175278114712
Average Performance in Validation = 0.7098256502426883

M Value = 8
Lambda Value = 0.5700000000000001

Fold 1
Performance Training = 0.7976623669015276
Performance Validation = 0.6453447788632479

Fold 2
Performance Training = 0.7854365764402028
Performance Validation = 0.7360050113009184

Fold 3
Performance Training = 0.7929382065390361
Performance Validation = 0.6786496903091872

Fold 4
Performance Training = 0.7814802025049699
Performance Validation = 0.7778038545632224

Average Performance in Training = 0.7893793380964341
Average Performance in Validation = 0.709450833759144

M Value = 8
Lambda Value = 0.58

Fold 1

Performance Training = 0.797437603371729
Performance Validation = 0.6444238694337152

Fold 2

Performance Training = 0.7852140098453397
Performance Validation = 0.7357369719939089

Fold 3

Performance Training = 0.7927087704887635
Performance Validation = 0.6783044186471265

Fold 4

Performance Training = 0.7812119444107903
Performance Validation = 0.7778359561342439

Average Performance in Training = 0.7891430820291556
Average Performance in Validation = 0.7090753040522486

M Value = 8
Lambda Value = 0.59

Fold 1

Performance Training = 0.7972147705993662
Performance Validation = 0.6435010991733732

Fold 2

Performance Training = 0.7849929553061152
Performance Validation = 0.7354705355967452

Fold 3

Performance Training = 0.7924811710527843
Performance Validation = 0.6779577082994207

Fold 4

Performance Training = 0.7809457334747064
Performance Validation = 0.7778670746151871

Average Performance in Training = 0.788908657608243
Average Performance in Validation = 0.7086991044211816

M Value = 8
Lambda Value = 0.6

Fold 1

Performance Training = 0.796993772728504
Performance Validation = 0.642576661011599

Fold 2

Performance Training = 0.7847733257330617

Performance Validation = 0.7352055974645861

Fold 3

Performance Training = 0.7922553135588665

Performance Validation = 0.677609651015705

Fold 4

Performance Training = 0.780681462294242

Performance Validation = 0.7778971904250984

Average Performance in Training = 0.7886759685786685

Average Performance in Validation = 0.708322274979247

M Value = 8

Lambda Value = 0.61

Fold 1

Performance Training = 0.7967745193546338

Performance Validation = 0.6416507357840913

Fold 2

Performance Training = 0.7845550389911102

Performance Validation = 0.7349420586995596

Fold 3

Performance Training = 0.7920311085709736

Performance Validation = 0.6772603322647848

Fold 4

Performance Training = 0.780419029371356

Performance Validation = 0.7779262852189196

Average Performance in Training = 0.7884449240720184

Average Performance in Validation = 0.7079448529918388

M Value = 8

Lambda Value = 0.62

Fold 1

Performance Training = 0.7965569251750433

Performance Validation = 0.6407234932322685

Fold 2

Performance Training = 0.7843380175918928

Performance Validation = 0.7346798258234599

Fold 3

Performance Training = 0.7918084715693732

Performance Validation = 0.6769098317835125

Fold 4

Performance Training = 0.7801583387528912

Performance Validation = 0.7779543418596753

Average Performance in Training = 0.7882154382723001
Average Performance in Validation = 0.7075668731747291

M Value = 8
Lambda Value = 0.63

Fold 1
Performance Training = 0.7963409096647006
Performance Validation = 0.6397950929039994

Fold 2
Performance Training = 0.7841221884076751
Performance Validation = 0.7344188104705978

Fold 3
Performance Training = 0.7915873226528778
Performance Validation = 0.6765582240707169

Fold 4
Performance Training = 0.7798992996959391
Performance Validation = 0.7779813443871358

Average Performance in Training = 0.7879874301052981
Average Performance in Validation = 0.7071883679581125

M Value = 8
Lambda Value = 0.64

Fold 1
Performance Training = 0.7961263967756028
Performance Validation = 0.6388656849649524

Fold 2
Performance Training = 0.7839074824052245
Performance Validation = 0.7341589290992343

Fold 3
Performance Training = 0.7913675862613432
Performance Validation = 0.6762055788319752

Fold 4
Performance Training = 0.7796418263561162
Performance Validation = 0.7780072779838028

Average Performance in Training = 0.7877608229495717
Average Performance in Validation = 0.7068093677199911

M Value = 8
Lambda Value = 0.65

Fold 1
Performance Training = 0.7959133146576234
Performance Validation = 0.6379354109318602

Fold 2
Performance Training = 0.7836938343980432
Performance Validation = 0.7339001027209822

Fold 3
Performance Training = 0.7911491909170443
Performance Validation = 0.6758519613811049

Fold 4
Performance Training = 0.7793858374970929
Performance Validation = 0.7780321289390806

Average Performance in Training = 0.787535544367451
Average Performance in Validation = 0.7064299009932569

M Value = 8
Lambda Value = 0.66

Fold 1
Performance Training = 0.7957015953991943
Performance Validation = 0.6370044043354629

Fold 2
Performance Training = 0.7834811828155722
Performance Validation = 0.7336422566464291

Fold 3
Performance Training = 0.7909320689834104
Performance Validation = 0.6754974330026206

Fold 4
Performance Training = 0.7791312562196858
Performance Validation = 0.7780558846121075

Average Performance in Training = 0.7873115258544657
Average Performance in Validation = 0.706049994649155

M Value = 8
Lambda Value = 0.67

Fold 1
Performance Training = 0.7954911747862132
Performance Validation = 0.6360727913215956

Fold 2
Performance Training = 0.7832694694880507
Performance Validation = 0.7333853202463114

Fold 3
Performance Training = 0.7907161564398543
Performance Validation = 0.6751420512795869

Fold 4

Performance Training = 0.7788780097089885
Performance Validation = 0.7780785333937942

Average Performance in Training = 0.7870887026057767
Average Performance in Validation = 0.705669674060322

M Value = 8
Lambda Value = 0.68

Fold 1
Performance Training = 0.7952819920777157
Performance Validation = 0.6351406911958363

Fold 2
Performance Training = 0.7830586394458129
Performance Validation = 0.7331292267271627

Fold 3
Performance Training = 0.7905013926714808
Performance Validation = 0.6747858703905654

Fold 4
Performance Training = 0.778626028998301
Performance Validation = 0.7781000646684978

Average Performance in Training = 0.7868670132983276
Average Performance in Validation = 0.7052889632455156

M Value = 8
Lambda Value = 0.6900000000000001

Fold 1
Performance Training = 0.7950739897970116
Performance Validation = 0.6342082169194154

Fold 2
Performance Training = 0.7828486407319725
Performance Validation = 0.7328739129205628

Fold 3
Performance Training = 0.7902877202725923
Performance Validation = 0.6744289413787325

Fold 4
Performance Training = 0.7783752487485001
Performance Validation = 0.7781204687756799

Average Performance in Training = 0.7866463998875192
Average Performance in Validation = 0.7049078849985977

M Value = 8
Lambda Value = 0.7000000000000001

Fold 1
Performance Training = 0.7948671135371387
Performance Validation = 0.6332754755595738

Fold 2
Performance Training = 0.7826394242274932
Performance Validation = 0.7326193190847272

Fold 3
Performance Training = 0.7900750848629778
Performance Validation = 0.6740713123962473

Fold 4
Performance Training = 0.7781256070417604
Performance Validation = 0.7781397369716816

Average Performance in Training = 0.7864268074173426
Average Performance in Validation = 0.7045264610030575

M Value = 8
Lambda Value = 0.71

Fold 1
Performance Training = 0.7946613117794386
Performance Validation = 0.6323425687008484

Fold 2
Performance Training = 0.78243094348767
Performance Validation = 0.7323653887182802

Fold 3
Performance Training = 0.7898634349160351
Performance Validation = 0.6737130289263188

Fold 4
Performance Training = 0.7778770451885485
Performance Validation = 0.7781578613920141

Average Performance in Training = 0.786208183842923
Average Performance in Validation = 0.7041447119343653

M Value = 8
Lambda Value = 0.72

Fold 1
Performance Training = 0.7944565357243064
Performance Validation = 0.6314095928196939

Fold 2
Performance Training = 0.7822231545892555
Performance Validation = 0.7321120683849225

Fold 3
Performance Training = 0.7896527215978644

Performance Validation = 0.6733541339851239

Fold 4

Performance Training = 0.7776295075469585

Performance Validation = 0.7781748350142307

Average Performance in Training = 0.7859904798645961

Average Performance in Validation = 0.7037626575509927

M Value = 8

Lambda Value = 0.73

Fold 1

Performance Training = 0.7942527391332159

Performance Validation = 0.6304766396274335

Fold 2

Performance Training = 0.7820160159873962

Performance Validation = 0.7318593075484489

Fold 3

Performance Training = 0.7894428986166471

Performance Validation = 0.6729946683058512

Fold 4

Performance Training = 0.7773829413534858

Performance Validation = 0.7781906516216736

Average Performance in Training = 0.7857736487726863

Average Performance in Validation = 0.7033803167758518

M Value = 8

Lambda Value = 0.74

Fold 1

Performance Training = 0.7940498781811149

Performance Validation = 0.6295437963840929

Fold 2

Performance Training = 0.7818094883816946

Performance Validation = 0.7316070584175628

Fold 3

Performance Training = 0.789233922081377

Performance Validation = 0.6726346705064897

Fold 4

Performance Training = 0.7771372965643673

Performance Validation = 0.7782053057677527

Average Performance in Training = 0.7855576463021385

Average Performance in Validation = 0.7029977077689745

M Value = 8
Lambda Value = 0.75

Fold 1

Performance Training = 0.7938479113184798
Performance Validation = 0.6286111461855197

Fold 2

Performance Training = 0.7816035345907497
Performance Validation = 0.731355275799654

Fold 3

Performance Training = 0.7890257503695327
Performance Validation = 0.6722741772428691

Fold 4

Performance Training = 0.776892525706871
Performance Validation = 0.7782187927415959

Average Performance in Training = 0.7853424304964083
Average Performance in Validation = 0.7026148479924097

M Value = 8
Lambda Value = 0.76

Fold 1

Performance Training = 0.793646799142287
Performance Validation = 0.6276787682278477

Fold 2

Performance Training = 0.7813981194345523
Performance Validation = 0.7311039169630973

Fold 3

Performance Training = 0.7888183440028305
Performance Validation = 0.6719132233486689

Fold 4

Performance Training = 0.7766485837396764
Performance Validation = 0.7782311085343109

Average Performance in Training = 0.7851279615798364
Average Performance in Validation = 0.7022317542684812

M Value = 8
Lambda Value = 0.77

Fold 1

Performance Training = 0.7934465042752331
Performance Validation = 0.6267467380492457

Fold 2

Performance Training = 0.7811932096241768
Performance Validation = 0.7308529415075748

Fold 3
Performance Training = 0.7886116655306671
Performance Validation = 0.6715518419633162

Fold 4
Performance Training = 0.7764054279218453
Performance Validation = 0.7782422498065313

Average Performance in Training = 0.7849142018379807
Average Performance in Validation = 0.7018484428316671

M Value = 8
Lambda Value = 0.78

Fold 1
Performance Training = 0.7932469912526919
Performance Validation = 0.6258151277536586

Fold 2
Performance Training = 0.7809887736582946
Performance Validation = 0.7306023112418125

Fold 3
Performance Training = 0.7884056794206414
Performance Validation = 0.6711900646492481

Fold 4
Performance Training = 0.7761630176897216
Performance Validation = 0.7782522138569744

Average Performance in Training = 0.7847011155053374
Average Performance in Validation = 0.7014649293754235

M Value = 8
Lambda Value = 0.79

Fold 1
Performance Training = 0.7930482264167015
Performance Validation = 0.6248840062160427

Fold 2
Performance Training = 0.7807847817259942
Performance Validation = 0.730351990068256

Fold 3
Performance Training = 0.7882003519557051
Performance Validation = 0.6708279214992675

Fold 4
Performance Training = 0.7759213145412072
Performance Validation = 0.7782609985919615

Average Performance in Training = 0.784488668659902

Average Performance in Validation = 0.7010812290938819

M Value = 8

Lambda Value = 0.8

Fold 1

Performance Training = 0.7928501778166729

Performance Validation = 0.6239534392721018

Fold 2

Performance Training = 0.7805812056155002

Performance Validation = 0.7301019438743024

Fold 3

Performance Training = 0.7879956511374169

Performance Validation = 0.670465441235109

Fold 4

Performance Training = 0.7756802819269913

Performance Validation = 0.7782686024962484

Average Performance in Training = 0.7842768291241453

Average Performance in Validation = 0.7006973567194403

M Value = 8

Lambda Value = 0.81

Fold 1

Performance Training = 0.7926528151161849

Performance Validation = 0.6230234898936031

Fold 2

Performance Training = 0.7803780186283786

Performance Validation = 0.7298521404297698

Fold 3

Performance Training = 0.7877915465950325

Performance Validation = 0.6701026512978758

Fold 4

Performance Training = 0.7754398851481596

Performance Validation = 0.7782750246047185

Average Performance in Training = 0.7840655663719389

Average Performance in Validation = 0.7003133265564918

M Value = 8

Lambda Value = 0.8200000000000001

Fold 1

Performance Training = 0.7924561095055163

Performance Validation = 0.6220942183499897

Fold 2
Performance Training = 0.78017519549884
Performance Validation = 0.7296025492900167

Fold 3
Performance Training = 0.7875880094998939
Performance Validation = 0.6697395779313131

Fold 4
Performance Training = 0.775200091259844
Performance Validation = 0.7782802644752747

Average Performance in Training = 0.7838548514410235
Average Performance in Validation = 0.6999291525116486

M Value = 8
Lambda Value = 0.8300000000000001

Fold 1
Performance Training = 0.7922600336195135
Performance Validation = 0.6211656823589476

Fold 2
Performance Training = 0.7799727123178609
Performance Validation = 0.7293531417044932

Fold 3
Performance Training = 0.7873850124848585
Performance Validation = 0.6693762462582971

Fold 4
Performance Training = 0.774960868980471
Performance Validation = 0.778284322162788

Average Performance in Training = 0.7836446568506759
Average Performance in Validation = 0.6995448481211315

M Value = 8
Lambda Value = 0.84

Fold 1
Performance Training = 0.7920645614604301
Performance Validation = 0.6202379372249924

Fold 2
Performance Training = 0.7797705464617067
Performance Validation = 0.7291038905303037

Fold 3
Performance Training = 0.7871825295683826
Performance Validation = 0.6690126803513103

Fold 4
Performance Training = 0.7747221886062439

Performance Validation = 0.778287198194052

Average Performance in Training = 0.7834349565241909

Average Performance in Validation = 0.6991604265751646

M Value = 8

Lambda Value = 0.85

Fold 1

Performance Training = 0.7918696683253386

Performance Validation = 0.6193110359688552

Fold 2

Performance Training = 0.7795686765246563

Performance Validation = 0.7288547701506759

Fold 3

Performance Training = 0.7869805360830415

Performance Validation = 0.6686489032974592

Fold 4

Performance Training = 0.7744840219305358

Performance Validation = 0.7782888935437607

Average Performance in Training = 0.783225725715893

Average Performance in Validation = 0.6987759007401877

M Value = 8

Lambda Value = 0.86

Fold 1

Performance Training = 0.7916753307379394

Performance Validation = 0.6183850294473059

Fold 2

Performance Training = 0.7793670822555796

Performance Validation = 0.7286057563977415

Fold 3

Performance Training = 0.7867790086080377

Performance Validation = 0.668284937258429

Fold 4

Performance Training = 0.7742463421678374

Performance Validation = 0.7782894096114475

Average Performance in Training = 0.7830169409423485

Average Performance in Validation = 0.698391283178731

M Value = 8

Lambda Value = 0.87

Fold 1

Performance Training = 0.7914815263843135
Performance Validation = 0.6174599664648049

Fold 2
Performance Training = 0.7791657444981859
Performance Validation = 0.7283568264796398

Fold 3
Performance Training = 0.786577924905645
Performance Validation = 0.6679208035258332

Fold 4
Performance Training = 0.7740091238820348
Performance Validation = 0.778288748199461

Average Performance in Training = 0.7828085799175448
Average Performance in Validation = 0.6980065861674347

M Value = 8
Lambda Value = 0.88

Fold 1
Performance Training = 0.7912882340524614
Performance Validation = 0.6165358938774341

Fold 2
Performance Training = 0.7789646451346585
Performance Validation = 0.7281079589114186

Fold 3
Performance Training = 0.7863772638611727
Performance Validation = 0.6675565225724033

Fold 4
Performance Training = 0.7737723429186864
Performance Validation = 0.7782869114918727

Average Performance in Training = 0.7826006214917448
Average Performance in Validation = 0.6976218217132821

M Value = 8
Lambda Value = 0.89

Fold 1
Performance Training = 0.7910954335753673
Performance Validation = 0.6156128566899475

Fold 2
Performance Training = 0.778763767032449
Performance Validation = 0.7278591334498383

Fold 3
Performance Training = 0.7861770054262984
Performance Validation = 0.6671921140993653

Fold 4

Performance Training = 0.7735359763410311

Performance Validation = 0.7782839020341779

Average Performance in Training = 0.7823930455937864

Average Performance in Validation = 0.6972370015683322

M Value = 8

Lambda Value = 0.9

Fold 1

Performance Training = 0.7909031057773105

Performance Validation = 0.6146908981458448

Fold 2

Performance Training = 0.7785630939940948

Performance Validation = 0.7276103310314944

Fold 3

Performance Training = 0.7859771305655674

Performance Validation = 0.6668275970802817

Fold 4

Performance Training = 0.7733000023695686

Performance Validation = 0.7782797227139461

Average Performance in Training = 0.7821858331766354

Average Performance in Validation = 0.6968521372428917

M Value = 8

Lambda Value = 0.91

Fold 1

Performance Training = 0.7907112324232797

Performance Validation = 0.6137700598125568

Fold 2

Performance Training = 0.7783626107098071

Performance Validation = 0.7273615337143154

Fold 3

Performance Training = 0.7857776212058231

Performance Validation = 0.6664629898016369

Fold 4

Performance Training = 0.7730644003249476

Performance Validation = 0.7782743767424463

Average Performance in Training = 0.7819789661659644

Average Performance in Validation = 0.6964672400177389

M Value = 8

Lambda Value = 0.92

Fold 1

Performance Training = 0.7905197961712247

Performance Validation = 0.6128503816600999

Fold 2

Performance Training = 0.7781623027126645

Performance Validation = 0.7271127246221805

Fold 3

Performance Training = 0.7855784601884279

Performance Validation = 0.6660983099006546

Fold 4

Performance Training = 0.7728291505739012

Performance Validation = 0.7782678676367849

Average Performance in Training = 0.7817724274115546

Average Performance in Validation = 0.6960823209549301

M Value = 8

Lambda Value = 0.93

Fold 1

Performance Training = 0.7903287805270021

Performance Validation = 0.6119319021351384

Fold 2

Performance Training = 0.7779621563363128

Performance Validation = 0.726863887892294

Fold 3

Performance Training = 0.7853796312240513

Performance Validation = 0.665733574400207

Fold 4

Performance Training = 0.7725942344781771

Performance Validation = 0.7782601992031211

Average Performance in Training = 0.7815662006413858

Average Performance in Validation = 0.6956973909076901

M Value = 8

Lambda Value = 0.9400000000000001

Fold 1

Performance Training = 0.7901381698018536

Performance Validation = 0.6110146582307971

Fold 2

Performance Training = 0.7777621586749095

Performance Validation = 0.7266150086254972

Fold 3
Performance Training = 0.7851811188499247
Performance Validation = 0.665368799741324

Fold 4
Performance Training = 0.7723596343461379
Performance Validation = 0.7782513755204542

Average Performance in Training = 0.7813602704182063
Average Performance in Validation = 0.6953124605295181

M Value = 8
Lambda Value = 0.9500000000000001

Fold 1
Performance Training = 0.7899479490722436
Performance Validation = 0.6100986855514641

Fold 2
Performance Training = 0.7775622975452746
Performance Validation = 0.726366072839044

Fold 3
Performance Training = 0.7849829083893842
Performance Validation = 0.6650040018135406

Fold 4
Performance Training = 0.7721253333869941
Performance Validation = 0.7782414009252628

Average Performance in Training = 0.7811546220984741
Average Performance in Validation = 0.6949275402823278

M Value = 8
Lambda Value = 0.96

Fold 1
Performance Training = 0.7897581041418869
Performance Validation = 0.6091840183742245

Fold 2
Performance Training = 0.777362561451058
Performance Validation = 0.7261170674218602

Fold 3
Performance Training = 0.7847849859135796
Performance Validation = 0.6646391959829789

Fold 4
Performance Training = 0.7718913156673954
Performance Validation = 0.7782302799966894

Average Performance in Training = 0.78094924179348
Average Performance in Validation = 0.6945426404439383

M Value = 8
Lambda Value = 0.97

Fold 1
Performance Training = 0.7895686215058534
Performance Validation = 0.6082706897064447

Fold 2
Performance Training = 0.7771629395488369
Performance Validation = 0.7258679800920249

Fold 3
Performance Training = 0.7845873382051645
Performance Validation = 0.664274397118568

Fold 4
Performance Training = 0.771657566070352
Performance Validation = 0.7782180175424983

Average Performance in Training = 0.7807441163325517
Average Performance in Validation = 0.694157771114884

M Value = 8
Lambda Value = 0.98

Fold 1
Performance Training = 0.7893794883166516
Performance Validation = 0.6073587313397988

Fold 2
Performance Training = 0.7769634216159681
Performance Validation = 0.7256187993565509

Fold 3
Performance Training = 0.7843899527239504
Performance Validation = 0.6639096196165495

Fold 4
Performance Training = 0.7714240702562728
Performance Validation = 0.7782046185856972

Average Performance in Training = 0.7805392332282107
Average Performance in Validation = 0.693772942224649

M Value = 8
Lambda Value = 0.99

Fold 1
Performance Training = 0.7891906923521329
Performance Validation = 0.606448173901472

Fold 2

Performance Training = 0.7767639980202066
Performance Validation = 0.7253695144730941

Fold 3
Performance Training = 0.7841928175743418
Performance Validation = 0.6635448774233064

Fold 4
Performance Training = 0.7711908146260039
Performance Validation = 0.778190088351647

Average Performance in Training = 0.7803345806431712
Average Performance in Validation = 0.6933881635373799

M Value = 8
Lambda Value = 1.0

Fold 1
Performance Training = 0.7890022219850575
Performance Validation = 0.6055390469023216

Fold 2
Performance Training = 0.7765646596908321
Performance Validation = 0.7251201154136339

Fold 3
Performance Training = 0.7839959214744275
Performance Validation = 0.6631801840566094

Fold 4
Performance Training = 0.7709577862858208
Performance Validation = 0.7781744322558954

Average Performance in Training = 0.7801301473590345
Average Performance in Validation = 0.6930034446571152

M Value = 8
Lambda Value = 1.01

Fold 1
Performance Training = 0.7888140661543774
Performance Validation = 0.6046313787823074

Fold 2
Performance Training = 0.7763653980913197
Performance Validation = 0.7248705928300371

Fold 3
Performance Training = 0.7837992537267061
Performance Validation = 0.6628155526256239

Fold 4
Performance Training = 0.7707249730141497
Performance Validation = 0.7781576558924336

Average Performance in Training = 0.7799259227466382
Average Performance in Validation = 0.6926187950326005

M Value = 8
Lambda Value = 1.02

Fold 1
Performance Training = 0.7886262143379397
Performance Validation = 0.6037251969532382

Fold 2
Performance Training = 0.7761662051934325
Performance Validation = 0.7246209380211861

Fold 3
Performance Training = 0.783602804190298
Performance Validation = 0.6624509958494955

Fold 4
Performance Training = 0.7704923632300191
Performance Validation = 0.7781397650226622

Average Performance in Training = 0.7797218967379222
Average Performance in Validation = 0.6922342239616455

M Value = 8
Lambda Value = 1.03

Fold 1
Performance Training = 0.7884386565267016
Performance Validation = 0.6028205278397868

Fold 2
Performance Training = 0.7759670734526105
Performance Validation = 0.7243711429019151

Fold 3
Performance Training = 0.7834065632545433
Performance Validation = 0.6620865260747955

Fold 4
Performance Training = 0.7702599459630797
Performance Validation = 0.7781207655646745

Average Performance in Training = 0.7795180597992337
Average Performance in Validation = 0.6918497405952929

M Value = 8
Lambda Value = 1.04

Fold 1
Performance Training = 0.7882513832002249

Performance Validation = 0.6019173969175485

Fold 2

Performance Training = 0.775767995784639

Performance Validation = 0.7241211999733912

Fold 3

Performance Training = 0.7832105218140263

Performance Validation = 0.661722155291967

Fold 4

Performance Training = 0.7700277108251621

Performance Validation = 0.7781006635832017

Average Performance in Training = 0.7793144029060131

Average Performance in Validation = 0.6914653539415272

M Value = 8

Lambda Value = 1.05

Fold 1

Performance Training = 0.7880643853034933

Performance Validation = 0.6010158287495933

Fold 2

Performance Training = 0.7755689655434915

Performance Validation = 0.7238711022949507

Fold 3

Performance Training = 0.7830146712447255

Performance Validation = 0.6613578951505399

Fold 4

Performance Training = 0.7697956479831954

Performance Validation = 0.7780794652799048

Average Performance in Training = 0.7791109175187264

Average Performance in Validation = 0.6910810728687471

M Value = 8

Lambda Value = 1.06

Fold 1

Performance Training = 0.7878776542248798

Performance Validation = 0.6001158470210584

Fold 2

Performance Training = 0.7753699765003028

Performance Validation = 0.7236208434573026

Fold 3

Performance Training = 0.7828190033814659

Performance Validation = 0.6609937569736026

Fold 4
Performance Training = 0.769563748133526
Performance Validation = 0.7780571769841984

Average Performance in Training = 0.7789075955600436
Average Performance in Validation = 0.6906969061090404

M Value = 8
Lambda Value = 1.07

Fold 1
Performance Training = 0.7876911817752666
Performance Validation = 0.5992174745717344

Fold 2
Performance Training = 0.7751710228233599
Performance Validation = 0.7233704175571449

Fold 3
Performance Training = 0.7826235104964442
Performance Validation = 0.6606297517713244

Fold 4
Performance Training = 0.7693320024774544
Performance Validation = 0.7780338051444318

Average Performance in Training = 0.7787044293931313
Average Performance in Validation = 0.6903128622611588

M Value = 8
Lambda Value = 1.08

Fold 1
Performance Training = 0.7875049601682251
Performance Validation = 0.598320733427404

Fold 2
Performance Training = 0.7749720990591368
Performance Validation = 0.7231198191728478

Fold 3
Performance Training = 0.7824281852788285
Performance Validation = 0.6602658902536267

Fold 4
Performance Training = 0.7691004026980149
Performance Validation = 0.7780093563195756

Average Performance in Training = 0.7785014118010514
Average Performance in Validation = 0.6899289497933635

M Value = 8
Lambda Value = 1.09

Fold 1
Performance Training = 0.787318982001179
Performance Validation = 0.5974256448291098

Fold 2
Performance Training = 0.7747732001142394
Performance Validation = 0.7228690433414267

Fold 3
Performance Training = 0.7822330208153534
Performance Validation = 0.6599021828421954

Fold 4
Performance Training = 0.7688689409378417
Performance Validation = 0.7779838371711848

Average Performance in Training = 0.7782985359671535
Average Performance in Validation = 0.6895451770459792

M Value = 9
Lambda Value = 0.01

Fold 1
Performance Training = 0.8224784603284305
Performance Validation = 0.5632054261937756

Fold 2
Performance Training = 0.8124355602447503
Performance Validation = 0.761651893348154

Fold 3
Performance Training = 0.8230788005333435
Performance Validation = 0.6683640859480493

Fold 4
Performance Training = 0.8107606496383144
Performance Validation = 0.7658337703507357

Average Performance in Training = 0.8171883676862096
Average Performance in Validation = 0.6897637939601786

M Value = 9
Lambda Value = 0.02

Fold 1
Performance Training = 0.8214219486052787
Performance Validation = 0.5243773591028285

Fold 2
Performance Training = 0.8097010248322258
Performance Validation = 0.7604738715268274

Fold 3

Performance Training = 0.8192729192179369
Performance Validation = 0.6702108323456446

Fold 4
Performance Training = 0.8091359994181103
Performance Validation = 0.764405907510077

Average Performance in Training = 0.814882973018388
Average Performance in Validation = 0.6798669926213444

M Value = 9
Lambda Value = 0.03

Fold 1
Performance Training = 0.8207031651430754
Performance Validation = 0.5069021493921411

Fold 2
Performance Training = 0.8077680224159539
Performance Validation = 0.7588851560767049

Fold 3
Performance Training = 0.8167529308991309
Performance Validation = 0.6699322001187991

Fold 4
Performance Training = 0.807777237371202
Performance Validation = 0.7641389914393764

Average Performance in Training = 0.8132503389573406
Average Performance in Validation = 0.6749646242567553

M Value = 9
Lambda Value = 0.04

Fold 1
Performance Training = 0.8201644371321535
Performance Validation = 0.49884170700709984

Fold 2
Performance Training = 0.8063492400025465
Performance Validation = 0.757576867798416

Fold 3
Performance Training = 0.814990304577481
Performance Validation = 0.6695860218052267

Fold 4
Performance Training = 0.80665525303443
Performance Validation = 0.7643651326503434

Average Performance in Training = 0.8120398086866527
Average Performance in Validation = 0.6725924323152715

M Value = 9
Lambda Value = 0.05

Fold 1

Performance Training = 0.8197223280948612
Performance Validation = 0.4956037669852753

Fold 2

Performance Training = 0.805250499628444
Performance Validation = 0.7565492659153957

Fold 3

Performance Training = 0.8136781870418206
Performance Validation = 0.6694247075621123

Fold 4

Performance Training = 0.805706880746052
Performance Validation = 0.7648204867856219

Average Performance in Training = 0.8110894738777944
Average Performance in Validation = 0.6715995568121013

M Value = 9
Lambda Value = 0.06

Fold 1

Performance Training = 0.8193337704376286
Performance Validation = 0.49507161226914376

Fold 2

Performance Training = 0.8043586834914844
Performance Validation = 0.7557322440234651

Fold 3

Performance Training = 0.8126489651165476
Performance Validation = 0.6694385117383513

Fold 4

Performance Training = 0.8048836142583123
Performance Validation = 0.7653852898823821

Average Performance in Training = 0.8103062583259933
Average Performance in Validation = 0.6714069144783354

M Value = 9
Lambda Value = 0.07

Fold 1

Performance Training = 0.8189752817998265
Performance Validation = 0.4961442493248882

Fold 2

Performance Training = 0.8036061318070696

Performance Validation = 0.7550649078185582

Fold 3

Performance Training = 0.8118063017522437

Performance Validation = 0.6695825989771633

Fold 4

Performance Training = 0.8041514397692164

Performance Validation = 0.7659995696891475

Average Performance in Training = 0.8096347887820892

Average Performance in Validation = 0.6716978314524393

M Value = 9

Lambda Value = 0.08

Fold 1

Performance Training = 0.8186334035246475

Performance Validation = 0.4981983564305631

Fold 2

Performance Training = 0.8029505737826664

Performance Validation = 0.7545027229534161

Fold 3

Performance Training = 0.8110916629296132

Performance Validation = 0.6698165471314792

Fold 4

Performance Training = 0.8034866028800829

Performance Validation = 0.7666311386506347

Average Performance in Training = 0.8090405607792526

Average Performance in Validation = 0.6722871912915233

M Value = 9

Lambda Value = 0.09

Fold 1

Performance Training = 0.8183001356985046

Performance Validation = 0.5008586224260241

Fold 2

Performance Training = 0.8023645091155409

Performance Validation = 0.754014222137178

Fold 3

Performance Training = 0.8104677366713996

Performance Validation = 0.6701095843989803

Fold 4

Performance Training = 0.8028722848499938

Performance Validation = 0.7672618844724608

Average Performance in Training = 0.8085011665838597
Average Performance in Validation = 0.6730610783586608

M Value = 9
Lambda Value = 0.1

Fold 1
Performance Training = 0.817970613981059
Performance Validation = 0.5038884932704255

Fold 2
Performance Training = 0.8018294193375536
Performance Validation = 0.7535772257653106

Fold 3
Performance Training = 0.8099097746417303
Performance Validation = 0.6704393099461348

Fold 4
Performance Training = 0.802296374761921
Performance Validation = 0.7678813208002384

Average Performance in Training = 0.808001545680566
Average Performance in Validation = 0.6739465874455273

M Value = 9
Lambda Value = 0.11

Fold 1
Performance Training = 0.8176418544402835
Performance Validation = 0.5071336888516422

Fold 2
Performance Training = 0.8013324708469581
Performance Validation = 0.7531760004419535

Fold 3
Performance Training = 0.8094008151086465
Performance Validation = 0.6707896159446295

Fold 4
Performance Training = 0.801750010069164
Performance Validation = 0.76848331759542

Average Performance in Training = 0.807531287616263
Average Performance in Validation = 0.6748956557084114

M Value = 9
Lambda Value = 0.12

Fold 1
Performance Training = 0.8173120396862988
Performance Validation = 0.5104910153242592

Fold 2
Performance Training = 0.8008645550340272
Performance Validation = 0.7527992949711937

Fold 3
Performance Training = 0.8089289136204165
Performance Validation = 0.6711489461866045

Fold 4
Performance Training = 0.8012266120969218
Performance Validation = 0.7690643394811563

Average Performance in Training = 0.8070830301094161
Average Performance in Validation = 0.6758758989908034

M Value = 9
Lambda Value = 0.13

Fold 1
Performance Training = 0.8169800949772443
Performance Validation = 0.5138902024292568

Fold 2
Performance Training = 0.8004190775421738
Performance Validation = 0.7524390059633299

Fold 3
Performance Training = 0.808485466979796
Performance Validation = 0.6715090104354751

Fold 4
Performance Training = 0.8007212373293481
Performance Validation = 0.7696224476271745

Average Performance in Training = 0.8066514692071405
Average Performance in Validation = 0.676865166613809

M Value = 9
Lambda Value = 0.14

Fold 1
Performance Training = 0.8166454272911587
Performance Validation = 0.5172828514878705

Fold 2
Performance Training = 0.7999911846135808
Performance Validation = 0.7520892677952522

Fold 3
Performance Training = 0.8080641604757178
Performance Validation = 0.671863869035073

Fold 4

Performance Training = 0.8002301325131569
Performance Validation = 0.7701567100148882

Average Performance in Training = 0.8062327262234036
Average Performance in Validation = 0.677848174583271

M Value = 9
Lambda Value = 0.15

Fold 1
Performance Training = 0.8163077598361607
Performance Validation = 0.5206354588920753

Fold 2
Performance Training = 0.7995772539335785
Performance Validation = 0.751745825010332

Fold 3
Performance Training = 0.8076602849869846
Performance Validation = 0.6722092859760193

Fold 4
Performance Training = 0.7997504237009148
Performance Validation = 0.7706668405054182

Average Performance in Training = 0.8058239306144096
Average Performance in Validation = 0.6788143525959611

M Value = 9
Lambda Value = 0.16

Fold 1
Performance Training = 0.8159670244651003
Performance Validation = 0.5239248735611559

Fold 2
Performance Training = 0.7991745508084834
Performance Validation = 0.7514055936121995

Fold 3
Performance Training = 0.8072702814481082
Performance Validation = 0.6725422699817587

Fold 4
Performance Training = 0.7992798950458925
Performance Validation = 0.7711529713088757

Average Performance in Training = 0.8054229379418961
Average Performance in Validation = 0.6797564271159975

M Value = 9
Lambda Value = 0.17

Fold 1
Performance Training = 0.8156232903068329
Performance Validation = 0.5271352600028779

Fold 2
Performance Training = 0.7987809907000802
Performance Validation = 0.7510663501372835

Fold 3
Performance Training = 0.806891429463961
Performance Validation = 0.6728607462720528

Fold 4
Performance Training = 0.798816828895713
Performance Validation = 0.7716155059520318

Average Performance in Training = 0.8050281348416468
Average Performance in Validation = 0.6806694655910615

M Value = 9
Lambda Value = 0.18

Fold 1
Performance Training = 0.8152767156513854
Performance Validation = 0.5302560214375158

Fold 2
Performance Training = 0.798394971939737
Performance Validation = 0.7507265083618548

Fold 3
Performance Training = 0.806521629840216
Performance Validation = 0.6731633194476303

Fold 4
Performance Training = 0.7983598885079285
Performance Validation = 0.7720550222915039

Average Performance in Training = 0.8046383014848167
Average Performance in Validation = 0.6815502178846262

M Value = 9
Lambda Value = 0.19

Fold 1
Performance Training = 0.8149275151067594
Performance Validation = 0.5332803515063662

Fold 2
Performance Training = 0.798015255809851
Performance Validation = 0.7503849569762447

Fold 3
Performance Training = 0.806159249798069

Performance Validation = 0.6734491004330578

Fold 4

Performance Training = 0.7979080308977742

Performance Validation = 0.7724722074687765

Average Performance in Training = 0.8042525129031134

Average Performance in Validation = 0.6823966540961114

M Value = 9

Lambda Value = 0.2

Fold 1

Performance Training = 0.8145759369817875

Performance Validation = 0.5362042071763058

Fold 2

Performance Training = 0.797640879245602

Performance Validation = 0.7500409402674215

Fold 3

Performance Training = 0.8058030109386862

Performance Validation = 0.6737175789155041

Fold 4

Performance Training = 0.7974604413122748

Performance Validation = 0.7728678137118785

Average Performance in Training = 0.8038700671195876

Average Performance in Validation = 0.6832076350177775

M Value = 9

Lambda Value = 0.21

Fold 1

Performance Training = 0.8142222476314062

Performance Validation = 0.5390255696979138

Fold 2

Performance Training = 0.7972710904098887

Performance Validation = 0.749693969548701

Fold 3

Performance Training = 0.8054519069277846

Performance Validation = 0.673968528442722

Fold 4

Performance Training = 0.7970164834445986

Performance Validation = 0.7732426280268128

Average Performance in Training = 0.8034904321034195

Average Performance in Validation = 0.6839826739290373

M Value = 9
Lambda Value = 0.22

Fold 1
Performance Training = 0.8138667206093607
Performance Validation = 0.5417439061474849

Fold 2
Performance Training = 0.796905300568465
Performance Validation = 0.7493437568336054

Fold 3
Performance Training = 0.8051051422043135
Performance Validation = 0.6742019352265389

Fold 4
Performance Training = 0.7965756612487921
Performance Validation = 0.7735974512934429

Average Performance in Training = 0.8031132061577327
Average Performance in Validation = 0.684721762375268

M Value = 9
Lambda Value = 0.23

Fold 1
Performance Training = 0.813509629179894
Performance Validation = 0.5443597729063094

Fold 2
Performance Training = 0.7965430477512756
Performance Validation = 0.7489901647934548

Fold 3
Performance Training = 0.8047620857937295
Performance Validation = 0.6744179443271393

Fold 4
Performance Training = 0.7961375894044438
Performance Validation = 0.7739330838346747

Average Performance in Training = 0.8027380880323357
Average Performance in Validation = 0.6854252414653945

M Value = 9
Lambda Value = 0.24

Fold 1
Performance Training = 0.8131512411985473
Performance Validation = 0.5468745210142488

Fold 2
Performance Training = 0.7961839690489689
Performance Validation = 0.7486331687539749

Fold 3
Performance Training = 0.8044222361289617
Performance Validation = 0.6746168187112858

Fold 4
Performance Training = 0.795701970298632
Performance Validation = 0.774250315492046

Average Performance in Training = 0.8023648541687775
Average Performance in Validation = 0.6860937059928889

M Value = 9
Lambda Value = 0.25

Fold 1
Performance Training = 0.8127918156768801
Performance Validation = 0.5492900755777718

Fold 2
Performance Training = 0.7958277793109433
Performance Validation = 0.7482728276819319

Fold 3
Performance Training = 0.8040851939954674
Performance Validation = 0.6747989079315273

Fold 4
Performance Training = 0.795268575965952
Performance Validation = 0.7745499188743741

Average Performance in Training = 0.8019933412373107
Average Performance in Validation = 0.6867279325164013

M Value = 9
Lambda Value = 0.26

Fold 1
Performance Training = 0.8124316005498622
Performance Validation = 0.5516087696263645

Fold 2
Performance Training = 0.795474254641583
Performance Validation = 0.7479092619359693

Fold 3
Performance Training = 0.8037506415437902
Performance Validation = 0.6749646240597172

Fold 4
Performance Training = 0.7948372338340914
Performance Validation = 0.774832644859194

Average Performance in Training = 0.8016234326423317

Average Performance in Validation = 0.6873288251203112

M Value = 9

Lambda Value = 0.27

Fold 1

Performance Training = 0.812070831305169

Performance Validation = 0.5538332184018502

Fold 2

Performance Training = 0.7951232195270332

Performance Validation = 0.7475426361526529

Fold 3

Performance Training = 0.8034183258809022

Performance Validation = 0.675114423129279

Fold 4

Performance Training = 0.7944078154133004

Performance Validation = 0.7750992197061282

Average Performance in Training = 0.8012550480316013

Average Performance in Validation = 0.6878973743474777

M Value = 9

Lambda Value = 0.28

Fold 1

Performance Training = 0.8117097302300283

Performance Validation = 0.5559662239450389

Fold 2

Performance Training = 0.7947745367342582

Performance Validation = 0.7471731460532516

Fold 3

Performance Training = 0.8030880461514832

Performance Validation = 0.6752487907941148

Fold 4

Performance Training = 0.7939802272801639

Performance Validation = 0.7753503433241047

Average Performance in Training = 0.8008881350989835

Average Performance in Validation = 0.6884346260291275

M Value = 9

Lambda Value = 0.29

Fold 1

Performance Training = 0.8113485060998229

Performance Validation = 0.5580107025608644

Fold 2
Performance Training = 0.794428099343913
Performance Validation = 0.7468010082674394

Fold 3
Performance Training = 0.8027596433028724
Performance Validation = 0.6753682312335119

Fold 4
Performance Training = 0.7935544038616865
Performance Validation = 0.7755866883713547

Average Performance in Training = 0.8005226631520738
Average Performance in Validation = 0.6889416576082925

M Value = 9
Lambda Value = 0.3

Fold 1
Performance Training = 0.8109873541804793
Performance Validation = 0.5599696296852967

Fold 2
Performance Training = 0.7940838244378761
Performance Validation = 0.7464264524828461

Fold 3
Performance Training = 0.802432991929938
Performance Validation = 0.6754732585713292

Fold 4
Performance Training = 0.7931303016404323
Performance Validation = 0.7758088999522637

Average Performance in Training = 0.8001586180471814
Average Performance in Validation = 0.6894195601729339

M Value = 9
Lambda Value = 0.31

Fold 1
Performance Training = 0.8106264564509833
Performance Validation = 0.5618459980623101

Fold 2
Performance Training = 0.7937416480780793
Performance Validation = 0.7460497154053416

Fold 3
Performance Training = 0.8021079937447373
Performance Validation = 0.6755643902508877

Fold 4
Performance Training = 0.7927078944873771

Performance Validation = 0.7760175957412028

Average Performance in Training = 0.7997959981902943

Average Performance in Validation = 0.6898694248649355

M Value = 9

Lambda Value = 0.32

Fold 1

Performance Training = 0.8102659819782662

Performance Validation = 0.5636427861572915

Fold 2

Performance Training = 0.7934015212998713

Performance Validation = 0.7456710361285503

Fold 3

Performance Training = 0.801784572324133

Performance Validation = 0.6756421419413731

Fold 4

Performance Training = 0.7922871698951166

Performance Validation = 0.7762133664097127

Average Performance in Training = 0.7994348113743468

Average Performance in Validation = 0.6902923326592318

M Value = 9

Lambda Value = 0.33

Fold 1

Performance Training = 0.8099060873936141

Performance Validation = 0.5653629344704592

Fold 2

Performance Training = 0.7930634069059856

Performance Validation = 0.7452906526055074

Fold 3

Performance Training = 0.8014626688682445

Performance Validation = 0.6757070236428786

Fold 4

Performance Training = 0.7918681259320735

Performance Validation = 0.776396776264719

Average Performance in Training = 0.7990750722749795

Average Performance in Validation = 0.690689346745891

M Value = 9

Lambda Value = 0.34

Fold 1

Performance Training = 0.8095469174341702
Performance Validation = 0.5670093279805442

Fold 2
Performance Training = 0.7927272768963893
Performance Validation = 0.7449087989843322

Fold 3
Performance Training = 0.8011422387638543
Performance Validation = 0.675759536737339

Fold 4
Performance Training = 0.7914507687775104
Performance Validation = 0.7765683640321863

Average Performance in Training = 0.798716800467981
Average Performance in Validation = 0.6910615069336004

M Value = 9
Lambda Value = 0.35000000000000003

Fold 1
Performance Training = 0.8091886055220371
Performance Validation = 0.5685847833384876

Fold 2
Performance Training = 0.7923931104044877
Performance Validation = 0.7445257036236285

Fold 3
Performance Training = 0.8008232487918223
Performance Validation = 0.6758001717846648

Fold 4
Performance Training = 0.7910351107253115
Performance Validation = 0.7767286437332395

Average Performance in Training = 0.7983600188609146
Average Performance in Validation = 0.6914098256200051

M Value = 9
Lambda Value = 0.36

Fold 1
Performance Training = 0.8088312743617603
Performance Validation = 0.5700920397669085

Fold 2
Performance Training = 0.7920608920391707
Performance Validation = 0.7441415876397945

Fold 3
Performance Training = 0.8005056748521138
Performance Validation = 0.675829406907542

Fold 4

Performance Training = 0.7906211685682082

Performance Validation = 0.7768781056200876

Average Performance in Training = 0.7980047524553132

Average Performance in Validation = 0.6917352849835832

M Value = 9

Lambda Value = 0.37

Fold 1

Performance Training = 0.808475036541142

Performance Validation = 0.5715337528372579

Fold 2

Performance Training = 0.7917306105520708

Performance Validation = 0.7437566638767639

Fold 3

Performance Training = 0.8001895001074115

Performance Validation = 0.6758477066427959

Fold 4

Performance Training = 0.790208962290693

Performance Validation = 0.7770172171401957

Average Performance in Training = 0.7976510273728293

Average Performance in Validation = 0.6920388351242533

M Value = 9

Lambda Value = 0.38

Fold 1

Performance Training = 0.8081199951256979

Performance Validation = 0.5729124904899738

Fold 2

Performance Training = 0.7914022577677544

Performance Validation = 0.7433711362031146

Fold 3

Performance Training = 0.7998747134652352

Performance Validation = 0.6758555211605137

Fold 4

Performance Training = 0.7897985140141163

Performance Validation = 0.7771464239120853

Average Performance in Training = 0.797298870093201

Average Performance in Validation = 0.6923213929414218

M Value = 9

Lambda Value = 0.39

Fold 1

Performance Training = 0.8077662442390847

Performance Validation = 0.574230730793246

Fold 2

Performance Training = 0.7910758277253958

Performance Validation = 0.7429851990690073

Fold 3

Performance Training = 0.7995613083366495

Performance Validation = 0.6758532857750051

Fold 4

Performance Training = 0.7893898471472537

Performance Validation = 0.7772661506956868

Average Performance in Training = 0.796948306862096

Average Performance in Validation = 0.6925838415832364

M Value = 9

Lambda Value = 0.4

Fold 1

Performance Training = 0.8074138696245571

Performance Validation = 0.5754908610524054

Fold 2

Performance Training = 0.7907513159925729

Performance Validation = 0.742599037265137

Fold 3

Performance Training = 0.7992492816199517

Performance Validation = 0.6758414206849586

Fold 4

Performance Training = 0.7889829857053561

Performance Validation = 0.7773768023483465

Average Performance in Training = 0.7965993632356094

Average Performance in Validation = 0.6928270303377119

M Value = 9

Lambda Value = 0.41000000000000003

Fold 1

Performance Training = 0.8070629491841828

Performance Validation = 0.5766951779520048

Fold 2

Performance Training = 0.7904287191180155

Performance Validation = 0.7422128258401814

Fold 3
Performance Training = 0.798938632868869
Performance Validation = 0.6758203308937718

Fold 4
Performance Training = 0.7885779537670963
Performance Validation = 0.7774787647576488

Average Performance in Training = 0.7962520637345409
Average Performance in Validation = 0.6930517748609016

M Value = 9
Lambda Value = 0.42

Fold 1
Performance Training = 0.8067135534937466
Performance Validation = 0.57784588849636

Fold 2
Performance Training = 0.7901080341974017
Performance Validation = 0.7418267301382444

Fold 3
Performance Training = 0.798629363612585
Performance Validation = 0.6757904062713089

Fold 4
Performance Training = 0.7881747750445418
Performance Validation = 0.7775724057463957

Average Performance in Training = 0.7959064315870688
Average Performance in Validation = 0.6932588576630773

M Value = 9
Lambda Value = 0.43

Fold 1
Performance Training = 0.8063657462920725
Performance Validation = 0.5789451115458554

Fold 2
Performance Training = 0.7897892585315716
Performance Validation = 0.7414409059313793

Fold 3
Performance Training = 0.7983214768002918
Performance Validation = 0.6757520217252888

Fold 4
Performance Training = 0.7877734725461562
Performance Validation = 0.7776580759463383

Average Performance in Training = 0.795562488542523
Average Performance in Validation = 0.6934490287872155

M Value = 9
Lambda Value = 0.44

Fold 1

Performance Training = 0.8060195849444932
Performance Validation = 0.5799948798010062

Fold 2

Performance Training = 0.7894723893593721
Performance Validation = 0.7410554996205831

Fold 3

Performance Training = 0.7980149763485046
Performance Validation = 0.675705537455746

Fold 4

Performance Training = 0.7873740683158374
Performance Validation = 0.7777361096381056

Average Performance in Training = 0.7952202547420519
Average Performance in Validation = 0.6936230066288602

M Value = 9
Lambda Value = 0.45

Fold 1

Performance Training = 0.8056751208805644
Performance Validation = 0.5809971421128327

Fold 2

Performance Training = 0.7891574236512438
Performance Validation = 0.74067064849244

Fold 3

Performance Training = 0.7977098667734336
Performance Validation = 0.6756512992743323

Fold 4

Performance Training = 0.7869765832346598
Performance Validation = 0.7778068255561569

Average Performance in Training = 0.7948797486349753
Average Performance in Validation = 0.6937814788589405

M Value = 9
Lambda Value = 0.46

Fold 1

Performance Training = 0.8053324000061775
Performance Validation = 0.581953766022129

Fold 2

Performance Training = 0.7888443579528169
Performance Validation = 0.7402864810102322

Fold 3
Performance Training = 0.7974061528938015
Performance Validation = 0.6755896389704581

Fold 4
Performance Training = 0.7865810368736754
Performance Validation = 0.7778705276579883

Average Performance in Training = 0.7945409869316178
Average Performance in Validation = 0.6939251034152019

M Value = 9
Lambda Value = 0.47000000000000003

Fold 1
Performance Training = 0.8049914630911759
Performance Validation = 0.582866540452599

Fold 2
Performance Training = 0.788533188268348
Performance Validation = 0.7399031171344459

Fold 3
Performance Training = 0.7971038395913468
Performance Validation = 0.6755208747116119

Fold 4
Performance Training = 0.7861874473890087
Performance Validation = 0.7779275058593295

Average Performance in Training = 0.7942039845849698
Average Performance in Validation = 0.6940545095394965

M Value = 9
Lambda Value = 0.48

Fold 1
Performance Training = 0.8046523461330906
Performance Validation = 0.5837371784986372

Fold 2
Performance Training = 0.7882239099772298
Performance Validation = 0.7395206686588045

Fold 3
Performance Training = 0.7968029316200931
Performance Validation = 0.675445311467338

Fold 4
Performance Training = 0.7857958314508913
Performance Validation = 0.7779780367330681

Average Performance in Training = 0.7938687547953263
Average Performance in Validation = 0.6941702988394619

M Value = 9
Lambda Value = 0.49

Fold 1
Performance Training = 0.804315080697967
Performance Validation = 0.5845673202601546

Fold 2
Performance Training = 0.7879165177766887
Performance Validation = 0.739139239557089

Fold 3
Performance Training = 0.7965034334550154
Performance Validation = 0.6753632414483404

Fold 4
Performance Training = 0.7854062042009355
Performance Validation = 0.7780223841751923

Average Performance in Training = 0.7935353090326516
Average Performance in Validation = 0.694273046360194

M Value = 9
Lambda Value = 0.5

Fold 1
Performance Training = 0.8039796942396107
Performance Validation = 0.5853585356881867

Fold 2
Performance Training = 0.7876110056458535
Performance Validation = 0.7387589263349722

Fold 3
Performance Training = 0.7962053491742709
Performance Validation = 0.6752749445542268

Fold 4
Performance Training = 0.7850185792318329
Performance Validation = 0.778060800037011

Average Performance in Training = 0.793203657072892
Average Performance in Validation = 0.6943633016535992

M Value = 9
Lambda Value = 0.51

Fold 1
Performance Training = 0.8036462103980595

Performance Validation = 0.5861123274122149

Fold 2

Performance Training = 0.7873073668270332

Performance Validation = 0.7383798183808791

Fold 3

Performance Training = 0.7959086823685037

Performance Validation = 0.6751806888239225

Fold 4

Performance Training = 0.7846329685855344

Performance Validation = 0.7780935247261228

Average Performance in Training = 0.7928738070447827

Average Performance in Validation = 0.6944415898357847

M Value = 9

Lambda Value = 0.52

Fold 1

Performance Training = 0.8033146492785513

Performance Validation = 0.5868301335293016

Fold 2

Performance Training = 0.7870055938205861

Performance Validation = 0.738001998314067

Fold 3

Performance Training = 0.7956134360729421

Performance Validation = 0.6750807308850002

Fold 4

Performance Training = 0.7842493827660173

Performance Validation = 0.7781207877763475

Average Performance in Training = 0.7925457654845243

Average Performance in Validation = 0.694508412626179

M Value = 9

Lambda Value = 0.53

Fold 1

Performance Training = 0.8029850277120727

Performance Validation = 0.587513330334871

Fold 2

Performance Training = 0.7867056783907096

Performance Validation = 0.737625542326205

Fold 3

Performance Training = 0.7953196127182551

Performance Validation = 0.674975316398434

Fold 4
Performance Training = 0.783867830763645
Performance Validation = 0.7781428083883688

Average Performance in Training = 0.7922195373961706
Average Performance in Validation = 0.6945642493619697

M Value = 9
Lambda Value = 0.54

Fold 1
Performance Training = 0.8026573594985652
Performance Validation = 0.5881632349868566

Fold 2
Performance Training = 0.7864076115796834
Performance Validation = 0.737250520514606

Fold 3
Performance Training = 0.7950272140968331
Performance Validation = 0.6748646804964216

Fold 4
Performance Training = 0.7834883200888433
Performance Validation = 0.7781597959421788

Average Performance in Training = 0.7918951263159811
Average Performance in Validation = 0.6946095579850158

M Value = 9
Lambda Value = 0.55

Fold 1
Performance Training = 0.8023316556337827
Performance Validation = 0.588781108091563

Fold 2
Performance Training = 0.7861113837285896
Performance Validation = 0.7368769972068413

Fold 3
Performance Training = 0.7947362413415877
Performance Validation = 0.6747490482102362

Fold 4
Performance Training = 0.7831108568127532
Performance Validation = 0.7781719504832244

Average Performance in Training = 0.7915725343791782
Average Performance in Validation = 0.6946447759979663

M Value = 9
Lambda Value = 0.56

Fold 1
Performance Training = 0.802007924520914
Performance Validation = 0.5893681562051595

Fold 2
Performance Training = 0.7858169845033568
Performance Validation = 0.7365050312730046

Fold 3
Performance Training = 0.794446694915512
Performance Validation = 0.6746286348885852

Fold 4
Performance Training = 0.7827354456133712
Performance Validation = 0.7781794631824099

Average Performance in Training = 0.7912517623882885
Average Performance in Validation = 0.6946703213872898

M Value = 9
Lambda Value = 0.5700000000000001

Fold 1
Performance Training = 0.8016861721677666
Performance Validation = 0.5899255342485233

Fold 2
Performance Training = 0.7855244029240898
Performance Validation = 0.7361346764281033

Fold 3
Performance Training = 0.7941585746091957
Performance Validation = 0.6745036466030253

Fold 4
Performance Training = 0.7823620898255705
Performance Validation = 0.7781825167725314

Average Performance in Training = 0.7909328098816556
Average Performance in Validation = 0.6946865935130457

M Value = 9
Lambda Value = 0.58

Fold 1
Performance Training = 0.8013664023707467
Performance Validation = 0.59045434783052

Fold 2
Performance Training = 0.7852336273973717
Performance Validation = 0.7357659815215349

Fold 3

Performance Training = 0.793871879545748
Performance Validation = 0.6743742805416533

Fold 4
Performance Training = 0.7819907914940788
Performance Validation = 0.7781812859616617

Average Performance in Training = 0.7906156752019863
Average Performance in Validation = 0.6946939739638425

M Value = 9
Lambda Value = 0.59

Fold 1
Performance Training = 0.8010486168859472
Performance Validation = 0.5909556554811075

Fold 2
Performance Training = 0.784944645750341
Performance Validation = 0.735398990814965

Fold 3
Performance Training = 0.7935866081907098
Performance Validation = 0.6742407253893358

Fold 4
Performance Training = 0.7816215514280629
Performance Validation = 0.7781759378241777

Average Performance in Training = 0.7903003555637651
Average Performance in Validation = 0.6946928273773965

M Value = 9
Lambda Value = 0.6

Fold 1
Performance Training = 0.8007328155888809
Performance Validation = 0.5914304707922189

Fold 2
Performance Training = 0.7846574452655812
Performance Validation = 0.7350337442476084

Fold 3
Performance Training = 0.7933027583666992
Performance Validation = 0.6741031616944171

Fold 4
Performance Training = 0.7812543692569657
Performance Validation = 0.7781666321718235

Average Performance in Training = 0.7899868471195317
Average Performance in Validation = 0.694683502226517

M Value = 9
Lambda Value = 0.61

Fold 1

Performance Training = 0.8004189966230318
Performance Validation = 0.5918797644693066

Fold 2

Performance Training = 0.7843720127170686
Performance Validation = 0.7346702776896236

Fold 3

Performance Training = 0.7930203272714015
Performance Validation = 0.6739617622224312

Fold 4

Performance Training = 0.7808892434867696
Performance Validation = 0.7781535219050908

Average Performance in Training = 0.7896751450245678
Average Performance in Validation = 0.6946663315716131

M Value = 9
Lambda Value = 0.62

Fold 1

Performance Training = 0.8001071565381614
Performance Validation = 0.5923044662926025

Fold 2

Performance Training = 0.7840883344056082
Performance Validation = 0.7343086231832463

Fold 3

Performance Training = 0.7927393114980643
Performance Validation = 0.6738166922957072

Fold 4

Performance Training = 0.7805261715559431
Performance Validation = 0.7781367533460644

Average Performance in Training = 0.7893652434994443
Average Performance in Validation = 0.694641633779405

M Value = 9
Lambda Value = 0.63

Fold 1

Performance Training = 0.799797290419313
Performance Validation = 0.5927054669928189

Fold 2

Performance Training = 0.7838063961944075

Performance Validation = 0.7339488091725869

Fold 3

Performance Training = 0.7924597070582204

Performance Validation = 0.6736681101200399

Fold 4

Performance Training = 0.7801651498911457

Performance Validation = 0.7781164665538738

Average Performance in Training = 0.7890571358907716

Average Performance in Validation = 0.6946097132098299

M Value = 9

Lambda Value = 0.64

Fold 1

Performance Training = 0.7994893920066185

Performance Validation = 0.5930836200412517

Fold 2

Performance Training = 0.7835261835439343

Performance Validation = 0.7335908607213009

Fold 3

Performance Training = 0.7921815094056412

Performance Validation = 0.6735161670975709

Fold 4

Performance Training = 0.779806173961777

Performance Validation = 0.7780927956233861

Average Performance in Training = 0.7887508147294928

Average Performance in Validation = 0.6945708608708774

M Value = 9

Lambda Value = 0.65

Fold 1

Performance Training = 0.7991834538069907

Performance Validation = 0.5934397433586296

Fold 2

Performance Training = 0.7832476815459385

Performance Validation = 0.7332347997197608

Fold 3

Performance Training = 0.7919047134614497

Performance Validation = 0.6733610081275365

Fold 4

Performance Training = 0.7794492383336389

Performance Validation = 0.7780658689687012

Average Performance in Training = 0.7884462717870044
Average Performance in Validation = 0.694525355043657

M Value = 9
Lambda Value = 0.66

Fold 1
Performance Training = 0.7988794671980692
Performance Validation = 0.5937746209448124

Fold 2
Performance Training = 0.7829708749566703
Performance Validation = 0.7328806450809768

Fold 3
Performance Training = 0.7916293136397746
Performance Validation = 0.6732027718936053

Fold 4
Performance Training = 0.7790943367210845
Performance Validation = 0.7780358095918108

Average Performance in Training = 0.7881434981288996
Average Performance in Validation = 0.6944734618778013

M Value = 9
Lambda Value = 0.67

Fold 1
Performance Training = 0.7985774225248125
Performance Validation = 0.5940890044312703

Fold 2
Performance Training = 0.7826957482286465
Performance Validation = 0.7325284129270937

Fold 3
Performance Training = 0.7913553038735635
Performance Validation = 0.6730415911390406

Fold 4
Performance Training = 0.7787414620377717
Performance Validation = 0.7780027353368862

Average Performance in Training = 0.7878424841661985
Average Performance in Validation = 0.6944154359585727

M Value = 9
Lambda Value = 0.68

Fold 1
Performance Training = 0.79827730918961
Performance Validation = 0.5943836145618915

Fold 2
Performance Training = 0.7824222855417138
Performance Validation = 0.7321781167646363

Fold 3
Performance Training = 0.7910826776408342
Performance Validation = 0.6728775929304885

Fold 4
Performance Training = 0.7783906064457152
Performance Validation = 0.7779667591319549

Average Performance in Training = 0.7875432197044684
Average Performance in Validation = 0.6943515208472427

M Value = 9
Lambda Value = 0.6900000000000001

Fold 1
Performance Training = 0.7979791157358653
Performance Validation = 0.5946591426018073

Fold 2
Performance Training = 0.7821504708325152
Performance Validation = 0.7318297676510988

Fold 3
Performance Training = 0.790811427990394
Performance Validation = 0.6727108989095913

Fold 4
Performance Training = 0.7780417614027607
Performance Validation = 0.777927989217835

Average Performance in Training = 0.7872456939903838
Average Performance in Validation = 0.6942819495950832

M Value = 9
Lambda Value = 0.7000000000000001

Fold 1
Performance Training = 0.7976828299258959
Performance Validation = 0.5949162516804883

Fold 2
Performance Training = 0.7818802878227139
Performance Validation = 0.7314833743525808

Fold 3
Performance Training = 0.7905415475672763
Performance Validation = 0.6725416255335621

Fold 4

Performance Training = 0.7776949177081987
Performance Validation = 0.7778865293656411

Average Performance in Training = 0.7869498957560213
Average Performance in Validation = 0.6942069452330681

M Value = 9
Lambda Value = 0.71

Fold 1
Performance Training = 0.7973884388133647
Performance Validation = 0.5951555780690472

Fold 2
Performance Training = 0.781611720045946
Performance Validation = 0.7311389434922595

Fold 3
Performance Training = 0.7902730286380018
Performance Validation = 0.6723698843054393

Fold 4
Performance Training = 0.7773500655466214
Performance Validation = 0.7778424790825006

Average Performance in Training = 0.7866558132609835
Average Performance in Validation = 0.6941267212373117

M Value = 9
Lambda Value = 0.72

Fold 1
Performance Training = 0.797095928810597
Performance Validation = 0.5953777323971672

Fold 2
Performance Training = 0.7813447508736104
Performance Validation = 0.7307964796906605

Fold 3
Performance Training = 0.7900058631149586
Performance Validation = 0.6721957819936166

Fold 4
Performance Training = 0.7770071945300319
Performance Validation = 0.7777959338076031

Average Performance in Training = 0.7863634343322995
Average Performance in Validation = 0.6940414819722619

M Value = 9
Lambda Value = 0.73

Fold 1
Performance Training = 0.7968052857512022
Performance Validation = 0.5955833008114719

Fold 2
Performance Training = 0.7810793635389113
Performance Validation = 0.7304559856986264

Fold 3
Performance Training = 0.7897400425802642
Performance Validation = 0.6720194208416015

Fold 4
Performance Training = 0.776666293738091
Performance Validation = 0.7777469850977513

Average Performance in Training = 0.7860727464021171
Average Performance in Validation = 0.6939514231123628

M Value = 9
Lambda Value = 0.74

Fold 1
Performance Training = 0.7965164949483219
Performance Validation = 0.5957728460764098

Fold 2
Performance Training = 0.7808155411600983
Performance Validation = 0.7301174625220459

Fold 3
Performance Training = 0.7894755583089077
Performance Validation = 0.6718408987682691

Fold 4
Performance Training = 0.7763273517566877
Performance Validation = 0.7776957208039843

Average Performance in Training = 0.7857837365435039
Average Performance in Validation = 0.6938567320426773

M Value = 9
Lambda Value = 0.75

Fold 1
Performance Training = 0.7962295412486565
Performance Validation = 0.59594690862429

Fold 2
Performance Training = 0.7805532667622348
Performance Validation = 0.7297809095397525

Fold 3
Performance Training = 0.7892124012911281

Performance Validation = 0.6716603095591483

Fold 4

Performance Training = 0.7759903567145995

Performance Validation = 0.7776422252386167

Average Performance in Training = 0.7854963915041546

Average Performance in Validation = 0.6937575882404519

M Value = 9

Lambda Value = 0.76

Fold 1

Performance Training = 0.7959444090828165

Performance Validation = 0.5961060075523591

Fold 2

Performance Training = 0.7802925232974139

Performance Validation = 0.7294463246154761

Fold 3

Performance Training = 0.7889505622539656

Performance Validation = 0.671477743048742

Fold 4

Performance Training = 0.7756552963186331

Performance Validation = 0.7775865793348841

Average Performance in Training = 0.7852106977382074

Average Performance in Validation = 0.6936541636378653

M Value = 9

Lambda Value = 0.77

Fold 1

Performance Training = 0.7956610825119337

Performance Validation = 0.5962506415733666

Fold 2

Performance Training = 0.7800332936644576

Performance Validation = 0.7291137042018906

Fold 3

Performance Training = 0.7886900316820753

Performance Validation = 0.6712932852947124

Fold 4

Performance Training = 0.7753221578869413

Performance Validation = 0.7775288607979997

Average Performance in Training = 0.7849266414363519

Average Performance in Validation = 0.6935466229669923

M Value = 9
Lambda Value = 0.78

Fold 1

Performance Training = 0.7953795452711585
Performance Validation = 0.5963812899197416

Fold 2

Performance Training = 0.7797755607267259
Performance Validation = 0.7287830434401126

Fold 3

Performance Training = 0.788430799837591
Performance Validation = 0.67110701874399

Fold 4

Performance Training = 0.7749909283807354
Performance Validation = 0.777469144248586

Average Performance in Training = 0.7846442085540527
Average Performance in Validation = 0.6934351240881076

M Value = 9
Lambda Value = 0.79

Fold 1

Performance Training = 0.7950997808098199
Performance Validation = 0.5964984132033662

Fold 2

Performance Training = 0.7795193073293265
Performance Validation = 0.7284543362525567

Fold 3

Performance Training = 0.7881728567792468
Performance Validation = 0.6709190223914674

Fold 4

Performance Training = 0.7746615944345242
Performance Validation = 0.7774075013597588

Average Performance in Training = 0.7843633848382293
Average Performance in Validation = 0.6933198183017872

M Value = 9
Lambda Value = 0.8

Fold 1

Performance Training = 0.7948217723288891
Performance Validation = 0.5966024542349136

Fold 2

Performance Training = 0.7792645163149711
Performance Validation = 0.7281275754309131

Fold 3
Performance Training = 0.7879161923806974
Performance Validation = 0.670729371931071

Fold 4
Performance Training = 0.7743341423846579
Performance Validation = 0.7773440009867545

Average Performance in Training = 0.7840841558523038
Average Performance in Validation = 0.6932008506459131

M Value = 9
Lambda Value = 0.81

Fold 1
Performance Training = 0.7945455028156431
Performance Validation = 0.5966938388042646

Fold 2
Performance Training = 0.7790111705389334
Performance Validation = 0.7278027527184174

Fold 3
Performance Training = 0.7876607963478757
Performance Validation = 0.6705381399002867

Fold 4
Performance Training = 0.7740085582965422
Performance Validation = 0.7772787092907214

Average Performance in Training = 0.7838065069997485
Average Performance in Validation = 0.6930783601784225

M Value = 9
Lambda Value = 0.8200000000000001

Fold 1
Performance Training = 0.7942709550758393
Performance Validation = 0.5967729764215193

Fold 2
Performance Training = 0.7787592528829168
Performance Validation = 0.7274798588878777

Fold 3
Performance Training = 0.7874066582357163
Performance Validation = 0.6703453958178034

Fold 4
Performance Training = 0.7736848279903381
Performance Validation = 0.7772116898560889

Average Performance in Training = 0.7835304235462026

Average Performance in Validation = 0.6929524802458223

M Value = 9

Lambda Value = 0.8300000000000001

Fold 1

Performance Training = 0.7939981117635642

Performance Validation = 0.5968402610249448

Fold 2

Performance Training = 0.7785087462679615

Performance Validation = 0.727158883815135

Fold 3

Performance Training = 0.7871537674639679

Performance Validation = 0.6701512063150954

Fold 4

Performance Training = 0.7733629370653344

Performance Validation = 0.7771430038025454

Average Performance in Training = 0.783255890640207

Average Performance in Validation = 0.6928233387394301

M Value = 9

Lambda Value = 0.84

Fold 1

Performance Training = 0.7937269554088995

Performance Validation = 0.5968960716535592

Fold 2

Performance Training = 0.7782596336666638

Performance Validation = 0.7268398165475809

Fold 3

Performance Training = 0.7869021133322293

Performance Validation = 0.6699556352616136

Fold 4

Performance Training = 0.7730428709230073

Performance Validation = 0.7770727098915937

Average Performance in Training = 0.7829828933327

Average Performance in Validation = 0.6926910583385868

M Value = 9

Lambda Value = 0.85

Fold 1

Performance Training = 0.7934574684434563

Performance Validation = 0.5969407730880735

Fold 2
Performance Training = 0.7780118981142645
Performance Validation = 0.7265226453692748

Fold 3
Performance Training = 0.7866516850342506
Performance Validation = 0.6697587438845337

Fold 4
Performance Training = 0.7727246147887646
Performance Validation = 0.777000864627679

Average Performance in Training = 0.782711416595184
Average Performance in Validation = 0.6925557567423902

M Value = 9
Lambda Value = 0.86

Fold 1
Performance Training = 0.7931896332242312
Performance Validation = 0.5969747164618175

Fold 2
Performance Training = 0.7777655227190957
Performance Validation = 0.7262073578617969

Fold 3
Performance Training = 0.7864024716716284
Performance Validation = 0.6695605908831381

Fold 4
Performance Training = 0.7724081537327131
Performance Validation = 0.7769275223554368

Average Performance in Training = 0.7824414453369171
Average Performance in Validation = 0.6924175468905474

M Value = 9
Lambda Value = 0.87

Fold 1
Performance Training = 0.7929234320553977
Performance Validation = 0.596998239842641

Fold 2
Performance Training = 0.7775204906721622
Performance Validation = 0.7258939409616958

Fold 3
Performance Training = 0.7861544622666122
Performance Validation = 0.6693612325376795

Fold 4
Performance Training = 0.7720934726888561

Performance Validation = 0.7768527353506464

Average Performance in Training = 0.782172964420757

Average Performance in Validation = 0.6922765371731657

M Value = 9

Lambda Value = 0.88

Fold 1

Performance Training = 0.7926588472086963

Performance Validation = 0.5970116687869488

Fold 2

Performance Training = 0.7772767852560979

Performance Validation = 0.725582381014519

Fold 3

Performance Training = 0.7859076457741707

Performance Validation = 0.6691607228133227

Fold 4

Performance Training = 0.7717805564736168

Performance Validation = 0.7767765539087546

Average Performance in Training = 0.7819059586781454

Average Performance in Validation = 0.6921328316308863

M Value = 9

Lambda Value = 0.89

Fold 1

Performance Training = 0.7923958609421236

Performance Validation = 0.5970153168682859

Fold 2

Performance Training = 0.7770343898532893

Performance Validation = 0.7252726638253302

Fold 3

Performance Training = 0.7856620110937613

Performance Validation = 0.6689591134597408

Fold 4

Performance Training = 0.771469389803067

Performance Validation = 0.776699026427919

Average Performance in Training = 0.7816404129230603

Average Performance in Validation = 0.691986530145319

M Value = 9

Lambda Value = 0.9

Fold 1

Performance Training = 0.7921344555172737
Performance Validation = 0.5970094861807843

Fold 2
Performance Training = 0.7767932879535
Performance Validation = 0.7249647747065529

Fold 3
Performance Training = 0.7854175470799716
Performance Validation = 0.6687564541056275

Fold 4
Performance Training = 0.7711599573092335
Performance Validation = 0.77662019948872

Average Performance in Training = 0.7813763119649948
Average Performance in Validation = 0.6918377286204211

M Value = 9
Lambda Value = 0.91

Fold 1
Performance Training = 0.7918746132152868
Performance Validation = 0.5969944678197245

Fold 2
Performance Training = 0.7765534631607345
Performance Validation = 0.7246586985225587

Fold 3
Performance Training = 0.7851742425530281
Performance Validation = 0.6685527923494607

Fold 4
Performance Training = 0.7708522435554759
Performance Validation = 0.7765401179302379

Average Performance in Training = 0.7811136406211314
Average Performance in Validation = 0.6916865191554955

M Value = 9
Lambda Value = 0.92

Fold 1
Performance Training = 0.7916163163515952
Performance Validation = 0.5969705423387864

Fold 2
Performance Training = 0.7763148991995485
Performance Validation = 0.7243544197320755

Fold 3
Performance Training = 0.7849320863083005
Performance Validation = 0.6683481738457414

Fold 4
Performance Training = 0.7705462330508398
Performance Validation = 0.7764588249221519

Average Performance in Training = 0.780852383727571
Average Performance in Validation = 0.6915329902096887

M Value = 9
Lambda Value = 0.93

Fold 1
Performance Training = 0.7913595472894185
Performance Validation = 0.596937980187425

Fold 2
Performance Training = 0.7760775799209529
Performance Validation = 0.7240519224273568

Fold 3
Performance Training = 0.784691067125537
Performance Validation = 0.6681426423876659

Fold 4
Performance Training = 0.7702419102637514
Performance Validation = 0.7763763620340861

Average Performance in Training = 0.780592526149915
Average Performance in Validation = 0.6913772267591335

M Value = 9
Lambda Value = 0.9400000000000001

Fold 1
Performance Training = 0.7911042884523043
Performance Validation = 0.596897042126903

Fold 2
Performance Training = 0.7758414893075827
Performance Validation = 0.7237511903713927

Fold 3
Performance Training = 0.7844511737774301
Performance Validation = 0.6679362399858071

Fold 4
Performance Training = 0.7699392596346462
Performance Validation = 0.7762927693012043

Average Performance in Training = 0.7803340527929908
Average Performance in Validation = 0.6912193104463267

M Value = 9

Lambda Value = 0.9500000000000001

Fold 1

Performance Training = 0.7908505223355047

Performance Validation = 0.5968479796279956

Fold 2

Performance Training = 0.7756066114785397

Performance Validation = 0.7234522070327342

Fold 3

Performance Training = 0.7842123950376998

Performance Validation = 0.6677290069435033

Fold 4

Performance Training = 0.7696382655880201

Performance Validation = 0.7762080852873112

Average Performance in Training = 0.7800769486099411

Average Performance in Validation = 0.691059319722886

M Value = 9

Lambda Value = 0.96

Fold 1

Performance Training = 0.7905982315164712

Performance Validation = 0.5967910352501611

Fold 2

Performance Training = 0.7753729306936648

Performance Validation = 0.7231549556183667

Fold 3

Performance Training = 0.7839747196886923

Performance Validation = 0.6675209819287564

Fold 4

Performance Training = 0.7693389125436716

Performance Validation = 0.7761223471447729

Average Performance in Training = 0.7798211986106249

Average Performance in Validation = 0.6908973299855143

M Value = 9

Lambda Value = 0.97

Fold 1

Performance Training = 0.7903473986646019

Performance Validation = 0.59672644300381

Fold 2

Performance Training = 0.7751404313574481

Performance Validation = 0.7228594191039094

Fold 3

Performance Training = 0.7837381365285033

Performance Validation = 0.667312202042915

Fold 4

Performance Training = 0.7690411849271626

Performance Validation = 0.7760355906719358

Average Performance in Training = 0.779566787869429

Average Performance in Validation = 0.6907334137056426

M Value = 9

Lambda Value = 0.98

Fold 1

Performance Training = 0.7900980065498168

Performance Validation = 0.5966544286957116

Fold 2

Performance Training = 0.7749090980225498

Performance Validation = 0.7225655802628652

Fold 3

Performance Training = 0.7835026343776236

Performance Validation = 0.6671027028863121

Fold 4

Performance Training = 0.7687450671797661

Performance Validation = 0.7759478503676951

Average Performance in Training = 0.7793137015324391

Average Performance in Validation = 0.690567640553146

M Value = 9

Lambda Value = 0.99

Fold 1

Performance Training = 0.7898500380507981

Performance Validation = 0.5965752102590884

Fold 2

Performance Training = 0.7746789153928904

Performance Validation = 0.7222734216930271

Fold 3

Performance Training = 0.7832682020852888

Performance Validation = 0.6668925186210848

Fold 4

Performance Training = 0.7684505437676378

Performance Validation = 0.7758591594838604

Average Performance in Training = 0.7790619248241537

Average Performance in Validation = 0.6904000775142651

M Value = 9
Lambda Value = 1.0

Fold 1
Performance Training = 0.7896034761621559
Performance Validation = 0.5964889980688919

Fold 2
Performance Training = 0.774449868326301
Performance Validation = 0.7219829258423383

Fold 3
Performance Training = 0.7830348285351937
Performance Validation = 0.6666816820309613

Fold 4
Performance Training = 0.7681575991904392
Performance Validation = 0.7757695500751016

Average Performance in Training = 0.7788114430535225
Average Performance in Validation = 0.6902307890043233

M Value = 9
Lambda Value = 1.01

Fold 1
Performance Training = 0.7893583040011454
Performance Validation = 0.5963959952428204

Fold 2
Performance Training = 0.7742219418370849
Performance Validation = 0.7216940750320687

Fold 3
Performance Training = 0.7828025026510174
Performance Validation = 0.6664702245787439

Fold 4
Performance Training = 0.7678662179894915
Performance Validation = 0.7756790530467423

Average Performance in Training = 0.7785622416196848
Average Performance in Validation = 0.6900598369750939

M Value = 9
Lambda Value = 1.02

Fold 1
Performance Training = 0.7891145048137042
Performance Validation = 0.59629639792938

Fold 2

Performance Training = 0.7739951210980724
Performance Validation = 0.7214068514789432

Fold 3
Performance Training = 0.7825712134015868
Performance Validation = 0.6662581764610354

Fold 4
Performance Training = 0.7675763847551554
Performance Validation = 0.7755876982002385

Average Performance in Training = 0.7783143060171297
Average Performance in Validation = 0.6898872810173993

M Value = 9
Lambda Value = 1.03

Fold 1
Performance Training = 0.7888720619797953
Performance Validation = 0.5961903955821286

Fold 2
Performance Training = 0.773769391442308
Performance Validation = 0.72112123731634

Fold 3
Performance Training = 0.7823409498054916
Performance Validation = 0.6660455666605409

Fold 4
Performance Training = 0.767288084133977
Performance Validation = 0.7754955142769256

Average Performance in Training = 0.778067621840393
Average Performance in Validation = 0.6897131784589838

M Value = 9
Lambda Value = 1.04

Fold 1
Performance Training = 0.7886309590184355
Performance Validation = 0.596078171223361

Fold 2
Performance Training = 0.7735447383647486
Performance Validation = 0.7208372146130098

Fold 3
Performance Training = 0.7821117009355846
Performance Validation = 0.6658324229962284

Fold 4
Performance Training = 0.767001300835126
Performance Validation = 0.7754025289996709

Average Performance in Training = 0.7778221747884737
Average Performance in Validation = 0.6895375844580675

M Value = 9
Lambda Value = 1.05

Fold 1
Performance Training = 0.7883911795919781
Performance Validation = 0.5959599016945689

Fold 2
Performance Training = 0.7733211475234044
Performance Validation = 0.7205547653918435

Fold 3
Performance Training = 0.7818834559231039
Performance Validation = 0.6656187721712763

Fold 4
Performance Training = 0.7667160196365276
Performance Validation = 0.7753087691128875

Average Performance in Training = 0.7775779506687536
Average Performance in Validation = 0.689360552092644

M Value = 9
Lambda Value = 1.06

Fold 1
Performance Training = 0.7881527075101652
Performance Validation = 0.5958357578973921

Fold 2
Performance Training = 0.7730986047405028
Performance Validation = 0.720273871646321

Fold 3
Performance Training = 0.7816562039614764
Performance Validation = 0.6654046398188489

Fold 4
Performance Training = 0.7664322253904653
Performance Validation = 0.7752142604204775

Average Performance in Training = 0.7773349354006525
Average Performance in Validation = 0.6891821324457599

M Value = 9
Lambda Value = 1.07

Fold 1
Performance Training = 0.787915526733517

Performance Validation = 0.5957059050228969

Fold 2

Performance Training = 0.7728770960031528

Performance Validation = 0.7199945153564866

Fold 3

Performance Training = 0.781429934309711

Performance Validation = 0.6651900505457996

Fold 4

Performance Training = 0.7661499030287741

Performance Validation = 0.7751190278224651

Average Performance in Training = 0.7770931150187886

Average Performance in Validation = 0.6890023746869121

M Value = 9

Lambda Value = 1.08

Fold 1

Performance Training = 0.7876796213764841

Performance Validation = 0.5955705027719821

Fold 2

Performance Training = 0.7726566074640405

Performance Validation = 0.719716678503725

Fold 3

Performance Training = 0.7812046362958953

Performance Validation = 0.6649750279748279

Fold 4

Performance Training = 0.7658690375677846

Performance Validation = 0.77502309535008

Average Performance in Training = 0.7768524756760511

Average Performance in Validation = 0.6888213261501537

M Value = 9

Lambda Value = 1.09

Fold 1

Performance Training = 0.787444975710111

Performance Validation = 0.595429705565081

Fold 2

Performance Training = 0.7724371254418065

Performance Validation = 0.7194403430844669

Fold 3

Performance Training = 0.7809802993199926

Performance Validation = 0.6647595947844573

Fold 4
Performance Training = 0.7655896141127145
Performance Validation = 0.7749264861990329

Average Performance in Training = 0.7766130036461563
Average Performance in Validation = 0.6886390324082595

M Value = 10
Lambda Value = 0.01

Fold 1
Performance Training = 0.8226367431330823
Performance Validation = 0.5190900525437593

Fold 2
Performance Training = 0.812403138598893
Performance Validation = 0.7633811180377378

Fold 3
Performance Training = 0.8285323113493352
Performance Validation = 0.6264741925172885

Fold 4
Performance Training = 0.8117340659538514
Performance Validation = 0.7634800678656928

Average Performance in Training = 0.8188265647587905
Average Performance in Validation = 0.6681063577411196

M Value = 10
Lambda Value = 0.02

Fold 1
Performance Training = 0.8215309927470467
Performance Validation = 0.4938997448841791

Fold 2
Performance Training = 0.8096600574376647
Performance Validation = 0.7636796759342939

Fold 3
Performance Training = 0.8238696118882746
Performance Validation = 0.6314690906358378

Fold 4
Performance Training = 0.8104912975675838
Performance Validation = 0.7595440363749489

Average Performance in Training = 0.8163879899101424
Average Performance in Validation = 0.6621481369573149

M Value = 10
Lambda Value = 0.03

Fold 1
Performance Training = 0.8207757801365227
Performance Validation = 0.486696148499992

Fold 2
Performance Training = 0.8077522497621445
Performance Validation = 0.7629210622339

Fold 3
Performance Training = 0.8209046276566381
Performance Validation = 0.6328891617421384

Fold 4
Performance Training = 0.8095121724026121
Performance Validation = 0.7576836232534154

Average Performance in Training = 0.8147362074894793
Average Performance in Validation = 0.6600474989323615

M Value = 10
Lambda Value = 0.04

Fold 1
Performance Training = 0.8202068700925346
Performance Validation = 0.48717128136820376

Fold 2
Performance Training = 0.8063704351980314
Performance Validation = 0.7621789546785134

Fold 3
Performance Training = 0.8188676257422352
Performance Validation = 0.6337285621002153

Fold 4
Performance Training = 0.8087323109989663
Performance Validation = 0.7568393679524197

Average Performance in Training = 0.8135443105079418
Average Performance in Validation = 0.659979541524838

M Value = 10
Lambda Value = 0.05

Fold 1
Performance Training = 0.819738365162546
Performance Validation = 0.4912575043764241

Fold 2
Performance Training = 0.8053106558995771
Performance Validation = 0.7615753615353039

Fold 3

Performance Training = 0.8173682782961587
Performance Validation = 0.6344943757675315

Fold 4
Performance Training = 0.8080827203009803
Performance Validation = 0.7565480254558464

Average Performance in Training = 0.8126250049148155
Average Performance in Validation = 0.6609688167837765

M Value = 10
Lambda Value = 0.06

Fold 1
Performance Training = 0.8193261575573494
Performance Validation = 0.49710424841083645

Fold 2
Performance Training = 0.8044558016422045
Performance Validation = 0.7610929498405872

Fold 3
Performance Training = 0.8162022600071199
Performance Validation = 0.6352794545791928

Fold 4
Performance Training = 0.807517616132921
Performance Validation = 0.7565776359327763

Average Performance in Training = 0.8118754588348988
Average Performance in Validation = 0.6625135721908482

M Value = 10
Lambda Value = 0.07

Fold 1
Performance Training = 0.8189462931346799
Performance Validation = 0.5037804882153282

Fold 2
Performance Training = 0.8037367515638996
Performance Validation = 0.7606984646943742

Fold 3
Performance Training = 0.8152545529089099
Performance Validation = 0.6360892797799682

Fold 4
Performance Training = 0.8070079353848948
Performance Validation = 0.7568005690084765

Average Performance in Training = 0.8112363832480961
Average Performance in Validation = 0.6643422004245367

M Value = 10
Lambda Value = 0.08

Fold 1

Performance Training = 0.8185850833381891
Performance Validation = 0.5107855109625737

Fold 2

Performance Training = 0.8031108770841348
Performance Validation = 0.760363790814334

Fold 3

Performance Training = 0.8144561326850936
Performance Validation = 0.6369115156831419

Fold 4

Performance Training = 0.8065350297846743
Performance Validation = 0.7571416678521725

Average Performance in Training = 0.810671780723023
Average Performance in Validation = 0.6663006213280556

M Value = 10
Lambda Value = 0.09

Fold 1

Performance Training = 0.8182344028481225
Performance Validation = 0.517838593617558

Fold 2

Performance Training = 0.8025507927193286
Performance Validation = 0.7600681045574369

Fold 3

Performance Training = 0.8137633572789095
Performance Validation = 0.6377326437610463

Fold 4

Performance Training = 0.8060866883871498
Performance Validation = 0.757554405178378

Average Performance in Training = 0.8101588103083777
Average Performance in Validation = 0.6682984367786049

M Value = 10
Lambda Value = 0.1

Fold 1

Performance Training = 0.8178893055756931
Performance Validation = 0.52477907736983

Fold 2

Performance Training = 0.8020382828023127

Performance Validation = 0.7597965708159156

Fold 3

Performance Training = 0.8131474413960361

Performance Validation = 0.6385416827586218

Fold 4

Performance Training = 0.8056547577238955

Performance Validation = 0.7580088849876655

Average Performance in Training = 0.8096824468744843

Average Performance in Validation = 0.6702815539830083

M Value = 10

Lambda Value = 0.11

Fold 1

Performance Training = 0.8175467420597552

Performance Validation = 0.5315150983485194

Fold 2

Performance Training = 0.8015608746339664

Performance Validation = 0.7595387629152988

Fold 3

Performance Training = 0.8125887324113223

Performance Validation = 0.6393305459899357

Fold 4

Performance Training = 0.8052337059586276

Performance Validation = 0.758485362559189

Average Performance in Training = 0.809232513765918

Average Performance in Validation = 0.6722174424532357

M Value = 10

Lambda Value = 0.12

Fold 1

Performance Training = 0.8172048334499763

Performance Validation = 0.5379956523757913

Fold 2

Performance Training = 0.8011098200610988

Performance Validation = 0.7592874001790866

Fold 3

Performance Training = 0.8120734258106419

Performance Validation = 0.6400935867440509

Fold 4

Performance Training = 0.8048197362695034

Performance Validation = 0.7589705400602602

Average Performance in Training = 0.8088019538978051
Average Performance in Validation = 0.6740867948397973

M Value = 10
Lambda Value = 0.13

Fold 1
Performance Training = 0.8168624426434581
Performance Validation = 0.5441946306116048

Fold 2
Performance Training = 0.8006788597994215
Performance Validation = 0.7590374317736857

Fold 3
Performance Training = 0.8115915927109989
Performance Validation = 0.640827039240488

Fold 4
Performance Training = 0.804410224925155
Performance Validation = 0.759455347947057

Average Performance in Training = 0.8083857800197585
Average Performance in Validation = 0.6758786123932089

M Value = 10
Lambda Value = 0.14

Fold 1
Performance Training = 0.8165189117696932
Performance Validation = 0.550101341972141

Fold 2
Performance Training = 0.8002634408143733
Performance Validation = 0.7587853902352575

Fold 3
Performance Training = 0.8111359484948377
Performance Validation = 0.6415285371132418

Fold 4
Performance Training = 0.8040033560818589
Performance Validation = 0.7599335637763349

Average Performance in Training = 0.8079804142901907
Average Performance in Validation = 0.6775872082742438

M Value = 10
Lambda Value = 0.15

Fold 1
Performance Training = 0.8161738966621087
Performance Validation = 0.5557147123297053

Fold 2
Performance Training = 0.7998602057097021
Performance Validation = 0.7585289376154551

Fold 3
Performance Training = 0.8107010580907262
Performance Validation = 0.6421967383348594

Fold 4
Performance Training = 0.8035978789057622
Performance Validation = 0.7604009238734024

Average Performance in Training = 0.8075832598420748
Average Performance in Validation = 0.6792103280383556

M Value = 10
Lambda Value = 0.16

Fold 1
Performance Training = 0.815827259883317
Performance Validation = 0.5610396471256845

Fold 2
Performance Training = 0.7994666508262971
Performance Validation = 0.7582665456261539

Fold 3
Performance Training = 0.8102828080883896
Performance Validation = 0.642831043945985

Fold 4
Performance Training = 0.8031929426859886
Performance Validation = 0.7608545359624641

Average Performance in Training = 0.8071924153709981
Average Performance in Validation = 0.6807479431650719

M Value = 10
Lambda Value = 0.17

Fold 1
Performance Training = 0.8154790001775154
Performance Validation = 0.5660847072991351

Fold 2
Performance Training = 0.799080892043522
Performance Validation = 0.7579972685523411

Fold 3
Performance Training = 0.8098780471927484
Performance Validation = 0.6434313902665028

Fold 4

Performance Training = 0.8027879828370604
Performance Validation = 0.7612924814941597

Average Performance in Training = 0.8068064805627115
Average Performance in Validation = 0.6822014619030347

M Value = 10
Lambda Value = 0.18

Fold 1
Performance Training = 0.8151292051794372
Performance Validation = 0.5708606036631014

Fold 2
Performance Training = 0.7987015011195465
Performance Validation = 0.7577205806571762

Fold 3
Performance Training = 0.8094843358897015
Performance Validation = 0.6439980961530374

Fold 4
Performance Training = 0.8023826408160645
Performance Validation = 0.761713540880065

Average Performance in Training = 0.8064244207511874
Average Performance in Validation = 0.683573205338345

M Value = 10
Lambda Value = 0.19

Fold 1
Performance Training = 0.8147780193060115
Performance Validation = 0.5753792126309984

Fold 2
Performance Training = 0.7983273892939761
Performance Validation = 0.7574362587722734

Fold 3
Performance Training = 0.8090997687370849
Performance Validation = 0.6445317507840066

Fold 4
Performance Training = 0.8019767070770085
Performance Validation = 0.7621170002913308

Average Performance in Training = 0.8060454711035203
Average Performance in Validation = 0.6848660556196523

M Value = 10
Lambda Value = 0.2

Fold 1
Performance Training = 0.8144256217531609
Performance Validation = 0.5796529302065099

Fold 2
Performance Training = 0.7979577232069616
Performance Validation = 0.757144296864916

Fold 3
Performance Training = 0.8087228460219997
Performance Validation = 0.6450331310969386

Fold 4
Performance Training = 0.8015700799448113
Performance Validation = 0.7625025137430415

Average Performance in Training = 0.8056690677317335
Average Performance in Validation = 0.6860832179778515

M Value = 10
Lambda Value = 0.21

Fold 1
Performance Training = 0.8140722113298537
Performance Validation = 0.5836942487438737

Fold 2
Performance Training = 0.7975918633207846
Performance Validation = 0.7568448434400142

Fold 3
Performance Training = 0.8083523796400948
Performance Validation = 0.6455031409332355

Fold 4
Performance Training = 0.8011627356620679
Performance Validation = 0.762870003359675

Average Performance in Training = 0.8052947974882002
Average Performance in Validation = 0.6872280591191996

M Value = 10
Lambda Value = 0.22

Fold 1
Performance Training = 0.8137179959848904
Performance Validation = 0.5875154822131126

Fold 2
Performance Training = 0.7972293182680525
Performance Validation = 0.756538155422851

Fold 3
Performance Training = 0.8079874231197317

Performance Validation = 0.6459427661180634

Fold 4

Performance Training = 0.8007547063853582

Performance Validation = 0.7632195864445176

Average Performance in Training = 0.8049223609395082

Average Performance in Validation = 0.6883039975496361

M Value = 10

Lambda Value = 0.23

Fold 1

Performance Training = 0.813363185591887

Performance Validation = 0.5911285913608368

Fold 2

Performance Training = 0.7968697106392759

Performance Validation = 0.756224564032818

Fold 3

Performance Training = 0.8076272189536923

Performance Validation = 0.6463530412831173

Fold 4

Performance Training = 0.8003460639091826

Performance Validation = 0.7635515216486235

Average Performance in Training = 0.8045515447735094

Average Performance in Validation = 0.6893144295813489

M Value = 10

Lambda Value = 0.24

Fold 1

Performance Training = 0.8130079870193845

Performance Validation = 0.5945450765005711

Fold 2

Performance Training = 0.7965127510962258

Performance Validation = 0.7559044494633267

Fold 3

Performance Training = 0.8072711585172916

Performance Validation = 0.6467350253890918

Fold 4

Performance Training = 0.7999369075614547

Performance Validation = 0.7638661689260017

Average Performance in Training = 0.8041822010485892

Average Performance in Validation = 0.6902626800697478

M Value = 10
Lambda Value = 0.25

Fold 1

Performance Training = 0.8126526008156375
Performance Validation = 0.5977759162065417

Fold 2

Performance Training = 0.7961582186184473
Performance Validation = 0.7555782220729346

Fold 3

Performance Training = 0.806918751259462
Performance Validation = 0.6470897837212075

Fold 4

Performance Training = 0.7995273551681281
Performance Validation = 0.7641639595569508

Average Performance in Training = 0.8038142314654186
Average Performance in Validation = 0.6911519703894086

M Value = 10
Lambda Value = 0.26

Fold 1

Performance Training = 0.8122972190403742
Performance Validation = 0.6008315371751225

Fold 2

Performance Training = 0.7958059453164493
Performance Validation = 0.7552463084264546

Fold 3

Performance Training = 0.806569600807929
Performance Validation = 0.6474183747211675

Fold 4

Performance Training = 0.799117536294752
Performance Validation = 0.7644453736026511

Average Performance in Training = 0.8034475753648761
Average Performance in Validation = 0.6919853984813489

M Value = 10
Lambda Value = 0.27

Fold 1

Performance Training = 0.811942023914283
Performance Validation = 0.6037218051329498

Fold 2

Performance Training = 0.7954558046793705
Performance Validation = 0.7549091409657117

Fold 3
Performance Training = 0.8062233862850618
Performance Validation = 0.6477218404482993

Fold 4
Performance Training = 0.7987075871903713
Performance Validation = 0.7647109228906064

Average Performance in Training = 0.8030822005172716
Average Performance in Validation = 0.6927659273593918

M Value = 10
Lambda Value = 0.28

Fold 1
Performance Training = 0.8115871870522358
Performance Validation = 0.6064560298010198

Fold 2
Performance Training = 0.7951077024282243
Performance Validation = 0.7545671504151316

Fold 3
Performance Training = 0.8058798475906691
Performance Validation = 0.6480011997706525

Fold 4
Performance Training = 0.7982976470127947
Performance Validation = 0.7649611381571337

Average Performance in Training = 0.802718096020981
Average Performance in Validation = 0.6934963795359844

M Value = 10
Lambda Value = 0.29

Fold 1
Performance Training = 0.8112328691115283
Performance Validation = 0.609042979081434

Fold 2
Performance Training = 0.7947615693623554
Performance Validation = 0.7542207602527908

Fold 3
Performance Training = 0.8055387737321764
Performance Validation = 0.6482574436167376

Fold 4
Performance Training = 0.7978878550231752
Performance Validation = 0.7651965593289082

Average Performance in Training = 0.8023552668073088

Average Performance in Validation = 0.6941794355699676

M Value = 10

Lambda Value = 0.3

Fold 1

Performance Training = 0.8108792197345697

Performance Validation = 0.6114908990889292

Fold 2

Performance Training = 0.7944173557415102

Performance Validation = 0.7538703827453112

Fold 3

Performance Training = 0.8051999935152052

Performance Validation = 0.6484915317797282

Fold 4

Performance Training = 0.797478348517948

Performance Validation = 0.7654177281971468

Average Performance in Training = 0.8019937293773083

Average Performance in Validation = 0.6948176354527789

M Value = 10

Lambda Value = 0.31

Fold 1

Performance Training = 0.8105263776973024

Performance Validation = 0.6138075376868476

Fold 2

Performance Training = 0.7940750268591479

Performance Validation = 0.7535164161721611

Fold 3

Performance Training = 0.8048633680770967

Performance Validation = 0.6487043908975025

Fold 4

Performance Training = 0.7970692613226744

Performance Validation = 0.7656251829172133

Average Performance in Training = 0.8016335084890553

Average Performance in Validation = 0.6954133819184312

M Value = 10

Lambda Value = 0.32

Fold 1

Performance Training = 0.8101744711999181

Performance Validation = 0.6160001698967548

Fold 2
Performance Training = 0.7937345595448719
Performance Validation = 0.7531592429545468

Fold 3
Performance Training = 0.8045287848689147
Performance Validation = 0.648896913316023

Fold 4
Performance Training = 0.7966607227164598
Performance Validation = 0.7658194539151316

Average Performance in Training = 0.8012746345825411
Average Performance in Validation = 0.6959689450206141

M Value = 10
Lambda Value = 0.33

Fold 1
Performance Training = 0.8098236182522318
Performance Validation = 0.6180756240590968

Fold 2
Performance Training = 0.7933959393964494
Performance Validation = 0.7527992284656477

Fold 3
Performance Training = 0.8041961527841599
Performance Validation = 0.6490699566192006

Fold 4
Performance Training = 0.7962528566857188
Performance Validation = 0.7660010608760598

Average Performance in Training = 0.8009171417796399
Average Performance in Validation = 0.6964864675050013

M Value = 10
Lambda Value = 0.34

Fold 1
Performance Training = 0.8094739271203256
Performance Validation = 0.6200403079759774

Fold 2
Performance Training = 0.7930591585859411
Performance Validation = 0.7524367203652432

Fold 3
Performance Training = 0.8038653981998898
Performance Validation = 0.6492243436534615

Fold 4
Performance Training = 0.7958457814301398

Performance Validation = 0.7661705105685124

Average Performance in Training = 0.8005610663340741

Average Performance in Validation = 0.6969679706407987

M Value = 10

Lambda Value = 0.35000000000000003

Fold 1

Performance Training = 0.8091254968087164

Performance Validation = 0.6219002345244593

Fold 2

Performance Training = 0.7927242141212381

Performance Validation = 0.7520720483230267

Fold 3

Performance Training = 0.8035364617484524

Performance Validation = 0.6493608629224448

Fold 4

Performance Training = 0.7954396090622102

Performance Validation = 0.7663282953187813

Average Performance in Training = 0.8002064454351543

Average Performance in Validation = 0.697415360272178

M Value = 10

Lambda Value = 0.36

Fold 1

Performance Training = 0.8087784175605155

Performance Validation = 0.6236610463979527

Fold 2

Performance Training = 0.7923911064687776

Performance Validation = 0.7517055240377502

Fold 3

Performance Training = 0.8032092956770032

Performance Validation = 0.6494802692487353

Fold 4

Performance Training = 0.7950344454540621

Performance Validation = 0.7664748919848785

Average Performance in Training = 0.7998533162900896

Average Performance in Validation = 0.6978304329173292

M Value = 10

Lambda Value = 0.37

Fold 1

Performance Training = 0.8084327713619177
Performance Validation = 0.6253280397758421

Fold 2
Performance Training = 0.7920598384646872
Performance Validation = 0.7513374414720706

Fold 3
Performance Training = 0.8028838616822725
Performance Validation = 0.6495832846306093

Fold 4
Performance Training = 0.7946303901962668
Performance Validation = 0.7666107613203015

Average Performance in Training = 0.799501715426286
Average Performance in Validation = 0.6982148817997059

M Value = 10
Lambda Value = 0.38

Fold 1
Performance Training = 0.8080886324419461
Performance Validation = 0.6269061867886365

Fold 2
Performance Training = 0.7917304144558417
Performance Validation = 0.750968077250264

Fold 3
Performance Training = 0.8025601291310361
Performance Validation = 0.6496705992332162

Fold 4
Performance Training = 0.7942275366413768
Performance Validation = 0.7667363476379558

Average Performance in Training = 0.7991516781675502
Average Performance in Validation = 0.6985703027275181

M Value = 10
Lambda Value = 0.39

Fold 1
Performance Training = 0.8077460677609001
Performance Validation = 0.6284001567394628

Fold 2
Performance Training = 0.7914028396258668
Performance Validation = 0.7505976911675354

Fold 3
Performance Training = 0.802238073594941
Performance Validation = 0.6497428724688101

Fold 4
Performance Training = 0.7938259720102047
Performance Validation = 0.7668520787041971

Average Performance in Training = 0.7988032382479782
Average Performance in Validation = 0.6988981997700013

M Value = 10
Lambda Value = 0.4

Fold 1
Performance Training = 0.8074051374825365
Performance Validation = 0.6298143360444097

Fold 2
Performance Training = 0.7910771194689019
Performance Validation = 0.7502265267801376

Fold 3
Performance Training = 0.80191767564273
Performance Validation = 0.6498007341344595

Fold 4
Performance Training = 0.7934257775447027
Performance Validation = 0.7669583658105247

Average Performance in Training = 0.7984564275347177
Average Performance in Validation = 0.6991999906923829

M Value = 10
Lambda Value = 0.41000000000000003

Fold 1
Performance Training = 0.8070658954275914
Performance Validation = 0.6311528469362135

Fold 2
Performance Training = 0.7907532593824674
Performance Validation = 0.7498548120479698

Fold 3
Performance Training = 0.8015989198432936
Performance Validation = 0.6498447855772582

Fold 4
Performance Training = 0.7930270286941123
Performance Validation = 0.7670556039798277

Average Performance in Training = 0.7981112758368663
Average Performance in Validation = 0.6994770121353173

M Value = 10

Lambda Value = 0.42

Fold 1

Performance Training = 0.8067283895062245

Performance Validation = 0.6324195649445908

Fold 2

Performance Training = 0.7904312643559698

Performance Validation = 0.749482760007891

Fold 3

Performance Training = 0.8012817939426962

Performance Validation = 0.6498756008701919

Fold 4

Performance Training = 0.7926297953242121

Performance Validation = 0.7671441722729231

Average Performance in Training = 0.7977678107822755

Average Performance in Validation = 0.6997305245238993

M Value = 10

Lambda Value = 0.43

Fold 1

Performance Training = 0.8063926621288374

Performance Validation = 0.6336181352141197

Fold 2

Performance Training = 0.7901111387360482

Performance Validation = 0.7491105694650924

Fold 3

Performance Training = 0.8009662881846639

Performance Validation = 0.6498937279825368

Fold 4

Performance Training = 0.7922341419418711

Performance Validation = 0.7672244341688338

Average Performance in Training = 0.7974260577478552

Average Performance in Validation = 0.6999617167076457

M Value = 10

Lambda Value = 0.44

Fold 1

Performance Training = 0.8060587505945446

Performance Validation = 0.6347519877187018

Fold 2

Performance Training = 0.7897928860537287

Performance Validation = 0.7487384256860818

Fold 3
Performance Training = 0.8006523947494658
Performance Validation = 0.6498996899315584

Fold 4
Performance Training = 0.7918401279286617
Performance Validation = 0.7672967380000983

Average Performance in Training = 0.7970860398316002
Average Performance in Validation = 0.7001717103341101

M Value = 10
Lambda Value = 0.45

Fold 1
Performance Training = 0.8057266874575334
Performance Validation = 0.6358243514221124

Fold 2
Performance Training = 0.7894765089012291
Performance Validation = 0.7483665010869376

Fold 3
Performance Training = 0.8003401072914865
Performance Validation = 0.6498939859115378

Fold 4
Performance Training = 0.7914478077789161
Performance Validation = 0.7673614174243938

Average Performance in Training = 0.7967477778572912
Average Performance in Validation = 0.7003615639612454

M Value = 10
Lambda Value = 0.46

Fold 1
Performance Training = 0.8053965008716857
Performance Validation = 0.6368382674615621

Fold 2
Performance Training = 0.7891620088484681
Performance Validation = 0.7479949559102451

Fold 3
Performance Training = 0.8000294205584902
Performance Validation = 0.64987709238977

Fold 4
Performance Training = 0.7910572313384899
Performance Validation = 0.767418791920088

Average Performance in Training = 0.7964112904042835
Average Performance in Validation = 0.7005322769204163

M Value = 10
Lambda Value = 0.47000000000000003

Fold 1
Performance Training = 0.8050682149140317
Performance Validation = 0.6377966014075307

Fold 2
Performance Training = 0.7888493863911636
Performance Validation = 0.7476239388837742

Fold 3
Performance Training = 0.7997203300785121
Performance Validation = 0.6498494641675983

Fold 4
Performance Training = 0.7906684440412681
Performance Validation = 0.7674691672963627

Average Performance in Training = 0.7960765938562437
Average Performance in Validation = 0.7006847929388165

M Value = 10
Lambda Value = 0.48

Fold 1
Performance Training = 0.8047418498879245
Performance Validation = 0.638702054665212

Fold 2
Performance Training = 0.7885386409241476
Performance Validation = 0.7472535878588904

Fold 3
Performance Training = 0.7994128319040275
Performance Validation = 0.6498115354041281

Fold 4
Performance Training = 0.7902814871417532
Performance Validation = 0.7675128362096649

Average Performance in Training = 0.7957437024644631
Average Performance in Validation = 0.7008200035344738

M Value = 10
Lambda Value = 0.49

Fold 1
Performance Training = 0.8044174226065437
Performance Validation = 0.6395571750735716

Fold 2

Performance Training = 0.7882297707347221
Performance Validation = 0.7468840304258173

Fold 3
Performance Training = 0.7991069224028436
Performance Validation = 0.6497637205993367

Fold 4
Performance Training = 0.7898963979417685
Performance Validation = 0.7675500786797342

Average Performance in Training = 0.7954126284214695
Average Performance in Validation = 0.7009387511946149

M Value = 10
Lambda Value = 0.5

Fold 1
Performance Training = 0.8040949466580938
Performance Validation = 0.6403643667606296

Fold 2
Performance Training = 0.7879227730115139
Performance Validation = 0.7465153845038996

Fold 3
Performance Training = 0.7988025980886394
Performance Validation = 0.6497064155371939

Fold 4
Performance Training = 0.7895132100103551
Performance Validation = 0.7675811626013314

Average Performance in Training = 0.7950833819421506
Average Performance in Validation = 0.7010418323507637

M Value = 10
Lambda Value = 0.51

Fold 1
Performance Training = 0.8037744326532628
Performance Validation = 0.6411258993097324

Fold 2
Performance Training = 0.7876176438652763
Performance Validation = 0.746147758906633

Fold 3
Performance Training = 0.7984998554843745
Performance Validation = 0.6496399981875186

Fold 4
Performance Training = 0.7891319533959645
Performance Validation = 0.7676063442475123

Average Performance in Training = 0.7947559713497195
Average Performance in Validation = 0.7011300001628491

M Value = 10
Lambda Value = 0.52

Fold 1
Performance Training = 0.803455888455963
Performance Validation = 0.6418439162778387

Fold 2
Performance Training = 0.787314378359385
Performance Validation = 0.7457812538792191

Fold 3
Performance Training = 0.7981986910130981
Performance Validation = 0.6495648295678191

Fold 4
Performance Training = 0.7887526548305646
Performance Validation = 0.7676258687616735

Average Performance in Training = 0.7944304031647527
Average Performance in Validation = 0.7012039671216376

M Value = 10
Lambda Value = 0.53

Fold 1
Performance Training = 0.8031393193987046
Performance Validation = 0.6425204431290898

Fold 2
Performance Training = 0.7870129705466997
Performance Validation = 0.7454159616123389

Fold 3
Performance Training = 0.7978991009118925
Performance Validation = 0.6494812545651172

Fold 4
Performance Training = 0.7883753379252072
Performance Validation = 0.7676399706371191

Average Performance in Training = 0.794106682195626
Average Performance in Validation = 0.7012644074859162

M Value = 10
Lambda Value = 0.54

Fold 1
Performance Training = 0.8028247284829512

Performance Validation = 0.6431573946076433

Fold 2

Performance Training = 0.7867134135120532

Performance Validation = 0.7450519667275829

Fold 3

Performance Training = 0.7976010811646197

Performance Validation = 0.6493896027190718

Fold 4

Performance Training = 0.7880000233570064

Performance Validation = 0.7676488741815629

Average Performance in Training = 0.7937848116291576

Average Performance in Validation = 0.7013119595589652

M Value = 10

Lambda Value = 0.55

Fold 1

Performance Training = 0.8025121165659458

Performance Validation = 0.6437565816056776

Fold 2

Performance Training = 0.7864156994180057

Performance Validation = 0.7446893467395785

Fold 3

Performance Training = 0.7973046274510407

Performance Validation = 0.6492901889674325

Fold 4

Performance Training = 0.7876267290473132

Performance Validation = 0.7676527939660155

Average Performance in Training = 0.7934647931205764

Average Performance in Validation = 0.7013472278196761

M Value = 10

Lambda Value = 0.56

Fold 1

Performance Training = 0.8022014825342589

Performance Validation = 0.6443197175589213

Fold 2

Performance Training = 0.7861198195528583

Performance Validation = 0.7443281724924569

Fold 3

Performance Training = 0.7970097351087361

Performance Validation = 0.6491833143546193

Fold 4
Performance Training = 0.7872554703314809
Performance Validation = 0.7676519352587438

Average Performance in Training = 0.7931466268818336
Average Performance in Validation = 0.7013707849161853

M Value = 10
Lambda Value = 0.5700000000000001

Fold 1
Performance Training = 0.8018928234655365
Performance Validation = 0.6448484244000252

Fold 2
Performance Training = 0.7858257643803557
Performance Validation = 0.7439685085729852

Fold 3
Performance Training = 0.7967163991064152
Performance Validation = 0.6490692667058091

Fold 4
Performance Training = 0.7868862601202214
Performance Validation = 0.767646494442167

Average Performance in Training = 0.7928303117681321
Average Performance in Validation = 0.7013831735302467

M Value = 10
Lambda Value = 0.58

Fold 1
Performance Training = 0.8015861347787174
Performance Validation = 0.64534423811461

Fold 2
Performance Training = 0.7855335235896286
Performance Validation = 0.7436104137006332

Fold 3
Performance Training = 0.796424614026068
Performance Validation = 0.6489483212673215

Fold 4
Performance Training = 0.7865191090523141
Performance Validation = 0.767636659411867

Average Performance in Training = 0.7925158453616821
Average Performance in Validation = 0.7013849081236079

M Value = 10
Lambda Value = 0.59

Fold 1
Performance Training = 0.8012814103738174
Performance Validation = 0.6458086139112

Fold 2
Performance Training = 0.7852430861452069
Performance Validation = 0.7432539410958203

Fold 3
Performance Training = 0.7961343740528525
Performance Validation = 0.6488207413142111

Fold 4
Performance Training = 0.7861540256400196
Performance Validation = 0.7676226099623822

Average Performance in Training = 0.7922032240529742
Average Performance in Validation = 0.7013764765709034

M Value = 10
Lambda Value = 0.6

Fold 1
Performance Training = 0.8009786427617993
Performance Validation = 0.6462429310534119

Fold 2
Performance Training = 0.7849544403365548
Performance Validation = 0.7428991388270112

Fold 3
Performance Training = 0.7958456729709379
Performance Validation = 0.6486867787276662

Fold 4
Performance Training = 0.7857910164059867
Performance Validation = 0.767604518152873

Average Performance in Training = 0.7918924431188197
Average Performance in Validation = 0.7013583416902406

M Value = 10
Lambda Value = 0.61

Fold 1
Performance Training = 0.8006778231851884
Performance Validation = 0.6466484973624209

Fold 2
Performance Training = 0.7846675738265676
Performance Validation = 0.7425460501380793

Fold 3

Performance Training = 0.7955585041646042
Performance Validation = 0.6485466745427357

Fold 4
Performance Training = 0.7854300860133671
Performance Validation = 0.7675825486606844

Average Performance in Training = 0.7915834967974318
Average Performance in Validation = 0.7013309426759801

M Value = 10
Lambda Value = 0.62

Fold 1
Performance Training = 0.80037894173012
Performance Validation = 0.6470265534256741

Fold 2
Performance Training = 0.7843824736987993
Performance Validation = 0.7421947137567539

Fold 3
Performance Training = 0.7952728606233552
Performance Validation = 0.6484006594687249

Fold 4
Performance Training = 0.785071237388377
Performance Validation = 0.7675568591158632

Average Performance in Training = 0.7912763783601628
Average Performance in Validation = 0.701294696441754

M Value = 10
Lambda Value = 0.63

Fold 1
Performance Training = 0.8000819874305587
Performance Validation = 0.6473782765252838

Fold 2
Performance Training = 0.7840991265033795
Performance Validation = 0.7418451641844865

Fold 3
Performance Training = 0.794988734950129
Performance Validation = 0.6482489543817429

Fold 4
Performance Training = 0.7847144718363479
Performance Validation = 0.7675276004236918

Average Performance in Training = 0.7909710801801038
Average Performance in Validation = 0.7012499988788014

M Value = 10
Lambda Value = 0.64

Fold 1

Performance Training = 0.7997869483647485
Performance Validation = 0.6477047843103227

Fold 2

Performance Training = 0.7838175183013338
Performance Validation = 0.7414974319694521

Fold 3

Performance Training = 0.7947061193725857
Performance Validation = 0.6480917707939244

Fold 4

Performance Training = 0.7843597891508951
Performance Validation = 0.7674949170700314

Average Performance in Training = 0.7906675937973908
Average Performance in Validation = 0.7011972260359327

M Value = 10
Lambda Value = 0.65

Fold 1

Performance Training = 0.7994938117451811
Performance Validation = 0.6480071382315484

Fold 2

Performance Training = 0.7835376347068598
Performance Validation = 0.7411515439642382

Fold 3

Performance Training = 0.794425005755817
Performance Validation = 0.6479293112963636

Fold 4

Performance Training = 0.7840071877173113
Performance Validation = 0.76745894741434

Average Performance in Training = 0.7903659099812923
Average Performance in Validation = 0.7011367352266226

M Value = 10
Lambda Value = 0.66

Fold 1

Performance Training = 0.79920256400161
Performance Validation = 0.6482863467521189

Fold 2

Performance Training = 0.7832594609282544

Performance Validation = 0.7408075235667613

Fold 3

Performance Training = 0.7941453856172493

Performance Validation = 0.6477617699814633

Fold 4

Performance Training = 0.7836566646097133

Performance Validation = 0.7674198239681815

Average Performance in Training = 0.7900660187892067

Average Performance in Validation = 0.7010688660671313

M Value = 10

Lambda Value = 0.67

Fold 1

Performance Training = 0.7989131908583633

Performance Validation = 0.6485433683558002

Fold 2

Performance Training = 0.7829829818066949

Performance Validation = 0.7404653909476395

Fold 3

Performance Training = 0.7938672501421999

Performance Validation = 0.647589332842352

Fold 4

Performance Training = 0.7833082156827662

Performance Validation = 0.7673776736612343

Average Performance in Training = 0.7897679096225061

Average Performance in Validation = 0.7009939414517565

M Value = 10

Lambda Value = 0.68

Fold 1

Performance Training = 0.7986256774061098

Performance Validation = 0.6487791143614862

Fold 2

Performance Training = 0.7827081818532653

Performance Validation = 0.7401251632638505

Fold 3

Performance Training = 0.7935905902006677

Performance Validation = 0.6474121781526383

Fold 4

Performance Training = 0.7829618356580017

Performance Validation = 0.7673326180940855

Average Performance in Training = 0.7894715712795111
Average Performance in Validation = 0.700912268468015

M Value = 10
Lambda Value = 0.6900000000000001

Fold 1
Performance Training = 0.7983400081681495
Performance Validation = 0.6489944515611548

Fold 2
Performance Training = 0.7824350452840714
Performance Validation = 0.7397868548596924

Fold 3
Performance Training = 0.7933153963642893
Performance Validation = 0.6472304768263548

Fold 4
Performance Training = 0.7826175182051689
Performance Validation = 0.7672847737800782

Average Performance in Training = 0.7891769920054197
Average Performance in Validation = 0.7008241392568201

M Value = 10
Lambda Value = 0.7000000000000001

Fold 1
Performance Training = 0.7980561671621766
Performance Validation = 0.6491902046914946

Fold 2
Performance Training = 0.782163556053669
Performance Validation = 0.7394504774545236

Fold 3
Performance Training = 0.7930416589240059
Performance Validation = 0.6470443927596314

Fold 4
Performance Training = 0.7822752560186932
Performance Validation = 0.7672342523743423

Average Performance in Training = 0.7888841595396362
Average Performance in Validation = 0.7007298318199979

M Value = 10
Lambda Value = 0.71

Fold 1
Performance Training = 0.7977741379572081
Performance Validation = 0.6493671587537467

Fold 2

Performance Training = 0.7818936978864498

Performance Validation = 0.739116040321216

Fold 3

Performance Training = 0.792769367907608

Performance Validation = 0.6468540831555046

Fold 4

Performance Training = 0.7819350408898554

Performance Validation = 0.7671811608937108

Average Performance in Training = 0.7885930611602804

Average Performance in Validation = 0.7006296107810445

M Value = 10

Lambda Value = 0.72

Fold 1

Performance Training = 0.797493903726596

Performance Validation = 0.6495260611872029

Fold 2

Performance Training = 0.7816254543065759

Performance Validation = 0.7387835504513915

Fold 3

Performance Training = 0.7924985130973213

Performance Validation = 0.6466596988318205

Fold 4

Performance Training = 0.7815968637743825

Performance Validation = 0.7671256019244248

Average Performance in Training = 0.788303683726219

Average Performance in Validation = 0.70052372809871

M Value = 10

Lambda Value = 0.73

Fold 1

Performance Training = 0.7972154472970483

Performance Validation = 0.6496676239117547

Fold 2

Performance Training = 0.7813588086658869

Performance Validation = 0.7384530127127062

Fold 3

Performance Training = 0.7922290840472112

Performance Validation = 0.6464613845134585

Fold 4

Performance Training = 0.7812607148563642
Performance Validation = 0.7670676738210344

Average Performance in Training = 0.7880160137166277
Average Performance in Validation = 0.7004124237397384

M Value = 10
Lambda Value = 0.74

Fold 1
Performance Training = 0.7969387511939651
Performance Validation = 0.6497925252451849

Fold 2
Performance Training = 0.7810937441703958
Performance Validation = 0.7381244299949818

Fold 3
Performance Training = 0.7919610701003339
Performance Validation = 0.646259279110563

Fold 4
Performance Training = 0.78092658360795
Performance Validation = 0.7670074708952003

Average Performance in Training = 0.7877300372681612
Average Performance in Validation = 0.7002959263114824

M Value = 10
Lambda Value = 0.75

Fold 1
Performance Training = 0.7966637976833766
Performance Validation = 0.6499014117042132

Fold 2
Performance Training = 0.7808302439051058
Performance Validation = 0.7377978033485822

Fold 3
Performance Training = 0.7916944604056473
Performance Validation = 0.6460535159820548

Fold 4
Performance Training = 0.7805944588458569
Performance Validation = 0.7669450835958777

Average Performance in Training = 0.7874457402099966
Average Performance in Validation = 0.7001744536576819

M Value = 10
Lambda Value = 0.76

Fold 1
Performance Training = 0.7963905688109608
Performance Validation = 0.6499948996981209

Fold 2
Performance Training = 0.7805682908572753
Performance Validation = 0.7374731321136994

Fold 3
Performance Training = 0.7914292439343152
Performance Validation = 0.645844223186501

Fold 4
Performance Training = 0.7802643287842942
Performance Validation = 0.7668805986810163

Average Performance in Training = 0.7871631080967114
Average Performance in Validation = 0.7000482134198344

M Value = 10
Lambda Value = 0.77

Fold 1
Performance Training = 0.7961190464377564
Performance Validation = 0.6500735771194932

Fold 2
Performance Training = 0.7803078679383779
Performance Validation = 0.7371504140418403

Fold 3
Performance Training = 0.7911654094957022
Performance Validation = 0.6456315237204686

Fold 4
Performance Training = 0.7799361810845686
Performance Validation = 0.7668140993802393

Average Performance in Training = 0.7868821262391013
Average Performance in Validation = 0.6999174035655104

M Value = 10
Lambda Value = 0.78

Fold 1
Performance Training = 0.7958492122735331
Performance Validation = 0.6501380048456524

Fold 2
Performance Training = 0.7800489580042089
Performance Validation = 0.7368296454102348

Fold 3
Performance Training = 0.7909029457526725

Performance Validation = 0.6454155357443894

Fold 4

Performance Training = 0.7796100029019606

Performance Validation = 0.7667456655506095

Average Performance in Training = 0.7866027797330939

Average Performance in Validation = 0.6997822128877216

M Value = 10

Lambda Value = 0.79

Fold 1

Performance Training = 0.7955810479074673

Performance Validation = 0.6501887181429018

Fold 2

Performance Training = 0.7797915438744472

Performance Validation = 0.7365108211285949

Fold 3

Performance Training = 0.7906418412367121

Performance Validation = 0.6451963727987093

Fold 4

Performance Training = 0.7792857809294804

Performance Validation = 0.7666753738254097

Average Performance in Training = 0.7863250534870267

Average Performance in Validation = 0.6996428214739039

M Value = 10

Lambda Value = 0.8

Fold 1

Performance Training = 0.7953145348363702

Performance Validation = 0.6502262279983292

Fold 2

Performance Training = 0.7795356083500371

Performance Validation = 0.7361939348401821

Fold 3

Performance Training = 0.7903820843620278

Performance Validation = 0.644974144007509

Fold 4

Performance Training = 0.7789635014392009

Performance Validation = 0.7666032977541997

Average Performance in Training = 0.786048932246909

Average Performance in Validation = 0.699499401150055

M Value = 10
Lambda Value = 0.81

Fold 1

Performance Training = 0.7950496544908401
Performance Validation = 0.6502510223695189

Fold 2

Performance Training = 0.7792811342301349
Performance Validation = 0.7358789790157033

Fold 3

Performance Training = 0.7901236634395222
Performance Validation = 0.6447489542741119

Fold 4

Performance Training = 0.7786431503207533
Performance Validation = 0.766529507937667

Average Performance in Training = 0.7857744006203127
Average Performance in Validation = 0.6993521158992504

M Value = 10
Lambda Value = 0.8200000000000001

Fold 1

Performance Training = 0.7947863882594666
Performance Validation = 0.6502635673611481

Fold 2

Performance Training = 0.7790281043273539
Performance Validation = 0.7355659450422252

Fold 3

Performance Training = 0.7898665666897385
Performance Validation = 0.6445209044652525

Fold 4

Performance Training = 0.7783247131177753
Performance Validation = 0.7664540721558997

Average Performance in Training = 0.7855014430985836
Average Performance in Validation = 0.6992011222561314

M Value = 10
Lambda Value = 0.8300000000000001

Fold 1

Performance Training = 0.7945247175108382
Performance Validation = 0.650264308337234

Fold 2

Performance Training = 0.7787765014822865
Performance Validation = 0.7352548233062799

Fold 3
Performance Training = 0.7896107822557513
Performance Validation = 0.6442900915883891

Fold 4
Performance Training = 0.7780081750618604
Performance Validation = 0.7663770554904293

Average Performance in Training = 0.7852300440776842
Average Performance in Validation = 0.6990465696805831

M Value = 10
Lambda Value = 0.84

Fold 1
Performance Training = 0.794264623614169
Performance Validation = 0.6502536709686587

Fold 2
Performance Training = 0.778526308576743
Performance Validation = 0.7349456032720245

Fold 3
Performance Training = 0.789356298215021
Performance Validation = 0.644056608958261

Fold 4
Performance Training = 0.7776935211045558
Performance Validation = 0.7662985204399254

Average Performance in Training = 0.7849601878776221
Average Performance in Validation = 0.6988886009097174

M Value = 10
Lambda Value = 0.85

Fold 1
Performance Training = 0.7940060879578855
Performance Validation = 0.6502320622217959

Fold 2
Performance Training = 0.7782775085462162
Performance Validation = 0.7346382735540051

Fold 3
Performance Training = 0.7891031025908595
Performance Validation = 0.6438205463571849

Fold 4
Performance Training = 0.7773807359472495
Performance Validation = 0.7662185270318893

Average Performance in Training = 0.7846918587605527

Average Performance in Validation = 0.6987273522912187

M Value = 10

Lambda Value = 0.86

Fold 1

Performance Training = 0.7937490919670833

Performance Validation = 0.6501998712913148

Fold 2

Performance Training = 0.7780300843913073

Performance Validation = 0.734332821986464

Fold 3

Performance Training = 0.7888511833634344

Performance Validation = 0.643581990186741

Fold 4

Performance Training = 0.7770698040693241

Performance Validation = 0.7661371329280322

Average Performance in Training = 0.7844250409477873

Average Performance in Validation = 0.6985629540981381

M Value = 10

Lambda Value = 0.87

Fold 1

Performance Training = 0.7934936171192315

Performance Validation = 0.6501574704814078

Fold 2

Performance Training = 0.7777840191882964

Performance Validation = 0.7340292356871783

Fold 3

Performance Training = 0.7886005284801545

Performance Validation = 0.6433410236124374

Fold 4

Performance Training = 0.7767607097543356

Performance Validation = 0.7660543935246477

Average Performance in Training = 0.7841597186355045

Average Performance in Validation = 0.6983955308264177

M Value = 10

Lambda Value = 0.88

Fold 1

Performance Training = 0.7932396449587162

Performance Validation = 0.6501052160381966

Fold 2
Performance Training = 0.7775392960989099
Performance Validation = 0.733727501118425

Fold 3
Performance Training = 0.7883511258652918
Performance Validation = 0.6430977267017179

Fold 4
Performance Training = 0.7764534371148604
Performance Validation = 0.7659703620494696

Average Performance in Training = 0.7838958760094447
Average Performance in Validation = 0.6982252014769523

M Value = 10
Lambda Value = 0.89

Fold 1
Performance Training = 0.7929871571101454
Performance Validation = 0.6500434489368254

Fold 2
Performance Training = 0.7772958983793148
Performance Validation = 0.7334276041433568

Fold 3
Performance Training = 0.7881029634295417
Performance Validation = 0.6428521765551729

Fold 4
Performance Training = 0.776147970115304
Performance Validation = 0.7658850896526592

Average Performance in Training = 0.7836334972585765
Average Performance in Validation = 0.6980520798220036

M Value = 10
Lambda Value = 0.9

Fold 1
Performance Training = 0.7927361352904252
Performance Validation = 0.6499724956234261

Fold 2
Performance Training = 0.7770538093883887
Performance Validation = 0.7331295300791534

Fold 3
Performance Training = 0.7878560290786285
Performance Validation = 0.642604447431729

Fold 4
Performance Training = 0.7758442925937775

Performance Validation = 0.7657986254947621

Average Performance in Training = 0.783372566587805

Average Performance in Validation = 0.6978762746572675

M Value = 10

Lambda Value = 0.91

Fold 1

Performance Training = 0.7924865613199517

Performance Validation = 0.6498926687202484

Fold 2

Performance Training = 0.7768130125952778

Performance Validation = 0.7328332637470016

Fold 3

Performance Training = 0.7876103107216926

Performance Validation = 0.6423546108678145

Fold 4

Performance Training = 0.775542388282134

Performance Validation = 0.7657110168297698

Average Performance in Training = 0.783113068229764

Average Performance in Validation = 0.6976978900412085

M Value = 10

Lambda Value = 0.92

Fold 1

Performance Training = 0.7922384171325934

Performance Validation = 0.6498042676886931

Fold 2

Performance Training = 0.7765734915863423

Performance Validation = 0.7325387895187817

Fold 3

Performance Training = 0.7873657962791496

Performance Validation = 0.6421027357912567

Fold 4

Performance Training = 0.7752422408249692

Performance Validation = 0.765622309084864

Average Performance in Training = 0.7828549864557637

Average Performance in Validation = 0.6975170255208989

M Value = 10

Lambda Value = 0.93

Fold 1

Performance Training = 0.7919916847848654
Performance Validation = 0.6497075794610285

Fold 2
Performance Training = 0.7763352300715194
Performance Validation = 0.7322460913606813

Fold 3
Performance Training = 0.787122473690013
Performance Validation = 0.6418488886298019

Fold 4
Performance Training = 0.7749438337972828
Performance Validation = 0.7655325459365755

Average Performance in Training = 0.7825983055859201
Average Performance in Validation = 0.6973337763470218

M Value = 10
Lambda Value = 0.9400000000000001

Fold 1
Performance Training = 0.7917463464642874
Performance Validation = 0.6496028790341825

Fold 2
Performance Training = 0.776098211889971
Performance Validation = 0.7319551528745178

Fold 3
Performance Training = 0.7868803309188412
Performance Validation = 0.6415931334141953

Fold 4
Performance Training = 0.774647150721004
Performance Validation = 0.7654417693831016

Average Performance in Training = 0.7823430099985259
Average Performance in Validation = 0.6971482336764993

M Value = 10
Lambda Value = 0.9500000000000001

Fold 1
Performance Training = 0.7915023844968192
Performance Validation = 0.6494904300356581

Fold 2
Performance Training = 0.7758624210154186
Performance Validation = 0.7316659573359374

Fold 3
Performance Training = 0.7866393559622703
Performance Validation = 0.6413355318773791

Fold 4

Performance Training = 0.7743521750804276

Performance Validation = 0.7653500198136424

Average Performance in Training = 0.782089084138734

Average Performance in Validation = 0.6969604847656543

M Value = 10

Lambda Value = 0.96

Fold 1

Performance Training = 0.7912597813537593

Performance Validation = 0.6493704852579489

Fold 2

Performance Training = 0.775627841560887

Performance Validation = 0.7313784877305214

Fold 3

Performance Training = 0.7863995368551727

Performance Validation = 0.6410761435482183

Fold 4

Performance Training = 0.774058890336616

Performance Validation = 0.7652573360744982

Average Performance in Training = 0.7818365125266088

Average Performance in Validation = 0.6967706131527966

M Value = 10

Lambda Value = 0.97

Fold 1

Performance Training = 0.7910185196576581

Performance Validation = 0.6492432871636846

Fold 2

Performance Training = 0.7753944577827894

Performance Validation = 0.731092726787349

Fold 3

Performance Training = 0.7861608616763096

Performance Validation = 0.6408150258418993

Fold 4

Performance Training = 0.7737672799408831

Performance Validation = 0.7651637555324982

Average Performance in Training = 0.7815852797644101

Average Performance in Validation = 0.6965786988313578

M Value = 10

Lambda Value = 0.98

Fold 1

Performance Training = 0.790778582187868

Performance Validation = 0.6491090683678664

Fold 2

Performance Training = 0.7751622540848901

Performance Validation = 0.7308086570108437

Fold 3

Performance Training = 0.7859233185536714

Performance Validation = 0.6405522341451362

Fold 4

Performance Training = 0.7734773273472806

Performance Validation = 0.7650693141349932

Average Performance in Training = 0.7813353705434275

Average Performance in Validation = 0.6963848184147099

M Value = 10

Lambda Value = 0.99

Fold 1

Performance Training = 0.7905399518854346

Performance Validation = 0.6489680520919003

Fold 2

Performance Training = 0.7749312150216545

Performance Validation = 0.7305262607101208

Fold 3

Performance Training = 0.7856868956695766

Performance Validation = 0.640287821898567

Fold 4

Performance Training = 0.7731890160243136

Performance Validation = 0.7649740464677478

Average Performance in Training = 0.7810867696502448

Average Performance in Validation = 0.696189045292084

M Value = 10

Lambda Value = 1.0

Fold 1

Performance Training = 0.790302611857293

Performance Validation = 0.648820452596361

Fold 2

Performance Training = 0.7747013253013126

Performance Validation = 0.7302455200263973

Fold 3
Performance Training = 0.7854515812652707
Performance Validation = 0.6400218406750332

Fold 4
Performance Training = 0.7729023294658109
Performance Validation = 0.7648779858100598

Average Performance in Training = 0.7808394619724218
Average Performance in Validation = 0.6959914497769628

M Value = 10
Lambda Value = 1.01

Fold 1
Performance Training = 0.7900665453801272
Performance Validation = 0.648666475590884

Fold 2
Performance Training = 0.7744725697883396
Performance Validation = 0.7299664169592808

Fold 3
Performance Training = 0.7852173636453023
Performance Validation = 0.6397543402541681

Fold 4
Performance Training = 0.7726172512010094
Performance Validation = 0.7647811641873838

Average Performance in Training = 0.7805934325036945
Average Performance in Validation = 0.6957920992479292

M Value = 10
Lambda Value = 1.02

Fold 1
Performance Training = 0.7898317359036424
Performance Validation = 0.6485063186220779

Fold 2
Performance Training = 0.7742449335060192
Performance Validation = 0.7296889333906365

Fold 3
Performance Training = 0.784984231181409
Performance Validation = 0.6394853686937263

Fold 4
Performance Training = 0.7723337648040376
Performance Validation = 0.7646836124217655

Average Performance in Training = 0.7803486663487771
Average Performance in Validation = 0.6955910582820515

M Value = 10
Lambda Value = 1.03

Fold 1

Performance Training = 0.7895981670534968
Performance Validation = 0.6483401714437118

Fold 2

Performance Training = 0.7740184016382496
Performance Validation = 0.729413051107202

Fold 3

Performance Training = 0.7847521723165383
Performance Validation = 0.6392149723983178

Fold 4

Performance Training = 0.772051853902533
Performance Validation = 0.7645853601797251

Average Performance in Training = 0.7801051487277044
Average Performance in Validation = 0.6953883887822392

M Value = 10
Lambda Value = 1.04

Fold 1

Performance Training = 0.7893658226336797
Performance Validation = 0.6481682163683158

Fold 2

Performance Training = 0.7737929595313137
Performance Validation = 0.7291387518214989

Fold 3

Performance Training = 0.7845211755680657
Performance Validation = 0.6389431961842218

Fold 4

Performance Training = 0.7717715021858774
Performance Validation = 0.7644864360188988

Average Performance in Training = 0.7798628649797341
Average Performance in Validation = 0.6951841500982339

M Value = 10
Lambda Value = 1.05

Fold 1

Performance Training = 0.7891346866287468
Performance Validation = 0.6479906285991675

Fold 2

Performance Training = 0.7735685926953141
Performance Validation = 0.728866017191885

Fold 3
Performance Training = 0.7842912295311056
Performance Validation = 0.6386700833421122

Fold 4
Performance Training = 0.7714926934125985
Performance Validation = 0.7643868674314523

Average Performance in Training = 0.7796218005669413
Average Performance in Validation = 0.6949783991411543

M Value = 10
Lambda Value = 1.06

Fold 1
Performance Training = 0.7889047432053942
Performance Validation = 0.6478075765474334

Fold 2
Performance Training = 0.7733452868053446
Performance Validation = 0.728594828840623

Fold 3
Performance Training = 0.7840623228814346
Performance Validation = 0.6383956756966006

Fold 4
Performance Training = 0.7712154114173293
Performance Validation = 0.764286680886543

Average Performance in Training = 0.7793819410773757
Average Performance in Validation = 0.6947711904928

M Value = 10
Lambda Value = 1.07

Fold 1
Performance Training = 0.7886759767139887
Performance Validation = 0.647619222132918

Fold 2
Performance Training = 0.7731230277023668
Performance Validation = 0.7283251683711216

Fold 3
Performance Training = 0.7838344443781369
Performance Validation = 0.6381200136632276

Fold 4
Performance Training = 0.7709396401173272
Performance Validation = 0.7641859018707805

Average Performance in Training = 0.779143272227955
Average Performance in Validation = 0.6945625765095119

M Value = 10
Lambda Value = 1.08

Fold 1
Performance Training = 0.7884483716895926
Performance Validation = 0.6474257210708931

Fold 2
Performance Training = 0.7729018013940454
Performance Validation = 0.7280570173835503

Fold 3
Performance Training = 0.7836075828662526
Performance Validation = 0.6378431363032335

Fold 4
Performance Training = 0.7706653635183023
Performance Validation = 0.7640845549266765

Average Performance in Training = 0.7789057798670482
Average Performance in Validation = 0.6943526074210884

M Value = 10
Lambda Value = 1.09

Fold 1
Performance Training = 0.7882219128529061
Performance Validation = 0.6472272231442633

Fold 2
Performance Training = 0.7726815940550545
Performance Validation = 0.7277903574904188

Fold 3
Performance Training = 0.7833817272788216
Performance Validation = 0.6375650813759037

Fold 4
Performance Training = 0.7703925657198968
Performance Validation = 0.7639826636894453

Average Performance in Training = 0.7786694499766698
Average Performance in Validation = 0.6941413314250078

In [28]: #####

Note that we should be looking for the minimum MSE (not the maximum!)

```
best_m_val, best_lam_val = np.where(avg_valid==np.max(avg_valid))

print('Best M value = ', M_vals[best_m_val])
print('Best lambda value = ', lam_vals[best_lam_val])
```

```
Best M value = [3]
Best lambda value = [0.37]
```

```
In [29]: M_vals
```

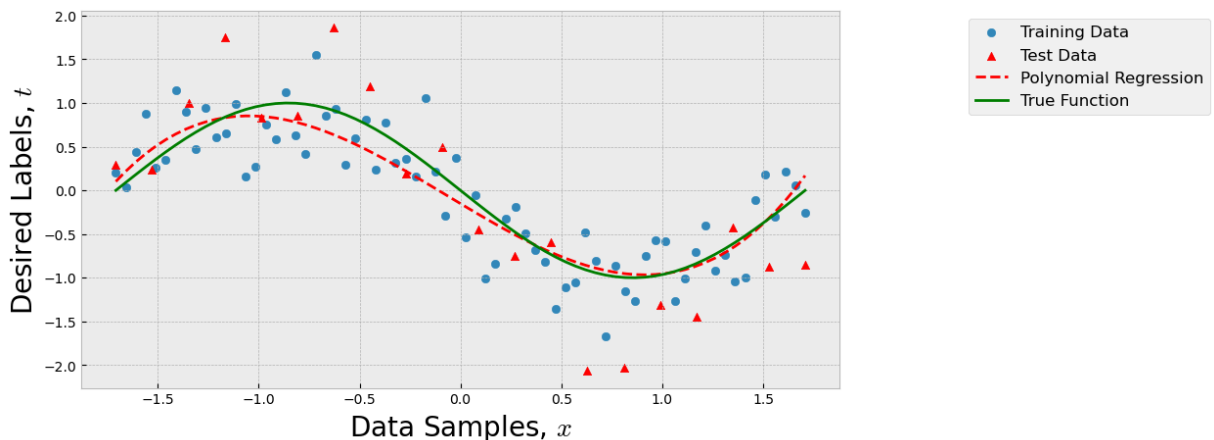
```
Out[29]: array([ 2,  3,  4,  5,  6,  7,  8,  9, 10])
```

Refit the Model

Once the optimal hyperparameters have been identified, the final model is trained using **all of the training data** (including the samples previously used for validation).

```
In [30]: w, y_train = PolynomialRegression_reg(x_train, t_train, M_vals[best_m_val])
y_test = PolynomialRegression_test(x_test, w)
```

```
fig=plt.figure(figsize=(10,5))
plt.scatter(x_train,t_train, label='Training Data')
plt.scatter(x_test,t_test, c='r', marker='^', label='Test Data')
plt.plot(x_train,y_train,'--r', label = 'Polynomial Regression')
plt.plot(x_true,t_true,'g', label = 'True Function')
plt.legend(bbox_to_anchor=(1.5, 1),fontsize=12,ncol=1)
plt.xlabel('Data Samples, $x$', fontsize=20)
plt.ylabel('Desired Labels, $t$', fontsize=20);
```



```
In [31]: print('Training r2-score:', r2_score(t_train, y_train))
print('Test r2-score:', r2_score(t_test, y_test))
```

```
Training r2-score: 0.7802660790687206
Test r2-score: 0.6902502632554288
```

Discussion

- We select the set of hyperparameters $\{M, \lambda\}$ with the best performance score (e.g. smallest MSE or largest r^2).

- If the value of the hyperparameter falls at the edge of provided range of values, it is appropriate to expand the range of values for further exploration.
- Other performance measures can be used, for example, the coefficient of determination of the Q-Q plot for regression tasks, or accuracy score for classification tasks.

Leave-One-Out Cross-Validation

One extreme case of cross-validation is the **leave-one-out** where given a dataset of N instances, only one instance is left out as the validation set (instance) and training uses the $N - 1$ instances. We then get N separate pairs by leaving out a different instance at each iteration. This is typically used in applications such as medical diagnosis, where labeled data is hard to find. Leave-one-out **does not permit stratification**.

The No Free Lunch Theorem

The No Free Lunch Theorem states that there is no single learning algorithm that in any domain always induces the most accurate learner. The usual approach is to try many and choose the one that performs the best on a separate validation set.

For any learning algorithm, there is a dataset where it is very accurate and another dataset where it is very poor. When we say that a learning algorithm is good, we only quantify how well its inductive bias matches the properties of the data.
